GMAT™

GMAT Official Guide Quantitative Review 2025–2026

From the makers of the GMAT exam

- Only study guide that features **real exam questions**
- **230+ practice questions** with detailed answer explanations
- Digital flashcards, games, a **customizable question bank**, and more
- Exclusive **tips and tricks** for exam success from the exam creators

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GMAT™ Official Guide Quantitative Review 2025–2026

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Table of Contents

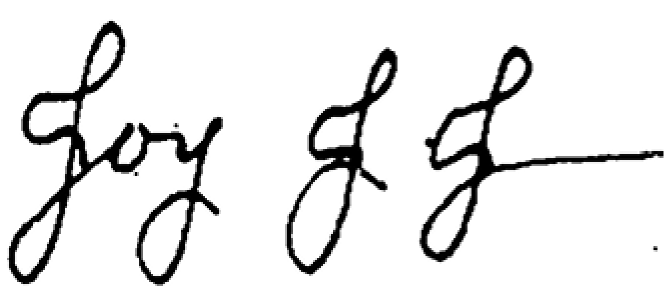
Dear GMAT™ Test Taker, Thank you for taking this important step toward your graduate management education. The GMAT exam stands as the global standard in business school admissions, used by more than 7,700 graduate programs worldwide to identify excellence in candidates for MBA, business master's, and other graduate-level management degrees. Seven out of ten business school applicants choose to differentiate themselves with a GMAT exam score.*

By choosing the GMAT™ Official Guide, you're preparing with the most authoritative resource available—the only guide featuring real GMAT questions published by the Graduate Management Admission Council (GMAC™), the makers of the GMAT exam. Combined with our prep tools at **www.mba.com**, this guide will build your confidence and help you achieve your personal best on exam day.

For more than 70 years, the GMAT exam has empowered candidates like you to demonstrate their readiness for academic success and signal their commitment to business education. Schools use and trust the GMAT because it consistently predicts classroom performance and indicates your potential to excel in your chosen program.

At GMAC, our purpose is to ensure no talent goes undiscovered. We're committed to providing the tools and insights you need throughout your business school journey, continuously enhancing the GMAT exam, and helping you connect with programs that match your aspirations. We applaud your commitment to advancing your education and wish you every success in your future educational and professional endeavors.

Sincerely,



Joy J. Jones
CEO, Graduate Management Admission Council

* Top 100 *Financial Times* full-time MBA programs

GMAT™ Official Guide Quantitative Review 2025–2026

1.0 What Is the GMAT™ Exam?

1.1 What Is the GMAT™ Exam?

The Graduate Management Admission Test™ (GMAT™) is used in admissions decisions by more than 7,700 graduate management programs at over 2,400 business schools worldwide. Unlike undergraduate grades and courses, whose meanings vary across regions and institutions, your GMAT scores are a standardized, statistically valid, and reliable measure for both you and these schools to predict your future performance and success in core courses of graduate-level management programs.

Hundreds of studies across hundreds of schools have demonstrated the validity of GMAT scores as being an accurate indicator of business school success. Together, these studies have shown that performance on the GMAT predicts success in business school even better than undergraduate grades.

The exam tests you on skills expected by management faculty and admission professionals for incoming graduate students. These skills include problem-solving, data analysis, and critical thinking, which all require complex judgments and are tested in the three sections of the GMAT exam: Quantitative Reasoning, Data Insights, and Verbal Reasoning. These three sections feature content relevant to today's business challenges and opportunities, ensuring you are prepared for graduate business school and beyond.

Your GMAT Official Score is meant to be an objective, numeric measure of your ability and potential for success. Business schools will use it as part of their holistic admissions processes, which may also consider recommendation letters, essays, interviews, work experiences, and other signs of social and emotional intelligence, as well as leadership. Even if your program does not require a GMAT score, you can stand out from the crowd by doing well on the exam to show you are serious about business school and have the skills to succeed.

The exam is always delivered in English on a computer, either online (such as at home) or at a test center. The exam tests your ability to apply foundational knowledge in the following areas: algebra and arithmetic, analyzing and interpreting data, reading and comprehending written material, and reasoning and evaluating arguments.

***Myth* -vs- FACT**

***M* – My GMAT score does not predict my success in business school.**

F – The GMAT exam measures your critical thinking skills, which you will need in business school and your career.

1.2 Why Take the GMAT™ Exam?

Taking the exam helps you stand out as an applicant and shows you're ready for and committed to a graduate management education. Schools use GMAT scores in choosing the most qualified applicants. They know an applicant who has taken the exam is serious about earning a graduate business degree, and they know the exam scores reliably predict how well applicants can do in graduate business programs.

No matter how you do on the exam, you should contact schools that interest you to learn more about them, and to ask how they use GMAT scores and other criteria in admissions decisions. School admissions offices, websites, and publications are key sources of information when you are researching business schools. Note that schools' published GMAT scores are *averages* of the scores of their admitted students, not minimum scores needed for admission.

While you might aim to get a high or perfect score, such a score is not required to get into top business school programs around the world. You should try your best to achieve a competitive score that aligns with the ranges provided by the schools of your choice. Admissions officers will use GMAT scores as one factor in admissions decisions along with undergraduate records, application essays, interviews, letters of recommendation, and other information.

To learn more about the exam, test preparation materials, registration, and how to use your GMAT Official Score in applying to business schools, please visit **www.mba.com/gmat**.

Myth -vs- FACT

M **– If I don't get a high GMAT score, I won't get into my top-choice schools.**

F – Schools use your GMAT score as a part of their holistic evaluation process.

1.3 GMAT™ Exam Format

The GMAT exam has three separately timed sections (see the table on the following page). The Quantitative Reasoning section and the Verbal Reasoning section consist of only multiple-choice questions. The Data Insights section includes multiple-choice questions along with other kinds of graphical and data analysis questions. Before you start the exam, you can choose any order in which you will take the three sections. For example, you can choose to start with Verbal Reasoning, then do Quantitative Reasoning, and end with Data Insights. Or you can choose to do Data Insights first, followed by Verbal Reasoning, and then Quantitative Reasoning. You can take one optional ten-minute break after either the first or second section.

All three GMAT sections are computer adaptive. This means the test chooses from a large bank of questions to adjust itself to your ability level, so you will not get many questions that are too hard or too easy for you. The first question will be of medium difficulty. As you answer each question, the computer uses your answer, along with your responses to earlier questions, to choose the next question with the right level of difficulty. Because the computer uses your answers to choose your next question, you cannot skip questions.

Computer adaptive tests get harder as you answer more questions correctly. But getting a question that seems easier than the last one doesn't always mean your last answer was wrong. At the end of each section, you can review any question(s) and edit up to three answers within the allotted section time.

Though each test taker gets different questions, the mix of question types is always consistent. Your score depends on the difficulty and statistical traits of the questions you answer, as well as on which of your answers are correct. If you don’t know how to answer a question, try to rule out as many wrong answer choices as possible. Then pick the answer choice you think is best. By adapting to each test taker, the exam can accurately and efficiently gauge a full range of skill levels, from very high to very low. Many factors may make the questions easier or harder, so don’t waste time worrying if some questions seem easy.

To make sure every test taker gets equivalent content, the test gives specific numbers of questions of each type. While the test covers the same kinds of questions for everyone, some questions may seem harder or easier for you because you may be stronger in some questions than in others.

At the end of the exam, you will see your unofficial score displayed on the screen. A few days after your exam, you will receive your Official Score Report, which includes detailed performance insights. Once you receive your report, you can select to send your Official Score to schools of your choice.

Format of the GMAT™ Exam		
	Questions	Timing
Quantitative Reasoning Problem-Solving	21	45 min.
Data Insights Data Sufficiency Multi-Source Reasoning Table Analysis Graphics Interpretation Two-Part Analysis	20	45 min.
Verbal Reasoning Reading Comprehension Critical Reasoning	23	45 min.
	Total Time	135 min.

A ten-minute optional break can be taken after the first or the second section.

Each section of the GMAT exam contains the following features:

- **Bookmarking:** Mark any questions you are unsure about so you can easily get back to them after you complete the section. Bookmarking can make the Question Review & Edit process more efficient.
- **Question Review & Edit:** Review as many questions as you would like (whether or not they're bookmarked) and change or edit up to three answers per section, within the section's allotted time.

Myth -vs- FACT

M **– Getting an easier question means I answered the previous one wrong.**

F – Many factors may make the questions easier or harder, so don't waste time worrying if some questions seem easy.

1.4 What Is the Testing Experience Like?

You can take the exam either online (such as at home) or at a test center—whichever you prefer. You may feel more comfortable at home with the online delivery format. Or you may prefer the uninterrupted, structured environment of a test center. It is your choice. Both options have the same content, structure, features, optional ten-minute break, scores, and score scales.

At the Test Center: Over 700 test centers worldwide administer the GMAT exam under standardized conditions. Each test center has proctored testing rooms with individual computer workstations that allow you to take the exam in a peaceful, quiet setting, with some privacy. To learn more about exam day, visit **www.mba.com/gmat**.

Online: In available regions, the GMAT exam is delivered online and is proctored remotely, so you can take it in the comfort of your home or office. You will need a quiet workspace with a desktop or laptop computer that meets minimum system requirements, a webcam, microphone, and a reliable internet connection. For more information about taking the exam online, visit **www.mba.com/gmat**.

Whether you're taking the GMAT exam online or at a test center, there are several accommodations available. To learn more about available accommodations for the exam, visit **www.mba.com/accommodations**.

1.5 What Is the Exam Content Like?

The GMAT exam measures several types of analytical reasoning skills. The Quantitative Reasoning section gives you basic arithmetic and algebra problems. The questions present you with a mix of word or pure math problems. The Data Insights section asks you to use diverse reasoning skills to solve real-world problems involving data. It also asks you to interpret and combine data from different sources and in different formats to reach conclusions. The Verbal Reasoning section tests your ability to read and comprehend written material and to reason through and evaluate arguments.

The test questions are contextualized in various subject areas, but each question provides you everything you need to know to answer it correctly. In other words, you do not need detailed outside knowledge of the subject areas.

1.6 Quantitative Reasoning Section

The GMAT Quantitative Reasoning section measures how well you solve math problems and interpret graphs. All questions in this section require solving problems using basic arithmetic, algebra, or both. Some are practical word problems, while others are pure math. Answering these questions correctly relies on logic, analytical skills, and basic knowledge of algebra and arithmetic skills. You **cannot** use a calculator while working on this section.

Chapter 3 of this book, “Math Review,” reviews the basic math you need to know to answer questions in the Quantitative Reasoning section. Chapter 4, “Quantitative Reasoning,” includes test-taking tips, as well as practice questions and answer explanations both in this book and in the Online Question Bank.

1.7 How Are Scores Calculated?

The Quantitative Reasoning, Data Insights, and Verbal Reasoning sections are each scored on a scale from 60 to 90, in 1-point increments. You will get four scores: a Section Score each for Quantitative Reasoning, Data Insights, and Verbal Reasoning, along with a Total Score based on your three section scores. The Total Score ranges from 205 to 805. Your scores depend on:

- Which questions you answered correctly.
- How many questions you answered.
- Each question's difficulty and other statistical characteristics.

There is a penalty for not completing each section of the exam. If you do not finish in the allotted time, your score will be penalized, reflecting the number of unanswered questions. Your GMAT exam score will be the best reflection of your performance when all questions are answered within the time limit.

Immediately after completing the exam, your unofficial scores and percentile for the Quantitative Reasoning, Data Insights, and Verbal Reasoning, as well as your Total Score, are displayed on-screen. You are **not** allowed to record, save, screenshot, or print your unofficial score. You will receive an email notification when your Official Score Report is available in your **www.mba.com** account.

The following table summarizes the different types of scores and their scale properties.

Score Type	Scale	Increment
Quantitative Reasoning	60–90	1
Data Insights	60–90	1
Verbal Reasoning	60–90	1
Total	205–805	10

Your GMAT Official Scores are valid for five years from your exam date. Your Total GMAT Score includes a percentile ranking, which shows the percentage of tests taken with scores lower than your score.

In addition to reviewing your Total and Section Scores, it's important to pay attention to your percentile ranking. Percentile rankings indicate what percentage of test takers you performed better than. For example, a percentile ranking of 75% means that you performed better than 75% of other test takers, and 25% of test takers performed better than you. Percentile ranks are calculated using scores from the most recent five years. Visit **www.mba.com/scores** to view the most recent predicted percentile rankings tables.

To better understand the exam experience and view score reports before exam day, we recommend taking at least one GMAT official practice exam to simulate the test-taking experience and gauge your potential score. The more practice exams you take, the better prepared you will be on your actual testing day. Visit **www.mba.com/examprep** to learn more about the practice exams offered by GMAC.

To register for the GMAT™ exam, go to www.mba.com/register

2.0 How to Prepare

2.1 How Should I Prepare for the GMAT™ Exam?

T he GMAT™ exam has several unique question formats. We recommend that you familiarize yourself with the test format and the different question types before you take the test. The key to prepping for any exam is setting a pace that works for you and your lifestyle. That might be easier said than done, but the ***GMAT™ Official 6-Week Study Planner*** does the planning for you! Our step-by-step planner will help you stick to a schedule, inform your activities, and track your progress. Go to **www.mba.com/examprep** to download the planner.

Here are our recommended steps to starting your prep journey with your best foot forward.

1. **Study the structure.** Use the study plan in our **free GMAT™ Official Starter Kit** to become familiar with the exam format and structure. The study plan will guide you through each question type and give you sample questions. This will boost your confidence come test day when you know what to expect.
2. **Understand the question types.** Beyond knowing how to answer questions correctly, learn what each type of question is asking of you. GMAT questions rely on logic and analytical skills, not underlying subject matter mastery, as detailed in the ***GMAT™ Official Guide 2025–2026*** **and Online Question Bank**.
3. **Establish your baseline.** Take the **GMAT™ Official Practice Exam 1 (FREE)** to establish your baseline. It uses the same format and scoring algorithm as the real test, so you can use the Official Score Report to accurately assess your strengths and growth areas.
4. **Study the answer explanations.** Take advantage of each question you get wrong by studying the correct answers, so you know how to get it right the next time. **GMAT™ Official Practice Questions** provide detailed answer explanations for hundreds of real GMAT questions. T his will help you understand why you got a question right or wrong.

5. **Simulate the test-taking experience.** Take the **GMAT™ Official Practice Exams**. All GMAT™ Official Practice Exams use the same algorithm, scoring, and timing as the real exam, so take them with test-day-like conditions (e.g., quiet space, use the tools allowed on test day) for the truest prep experience.

Remember, the exam is timed, so learning to pace yourself and understanding the question formats and the skills you need can be a stepping stone to achieving your desired score. The timed practice in the **Online Question Bank** can help you prepare for this. T he time management performance chart provided in practice exam score reports can also help you practice your pacing.

Because the exam assesses reasoning rather than knowledge, memorizing facts probably won't help you. You don't need to study advanced math, but you should know basic arithmetic and algebra. Likewise, you don't need to study advanced vocabulary words, but you should know English well enough to understand writing at an undergraduate level.

> **"The GMAT is not a math test. 'Textbook' math will work—but you'll take longer than you need to. Approach the GMAT from a business mindset: What's the least-effort path to a legitimate answer? Estimate, logic it out, test real numbers—whatever works for each problem."**
>
> **—A test instructor from Manhattan Prep Powered By Kaplan**

2.2 Getting Ready for Exam Day

Whether you take the exam online or in a test center, knowing what to expect will help you feel confident and succeed. To understand which exam delivery is right for you, visit **www.mba.com/plan-for-exam-day**.

Our Top Exam Day Strategies:

1. Get a good night's sleep the night before.
2. Pacing is key. Consult your on-screen timer periodically to avoid having to rush through sections.

3. Read each question carefully to fully understand what is being asked.
4. Don’t waste time trying to solve a problem you recognize as too difficult or time-consuming. Instead, eliminate answers you know are wrong and select the best from the remaining choices.
5. Leverage the bookmarking tool to make your question review and edit process more efficient.

> **“Don’t take a ‘brute force’ approach to GMAT questions—think strategically instead.”**
>
> **—A test instructor from GMAT Genius**

> **“Compile summarized notes that can be reviewed on the morning of the test. Awareness of the key points and types of mistakes will boost your scores.”**
>
> **—A test instructor from LEADERSMBA**

Myth -vs- FACT

***M* – You need advanced math skills to get a high GMAT score.**

F – The exam measures your reasoning ability rather than your advanced math skills.

2.3 How to Use the *GMAT™ Official Guide Quantitative Review 2025–2026*

The GMAT™ Official Guide series is the largest official source of actual GMAT questions. You can use this series of books and the included Online Question Bank to practice answering the different types of questions. The

GMAT™ Official Guide Quantitative Review is designed for those who have completed the Quantitative Reasoning questions in the *GMAT™ Official Guide 2025–2026* and are looking for additional practice questions, as well as those who are interested in practicing only Quantitative Reasoning questions. Questions of each type are organized by difficulty level of easy, medium, and hard. Your rate of accuracy in each category might differ from what you expect. You might be able to answer the “hard” questions easily, while the “easy” ones are challenging. This is common and is not an indicator of exam performance. The questions in this book are not adaptive based on your performance but are meant to serve as exposure to the range of question types and formats you might encounter on the exam. Also, the proportions of questions about different content areas in this book don’t reflect the proportions in the actual exam. To find questions of a specific type and difficulty level (for example, easy arithmetic questions), use the index of questions in Chapter 5.

We recommend the steps below for how to best use this book:

1. Start with the review chapters to gain an overview of the required concepts.

   > **“Building a strong foundation is crucial to achieve a high score. If you struggle with fundamental questions, your progress on more advanced questions will be hindered.”**
   >
   > **—A test instructor from XY Education**

2. Go through the practice questions in this book. Once you’ve familiarized yourself with the concepts and question types, use the Online Question Bank to further customize your practice by choosing your preferred level of difficulty, category of concepts, or question types.

> **“The Official Guide offers in-depth answer explanations. After completing a question, reviewing it alongside the explanation helps you gain a deeper understanding of the question’s key concepts and solution strategies. On the result interface of the online question bank, it serves a dual purpose: on one hand, it encourages us to review incorrect answers; on the other hand, it provides detailed insights into the time spent on each question, aiding in optimizing our pace during exercises.”**
>
> **—A test instructor from JiangjiangGMAT**

3. Use the **Online Question Bank** to continue practicing based on your progress. To better customize and enhance your practice, use the Online Question Bank to:

    a. Review and retry practice questions to improve performance by using the untimed or timed features along with a study mode or an exam mode.

    b. Analyze key performance metrics to help assess focus area and track improvement.

    c. Use flashcards to master key concepts.

> **“The biggest mistake students make is completing too many new problems. Completing problems doesn’t move your score! Learning from problems does. Keep a list of questions you want to go back to and redo. Redo at least three of these questions every time you study. This book is one of the most important resources you can use to create the future you want. Make sure you understand every question you complete extremely well. You should be able to explain every problem you complete to someone who is new to the exam. Don’t focus on simply ‘getting through’ the book. That mindset will work against you and your dreams.”**
>
> **—A test instructor from The GMAT Strategy**

TIP

Since the exam is given on a computer, we suggest you practice the questions in this book using the **Online Question Bank** accessed via **www.mba.com/my-account**. It includes all the questions in this book, and it lets you create practice sets—both timed and untimed—and track your progress more easily. T he Online Question Bank is also available on your mobile device through the GMAT™ Official Practice mobile app. To access the Online Question Bank on your mobile device, first create an account at **www.mba.com**, and then sign into your account on the mobile app.

2.4 How to Use Other GMAT™ Official Prep Products

We recommend using our other GMAT™ Official Prep products along with this guidebook.

- **For a realistic simulation of the exam:** GMAT™ Official Practice Exams 1–6 are the only practice exams that use real exam questions along with the scoring algorithm and user interface from the actual exam. The first two practice exams are free to all test takers at **www.mba.com/gmatprep**.
- **For more practice questions:** *GMAT™ Official Guide Data Insights Review 2025–2026* and *GMAT™ Official Guide Verbal Review 2025–2026* offer over 225 additional practice questions not included in this book.
- **For focused practice:** GMAT™ Official Practice Questions for Quantitative, Data Insights, and Verbal products offer 100+ questions that are not included in the Official Guide series.

> **“Build teaching-level depth; don’t just finish content mindlessly. Even if you solve thousands of questions on a shaky foundation, you will remain stuck on a really low accuracy.”**
>
> **—A test instructor from Top One Percent**

2.5 Tips for Taking the Exam

Tips for answering questions of the different types are given later in this book. Here are some general tips to help you do your best on the test.

1. **Before the actual exam, decide in what order to take the sections.**

   The exam lets you choose in which order you’ll take the sections. Use the GMAT™ Official Practice Exams to practice and find your preferred order. No order is “wrong.” Some test takers prefer to complete the section that challenges them the most first, while others prefer to ease into the exam by starting with a section that they’re stronger in. Practice each order and see which one works best for you.

2. **Try the practice questions and practice exams.**

   Timing yourself as you answer the practice questions and taking the practice exams can give you a sense of how long you will have for each question on the actual test, and whether you are answering them fast enough to finish in time.

   > **TIP**
   >
   > After you’ve learned about all the question types, use the practice questions in this book and practice them online at **www.mba.com/my-account** to prepare for the actual test.

3. **Review all test directions ahead of time.**

   The directions explain exactly what you need to do to answer questions of each type. You can review the directions in the GMAT™

Official Practice Exams ahead of time so that you don't miss anything you need to know to answer properly. To review directions during the test, you can click on the Help icon. But note that your time spent reviewing directions counts against your available time for that section of the test.

4. **Study each question carefully.**

   Before you answer a question, understand exactly what it is asking, then pick the best answer choice. Never skim a question. Skimming may make you miss important details or nuances.

5. **Use your time wisely.**

   Although the exam stresses accuracy over speed, you should use your time wisely. On average, you have just about 2 minutes and 9 seconds per Quantitative Reasoning question; 2 minutes and 15 seconds per Data Insights question; and under 2 minutes per Verbal Reasoning question. Once you start the test, an on-screen clock shows how much time you have left. You can hide this display if you want, but by checking the clock periodically, you can make sure to finish in time.

6. **Do not spend too much time on any one question.**

   If finding the right answer is taking too long, try to rule out answer choices you know are wrong. Then pick the best of the remaining choices and move on to the next question.

   Not finishing sections or randomly guessing answers can lower your score significantly. As long as you've worked on each section, you will get a score even if you didn't finish one or more sections in time. You don't earn points for questions you never get to see.

   Pacing is important. If a question stumps you, pick the answer choice that seems best and move on. If you guess wrong, the computer will likely give you an easier question, which you're more likely to answer correctly. Soon the computer will return to giving you questions matched to your ability. You can bookmark questions you get stuck on, then return to change up to three of your answers if you still have time left at the end of the section. But if you don't finish the section, your score will be reduced.

7. **Confirm your answers ONLY when you are ready to move on.**

   In the GMAT Quantitative Reasoning, Data Insights, and Verbal Reasoning sections, once you choose your answer to a question, you are asked to confirm it. As soon as you confirm your response, the next question appears. You can't skip questions. In the Data Insights and Verbal Reasoning sections, several questions based on the same prompt may appear at once. When more than one question is on a single screen, you can change your answers to any questions on that screen before moving on to the next screen. But until you've reached the end of the section, you can't navigate back to a previous screen to change any answers.

Myth -vs- FACT

***M* – Avoiding wrong answers is more important than finishing the test.**

F – Not finishing can lower your score a lot.

Myth -vs- FACT

***M* – The first ten questions are critical, so you should spend the most time on them.**

F – All questions impact your score.

2.6 Quantitative Reasoning Section Strategies

Utilize the strategies below to better prepare for the exam. Creating a solid study plan and selecting the right prep materials are two key elements of getting accepted into your top business schools. But knowing how to strategically approach the exam is another crucial factor that can increase your confidence going into test day and help you perform your best.

> **"Don't just read explanations. Review your notes and learn from your mistakes."**
>
> **—A test instructor from Admit Master**

Problem-Solving Questions

- Familiarize yourself with the rules and concepts of arithmetic and algebra.
- For questions that require approximations, skim answer choices first. If you are unable to get some idea of how close the approximation should be, you may waste time on long computations when a short mental process would serve you better.
- Take advantage of your whiteboard or note board. Solving problems in writing may help you avoid errors. Make sure you understand how the whiteboard will work on test day. If you are taking the exam online, you also have the option to use an online whiteboard tool—make sure you understand how the online whiteboard works and practice using it before test day.

3.0 Math Review

3.0 Math Review

This chapter reviews the math you need to answer GMAT™ Quantitative Reasoning questions. This is only a brief overview. If you find unfamiliar terms, consult other resources to learn more.

Unlike some math problems you may have solved in school, GMAT math questions ask you to **apply** your math knowledge. For example, rather than asking you to list a number's prime factors to show you understand prime factorization, a GMAT question may ask you to **use** prime factorization and exponents to simplify an algebraic expression with a radical.

To prepare for the GMAT Quantitative Reasoning section, first review basic math to make sure you know enough to answer the questions. Then practice with GMAT questions from past exams.

Section 3.1, "Value, Order, and Factors," includes:

1. Numbers and the Number Line
2. Factors, Multiples, and Remainders
3. Exponents
4. Decimals and Place Value
5. Arithmetic Shortcuts and Properties of Operations

Section 3.2, "Algebra, Equalities, and Inequalities," includes:

1. Algebraic Expressions and Equations
2. Linear Equations
3. Factoring and Quadratic Equations
4. Inequalities
5. Functions
6. Graphing
7. Formulas and Measurement Conversion

Section 3.3, "Rates, Ratios, and Percents," includes:

1. Ratio and Proportion
2. Fractions
3. Percents
4. Converting Decimals, Fractions, and Percents
5. Working with Decimals, Fractions, and Percents
6. Rate, Work, and Mixture Problems

Section 3.4, "Statistics, Sets, Counting, Probability, Estimation, and Series," includes:

1. Statistics
2. Sets
3. Counting Methods
4. Probability
5. Estimation
6. Sequences and Series

Section 3.5, "Reference Sheets"

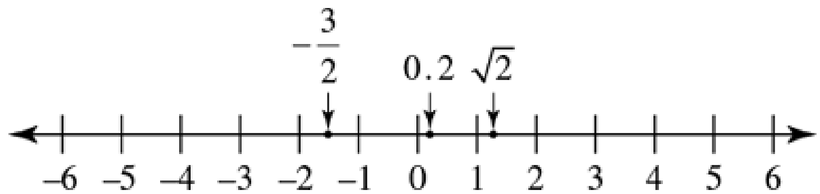
3.1 Value, Order, and Factors

1. Numbers and the Number Line

A. All ***real numbers*** match points on ***the number line***, and all points on the number line represent real numbers.

The figure below shows the number line with labeled points standing for the real numbers $-\frac{3}{2}$, 0.2, and $\sqrt{2}$.

The Number Line



B. On the number line, points to the left of zero stand for ***negative*** numbers, and points to the right of zero stand for ***positive*** numbers. Every real number except zero is either positive or negative.

C. On the number line, each number is less than any number to its right. So, as the figure above shows, $-4 < -3 < -\frac{3}{2} < -1$, and $1 < \sqrt{2} < 2$.

D. If a number n is between 1 and 4 on the number line, then $n > 1$ and $n < 4$; that is, $1 < n < 4$. If n is "between 1 and 4, inclusive," then $1 \leq n \leq 4$.

E. The ***absolute value*** of a real number x, written as $|x|$, is x if $x \geq 0$ and $-x$ if $x < 0$. A number's absolute value is the distance between that number and zero on the number line. Thus, −3 and 3 have the same absolute value since each is three units from zero on the number line. The absolute value of any nonzero number is positive.

Examples:

$|-5| = |5| = 5$, $|0| = 0$, and

$\left|-\frac{7}{2}\right| = \frac{7}{2}$.

For any real numbers x and y, $|x + y| \leq |x| + |y|$, and $|xy| = |x||y|$.

Examples:

If $x = 10$ and $y = 2$, then $|x + y| = |12| = 12 = |x| + |y|$.

And $|xy| = |20| = 20 = |x||y|$.

If $x = 10$ and $y = -2$, then $|x + y| = |8| = 8 < 12 = |x| + |y|$.

And $|xy| = |-20| = 20 = |x||y|$.

2. Factors, Multiples, and Remainders

A. An ***integer*** is any number in the set {… −3, −2, −1, 0, 1, 2, 3, …}. For any integer n, the numbers in the set $\{n, n+1, n+2, n+3, \ldots\}$ are ***consecutive integers***.

B. For integers x and y, if $x \neq 0$, x is a ***divisor*** or ***factor*** of y if $y = xn$ for some integer n. Then y is ***divisible*** by x and is a ***multiple*** of x.

Example:

Since 28 = (7)(4), both 4 and 7 are divisors or factors of 28.

But 8 isn't a divisor or factor of 28 since n isn't an integer if $28 = 8n$.

C. Dividing a positive integer y by a positive integer x and then rounding down to the nearest nonnegative integer gives the ***quotient*** of the division.

To find the ***remainder*** of the division, multiply x by the quotient, then subtract the result from y. The quotient and the remainder are the unique positive integers q and r, respectively, such that $y = xq + r$ and $0 \leq r < x$.

Example:

When 28 is divided by 8, the quotient is 3 and the remainder is 4 because 28 = (8)(3) + 4.

The remainder r is 0 if and only if y is ***divisible*** by x. Then x is a divisor or factor of y, and y is a multiple of x.

Example:

Since 32 divided by 8 has a remainder of 0, 32 is divisible by 8. So, 8 is a divisor of 32, and 32 is a multiple of 8.

When a smaller integer is divided by a larger integer, the quotient is 0 and the remainder is the smaller integer.

Example:

When 5 is divided by 7, the quotient is 0 and the remainder is 5 since 5 = (7)(0) + 5.

D. For any positive integers x, y, and z, suppose that r is the remainder when x is divided by z, and that s is the remainder when y is divided by z. Then the remainder when $x + y$ is divided by z equals the remainder when $r + s$ is divided by z. Similarly, the remainder when xy is divided by z equals the remainder when rs is divided by z.

Examples:

When 142 is divided by 13, the remainder is 12 because 132 = (13 × 10) + 12. And when 29 is divided by 13, the remainder is 3 because 29 = (13 × 2) + 3.

So, the remainder when 142 + 29 is divided by 13 equals the remainder when 12 + 3 = 15 is divided by 13; that is, it’s 2.

And the remainder when 142 × 29 is divided by 13 equals the remainder when 12 × 3 = 36 is divided by 13; that is, it’s 10.

E. Any integer divisible by 2 is ***even***. The set of even integers is {… −4, −2, 0, 2, 4, 6, 8, …}. Integers not divisible by 2 are ***odd***, so {… −3, −1, 1, 3, 5, …} is the set of odd integers. For any integer n, the numbers in the set $\{2n, 2n + 2, 2n + 4, \ \ldots\}$ are ***consecutive even integers***, and the numbers in $\{2n + 1, 2n + 3, 2n + 5, \ \ldots\}$ are ***consecutive odd integers***.

If a product of integers has at least one even factor, the product is even; otherwise, it’s odd. If two integers are both even or both odd, their sum and their difference are even. Otherwise, their sum and their difference are odd.

F. A ***prime*** number is a positive integer with exactly two positive divisors: 1 and itself. That is, a prime number is divisible by no integer but itself and 1.

> *Examples:*
>
> The first six prime numbers are 2, 3, 5, 7, 11, and 13.
>
> But 15 is not a prime number because it has four positive divisors: 1, 3, 5, and 15.
>
> Nor is 1 a prime number because it has only one positive divisor: itself.

3. Exponents

A. An expression of the form k^n means the n^{th} ***power*** of k, or k raised to the n^{th} power, where n is the ***exponent*** and k is the ***base***.

B. A positive integer exponent shows how many instances of the base are multiplied together. That is, when n is a positive integer, k^n is the product of n instances of k.

> *Examples:*
>
> x^5 is $(x)(x)(x)(x)(x)$; that is, the product in which x is a factor 5 times with no other factors. We can also say x^5 is the 5^{th} power of x, or x raised to the 5^{th} power.
>
> The second power of 2, also called 2 ***squared***, is $2^2 = 2 \times 2 = 4$. The third power of 2, also called 2 ***cubed***, is $2^3 = 2 \times 2 \times 2 = 8$.

Squaring a number greater than 1, or raising it to any power greater than 1, gives a larger number.

Squaring a number between 0 and 1 gives a smaller number.

Examples:

$3^2 = 9$, and $9 > 3$.

$(0.1)^2 = 0.01$, and $0.01 < 0.1$.

C. A ***composite number*** is an integer greater than 1 that's not prime. Any composite number is a product of a unique set of prime factors, each raised to some positive integer power. The ***prime factorization*** of a composite number shows it as such a product.

Examples:

Some prime factorizations of composite numbers are $14 = (2)(7)$, $81 = 3^4$, and $484 = (2^2)(11^2)$.

D. For any two positive integers x and y, y is divisible by x if and only if the prime factorization of y includes each prime number in the prime factorization of x, raised to at least as great a power as it is in the prime factorization of x.

Examples:

$1080 = (2^3)(3^3)(5)$ is divisible by $24 = (2^3)(3)$, because the prime factorization of 1080 includes both 2 and 3, which are the only prime factors in the prime factorization of 24, and each is raised to at least as great a power as it is in the prime factorization of 24.

But 1080 is not divisible by $16 = 2^4$, because the prime factor 2 in the prime factorization of 16 is raised to a higher power than it is in the prime factorization of 1080.

Nor is 1080 divisible by $21 = (3)(7)$, because its prime factor 7 isn't in the prime factorization of 1080.

E. A ***square root*** of a number n is a number x such that $x^2 = n$. Every positive number has two real square roots, one positive and the other negative. The positive square root of n is written as $\sqrt{n}$ or $n^{\frac{1}{2}}$.

> *Example:*
>
> The two square roots of 9 are $\sqrt{9} = 3$ and $-\sqrt{9} = -3$.

For any x, the nonnegative square root of x^2 equals the absolute value of x; that is, $\sqrt{x^2} = |x|$.

The square root of a negative number is not a real number. Only real numbers appear on the GMAT

F. Every real number r has exactly one real ***cube root***, the number s such that $s^3 = r$. The real cube root of r is written as $\sqrt[3]{r}$ or $r^{\frac{1}{3}}$.

> *Examples:*
>
> Since $2^3 = 8$, $\sqrt[3]{8} = 2$.
>
> Likewise, $\sqrt[3]{-8} = -2$ because $(-2)^3 = -8$.

4. Decimals and Place Value

A. A ***decimal*** is a real number written as a series of digits, often with a period called a ***decimal point.*** The decimal point's position sets the ***place values*** of the digits.

> *Example:*
>
> The digits in the decimal 7,654.321 have these place values:

Thousands		Hundreds	Tens	Ones or units		Tenths	Hundredths	Thousandths
7	,	6	5	4	.	3	2	1

B. Some decimal numbers extend rightward by infinitely many digits. In a ***repeating decimal***, a finite sequence of digits that are not all 0s repeats forever. A repeating decimal may be written showing the finite sequence of digits to the right of the decimal point only once, with a bar over them. There are never digits to the right of the repeating sequence.

> *Example:*
>
> The repeating decimal $12.34\overline{56}$ has the digits 12.3456565656…, with the digit sequence 56 repeating forever.

C. In ***scientific notation***, a decimal is written with only one nonzero digit to the decimal point's left, multiplied by an integer power of 10. To convert a number from scientific notation to regular decimal notation, move the decimal point by the number of places equal to the absolute value of the exponent on the 10. Move the decimal point to the right if the exponent is positive or to the left if the exponent is negative.

> *Examples:*
>
> In scientific notation, 231 is written as 2.31×10^2, and 0.0231 is written as 2.31×10^{-2}.
>
> To convert 2.013×10^4 to regular decimal notation, move the decimal point 4 places to the right, giving 20,130.
>
> Likewise, to convert 1.91×10^{-4} to regular decimal notation, move the decimal point 4 places to the left, giving 0.000191.

D. To add or subtract decimals with many finite digits, line up their decimal points. If one decimal has fewer digits to the right of its decimal point than another, insert zeros to the right of its last digit.

Examples:

To add 17.6512 and 653.27, insert zeros to the right of the last digit in 653.27 to line up the decimal points when the numbers are in a column:

$$\begin{array}{r} 17.6512 \\ \underline{+\ 653.2700} \\ 670.9212 \end{array}$$

Likewise for 653.27 minus 17.6512:

$$\begin{array}{r} 653.2700 \\ \underline{-\ 17.6512} \\ 635.6188 \end{array}$$

E. Multiply decimals with many finite digits as if they were integers. Then insert the decimal point in the product so that the number of digits to the right of the decimal point is the sum of the numbers of digits to the right of the decimal points in the numbers being multiplied.

Example:

To multiply 2.09 by 1.3, first multiply the integers 209 and 13 to get 2,717. Since 2 + 1 = 3 digits are to the right of the decimal points in 2.09 and 1.3, put 3 digits in 2,717 to the right of the decimal point to find the product:

$$\begin{array}{rl} 2.09 & \text{(2 digits to the right)} \\ \underline{\times\ 1.3} & \text{(l digit to the right)} \\ 627 & \\ \underline{2090} & \\ 2.717 & \text{(2 + 1 = 3 digits to the right)} \end{array}$$

F. To divide a number (the ***dividend***) by a decimal (the ***divisor***), when both have many finite digits, move the decimal points of the dividend and the divisor the same number of digits to the right until the divisor is an integer.

Then divide as you would integers. The decimal point in the quotient goes directly above the decimal point in the new dividend.

> *Example:*
>
> To divide 698.12 by 12.4, first move the decimal points in both the divisor 12.4 and the dividend 698.12 one place to the right to make the divisor an integer. That is, replace 698.12/12.4 with 6981.2/124. Then do the long division normally:
>
> $$\begin{array}{r} 56.3 \\ 124\overline{)6981.2} \\ \underline{620} \\ 781 \\ \underline{744} \\ 37\,2 \\ \underline{37\,2} \\ 0 \end{array}$$

5. Arithmetic Shortcuts and Properties of Operations

A. For any two nonnegative integers x and y, the final digit of $x + y$ is the final digit of the sum of x's final digit and y's final digit. And for any two integers x and y, the final digit of xy is the final digit of the product of x's final digit and y's final digit.

> *Examples:*
>
> The final digit of 345 + 789 is the final digit of 5 + 9 = 14, which is 4.
>
> The final digit of 345 × 789 is the final digit of 5 × 9 = 45, which is 5.

B. Here are some shortcuts to help tell whether one integer is divisible by another:

An integer is divisible by 2 if and only if its final digit is even.

An integer is divisible by 3 if and only if the sum of its digits is divisible by 3.

An integer with 3 or more digits is divisible by 4 if and only if its final 2 digits make a number divisible by 4.

An integer is divisible by 5 if and only if its final digit is 5 or 0.

An integer is divisible by 9 if and only if the sum of its digits is divisible by 9.

An integer is divisible by 10 if and only if its final digit is 0.

Examples:

The integer 180 is divisible by 2, 5, and 10, because its final digit is 0. It's also divisible by 3 and 9, because its digits sum to 1 + 8 + 0 = 9, which is divisible by both 3 and 9. And it's divisible by 4, because its final 2 digits, 80, make a number divisible by 4.

But the integer 121 is not divisible by 2, because its final digit is odd. Nor is it divisible by 3 or 9, because its digits sum to 1 + 2 + 1 = 4, which is divisible by neither 3 nor 9. And it's not divisible by 4, because its final two digits, 21, make a number not divisible by 4. Finally, because its final digit is neither 5 nor 0, it's not divisible by 5 or 10.

C. Here are some basic properties of arithmetical operations for any real numbers x, y, and z:

- Addition and Subtraction

  $x + 0 = x = x - 0$

  $x - x = 0$

  $x + y = y + x$

  $x - y = -(y - x) = x + (-y)$

  $(x + y) + z = x + (y + z)$

  If x and y are both positive, then $x + y$ is positive.

  If x and y are both negative, then $x + y$ is negative.

- Multiplication and Division

  $x \times 1 = x = \frac{x}{1}$

  $x \times 0 = 0$

  If $x \neq 0$, then $\frac{x}{x} = 1$.

  $\frac{x}{0}$ is undefined.

  $xy = yx$

  If $x \neq 0$ and $y \neq 0$, then $\frac{x}{y} = \frac{1}{\left(\frac{y}{x}\right)}$.

  $(xy)z = x(yz)$

  $xy + xz = x(y + z)$

  If $y \neq 0$, then $\left(\frac{x}{y}\right) + \left(\frac{z}{y}\right) = \frac{(x + z)}{y}$.

  If x and y are both positive, then xy is positive.

  If x and y are both negative, then xy is positive.

  If x is positive and y is negative, then xy is negative.

  If $xy = 0$, then $x = 0$ or $y = 0$, or both.

- Exponentiation

  $x^1 = x$

  If $x \neq 0$, then $x^0 = 1$.

  If $x \neq 0$, then $x^{-1} = \frac{1}{x}$.

  $(x^y)^z = x^{yz} = (x^z)^y$

  $x^{y+z} = x^y x^z$

  If $x \neq 0$, then $x^{y-z} = \frac{x^y}{x^z}$.

$(xz)^y = x^y z^y$

If $z \neq 0$, then $\left(\frac{x}{z}\right)^y = \frac{x^y}{z^y}$.

If $z \neq 0$, then $x^{\frac{y}{z}} = (x^y)^{\frac{1}{z}} = \left(x^{\frac{1}{z}}\right)^y$.

3.2 Algebra, Equalities, and Inequalities

1. Algebraic Expressions and Equations

A. Algebra is based on arithmetic and on the concept of an ***unknown quantity***. Letters like $\boldsymbol{x}$ or $\boldsymbol{n}$ are ***variables*** that stand for unknown quantities. Other numerical expressions called ***constants*** stand for known quantities. A combination of variables, constants, and arithmetical operations is an ***algebraic expression***.

Solving word problems often requires translating words into algebraic expressions. The table below shows how some words and phrases can be translated as math operations in algebraic expressions:

<table>
<tr><th colspan="5">3.2 Translating Words into Math Operations</th></tr>
<tr><th>$x+y$</th><th>$x-y$</th><th>xy</th><th>$\frac{x}{y}$</th><th>x^y</th></tr>
<tr>
<td rowspan="3">x added to y
x increased by y
x more than y
x plus y
the sum of x and y
the total of x and y</td>
<td rowspan="3">x decreased by y
difference of x and y
y fewer than x
y less than x
x minus y
x reduced by y
y subtracted from x</td>
<td>x multiplied by y
the product of x and y
x times y</td>
<td>x divided by y
x over y
the quotient of x and y
the ratio of x to y</td>
<td>x to the power of y
x to the y^{th} power</td>
</tr>
<tr>
<td>If $y=2$:
double x
twice x</td>
<td rowspan="2">If $y=2$:
half of x
x halved</td>
<td>If $y=2$:
x squared</td>
</tr>
<tr>
<td>If $y=3$:
triple x</td>
<td>If $y=3$:
x cubed</td>
</tr>
</table>

B. In an algebraic expression, a ***term*** is a constant, a variable, or a product of terms that are each a constant or a variable. A variable in a term may be raised to an exponent. A term with no variables is a ***constant term***. A constant in a term with one or more variables is a ***coefficient***.

> *Example:*
>
> Suppose Pam has 5 more pencils than Fred. If F is the number of pencils Fred has, then Pam has $F+5$ pencils. The algebraic expression $F+5$ has two terms: the variable F and the constant 5.

C. A ***polynomial*** is an algebraic expression that's a sum of terms and has exactly one variable. Each term in a polynomial is a variable raised to some power and multiplied by some coefficient. If the highest power a variable is raised to is 1, the expression is a ***first degree*** (or ***linear***) ***polynomial*** in that variable. If the highest power a variable is raised to is 2, the expression is a ***second degree*** (or ***quadratic***) ***polynomial*** in that variable.

Examples:

$F+5$ is a linear polynomial in F, since the highest power of F is 1.

$19x^2-6x+3$ is a quadratic polynomial in x, since the highest power of x is 2.

$\dfrac{3x^2}{(2x-5)}$ is not a polynomial because it's not a sum of powers of x multiplied by coefficients.

D. You can simplify many algebraic expressions by factoring or combining ***like*** terms.

Examples:

In the expression $6x+5x$, x is a factor common to both terms. So, $6x+5x$ is equivalent to $(6+5)x$, or $11x$.

In the expression $9x-3y$, 3 is a factor common to both terms: $9x-3y=3(3x-y)$.

The expression $5x^2+6y$ has no like terms and no common factors.

E. In a fraction $\dfrac{n}{d}$, n is the ***numerator*** and d is the ***denominator.*** In a numerator and denominator, you can divide out any common factors not equal to zero.

Example:

If $x \neq 3$, then $\dfrac{(x-3)}{(x-3)} = 1$.

So, $\dfrac{(3xy-9y)}{(x-3)} = \dfrac{3y\,(x-3)}{(x-3)} = 3y(1) = 3y$.

F. To multiply two algebraic expressions, multiply each term of one expression by each term of the other.

Example:

$$\begin{aligned}(3x-4)(9y+x) &= 3x(9y+x) - 4(9y+x)\\ &= 3x(9y) + 3x(x) - 4(9y) - 4(x)\\ &= 27xy + 3x^2 - 36y - 4x\end{aligned}$$

G. To ***evaluate*** an algebraic expression, replace its variables with constants.

Example:

If $x = 3$ and $y = -2$, we can evaluate $3xy - x^2 + y$ as

$3(3)(-2) - (3)^2 + (-2) = -18 - 9 - 2 = -29$.

H. An ***algebraic equation*** is an equation with only algebraic expression. An algebraic equation's ***solutions*** are the sets of assignments of constant values to its variables that make it true, or "satisfy the equation." An equation may have no solution, one solution, or more than one solution. For equations solved together, the solutions must satisfy all the equations at once. An equation's solutions are also called its ***roots***. To confirm the roots are correct, you can substitute them into the equation.

I. Two equations with the same solution or solutions are ***equivalent***.

Examples:

The equations $2 + x = 3$ and $4 + 2x = 6$ are equivalent, because each has the unique solution $x = 1$. Notice the second equation is the first equation multiplied by 2.

Likewise, the equations $3x - y = 6$ and $6x - 2y = 12$ are equivalent, although each has infinitely many solutions. For any value of x, giving the value $3x - 6$ to y satisfies both these equations. For example, $x = 2$ with $y = 0$ is a solution to both equations, and so is $x = 5$ with $y = 9$.

2. Linear Equations

A. A ***linear equation*** has a linear polynomial on one side of the equals sign and either a linear polynomial or a constant on the other side—or can be converted to that form. A linear equation with only one variable is a ***linear equation with one unknown***. A linear equation with two variables is ***a linear equation with two unknowns***.

Examples:

$5x - 2 = 9 - x$ is a linear equation with one unknown.

$3x + 1 = y - 2$ is a linear equation with two unknowns.

B. To solve a linear equation with one unknown (that is, to find what value of the unknown satisfies the equation), isolate the unknown on one side of the equation by doing the same operations on both sides. Adding or subtracting the same number on both sides of the equation doesn't change the equality. Likewise, multiplying or dividing both sides by the same nonzero number doesn't change the equality.

Example:

To solve the equation $\frac{5x-6}{3} = 4$, isolate the variable x like this:

$$5x - 6 = 12 \quad \text{multiply both sides by 3}$$

$$5x = 18 \quad \text{add 6 to both sides}$$

$$x = \frac{18}{5} \quad \text{divide both sides by 5}$$

To check the answer $\frac{18}{5}$, substitute it for x in the original equation to confirm it satisfies that equation:

$$\frac{\left(5\left(\frac{18}{5}\right) - 6\right)}{3} = \frac{(18-6)}{3} = \frac{12}{3} = 4$$

So, $x = \frac{18}{5}$ is the solution.

C. Two equivalent linear equations with the same two unknowns have infinitely many solutions, as in the example of the equivalent equations $3x - y = 6$ and $6x - 2y = 12$ in Section 3.2.1.I above. But if two linear equations with the same two unknowns aren't equivalent, they have at most one solution.

When solving two linear equations with two unknowns, if you reach a trivial equation like 0 = 0, the equations are equivalent and have infinitely many solutions. But if you reach a contradiction, the equations have no solution.

Example:

Consider the two equations $3x + 4y = 17$ and $6x + 8y = 35$. Note that $3x + 4y = 17$ implies $6x + 8y = 34$, contradicting the second equation. So, no values of x and y can satisfy both equations at once. The two equations have no solution.

If neither a trivial equation nor a contradiction is reached, a unique solution can be found.

D. To solve two linear equations with two unknowns, you can use one of the equations to express one unknown in terms of the other unknown. Then substitute this result into the second equation to make a new equation with only one unknown. Next, solve this new equation. Substitute the value of its unknown into either original equation to solve for the remaining unknown.

Example:

Let's solve these two equations for x and y:

$$\begin{array}{lrcl} (1) & 3x + 2y & = & 11 \\ (2) & x - y & = & 2 \end{array}$$

In equation (2), $x = 2 + y$. So, in equation (1), substitute $2 + y$ for x:

$$\begin{aligned} 3(2 + y) + 2y &= 11 \\ 6 + 3y + 2y &= 11 \\ 6 + 5y &= 11 \\ 5y &= 5 \\ y &= 1 \end{aligned}$$

Since $y = 1$, we find $x - 1 = 2$, so $x = 2 + 1 = 3$.

E. Another way to remove one unknown and solve for x and y is to make the coefficients of one unknown the same in both equations (ignoring the sign). Then either add the equations or subtract one from the other.

Example:

Let's solve the equations:

$$\begin{aligned}&(1)\ \ 6x+5y=29 \text{ and}\\&(2)\ \ 4x-3y=-6\end{aligned}$$

Multiply equation (1) by 3 and equation (2) by 5 to get

$$\begin{aligned}&18x+15y=87 \text{ and}\\&20x-15y=-30\end{aligned}$$

Add the two equations to remove y. This gives us $38x=57$, or $x=\frac{3}{2}$.

Substituting $\frac{3}{2}$ for x in either original equation gives $y=4$. To check these answers, substitute both values into both the original equations.

3. Factoring and Quadratic Equations

A. Some equations can be solved by ***factoring***. To do this, first add or subtract to bring all the expressions to one side of the equation, with 0 on the other side. Then try to express the nonzero side as a product of factors that are algebraic expressions. When that's possible, setting any of these factors equal to 0 makes a simpler equation. The solutions of the simpler equations made this way are also solutions of the factored equation.

Example:

Factor to find the solutions of the equation $x^3 - 2x^2 + x = -5(x-1)^2$:

$$\begin{aligned} x^3 - 2x^2 + x + 5(x-1)^2 &= 0 \\ x(x^2 - 2x + 1) + 5(x-1)^2 &= 0 \\ x(x-1)^2 + 5(x-1)^2 &= 0 \\ (x+5)(x-1)^2 &= 0 \\ x + 5 = 0 \text{ or } x - 1 &= 0 \\ x = -5 \text{ or } x &= 1 \end{aligned}$$

So, $x = -5$ or $x = 1$.

B. When factoring to solve equations with algebraic fractions, note that a fraction equals 0 if and only if its numerator equals 0 and its denominator doesn't.

Example:

Find the solutions of the equation $\frac{x(x-3)\left(x^2+5\right)}{x-4} = 0$.

The denominator must not equal 0, so $x - 4 \neq 0$. Thus, $x \neq 4$.

But the numerator must equal 0: $x\left(x-3\right)\left(x^2+5\right) = 0$.

Thus, $x = 0$, or $x - 3 = 0$, or $x^2 + 5 = 0$. So, $x = 0$, or $x = 3$, or $x^2 + 5 = 0$.

But $x^2 + 5 = 0$ has no real solution because $x^2 + 5 \geq 0$ for every real number x. So, the original equation's solutions are 0 and 3.

C. Here are some common algebraic equations used for factoring. They hold for any real numbers x, y, and z:

Examples:

$$\left(x+\frac{1}{x}\right)^2=x^2+\frac{1}{x^2}+2$$

$$(x+y)^2=x^2+2xy+y^2$$

$$(x-y)^2=x^2-2xy+y^2$$

$$\big(x+y\big)\big(x-y\big)=x^2-y^2$$

$$x^2+y^2=(x+y)^2-2xy=(x-y)^2+2xy$$

$$(x+y+z)^2=x^2+y^2+z^2+2xy+2xz+2yz$$

$$(x+y)^3=x^3+3x^2y+3xy^2+y^3$$

$$(x-y)^3=x^3-3x^2y+3xy^2-y^3$$

D. A ***quadratic equation*** has the standard form $ax^2+bx+c=0$, where a, b, and c are real numbers and $a\neq 0$.

Examples:

$$\begin{aligned} x^2+6x+5 &= 0, \\ 3x^2-2x &= 0, \text{ and} \\ x^2+4 &= 0. \end{aligned}$$

E. Some quadratic equations are easily solved by factoring.

Example (1):

$$x^2 + 6x + 5 = 0$$

$$(x+5)(x+1) = 0$$

$x + 5 = 0$ or $x + 1 = 0$

$x = -5$ or $x = -1$

Example (2):

$$3x^2 - 3 = 8x$$

$$3x^2 - 8x - 3 = 0$$

$$(3x+1)(x-3) = 0$$

$3x + 1 = 0$ or $x - 3 = 0$

$x = -\frac{1}{3}$ or $x = 3$

F. A quadratic equation has at most two real roots but may have just one or even no real root.

Examples:

The equation $x^2 - 6x + 9 = 0$ can be written as $(x-3)^2 = 0$ or $(x-3)(x-3) = 0$. So, its only root is 3.

The equation $x^2 + 4 = 0$ has no real root. Since any real number squared is greater than or equal to zero, $x^2 + 4$ must be greater than or equal to 4 if x is a real number.

G. An expression of the form $a^2 - b^2$ can be factored as $(a-b)(a+b)$.

Example:

We can solve the quadratic equation $9x^2 - 25 = 0$ like this:

$$\begin{aligned}(3x-5)(3x+5) &= 0\\ 3x-5=0 \text{ or } x+5 &= 0\\ x=\frac{5}{3} \text{ or } x &= -\frac{5}{3}\end{aligned}$$

H. If a quadratic expression isn't easily factored, we can still find its roots with the ***quadratic formula:*** If $ax^2 + bx + c = 0$ and $a \neq 0$, the roots are

$$x = \frac{-b+\sqrt{b^2-4ac}}{2a} \text{ and } x = \frac{-b-\sqrt{b^2-4ac}}{2a}$$

These roots are two distinct real numbers unless $b^2 - 4ac \leq 0$.

If $b^2 - 4ac = 0$, the two root expressions both equal $-\frac{b}{2a}$, so the equation has only one root.

If $b^2 - 4ac < 0$, then $\sqrt{b^2 - 4ac}$ is not a real number, so the equation has no real root.

4. Inequalities

A. An ***inequality*** is a statement with one of these symbols:

≠ is not equal to

> is greater than

≥ is greater than or equal to

< is less than

≤ is less than or equal to

Examples:

$5x - 3 < 9$ and $6x \geq y$

B. Solve a linear inequality with one unknown the same way you solve a linear equation: isolate the unknown on one side. As in an equation, the same number can be added to or subtracted from both sides of the inequality. And you can multiply or divide both sides by a positive number without changing the order of the inequality. However, multiplying or dividing an inequality by a negative number reverses the order of the inequality. Thus, 6 > 2, but (−1)(6) < (−1)(2).

Example (*1*):

To solve the inequality $3x - 2 > 5$ for x, isolate x:

$3x - 2 > 5$

$3x > 7$ (add 2 to both sides)

$x > \frac{7}{3}$ (divide both sides by 3)

Example (*2*):

To solve the inequality $\frac{5x - 1}{-2} < 3$ for x, isolate x:

$\frac{5x - 1}{-2} < 3$

$5x - 1 > -6$ (multiply both sides by −2)

$5x > -5$ (add 1 to both sides)

$x > -1$ (divide both sides by 5)

5. Functions

A. An algebraic expression in one variable can define a ***function*** of that variable. A function is written as a letter like f or g along with the variable in the expression. Function notation is a short way to express that a value is being substituted for a variable.

Examples:

i. The expression $x^3 - 5x^2 + 2$ can define a function f written as $f(x) = x^3 - 5x^2 + 2$.

ii. The expression $\dfrac{2z+7}{\sqrt{z+1}}$ can define a function g written as $g(z) = \dfrac{2z+7}{\sqrt{z+1}}$.

In these examples, the symbols "$f(x)$" and "$g(z)$" don't stand for products. Each is just a symbol for a function. Pronounce "$f(x)$" as "$f \text{ of } x$" and "$g(z)$" as "$g \text{ of } z$."

The substitution of 1 for x in the first expression can be written as $f(1) = -2$. Then $f(1)$ is called the "value of f at $x = 1$."

Likewise, in the second expression the value of g at $z = 0$ is $g(0) = 7$.

B. Once a function $f(x)$ is defined, think of x as an input and $f(x)$ as the output. In any function, any one input gives at most one output. But different inputs can give the same output.

Example:

If $h(x) = |x+3|$, then $h(-4) = 1 = h(-2)$.

C. The set of all allowed inputs for a function is the function's ***domain***. In the examples in Section 3.2.5.A above, the domain of f is the set of all real numbers, and the domain of g is the set of all numbers greater than −1.

Any function's definition can restrict the function's domain. For example, the definition "$a(x) = 9x - 5$ for $0 \leq x \leq 10$" restricts the domain of a to real numbers greater than or equal to 0 but less than or equal to 10. If the definition has no restrictions, the domain is the set of all values of x that each give a real output when input into the function.

D. The set of a function's outputs is the function's ***range***.

Examples:

i. For the function $h(x) = |x + 3|$ in the example in Section 3.2.5.B above, the range is the set of all numbers greater than or equal to 0.

ii. For the function $a(x) = 9x - 5$ for $0 \le x \le 10$ defined in Section 3.2.5.C above, the range is the set of all real numbers y such that $-5 \le y \le 85$.

E. A ***quadratic function*** has the form $f(x) = ax^2 + bx + c$ where a, b, and c are constants and $a \neq 0$. If $a > 0$, then $f(x)$ has a unique minimum value but no maximum value. Otherwise, it has a unique maximum value but no minimum value. In either case, the unique minimum or maximum value of $f(x)$ is $c - \frac{b^2}{4a}$. This value occurs only when $x = -\frac{b}{2a}$.

Examples:

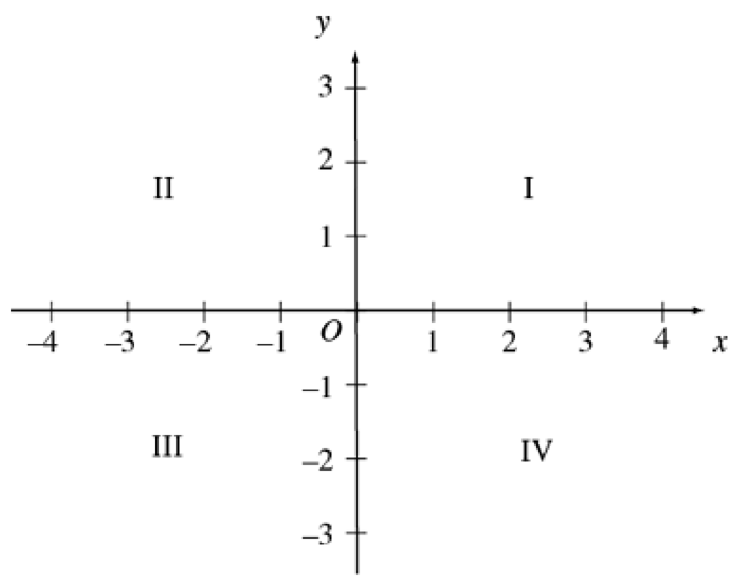
i. The quadratic function $f(x) = 2x^2 - 3x + 4$ has a unique minimum value of $4 - \frac{(-3)^2}{4 \times 2} = 4 - \frac{9}{8} = \frac{25}{8}$. This minimum value occurs only when $x = -\left(\frac{-3}{2 \times 2}\right) = \frac{3}{4}$.

ii. The quadratic function $g(x) = -3x^2 + 2x + 1$ has a unique maximum value of $1 - \frac{2^2}{4 \times -3} = 1 - \frac{-1}{3} = \frac{4}{3}$. This maximum value occurs only when $x = -\left(\frac{2}{2 \times -3}\right) = \frac{1}{3}$.

6. Graphing

A. The figure below shows the rectangular ***coordinate plane.*** The horizontal line is the ***x*-*axis*** and the vertical line is the ***y*-*axis***. These two axes intersect at the ***origin***, called O. The axes divide the plane into four quadrants: I, II, III, and IV, as shown.

The Coordinate Plane

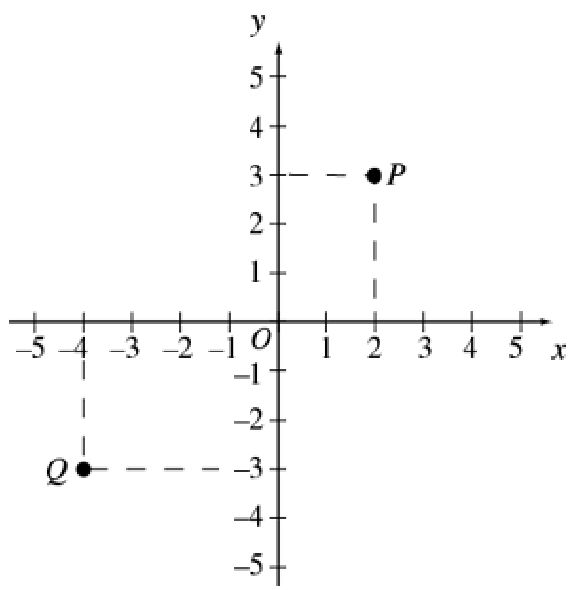


B. Any ordered pair (x, y) of real numbers defines a point in the coordinate plane. The point's ***x*-*coordinate*** is the first number in this pair. It shows how far the point is to the right or left of the y-axis. If the x-coordinate is positive, the point is to the right of the y-axis. If it's negative, the point is to the left of the y-axis. If it's 0, the point is on the y-axis. The point's ***y*-*coordinate*** is the second number in the ordered pair. It shows how far the point is above or below the x-axis. If the y-coordinate is positive, the point is above the x-axis. If it's negative, the point is below the x-axis. If it's 0, the point is on the x-axis.

Example:

In the graph below, the (x, y) coordinates of point P are (2, 3). P is 2 units to the right of the y-axis, so $x = 2$. Since P is 3 units above the x-axis, $y = 3$.

Likewise, the (x, y) coordinates of point Q are (−4, −3). The origin O has coordinates (0, 0).



C. The coordinates of the points on a straight line in the coordinate plane satisfy a linear equation of the form $y = mx + b$ (or the form $x = a$ if the line is vertical).

In the equation $y = mx + b$, the coefficient m is the line's ***slope***, and the constant b is the line's ***y-intercept***.

The y-intercept is the y-coordinate of the point where the line intersects the y-axis. Likewise, the ***x-intercept*** is the x-coordinate of the point where the line intersects the x-axis.

For any two points on the line, the line's slope is the ratio of the difference in their y-coordinates to the difference in their x-coordinates. To find the slope, subtract one point's y-coordinate from the other's. Then subtract the former point's x-coordinate from the latter's—not the other way around! Then divide the first difference by the second.

If a line’s slope is negative, the line slants down from left to right.

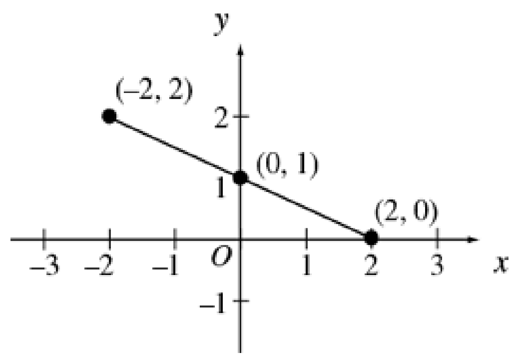
If the slope is positive, the line slants up.

If the slope is 0, the line is horizontal. A horizontal line’s equation has the form $y = b$, since $m = 0$.

A vertical line’s slope is undefined.

Example:

In the graph below, each point on the line satisfies the equation $y = -\frac{1}{2}x + 1$. To check this for the points (−2, 2), (2, 0), and (0, 1), substitute each point's coordinates for x and y in the equation.



You can use the points (−2, 2) and (2, 0) to find the line's slope:

$$\frac{\text{the difference in the } y\text{-coordinates}}{\text{the difference in the } x\text{-coordinates}} = \frac{0-2}{2-(-2)} = \frac{-2}{4} = -\frac{1}{2}.$$

The y-intercept is 1. That's y's value when x is 0 in $y = -\frac{1}{2}x + 1$.

To find the x-intercept, set y to 0 in the same equation:

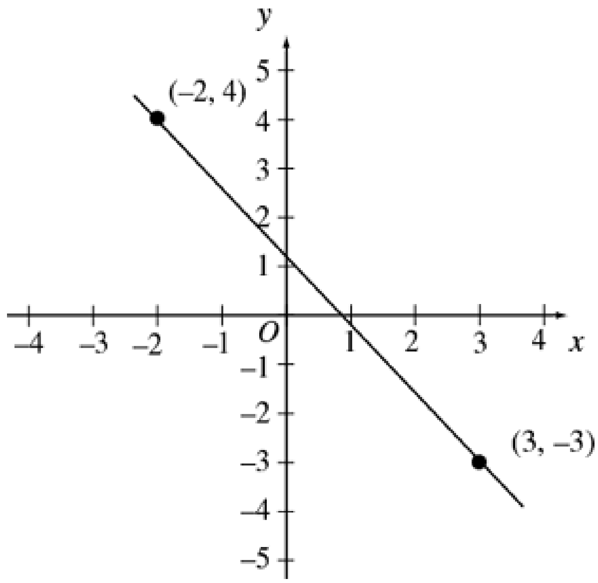
$$\begin{aligned} -\frac{1}{2}x + 1 &= 0 \\ -\frac{1}{2}x &= -1 \\ x &= 2 \end{aligned}$$

Thus, the x-intercept is 2.

D. You can use the definition of slope to find an equation of the line through two points (x_1, y_1) and (x_2, y_2) with $x_1 \neq x_2$. The slope is $m = \frac{y_2 - y_1}{x_2 - x_1}$. Given the known point (x_1, y_1) and the slope m, any other point (x, y) on the line satisfies the equation $m = \dfrac{(y - y_1)}{(x - x_1)}$, or equivalently $(y - y_1) = m(x - x_1)$. Using (x_2, y_2) instead of (x_1, y_1) as the known point gives an equivalent equation.

Example:

The graph below shows points (−2, 4) and (3, −3).



The line's slope is $\dfrac{(-3-4)}{(3-(-2))} = -\dfrac{7}{5}$. To find an equation of this line, let's use the point (3, −3):

$$\begin{aligned} y-(-3) &= \left(-\frac{7}{5}\right)(x-3) \\ y+3 &= \left(-\frac{7}{5}\right)x+\frac{21}{5} \\ y &= \left(-\frac{7}{5}\right)x+\frac{6}{5} \end{aligned}$$

So, the y-intercept is $\dfrac{6}{5}$.

Find the x-intercept like this:

$$\begin{aligned} 0 &= -\frac{7}{5}x + \frac{6}{5} \\ \frac{7}{5}x &= \frac{6}{5} \\ x &= \frac{6}{7} \end{aligned}$$

The graph shows both these intercepts.

E. If two linear equations with unknowns x and y have a unique shared solution, their graphs are two lines intersecting at the shared point that is the solution.

   If two linear equations are equivalent, they both stand for the same line and have infinitely many shared solutions.

   Two linear equations with no shared solution stand for two parallel lines.

F. Graph any function $f(x)$ in the coordinate plane by equating y with the function's value: $y = f(x)$. For any x in the function's domain, the point $(x, f(x))$ is on the function's graph. For every point on the graph, the y-coordinate is the function's value at the x-coordinate.

   Example:

   Consider the function $f(x) = -\frac{7}{5}x + \frac{6}{5}$.

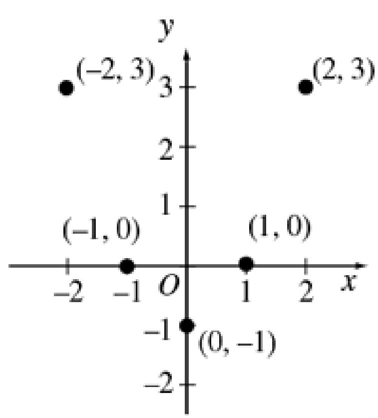
   If $f(x)$ is equated with the variable y, the function's graph is the graph of $y = -\frac{7}{5}x + \frac{6}{5}$ in the example above.

G. For any function f, the x-intercepts are the solutions of the equation $f(x) = 0$. The y-intercept is the value $f(0)$.

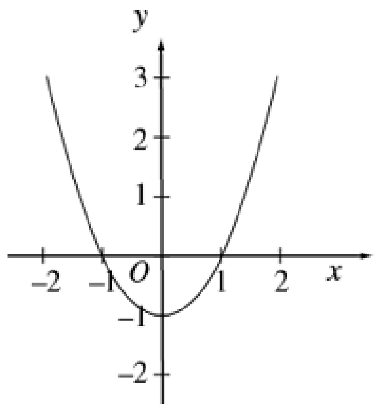
Example:

To see how the quadratic function $f(x) = x^2 - 1$ relates to its graph, let's plot some points $(x,\, f(x))$ in the coordinate plane:

x	$f(x)$
−2	3
−1	0
0	−1
1	0
2	3



The graph below shows all the points for $-2 \le x \le 2$:



The roots of this equation $f(x) = x^2 - 1 = 0$ are $x = 1$ and $x = -1$. They match the x-intercepts since x-intercepts are found by setting $y = 0$ and solving for x.

The y-intercept is $f(0) = -1$ because that's the value of y for $x = 0$.

7. Formulas and Measurement Conversion

A. A ***formula*** is an algebraic equation with variables that have specific meanings. To use a formula, assign quantities to its variables to match these meanings.

Example:

In the physics formula $F = ma$, the variable F stands for force, the variable m stands for mass, and the variable a stands for acceleration. The standard metric unit of force, the newton, is just enough force to accelerate a mass of 1 kilogram by 1 meter/second^2.

If we know a rock with a mass of 2 kilograms is accelerating at 5 meters/second^2, we can use the formula $F = ma$ by setting the variable m to 2 kilograms, and the variable a to 5 meters/second^2. Then we find that 10 newtons of force F are pushing the rock.

Note: You don't need to learn physics formulas or terms like this for the GMAT, but some specific GMAT questions may give you the formulas and terms you need to solve them.

B. Any quantitative relationship between units of measure can be written as a formula.

> *Examples:*
>
> i. Since 1 kilometer is 1,000 meters, the formula $m = 1000k$ can stand for the relationship between kilometers (k) and meters (m).
>
> ii. The formula $C = \frac{5}{9}(F - 32)$ can stand for the relationship between temperature measurements in degrees Celsius (C) and degrees Fahrenheit (F).

C. Except for units of time, a GMAT question that requires converting one unit of measure to another will give the relationship between those units.

> *Example:*
>
> A train travels at a constant 25 meters per second. How many kilometers does it travel in 5 minutes? (1 kilometer = 1,000 meters)
>
> *Solution:* In 1 minute the train travels (25)(60) = 1,500 meters, so in 5 minutes it travels 7,500 meters. Since 1 kilometer = 1,000 meters, we find 7,500 meters = 7.5 kilometers.

3.3 Rates, Ratios, and Percents

1. Ratio and Proportion

A. The ***ratio*** of a number x to a nonzero number y may be written as $x{:}y$, or $\frac{x}{y}$, or x to y. In each of these expressions, x is the numerator and y is the denominator. The order of a ratio's terms is important. Unless the absolute values of x and y are equal, $\frac{x}{y} \neq \frac{y}{x}$.

Examples:

The ratio of 2 to 3 may be written as 2:3, or $\frac{2}{3}$, or 2 to 3.

The ratio of the number of months with exactly 30 days to the number of months with exactly 31 days is 4:7, not 7:4.

B. A ***proportion*** is an equation between two ratios. Remember, the denominator on either side of the proportion must not equal zero.

Example:

2:3 = 8:12 is a proportion.

C. One way to solve for an unknown in a proportion is to cross multiply, then solve the resulting equation.

Example:

To solve for n in the proportion $\frac{2}{3} = \frac{n}{12}$, cross multiply to get $3n = 24$, then divide both sides by 3 to find $n = 8$.

D. Some word problems can be solved using ratios.

Example:

If 5 shirts cost a total of \$44, then what is the total cost of 8 shirts at the same cost per shirt?

Solution: If c is the cost of the 8 shirts, then $\frac{5}{44} = \frac{8}{c}$. Cross multiplying gives $5c = 8 \times 44 = 352$, so $c = \frac{352}{5} = 70.4$. Thus, the 8 shirts cost a total of \$70.40.

2. Fractions

A. In a fraction $\frac{n}{d}$, n is the ***numerator*** and d is the ***denominator***. A fraction's denominator can never be 0 because division by 0 is undefined.

B. ***Equivalent*** fractions stand for the same number. To check whether two fractions with numerators and denominators that are integers are equivalent, divide each fraction's numerator and denominator by the largest factor common to that numerator and that denominator, their ***greatest common divisor*** (GCD). This is called ***reducing each fraction to its lowest terms***. Two fractions are equivalent if and only if reducing each to its lowest terms makes them identical.

Example:

To check whether $\frac{8}{36}$ and $\frac{14}{63}$ are equivalent, first reduce each to its lowest terms. In the first fraction, 4 is the GCD of the numerator 8 and the denominator 36. Dividing both the numerator and the denominator of $\frac{8}{36}$ by 4 gives $\frac{2}{9}$. In the second fraction, 7 is the GCD of the numerator 14 and the denominator 63. Dividing both the numerator and the denominator of $\frac{14}{63}$ by 7 also gives $\frac{2}{9}$. Since reducing $\frac{8}{36}$ and $\frac{14}{63}$ to their lowest terms makes them identical, they're equivalent.

C. To add or subtract two fractions with the same denominator, just add or subtract the numerators, leaving the denominators the same.

Examples:

$\frac{3}{5} + \frac{4}{5} = \frac{3+4}{5} = \frac{7}{5}$ and

$\frac{5}{7} - \frac{2}{7} = \frac{5-2}{7} = \frac{3}{7}$

D. To add or subtract two fractions with different denominators, first express them as fractions with the same denominator.

Example:

To add $\frac{3}{5}$ and $\frac{4}{7}$, multiply the numerator and denominator of $\frac{3}{5}$ by 7 to get $\frac{21}{35}$. Then multiply the numerator and denominator of $\frac{4}{7}$ by 5 to get $\frac{20}{35}$. Since both fractions now have the same denominator 35, you can easily add them: $\frac{3}{5} + \frac{4}{7} = \frac{21}{35} + \frac{20}{35} = \frac{41}{35}$

E. To multiply two fractions, multiply their numerators, and also multiply their denominators.

Example:

$\frac{2}{3} \times \frac{4}{7} = \frac{2 \times 4}{3 \times 7} = \frac{8}{21}$

F. The ***reciprocal*** of a fraction $\frac{n}{d}$ is $\frac{d}{n}$ if neither n nor d is 0.

Example:

The reciprocal of $\frac{4}{7}$ is $\frac{7}{4}$.

G. To divide by a fraction, multiply by its reciprocal.

Example:

$$\frac{2}{3} \div \frac{4}{7} = \frac{2}{3} \times \frac{7}{4} = \frac{14}{12} = \frac{7}{6}$$

H. A ***mixed number*** is written as an integer next to a fraction. It equals the integer plus the fraction.

Example:

The mixed number $7\frac{2}{3} = 7 + \frac{2}{3}$

I. To write a mixed number as a fraction, multiply the integer part of the mixed number by the denominator of the fractional part. Add this product to the numerator. Then put this sum over the denominator.

Example:

$$7\frac{2}{3} = \frac{(7 \times 3) + 2}{3} = \frac{23}{3}$$

3. Percents

A. The word ***percent*** means ***per hundred*** or ***number out of 100.***

Example:

Saying that 37 percent, or 37%, of the houses in a city are painted blue means that 37 houses per 100 in the city are painted blue.

B. A percent may be greater than 100.

Example:

Saying that the number of blue houses in a city is 150% of the number of red houses means the city has 150 blue houses for every 100 red houses. Since 150:100 = 3:2, this means the city has 3 blue houses for every 2 red houses.

C. A percent need not be an integer.

Example:

Saying that the number of pink houses in a city is 0.5% of the number of blue houses means the city has 0.5 of a pink house for every 100 blue houses. Since 0.5:100 = 1:200, this means the city has 1 pink house for every 200 blue houses.

Likewise, saying that the number of orange houses is 12.5% of the number of blue houses means the ratio of orange houses to blue houses is 12.5:100 = 1:8. Therefore, there is 1 orange house for every 8 blue houses.

4. Converting Decimals, Fractions, and Percents

A. Decimals with many finite digits can be rewritten as fractions or sums of fractions.

Examples:

$$0.321 = \frac{3}{10} + \frac{2}{100} + \frac{1}{1{,}000} = \frac{321}{1{,}000}$$

$$0.0321 = \frac{0}{10} + \frac{3}{100} + \frac{2}{1{,}000} + \frac{1}{10{,}000} = \frac{321}{10{,}000}$$

$$1.56 = 1 + \frac{5}{10} + \frac{6}{100} = \frac{156}{100}$$

B. Repeating decimals can also be rewritten as fractions. To do this, set a variable x equal to the repeating decimal. Next multiply both sides by a power of 10 just large enough to make the repeating digits start right after the decimal if they don't already. Call this Equation A. Then make another equation by multiplying both sides of Equation A by a power of 10 just large enough to move one copy of the repeating digits from the right to the left of the decimal point. From this, subtract Equation A, and solve for x.

Examples:

i. To rewrite the repeating decimal $3.\bar{6}$ as a fraction, first let $x = 3.\bar{6}$. This equation already has the repeating digit 6 starting right after the decimal, so it's our Equation A. Next we multiply both sides by 10 to move one repeating 6 from the right to the left of the decimal point: $10x = 36.\bar{6}$. From this, subtract Equation A to get $9x = 36.\bar{6} - 3.\bar{6} = 33$. Finally divide both sides by 9 to solve for x: $x = \dfrac{33}{9} = 3\dfrac{2}{3}$.

ii. To rewrite the repeating decimal $1.2\overline{79}$ as a fraction, first let $x = 1.2\overline{79}$. Next multiply both sides by 10 to make the repeating digits start right after the decimal: $10x = 12.\overline{79}$. This is our Equation A. Then multiply both sides by 100 to move one copy of the repeating digits 79 from the right to the left of the decimal point: $1000x = 1279.\overline{79}$. From this, subtract Equation A to get $990x = 1279.\overline{79} - 12.\overline{79} = 1267$. Finally divide both sides by 990 to solve for x: $x = \dfrac{1267}{990} = 1\dfrac{277}{990}$.

C. To rewrite a percent as a fraction, write the percent number as the numerator over a denominator of 100. To rewrite a percent as a decimal, move the decimal point in the percent two places to the left and drop the percent sign. To rewrite a decimal as a percent, move the decimal point two places to the right, then add a percent sign.

Examples:

$$37\% = \frac{37}{100} = 0.37$$

$$300\% = \frac{300}{100} = 3$$

$$0.5\% = \frac{0.5}{100} = 0.005$$

D. To find a percent of a number, multiply the number by the percent written as a fraction or decimal.

Examples:

20% of $90 = 90\left(\frac{20}{100}\right) = 90\left(\frac{1}{5}\right) = \frac{90}{5} = 18$

20% of $90 = 90(0.2) = 18$

250% of $80 = 80\left(\frac{250}{100}\right) = 80(2.5) = 200$

0.5% of $12 = 12\left(\frac{0.5}{100}\right) = 12(0.005) = 0.06$

5. Working with Decimals, Fractions, and Percents

A. To find the percent increase or decrease from one positive quantity to another, first find the amount of increase or decrease. Then divide this amount by the original quantity. Write this quotient as a percent.

Examples:

Suppose a price increases from \$24 to \$30. To find the percent increase, first find the amount of increase: \$30 − \$24 = \$6. Divide this \$6 by the original price of \$24 to find the percent increase: $\frac{6}{24} = 0.25 = 25\%$.

Now suppose a price falls from \$30 to \$24. The amount of decrease is \$30 − \$24 = \$6. So, the percent decrease is $\frac{6}{30} = 0.20 = 20\%$.

Notice the percent **increase** from 24 to 30 (25%) doesn't equal the percent **decrease** from 30 to 24 (20%).

A percent increase may be greater than 100%, but a percent decrease cannot be.

Example:

Suppose a house's price in 2018 was 300% of its price in 2003, which was above 0. By what percent did the price increase?

Solution: If n is the price in 2003, the percent increase is $\left|\frac{(3n-n)}{n}\right| = \left|\frac{2n}{n}\right| = 2, \frac{(3n-n)}{n} = \frac{2n}{2} = 2$, or 200%.

B. A price discounted by n percent is $(100 - n)$ percent of the original price.

Example:

A customer paid $24 for a dress. If the customer got a 25% discount off the original price of the dress, what was the original price before the discount?

Solution: The discounted price is 100% − 25% = 75% of the original price. So, if p is the original price, $0.75p = \$24$ is the discounted price. Thus, $p = (\$24/0.75) = \$32$, the original price before the discount.

Two discounts can be combined to make a larger discount.

Example:

A price is discounted 20%. Then this reduced price is discounted another 30%. These two discounts together make an overall discount of what percent?

Solution: If p is the original price, then $0.8p$ is the price after the first discount. The price after the second discount is $(0.7)(0.8)p = 0.56p$. The overall discount is 100% − 56% = 44%.

C. ***Profit*** equals revenues minus expenses, or selling price minus total costs.

> *Example:*
>
> A certain appliance costs a merchant a total of $30 to acquire and sell. At what price should the merchant sell the appliance to make a profit of 50% of the appliance's total cost to the merchant?
>
> *Solution:* The merchant should sell the appliance for a price $\$s$ such that $s - 30 = (0.5)(30)$. So, $\$s = \$30 + \$15 = \$45$.

D. ***Simple annual interest*** on a loan or investment is based only on the original loan or investment amount (the ***principal***). It equals (principal) × (interest rate) × (time).

> *Example:*
>
> If $8,000 is invested at 6% simple annual interest, how much interest is earned in 3 months?
>
> *Solution:* Since the annual interest rate is 6%, the interest for 1 year is (0.06)($8,000) = $480. A year has 12 months, so the interest earned in 3 months is $\left(\frac{3}{12}\right)(\$480) = \$120$.

E. ***Compound interest*** is based on the principal plus any interest already earned.

Compound interest over $n \text{ periods} = (\text{principal}) \times (1 + \text{interest per period})^n - \text{principal}$.

Example:

If \$10,000 is invested at 10% annual interest, compounded every 6 months, what is the investment's value after 1 year?

Solution: Since the interest is compounded every 6 months, or twice a year, the interest rate for each 6-month period is 5%, half the 10% annual rate. So, the investment's value after the first 6 months is 10,000 + (10,000) (0.05) = \$10,500.

For the second 6 months, the interest is based on the \$10,500 balance after the first 6 months. So, the investment's value after 1 year is 10,500 + (10,500)(0.05) = \$11,025.

The investment's value after 1 year can also be written as $10{,}000 \times \left(1 + \frac{0.10}{2}\right)^2$ dollars.

F. To solve some word problems with percents and fractions, you can organize the information in a table.

Example:

In a production lot, 40% of the toys are red, and the rest are green. Half of the toys are small, and half are large. If 10% of the toys are red and small, and 40 toys are green and large, how many of the toys are red and large?

Solution: First make a table to organize the information:

	Red	Green	Total
Small	10%		50%
Large			50%
Total	40%	60%	100%

Then fill in the missing percents so that the “Red” and “Green” percents in each row add up to that row’s total, and the “Small” and “Large” percents in each column add up to that column’s total:

	Red	Green	Total
Small	10%	40%	50%
Large	30%	20%	50%
Total	40%	60%	100%

The number of large green toys, 40, is 20% of the total number of toys (n), so $0.20n = 40$. Thus, the total number of toys $n = 200$. So, 30% of the 200 toys are red and large. Since (0.3)(200) = 60, we find that 60 of the toys are red and large.

6. Rate, Work, and Mixture Problems

A. The distance an object travels is its average speed multiplied by the time it takes to travel that distance. That is, ***distance = rate × time***.

> *Example:*
>
> How many kilometers did a car travel in 4 hours at an average speed of 70 kilometers per hour?
>
> *Solution:* Since distance = rate × time, multiply 70 kilometers/hour × 4 hours to find that the car went 280 kilometers.

B. To find an object's average travel speed, divide the total travel distance by the total travel time.

> *Example:*
>
> On a 600-kilometer trip, a car went half the distance at an average speed of 60 kilometers per hour (kph), and the other half at an average speed of 100 kph. The car didn't stop between the two halves of the trip. What was the car's average speed over the whole trip?
>
> *Solution:* First find the total travel time. For the first 300 kilometers, the car went at 60 kph, taking $\frac{300\ \text{kilometers}}{60\ \text{kph}} = 5\ \text{hours}$. For the second 300 kilometers, the car went at 100 kph, taking $\frac{300\ \text{kilometers}}{100\ \text{kph}} = 3\ \text{hours}$. So, the total travel time was 5 + 3 = 8 hours. The car's average speed was $\frac{600\ \text{kilometers}}{8\ \text{hours}} = 75\ \text{kph}$. Notice the average speed was not $\frac{(60+100)\ \text{kilometers}}{2\ \text{hours}} = 80\ \text{kph}$.

C. A ***work problem*** usually says how fast certain individuals work alone, then asks you to find how fast they work together, or vice versa.

The basic formula for work problems is $\dfrac{1}{r}+\dfrac{1}{s}=\dfrac{1}{h}$, where r is how long an amount of work takes a certain individual, s is how long that much work takes a different individual, and h is how long that much work takes both individuals working at the same time.

Example:

Suppose one machine takes 4 hours to make 1,000 bolts, and a second machine takes 5 hours to make 1,000 bolts. How many hours do both machines working at the same time take to make 1,000 bolts?

Solution:

$$\begin{aligned}\frac{1}{4}+\frac{1}{5} &= \frac{1}{h}\\ \frac{5}{20}+\frac{4}{20} &= \frac{1}{h}\\ \frac{9}{20} &= \frac{1}{h}\\ 9h &= 20\\ h &= \frac{20}{9}=2\frac{2}{9}\end{aligned}$$

Working together, the two machines can make 1,000 bolts in $2\frac{2}{9}$ hours.

If a work problem says how long it takes two individuals to do an amount of work together, and how long it takes one of them to do that much work alone, you can use the same formula to find how long it takes the other individual to do that much work alone.

Example:

Suppose Art and Rita both working at the same time take 4 hours to do an amount of work, and Art alone takes 6 hours to do that much work. Then how many hours does Rita alone take to do that much work?

Solution:

$$\frac{1}{6} + \frac{1}{R} = \frac{1}{4}$$
$$\frac{1}{R} = \frac{1}{4} - \frac{1}{6} = \frac{1}{12}$$
$$R = 12$$

Rita alone takes 12 hours to do that much work.

D. In ***mixture problems***, substances with different properties are mixed, and you must find the mixture's properties.

Example:

If 6 kilograms of nuts that cost \$1.20 per kilogram are mixed with 2 kilograms of nuts that cost \$1.60 per kilogram, how much does the mixture cost per kilogram?

Solution: The 8 kilograms of nuts cost a total of 6(\$1.20) + 2(\$1.60) = \$10.40. So, the cost per kilogram of the mixture is $\frac{\$10.40}{8} = \$1.30$.

Some mixture problems use percents.

Example:

How many liters of a solution that is 15% salt must be added to 5 liters of a solution that is 8% salt to make a solution that is 10% salt?

Solution: Let n be the needed number of liters of the 15% solution. The amount of salt in n liters of 15% solution is $0.15n$. The amount of salt in the 5 liters of 8% solution is (0.08)(5). These amounts add up to the amount of salt in the 10% mixture, which is $0.10(n+5)$. So,

$$\begin{aligned} 0.15n + 0.08(5) &= 0.10(n+5) \\ 15n + 40 &= 10n + 50 \\ 5n &= 10 \\ n &= 2 \text{ liters} \end{aligned}$$

So, 2 liters of the 15% salt solution must be added to the 8% solution to make the 10% solution.

3.4 Statistics, Sets, Counting, Probability, Estimation, and Series

1. Statistics

A. A common statistical measure is the ***average*** or ***(arithmetic) mean***, a type of center for the numerical values in a data set. The average or mean of n values is the sum of the n values divided by n. If a number appears more than once in the data set, each instance of that number is a separate value and is added separately to find the sum. To learn more about values in data sets, see Chapter 5, “Data Insights Review.”

Example:

The average of the 5 values 6, 4, 7, 10, and 4 is $\dfrac{(6+4+7+10+4)}{5} = \dfrac{31}{5} = 6.2$. Notice that 4 appears twice in the list of 5 values, so it is added twice when finding the sum.

B. The ***median*** is another type of center for ordered values in a data set. As above, if a number appears more than once in the data set, each instance of that number is a separate value. To find the median, arrange the values from least to greatest. If the number of entries in the list is odd, the median is the middle entry in the list. But if the number of entries in the list is even, the median is the average of the two middle entries. The median may be less than, equal to, or greater than the mean.

Example:

To find the median of the 5 values 6, 4, 7, 10, and 4, arrange them from least to greatest: 4, 4, 6, 7, 10. The median is 6, the middle entry in this list.

The median of the 6 values 4, 6, 6, 8, 9, and 12 is $\dfrac{(6+8)}{2} = 7$. But the mean of these 6 values is $\dfrac{(4+6+6+8+9+12)}{6} = \dfrac{45}{6} = 7.5$.

Often about half the values in a data set are less than the median, and about half are greater than the median. But not always.

Example:

For the 15 values 3, 5, 7, 7, 7, 7, 7, 7, 8, 9, 9, 9, 9, 10, and 10, the median is 7. Only $\dfrac{2}{15}$ of the values are less than the median.

C. The ***mode*** of a data set is the value appearing most often in the list.

> *Example:*
>
> The mode of the list of values 1, 3, 6, 4, 3, and 5 is 3, since 3 is the only value appearing more than once in the list.

A list may have more than one mode.

> *Example:*
>
> The list 1, 2, 3, 3, 3, 5, 7, 10, 10, 10, and 20 has two modes, 3 and 10.

D. The ***dispersion*** of numerical data is how spread out the data is. The simplest measure of dispersion is the ***range,*** which is the greatest value in the data minus the least value.

> *Example:*
>
> The range of the values 11, 10, 5, 13, and 21 is $21 - 5 = 16$. Notice the range depends on only 2 of the values.

E. Another common measure of dispersion is the ***standard deviation.*** Generally, the farther the values are from the mean, the greater the standard deviation. To find the standard deviation of n values:

1. Find their mean
2. Find the differences between the mean and each of the n values
3. Square each difference
4. Find the average of the squared differences, and
5. Take the nonnegative square root of this average

Examples:

Let's use the table below to find the standard deviation of the 5 values 0, 7, 8, 10, and 10, which have the mean 7.

x	$x-7$	$(x-7)^2$
0	−7	49
7	0	0
8	1	1
10	3	9
10	3	9
	Total	68

The standard deviation is $\sqrt{\frac{68}{5}} \approx 3.7$

The standard deviation depends on every value in the data set, but more on those farther from the mean. This is why the standard deviation is smaller for a data set grouped closer around its mean.

As a second example, consider the values 6, 6, 6.5, 7.5, and 9, which also have the mean 7. These values are grouped closer around the mean 7 than are those in the first example. That makes the standard deviation in this second example only about 1.1, far below the standard deviation of 3.7 in the first example.

F. How many times a value occurs in a data set is its ***frequency*** in the set. When different values have different frequencies, a ***frequency distribution*** can help show how the values are distributed.

Example:

Consider this data set of 20 numbers:

-4	0	0	-3	-2	-1	-1	0	-1	-4
-1	-5	0	-2	0	-5	-2	0	0	-1

We can show its frequency distribution in a table showing each value x and x's frequency f:

Value x	Frequency f
-5	2
-4	2
-3	1
-2	3
-1	5
0	7
Total	20

This frequency distribution table makes finding statistical measures easier:

Mean:

$$= \frac{(-5)(2) + (-4)(2) + (-3)(1) + (-2)(3) + (-1)(5) + (0)(7)}{20} = -1.6$$

Median: -1 (the average of the 10^{th} and 11^{th} numbers)

Mode: 0 (the number that occurs most often)

Range: 0 − (−5) = 5

Standard deviation:

$$\sqrt{\frac{(-5+1.6)^2\,(2)+(-4+1.6)^2\,(2)+\ldots+(0+1.6)^2\,(7)}{20}}\approx 1.7$$

2. Sets

A. In math, a ***set*** is a collection of numbers or other things. The things in the set are its ***elements***. A list of a set's elements between braces stands for the set. The list's order doesn't matter.

Example:

{−5, 0, 1} is the same set as {0, 1, −5}. That is, {−5, 0, 1} = {0, 1, −5}.

B. The number of elements in a finite set S is written as $|S|$.

Example:

$S=\{-5,0,1\}$ is a set with $|S|=3$.

C. If all the elements in a set S are also in a set T, then S is a ***subset*** of T. This is written as $S\subseteq T$ or $T\supseteq S$.

Example:

{−5, 0, 1} is a subset of {−5, 0, 1, 4, 10}. That is, {−5, 0, 1} ⊆ {−5, 0, 1, 4, 10}.

D. The ***union*** of two sets A and B is the set of all elements that are each in A or in B or both. The union is written as $A\cup B$.

> *Example:*
>
> $\{3, 4\} \cup \{4, 5, 6\} = \{3, 4, 5, 6\}$

E. The ***intersection*** of two sets A and B is the set of all elements that are each in **both** A and B. The intersection is written as $A \cap B$.

> *Example:*
>
> $\{3, 4\} \cap \{4, 5, 6\} = \{4\}$

F. Two sets sharing no elements are ***disjoint*** or ***mutually exclusive.***

> *Example:*
>
> {−5, 0, 1} and {4, 10} are disjoint.

G. The ***empty set*** is the set with no elements. It's written as $\emptyset$ or { }.

H. A ***Venn diagram*** shows the unions and intersections of two or more sets. Suppose sets cap s and cap t aren't disjoint, and neither is a subset of the other. The Venn diagram below shows their intersection cap s intersection cap t as a shaded area.

> **A Venn Diagram of Two Intersecting Sets**
>
> image
>
> A Venn diagram of sets cap s and cap t, with their intersection cap s intersection cap t shaded.

I. The number of elements in the union of two finite sets cap s and cap t is the number of elements in cap s, plus the number of elements in cap t, minus the number of elements in the intersection of cap s and cap t. That is, absolute value of cap s union cap t equals absolute value of cap s plus absolute value of cap t minus absolute value of cap s intersection cap t . This is the ***general addition rule for two sets.***

Example:

multiline equation line 1 equation sequence part 1 absolute value of left curly bracket three comma four right curly bracket union left curly bracket four comma five comma six right curly bracket equals part 2 absolute value of left curly bracket three comma four right curly bracket plus absolute value of left curly bracket four comma five comma six right curly bracket minus absolute value of left curly bracket three comma four right curly bracket intersection left curly bracket four comma five comma six right curly bracket equals part 3 line 2 absolute value of left curly bracket three comma four right curly bracket plus absolute value of left curly bracket four comma five comma six right curly bracket minus one times left curly bracket four right curly bracket vertical line equation sequence part 1 equals part 2 two plus three minus one equals part 3 four full stop

If cap s and cap t are disjoint, then absolute value of cap s union cap t equals absolute value of cap s plus absolute value of cap t , since absolute value of cap s intersection cap t equals zero.

J. You can often solve word problems involving sets by using Venn diagrams and the general addition rule.

Example:

Each of 25 students is taking history, mathematics, or both. If 20 of them are taking history and 18 of them are taking mathematics, how many of them are taking both history and mathematics?

Solution: Separate the 25 students into three disjoint sets: the students taking history only, those taking mathematics only, and those taking both history and mathematics. This gives us the Venn diagram below, where n is the number of students taking both courses, $20 - n$ is the number taking history only, and $18 - n$ is the number taking mathematics only.

image

Since there are 25 students total, $(20 - n) + n + (18 - n) = 25$, so $n = 13$. So, 13 students are taking both history and mathematics. Notice that 20 + 18 − 13 = 25 uses the general addition rule for two sets.

3. Counting Methods

A. To count elements in sets without listing them, you can sometimes use this ***multiplication principle***:

If an object will be chosen from a set of m different objects, and another object will be chosen from a disjoint set of n different objects, then $m \times n$ different choices are possible.

Example:

Suppose a meal at a restaurant must include exactly 1 entree and 1 dessert. The entree can be any 1 of 5 options, and the dessert can be any 1 of 3 options. Then 5 × 3 = 15 different meals are available.

B. Here’s a more general version of the multiplication principle: The number of possible choices of 1 object apiece out of any number of sets is the product of the numbers of objects in those sets. For example, when choosing 1 object apiece out of 3 sets with x, y, and z elements, respectively, x times y times z different choices are possible. The general multiplication principle also means that when choosing 1 object apiece out of n sets of exactly m objects apiece, m super n different choices are possible.

> *Example:*
>
> Each time a coin is flipped, the 2 possible results are heads and tails. In a sequence of 8 consecutive coin flips, think of each flip as a set of those 2 possible results. The 8 flips give us a sequence of 8 of these 2-element sets. So, the sequence of 8 flips has 2^8 possible results.

C. A concept often used with the multiplication principle is the ***factorial.*** For any integer n greater than one, n factorial is written as n factorial and is the product of all the integers from 1 through n. Also, by definition, 0! = 1! = 1.

> *Examples:*
>
> $2! = 2 \times 1 = 2$
>
> $3! = 3 \times 2 \times 1 = 6$
>
> $4! = 4 \times 3 \times 2 \times 1 = 24$, etc.

A useful equation with factorials is
n factorial equals left parenthesis n minus one right parenthesis factorial left parenthesis n right parenthesis
.

D. Any sequential ordering of a set’s elements is a ***permutation*** of the set. A permutation is a way to choose elements one by one in a certain order.

The factorial is useful for finding how many permutations a set has. If a set of n objects is being ordered from 1st to n super th, there are n choices for the 1^{st} object, n minus one choices left for the 2^{nd} object, n minus two choices left for the 3^{rd} object, and so on, until only 1 choice is left for the n super th object. So, by the multiplication principle, a set of n objects has $n(n-1)(n-2)\ \ldots\ (3)(2)(1) = n!$ permutations.

> *Example:*
>
> The set of letters A, B, and C has 3! = 6 permutations: ABC, ACB, BAC, BCA, CAB, and CBA.

E. When $0 \leq k \leq n$, each possible choice of k objects out of n objects is a ***combination*** of n objects taken k at a time. The number of these combinations is written as $\begin{pmatrix} n \\ k \end{pmatrix}$. This is also the number of k-element subsets of a set with n elements, since the combinations simply are these subsets. It can be calculated as $\begin{pmatrix} n \\ k \end{pmatrix} = \dfrac{n!}{k!\,(n-k)!}$. Note that $\begin{pmatrix} n \\ k \end{pmatrix} = \begin{pmatrix} n \\ n-k \end{pmatrix}$.

Example:

The 2-element subsets of $S = \{\mathrm{A, B, C, D, E}\}$ are the combinations of the 5 letters in S taken 2 at a time. There are $\binom{5}{2} = \frac{5!}{2!3!} = \frac{120}{(2)(6)} = 10$ of these subsets: {A, B}, {A, C}, {A, D}, {A, E}, {B, C}, {B, D}, {B, E}, {C, D}, {C, E}, and {D, E}.

For each of its 2-element subsets, a 5-element set also has exactly one 3-element subset containing the elements not in that 2-element subset. For example, in S the 3-element subset {C, D, E} contains the elements not in the 2-element subset {A, B}, the 3-element subset {B, D, E} contains the elements not in the 2-element subset {A,C}, and so on. This shows that a 5-element set like S has exactly as many 2-element subsets as 3-element subsets, so $\binom{5}{2} = 10 = \binom{5}{3}$.

4. Probability

A. Sets and counting methods are also important to ***discrete probability.*** Discrete probability involves ***experiments*** with finitely many possible ***outcomes.*** An ***event*** is a set of an experiment's possible outcomes.

Example:

Rolling a 6-sided die with faces numbered 1 to 6 is an experiment with 6 possible outcomes. Let's call these outcomes 1, 2, 3, 4, 5, and 6, each number being the one facing up after the roll. Notice that no two outcomes can occur together.

One event in this experiment is that the outcome is 4. This event is written as {4}.

Another event in the experiment is that the outcome is an odd number. This event has the three outcomes 1, 3, and 5. It is written as {1, 3, 5}.

B. The probability of an event E is written as $P(E)$ and is a number between 0 and 1, inclusive. If E is the empty set of no possible outcomes, then E is ***impossible***, and $P(E) = 0$. If E is the set of all possible outcomes of the experiment, then E is ***certain***, and $P(E) = 1$. Otherwise, E is possible but not certain, and $0 < P(E) < 1$. If F is a subset of E, then $P(F) \leq P(E)$.

C. If the probabilities of two or more outcomes are equal, those outcomes are ***equally likely.*** For an experiment with outcomes that are all equally likely, the probability of an event E is
$$P(E) = \frac{\text{the number of outcomes in } E}{\text{the total number of possible outcomes}}.$$

> *Example:*
>
> In the earlier example of a 6-sided die rolled once, suppose all 6 outcomes are equally likely. Then each outcome's probability is $\frac{1}{6}$. The probability that the outcome is an odd number is
> $P(\{1,3,5\}) = \frac{|\{1,3,5\}|}{6} = \frac{3}{6} = \frac{1}{2}$.

D. Given two events E and F, these further events are defined:

i. "not E" is the set of outcomes not in E;

ii. "E or F" is the set of outcomes in E or F or both, that is, $E \cup F$;

iii. "E and F" is the set of outcomes in both E and F, that is, $E \cap F$.

The probability that E doesn't occur is $P(\text{not } E) = 1 - P(E)$.

The probability that "E or F" occurs is $P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F)$. This is based on the general addition rule for two sets, given above in Section 3.4.2.G.

Example:

In the example above of a 6-sided die rolled once, let E be the event {1, 3, 5} that the outcome is an odd number. Let F be the event {2, 3, 5} that the outcome is a prime number. Then

$$P(E \text{ and } F) = P(E \cap F) = P(\{3, 5\}) = \frac{|\{3, 5\}|}{6} = \frac{2}{6} = \frac{1}{3}.$$

So,

$$P(E \text{ or } F) = P(E) + P(F) - P(E \text{ and } F) = \frac{3}{6} + \frac{3}{6} - \frac{2}{6} = \frac{4}{6} = \frac{2}{3}.$$

We get the same result by noticing that the event "$E \text{ or } F$" is $E \cup F = \{1, 2, 3, 5\}$, so $P(E \text{ or } F) = \dfrac{|\{1, 2, 3, 5\}|}{6} = \dfrac{4}{6} = \dfrac{2}{3}$.

Events E and F are ***mutually exclusive*** if no outcomes are in $E \cap F$. Then the event "$E \text{ and } F$" is impossible: $P(E \text{ and } F) = 0$. The special addition rule for the probability of two mutually exclusive events is $P(E \text{ or } F) = P(E) + P(F)$.

E. Two events A and B are ***independent*** if neither changes the other's probability. The multiplication rule for independent events E and F is $P(E \text{ and } F) = P(E)P(F)$.

Example:

In the example above of the 6-sided die rolled once, let A be the event $\{2, 4, 6\}$ and B be the event $\{5, 6\}$. Then A's probability is $P(A) = \dfrac{|A|}{6} = \dfrac{3}{6} = \dfrac{1}{2}$. The probability of A occurring **if B occurs** is $\dfrac{|A \cap B|}{|B|} = \dfrac{|\{6\}|}{|\{5, 6\}|} = \dfrac{1}{2}$, the same as $P(A)$.

Likewise, B's probability is $P(B) = \dfrac{|B|}{6} = \dfrac{2}{6} = \dfrac{1}{3}$. The probability of B occurring **if A occurs** is $\dfrac{|B \cap A|}{|A|} = \dfrac{\{6\}}{|\{2, 4, 6\}|} = \dfrac{1}{3}$, the same as $P(B)$.

So, neither event changes the other's probability. Thus, A and B are independent. Therefore, by the multiplication rule for independent events, $P(A \text{ and } B) = P(A)P(B) = \left(\dfrac{1}{2}\right)\left(\dfrac{1}{3}\right) = \dfrac{1}{6}$.

Notice the event "A and B" is $A \cap B = \{6\}$, so $P(A \text{ and } B) = P(\{6\}) = \dfrac{1}{6}$.

The general addition rule and the multiplication rule discussed above imply that if E and F are independent,
$P(E \text{ or } F) = P(E) + P(F) - P(E)P(F)$.

F. An event cap a is ***dependent*** on a nonempty event cap b if cap b changes the probability of cap a.

The probability of cap a occurring if cap b occurs is written as cap p of cap a vertical line cap b. So, the statement that cap a is dependent on cap b can be written as cap p of cap a vertical line cap b not equals cap p of cap a.

A general multiplication rule for any dependent or independent events
cap a and cap b is
cap p of cap a and cap b equals cap p of cap a vertical line cap b times cap p of cap b
.

Example:

In the example of the 6-sided die rolled once, let cap a be the event {4, 6} and cap b be the event {4, 5, 6}.

Then the probability of cap a is equation sequence part 1 cap p of cap a equals part 2 absolute value of cap a divided by six equals part 3 two divided by six equals part 4 one divided by three
. But the probability that cap a occurs **if** bold-italic cap b **occurs** is equation sequence part 1 cap p of cap a vertical line cap b equals part 2 absolute value of cap a intersection cap b divided by absolute value of cap b equals part 3 absolute value of left curly bracket four comma six right curly bracket divided by absolute value of left curly bracket four comma five comma six right curly bracket equals part 4 two divided by three
. Thus, cap p of cap a vertical line cap b not equals cap p of cap a, so cap a is dependent on cap b.

Likewise, the probability that cap b occurs is equation sequence part 1 cap p of cap b equals part 2 absolute value of cap b divided by six equals part 3 three divided by six equals part 4 one divided by two
. But the probability that cap b occurs **if** bold-italic cap a **occurs** is equation sequence part 1 cap p of cap b vertical line cap a equals part 2 absolute value of cap b intersection cap a divided by absolute value of cap a equals part 3 absolute value of left curly bracket four comma six right curly bracket divided by absolute value of left curly bracket four comma six right curly bracket equals part 4 one
. Thus, cap p of cap b vertical line cap a not equals cap p of cap b, so cap b is dependent on cap a.

In this example, by the general multiplication rule for events, equation sequence part 1 cap p of cap a and cap b equals part 2 cap p of cap a vertical line cap b times cap p of cap b equals part 3 left parenthesis two divided by three right parenthesis times left parenthesis one divided by two right parenthesis equals part 4 one divided by three

. Likewise,
equation sequence part 1 cap p of cap a and cap b equals part 2 cap p of cap b vertical line cap a times cap p of cap a equals part 3 left parenthesis one right parenthesis times left parenthesis one divided by three right parenthesis equals part 4 one divided by three full stop

Notice the event “cap a and cap b” is
equation sequence part 1 cap a intersection cap b equals part 2 left curly bracket four comma six right curly bracket equals part 3 cap a
, so
equation sequence part 1 cap p of cap a and cap b equals part 2 cap p of left curly bracket four comma six right curly bracket equals part 3 one divided by three equals part 4 cap p of cap a
.

G. The rules above can be combined for more complex probability calculations.

Example:

In an experiment with events A, B, and C, suppose $P(A) = 0.23$, $P(B) = 0.40$, and $P(C) = 0.85$. Also suppose events A and B are mutually exclusive, and events B and C are independent. Since A and B are mutually exclusive,

$$P(A \text{ or } B) = P(A) + P(B) = 0.23 + 0.40 = 0.63.$$

Since B and C are independent,

$$P(B \text{ or } C) = P(B) + P(C) - P(B) \cdot P(C) = 0.40 + 0.85 - (0.40)(0.85) = 0.91.$$

$P(A \text{ or } C)$ and $P(A \text{ and } C)$ can't be found from the information given. But we can find that

$$P(A) + P(C) = 1.08 > 1.$$

So, $P(A) + P(C)$ can't equal $P(A \text{ or } C)$, which like any probability cannot be greater than 1. This means that A and C can't be mutually exclusive, and that $P(A \text{ and } C) \geq 0.08$.

Since $A \cap B$ is a subset of A, we can also find that

$$P(A \text{ and } C) \leq P(A) = 0.23.$$

And C is a subset of $A \cup C$, so

$$P(A \text{ or } C) \geq P(C) = 0.85.$$

Thus, we’ve found that
$0.85 \le P(A \text{ or } C) \le 1$
and that
$0.08 \le P(A \text{ and } C) < 0.23$
.

5. Estimation

A. Calculating exact answers to complex math questions is often too hard or slow. Estimating the answers by simplifying the questions may be easier and faster.

   One way to estimate is to ***round*** the numbers in the original question: replace each number with a nearby number that has fewer digits.

   For any integer n and real number m, you can ***round*** $\boldsymbol{m}$ ***down*** to a multiple of 10^n by deleting all of m’s digits to the right of the digit that stands for multiples of 10^n.

   To ***round*** $\boldsymbol{m}$ ***up*** to a multiple of 10^n, first add 10^n to m, then round the result down.

   To ***round*** $\boldsymbol{m}$ ***to the nearest*** 10^n, first find the digit in m that stands for a multiple of 10^{n-1}. If this digit is 5 or higher, round m up to a multiple of 10^n. Otherwise, round m down to a multiple of 10^n.

Examples:

i. To round 7651.4 to the nearest hundred (multiple of 10^2), first notice the digit standing for tens (multiples of 10^1) is 5.

   Since this digit is 5 or higher, round up:

   First add 100 to the original number: 7651.4 + 100 = 7751.4.

   Then drop all the digits to the right of the one standing for multiples of 100 to get 7700.

   Notice that 7700 is closer to 7651.4 than 7600 is, so 7700 is the nearest 100.

ii. To round 0.43248 to the nearest thousandth (multiple of 10^{-3}), first notice the digit standing for ten-thousandths (multiples of 10^{-4}) is 4. Since 4 < 5, round down: just drop all the digits to the right of the digit standing for thousandths to get 0.432.

B. Rounding can simplify complex calculations and give rough answers. If you keep more digits of the original numbers, the answers are usually more exact, but the calculations take longer.

Example:

You can estimate the value of
left parenthesis 298.534 plus 58.296 right parenthesis divided by sum with 3 summands 1.4822 plus 0.937 plus 0.014679
by rounding the numbers in the dividend to the nearest 10 and the numbers in the divisor to the nearest 0.1:

multirelation left parenthesis 298.534 plus 58.296 right parenthesis divided by sum with 3 summands 1.4822 plus 0.937 plus 0.014679 almost equals 300 plus 60 divided by sum with 3 summands 1.5 plus 0.9 plus zero equals 360 divided by 2.4 equals 150

C. Sometimes it's easier to estimate by rounding to a multiple of a number other than 10, like the nearest square or cube of an integer.

Examples:

i. You can estimate the value of 2447.13 divided by 11.9 by noting first that both the dividend and the divisor are near multiples of 12: 2448 and 12. So,
multirelation 2447.13 divided by 11.9 almost equals 2448 divided by 12 equals 204
.

ii. You can estimate the value of
Square root of 8.96 divided by 24.82 multiplication 4.057 by noting first that each decimal number in the expression is near the square of an integer: $8.96 \approx 9 = 3^2$, $24.82 \approx 25 = 5^2$, and $4.057 \approx 4 = 2^2$. So,
multirelation Square root of 8.96 divided by 24.82 multiplication 4.057 almost equals Square root of three squared divided by five squared multiplication two squared equals Square root of three squared divided by 10 squared equals three divided by 10
.

D. Sometimes finding a ***range*** of possible values for an expression is more useful than finding a single estimated value. The range's ***upper bound*** is the smallest number found to be greater than (or no less than) the expression's value. The range's ***lower bound*** is the largest number found to be less than (or no greater than) the expression's value.

Example:

In the equation
x equals 2.32 squared minus 2.536 divided by 2.68 squared plus 2.79, each decimal is greater than 2 and less than 3. So,
two squared minus three divided by three squared plus three less than x less than three squared minus two divided by two squared plus two
. Simplifying these fractions, we find that x is in the range
one divided by 12 less than x less than seven divided by six. The range's lower bound is one divided by 12, and the upper bound is
seven divided by six.

6. Sequences and Series

A. A ***sequence*** is an ordered list of elements, starting with a first element. Any sequence may be defined as the range of a function a of n with a domain that is the positive integers. Its value a sub i for a specific positive integer i is its bold-italic i times th **term**.

An ***arithmetic sequence*** has the form a sub n equals b plus c times n, where b and c are constants. The first term of an arithmetic sequence is a sub one equals b plus c. For each term a sub i, the next term a sub i plus one equals a sub i plus c.

A ***geometric sequence*** has the form a sub n equals b times c super n, where b and c are constants. The first term of an arithmetic sequence is a sub one equals b times c. For each term a sub i, the next term a sub i plus one equals a sub i times c.

Sometimes a sequence is defined only by giving the value of the first term or terms, then giving an equation for finding each subsequent term's value based on the value of one or more earlier terms.

Examples:

i. The function
a of n equals n squared plus left parenthesis n divided by five right parenthesis
with a domain that is the positive integers is an infinite sequence a sub one comma a sub two comma a sub three comma ellipsis. Its third term is its value at n equals three, which is
equation sequence part 1 a sub three equals part 2 three squared plus three divided by five equals part 3 9.6
.

ii. equation sequence part 1 a of n equals part 2 n squared plus left parenthesis n divided by five right parenthesis times b sub n equals part 3 left parenthesis negative one right parenthesis super n times left parenthesis n factorial right parenthesis
The function a of n equals two plus three times n with a domain that is the positive integers is an arithmetic sequence. Its first term a sub one equals five. For any positive integer i,
a sub i plus one equals a sub i plus three. Thus,
equation sequence part 1 a sub two equals part 2 a sub one plus three equals part 3 eight
.

iii. The function
b of n equals two times left parenthesis three super n right parenthesis
with a domain that is the positive integers is a geometric sequence. Its first term b sub one equals six. For any positive integer i,
b sub i plus one equals three times b sub i. Thus,
equation sequence part 1 b sub two equals part 2 three multiplication six equals part 3 18
.

iv. We can define a sequence c of n just by stating that its first term c sub one equals one, its second term c sub two equals one, and each subsequent term
c sub n plus two equals sum with 3 summands c sub n plus c sub n plus one

. Thus,
equation sequence part 1 c sub three equals part 2 c sub one plus c sub two equals part 3 one plus one equals part 4 two
.

B. A ***series*** is the sum of a sequence's terms.

For an infinite sequence a of n, the ***infinite series*** n ary summation from n equals one to infinity over a of n adds up the sequence's infinitely many terms,
sum with variable number of summands a sub one plus a sub two plus a sub three plus ellipsis
. Some infinite series add up to finite sums, but others don't.

Example:

The infinite series based on the function
a of n equals n squared plus left parenthesis n divided by five right parenthesis
is
n ary summation from i equals one to infinity over n squared plus left parenthesis n divided by five right parenthesis
. It adds up the infinitely many terms
sum with variable number of summands left parenthesis one squared plus one divided by five right parenthesis plus left parenthesis two squared plus two divided by five right parenthesis plus left parenthesis three squared plus three divided by five right parenthesis plus ellipsis full stop
This series has no finite sum.

The partial sum of the first three terms
a of n equals n squared plus left parenthesis n divided by five right parenthesis
is
equation sequence part 1 n ary summation from i equals one to three over a sub i equals part 2 sum with 3 summands left parenthesis one squared plus one divided by five right parenthesis plus left parenthesis two squared plus two divided by five right parenthesis plus left parenthesis three squared plus three divided by five right parenthesis equals part 3 sum with 3 summands 1.2 plus 4.4 plus 9.6 equals part 4 15.2 full stop

3.5 Reference Sheets

Arithmetic and Decimals

ABSOLUTE VALUE:

absolute value of x is x if x greater than or equals zero and negative x if x less than zero.

For any x and y comma absolute value of x plus y less than absolute value of x plus absolute value of y , and absolute value of x times y equals absolute value of x times absolute value of y .

Square root of x squared equals absolute value of x .

EVEN AND ODD NUMBERS:

Even × Even = Even	Even × Odd = Even
Odd × Odd = Odd	Even + Even = Even
Even + Odd = Odd	Odd + Odd = Even

ADDITION AND SUBTRACTION:

QUOTIENTS AND REMAINDERS:

The quotient q and the remainder r of dividing positive integer x by positive integer y are unique positive integers such that

y equals x times q plus r and zero less than r less than x.

The remainder r is 0 if and only if y is divisible by x. Then x is a factor of y.

MULTIPLICATION AND DIVISION:

equation sequence part 1 x multiplication one equals part 2 x equals part 3 x divided by one

x multiplication zero equals zero

If x not equals zero, then x divided by x equals one.

x divided by zero is undefined.

x times y equals y times x

equation sequence part 1 x plus zero equals part 2 x equals part 3 x minus zero

x minus x equals zero

x plus y equals y plus x

equation sequence part 1 x minus y equals part 2 negative left parenthesis y minus x right parenthesis equals part 3 negative y plus x

left parenthesis x plus y right parenthesis plus z equals x plus left parenthesis y plus z right parenthesis

If x and y are both positive, then x plus y is positive.

If x and y are both negative, then x plus y is negative.

DECIMALS WITH MANY FINITE DIGITS:

Add or subtract decimals by lining up their decimal points:

17.6512	653.2700
+ 653.2700	−17.6512
670.9212	635.6188

To multiply decimal cap a by decimal cap b:

First, ignore the decimal points and multiply cap a and cap b as if they were integers.

Next, if decimal cap a has n digits to the right of its decimal point and

If x not equals zero and y not equals zero, then x divided by y equals one divided by left parenthesis y divided by x right parenthesis

.

left parenthesis x times y right parenthesis times z equals x of y times z

x times y plus x times z equals x times left parenthesis y plus z right parenthesis

If y not equals zero, then left parenthesis x divided by y right parenthesis plus left parenthesis z divided by y right parenthesis equals left parenthesis x plus z right parenthesis divided by y

.

If x and y are both positive, then x times y is positive.

If x and y are both negative, then x times y is positive.

If x is positive and y is negative, then x times y is negative.

If x times y equals zero, then x equals zero or y equals zero, or both.

SCIENTIFIC NOTATION:

To convert a number in scientific notation cap a multiplication 10 super n into regular decimal notation, move cap a decimal cap b has m digits to the right of its decimal point, place the decimal point in cap a multiplication cap b so it has m plus n digits to its right.

To divide decimal cap a by decimal cap b, first move the decimal points of cap a and cap b equally many digits to the right until cap b is an integer. Then divide as you would integers.

’s decimal point n places to the right if n is positive, or absolute value of n places to the left if n is negative.

To convert a decimal to scientific notation, move the decimal point n spaces so that exactly one nonzero digit is to its left. Multiply the result by 10 super n if you moved the decimal point to the left or by 10 super negative n if you moved it to the right.

ARITHMETIC SHORTCUTS:

For any two nonnegative integers x and y, the final digit of x plus y is the final digit of the sum of x’s final digit and y’s final digit. And for any two integers x and y, the final digit of x times y is the final digit of the product of x’s final digit and y’s final digit.

An integer is divisible by	if and only if
2	its final digit is 0, 2, 4, 6, or 8.
3	the sum of its digits is divisible by 3.
4	its final two digits make a number divisible by 4.
5	its final digit is 5 or 0.
9	the sum of its digits is divisible by 9.
10	its final digit is 0.

Exponents

SQUARES, CUBES, AND SQUARE ROOTS:

Every positive number has two real square roots, one positive and the other negative. The table below shows the positive square roots rounded to the nearest hundredth.

n	n squared	n cubed	Square root of n
1	1	1	1
2	4	8	1.41
3	9	27	1.73
4	16	64	2
5	25	125	2.24
6	36	216	2.45
7	49	343	2.65
8	64	512	2.83
9	81	729	3
10	100	1,000	3.16

EXPONENTIATION:

Formula	Example
x super one equals x	$2^1 = 2$
If x not equals zero , then x super zero equals one .	$2^0 = 1$
If x not equals zero , then x super negative one equals one divided by x .	two super negative one equals one divided by two
If x greater than one and y greater than one , then x super y greater than x .	$2^3 = 8 > 2$

If zero less than x less than one and y greater than one , then x super y less than x .	$0.2^3 = 0.008 < 0.2$
equation sequence part 1 left parenthesis x super y right parenthesis super z equals part 2 x super y times z equals part 3 left parenthesis x super z right parenthesis super y	$(2^3)^4 = 2^{12} = (2^4)^3$
x super y plus z equals x super y times x super z	$2^7 = 2^3 2^4$
If x not equals zero , then x super y minus z equals x super y divided by x super z .	two super five minus three equals two super five divided by two cubed

left parenthesis x times z right parenthesis super y equals x super y times z super y	$6^4 = 2^4 3^4$
If z not equals zero , then left parenthesis x divided by z right parenthesis super y equals x super y divided by z super y .	equation sequence part 1 left parenthesis three divided by four right parenthesis squared equals part 2 three squared divided by four squared equals part 3 nine divided by 16
If z not equals zero , then equation sequence part 1 x super y divided by z equals part 2 left parenthesis x super y right parenthesis super one divided by z equals part 3 left parenthesis x super one divided by z right parenthesis super y .	equation sequence part 1 four super two divided by three equals part 2 left parenthesis four squared right parenthesis super one divided by three equals part 3 left parenthesis four super one divided by three right parenthesis squared

Algebraic Expressions and Linear Equations

TRANSLATING WORDS INTO MATH OPERATIONS:

x plus y	x minus y	x times y	x divided by y	x super y
x added to y x increased by y x more than y x plus y the sum of x and y the total of x and y	x decreased by y difference of x and y y fewer than x y less than x x minus y x reduced by y y subtracted from x	x multiplied by y the product of x and y x times y If y equals two: double x twice x If y equals three : triple x	x divided by y x over y the quotient of x and y the ratio of x to y If y equals two : half of x x halved	x to the power of y x to the y^{th} power If y equals two : x squared If y equals three : x cubed

MANIPULATING ALGEBRAIC EXPRESSIONS:

Technique	**Example**
Factor to combine like terms	$3xy - 9y = 3y(x - 3)$
Divide out common factors	equation sequence part 1 left parenthesis three times x times y minus nine times y right parenthesis divided by left parenthesis x minus three right parenthesis equals part 2 three times y times left parenthesis x minus three right parenthesis divided by left parenthesis x minus three right parenthesis equals part 3 three times y of one equals part 4 three times y
Multiply two expressions by multiplying each term of one expression by each term of the other	multiline equation row 1 left parenthesis three times x minus four right parenthesis times left parenthesis nine times y plus x right parenthesis equals three times x times left parenthesis nine times y plus x right parenthesis minus four times left parenthesis nine times y plus x right parenthesis row 2 equals three times x of nine times y plus three times x of x minus four times left parenthesis nine times y right parenthesis minus four times left parenthesis x right parenthesis row 3 equals 27 times x times y plus three times x squared minus 36 times y minus four times x
Substitute constants for variables	If $x = 3$ and y equals negative two, then $3xy - x^2 + y = 3(3)(-2) - (3)^2 + (-2) = -18 - 9 - 2 = -29.$

SOLVING LINEAR EQUATIONS:

Technique	Example
Isolate a variable on one side of an equation by doing the same operations on both sides of the equation.	Solve the equation left parenthesis five times x minus six right parenthesis divided by three equals four like this: (1) Multiply both sides by 3 to get $5x - 6 = 12$. (2) Add 6 to both sides to get five times x equals 18. (3) Divide both sides by 5 to get $x = \frac{18}{5}$.
To solve two equations with two variables x and y: (1) Express x in terms of y using one of the equations. (2) Substitute that expression for x to make the second equation have only the variable y. (3) Solve the second equation for y. (4) Substitute the solution for y into the first equation to solve for x.	Solve the equations A: x minus y equals two and B: $3x + 2y = 11$: (1) From A, x equals two plus y. (2) In B, substitute $2 + y$ for x to get $3(2 + y) + 2y = 11$. (3) Solve B for y: $6 + 3y + 2y = 11$ multiline equation row 1 six plus five times y equals 11 row 2 five times y equals five row 3 y equals one (4) Since $y = 1$, from A we find $x = 2 + 1 = 3$.
Alternative technique:	Solve the equations A: $x - y = 2$ and B:

(1) Multiply both sides of one equation or both equations so that the coefficients on y have the same absolute value in both equations. (2) Add or subtract the two equations to remove y and solve for x. (3) Substitute the solution for x into the first equation to find the value of y .	three times x plus two times y equals 11 : (1) Multiply both sides of A by 2 to get $2x - 2y = 4$. (2) Add the equation in (1) to equation B: multiline equation row 1 sum with 3 summands two times x minus two times y plus three times x plus two times y equals four plus 11 row 2 five times x equals 15 row 3 x equals three (3) Since x equals three, from A we find three minus y equals two, so y equals one.

Factoring, Quadratic Equations, and Inequalities

SOLVING EQUATIONS BY FACTORING:

Techniques	**Example**
(1) Start with a polynomial equation. (2) Add or subtract from both sides until 0 is on one side of the equation. (3) Write the nonzero side as a product of factors. (4) Set each factor to 0 to find simple equations giving the solutions to the original equation.	multiline equation row 1 x cubed minus two times x squared plus x equals negative five times left parenthesis x minus one right parenthesis squared row 2 sum with 3 summands x cubed minus two times x squared plus x plus five times left parenthesis x minus one right parenthesis squared equals zero row 3 left parenthesis i right parenthesis x times left parenthesis x squared minus two times x plus one right parenthesis plus five times left parenthesis x minus one right parenthesis squared equals zero row 4 left parenthesis ii right parenthesis x times left parenthesis x minus one right parenthesis squared plus five times left parenthesis x minus one right parenthesis squared equals zero row 5 left parenthesis iii right parenthesis left parenthesis x plus five right parenthesis times left parenthesis x minus one right parenthesis squared equals zero x plus five equals zero or x minus one equals zero. So, x equals negative five or x equals one.

FORMULAS FOR FACTORING:	**THE QUADRATIC FORMULA:**

left parenthesis a plus one divided by a right parenthesis squared equals sum with 3 summands a squared plus one divided by a squared plus two

a squared minus b squared equals left parenthesis a minus b right parenthesis times left parenthesis a plus b right parenthesis

sum with 3 summands a squared plus two times a times b plus b squared equals left parenthesis a plus b right parenthesis squared

a squared minus two times a times b plus b squared equals left parenthesis a minus b right parenthesis squared

equation sequence part 1 a squared plus b squared equals part 2 left parenthesis a plus b right parenthesis squared minus two times a times b equals part 3 left parenthesis a minus b right parenthesis squared plus two times a times b

left parenthesis sum with 3 summands a plus b plus c right parenthesis squared equals sum with 6 summands a squared plus b squared plus c squared plus two times a times b plus two times a times c plus two times b times c

left parenthesis a plus b right parenthesis cubed equals sum with 4 summands a cubed plus three times a squared times b plus three times a times b squared plus b cubed

For any quadratic equation
sum with 3 summands a times x squared plus b times x plus c equals zero
with a not equals zero, the roots are

x equals negative b plus Square root of b squared minus four times a times c divided by two times a
and
x equals negative b minus Square root of b squared minus four times a times c divided by two times a

These roots are two distinct real numbers if
b squared minus four times a times c greater than or equals zero
.

If
b squared minus four times a times c equals zero
, the equation has only one root:
negative b divided by two times a.

If
b squared minus four times a times c less than zero
, the equation has no real roots.

SOLVING INEQUALITIES:

Explanation	**Example**
As in solving an equation, the same number can be added to or subtracted from both sides of the inequality, or both sides can be multiplied or divided by a positive number, without changing the order of the inequality. But multiplying or dividing an inequality by a negative number reverses the order of the inequality. Thus, 6 > 2, but (−1)(6) < (−1)(2).	To solve the inequality left parenthesis five times x minus one right parenthesis divided by negative two less than three for x, isolate x: (1) five times x minus one greater than negative s (multiplying both sides by −2, reversing the order of the inequality) (2) five times x greater than negative five (add 1 to both sides) (3) x greater than negative one (divide both sides by 5)

LINES IN THE COORDINATE PLANE:

An equation y equals m times x plus b defines a line with slope m which has a y-intercept that is b.

image

For a line through two points left parenthesis x sub one comma y sub one right parenthesis and left parenthesis x sub two comma y sub two right parenthesis with x sub one not equals x sub two, the slope is m equals left parenthesis y sub two minus y sub one right parenthesis divided by left parenthesis x sub two minus x sub one right parenthesis. Given the known point (x_1, y_1) and the slope m, any other point (x, y) on the line satisfies the equation $m = \dfrac{(y - y_1)}{(x - x_1)}$.

Above, the line's slope is $\dfrac{(-3-4)}{(3-(-2))} = -\dfrac{7}{5}$. To find an equation of the line, we can use the point (3, −3):

$$\begin{aligned} y - (-3) &= \left(-\frac{7}{5}\right)(x-3) \\ y + 3 &= \left(-\frac{7}{5}\right)x + \frac{21}{5} \\ y &= \left(-\frac{7}{5}\right)x + \frac{6}{5} \end{aligned}$$

So, the y-intercept is $\dfrac{6}{5}$.

Find the x-intercept like this:

$$
\begin{aligned}
0 &= \left(-\frac{7}{5}\right)x + \frac{6}{5} \\
\left(\frac{7}{5}\right)x &= \frac{6}{5} \\
x &= \frac{6}{7}
\end{aligned}
$$

The graph shows both these intercepts.

Rates, Ratios, and Percents

FRACTIONS:

Equivalent or Equal Fractions:

Two fractions stand for the same number if dividing each fraction's numerator and denominator by their greatest common divisor makes the fractions identical.

Adding, Subtracting, Multiplying, and Dividing Fractions:

$$\frac{a}{b} + \frac{c}{d} = \frac{ad}{bd} + \frac{bc}{bd};$$
$$\frac{a}{b} - \frac{c}{d} = \frac{ad}{bd} - \frac{bc}{bd}$$
$$\frac{a}{b} \times \frac{c}{d} = \frac{ac}{bd}; \frac{a}{b} \div \frac{c}{d} = \frac{ad}{bc}$$

MIXED NUMBERS:

A mixed number of the form a times b divided by c equals the fraction $\frac{ac+b}{c}$.

RATE:

distance = rate × time

PROFIT:

Profit = Revenues − Expenses, or

Profit = Selling price − Cost.

PERCENTS:

$x\% = \frac{x}{100}$.

$x\%$ of y equals $\frac{xy}{100}$.

To convert a percent to a decimal, drop the percent sign, then move the decimal point two digits left.

To convert a decimal to a percent, add a percent sign, then move the decimal point two digits right.

PERCENT INCREASE OR DECREASE:

The percent increase from x to y is $100\left(\frac{y-x}{x}\right)\%$.

The percent decrease from x to y is 100 times left parenthesis x minus y divided by x right parenthesis percent .

DISCOUNTS:

A price discounted by n percent becomes left parenthesis 100 minus n right parenthesis percent of the original price.

A price discounted by n percent and then by m percent becomes $(100 - n)(100 - m)$ percent of the original price.

INTEREST:

Simple annual interest = (principal) × (interest rate) × (time)

Compound interest over n periods equals left parenthesis principal right parenthesis multiplication left parenthesis one plus interest per period right parenthesis super n minus principal

WORK:

one divided by r plus one divided by s equals one divided by h
, where r is how long one individual takes to do an amount of work, s is how long a second individual takes to do that much work, and h is how long they take to do that much work when both are working at the same time.

MIXTURES:

	Number of units of a substance or mixture	Amount of an ingredient per unit of the substance or mixture	Total amount of that ingredient in the substance or mixture
Substance A	X	M	X × M
Substance B	Y	N	Y × N
Mixture of A and B	X + Y	left parenthesis normal cap x multiplication normal cap m right parenthesis plus left parenthesis normal cap y multiplication normal cap n right parenthesis divided by normal cap x plus normal cap y	(X × M) + (Y × N)

Statistics, Sets, and Counting Methods

STATISTICS:

Concept	**Definition for a data set of n values ordered from least to greatest**	**Example with data set {4, 4, 5, 7, 10}**
Mean	The sum of the n values, divided by n	equation sequence part 1 left parenthesis sum with 5 summands four plus four plus five plus seven plus 10 right parenthesis divided by five equals part 2 30 divided by five equals part 3 six
Median	The middle value if n is odd; The mean of the two middle values if n is even.	5 is the middle value in {4, 4, 5, 7, 10}.
Mode	The value that appears most often in the data set.	4 is the only value that appears more than once in {4, 4, 5, 7, 10}.
Range	The largest value in the data set minus the smallest	10 − 4 = 6
Standard Deviation	Calculate like this: (1) Find the arithmetic mean (2) Find the differences between each of the n values and the mean (3) Square each of the differences	(1) The mean is 6. (2) −2, −2, −1, 1, 4 (3) 4, 4, 1, 1, 16 (4) $\frac{26}{5} = 5.2$ (5) Square root of 5.2

	(4) Find the average of the squared differences, and (5) Take the nonnegative square root of this average	

SETS:

Concept	**Notation for finite sets $\boldsymbol{S}$ and** bold-italic cap t	**Example**
Number of elements	absolute value of cap s	$S = \{-5, 0, 1\}$ is a set with absolute value of cap s equals three.
Subset	cap s subset of or equals cap t (cap s is a subset of T); cap s superset of or equals cap t (cap t is a subset of cap s)	{−5, 0, 1} is a subset of {−5, 0, 1, 4, 10}.
Union	$S \cup T$	left curly bracket three comma four right curly bracket union left curly bracket four comma five comma six right curly bracket equals left curly bracket three comma four comma five comma six right curly bracket
Intersection	cap s intersection cap t	$\{3, 4\} \cap \{4, 5, 6\} = \{4\}$

The general addition rule for two sets	absolute value of cap s union normal cap t equals absolute value of cap s plus absolute value of normal cap t minus absolute value of cap s intersection normal cap t	multiline equation line 1 absolute value of left curly bracket three comma four right curly bracket union left curly bracket four comma five comma six right curly bracket equals line 2 absolute value of left curly bracket three comma four right curly bracket plus absolute value of left curly bracket four comma five comma six right curly bracket minus absolute value of left curly bracket three comma four right curly bracket intersection left curly bracket four comma five comma six right curly bracket equals line 3 equation sequence part 1 absolute value of left curly bracket three comma four right curly bracket plus absolute value of left curly bracket four comma five comma six right curly bracket minus absolute value of left curly bracket four right curly bracket equals part 2 two plus three minus one equals part 3 four full stop

COUNTING METHODS:

Concept and Equations	Examples
Multiplication Principle: The number of possible choices of 1 element apiece from the data sets cap a sub one comma cap a sub two comma ellipsis comma cap a sub n is absolute value of cap a sub one multiplication absolute value of cap a sub two multiplication ellipsis multiplication absolute value of cap a sub n .	The number of possible choices of 1 element apiece from the data sets $S = \{-5, 0, 1\}$, cap t equals left curly bracket three comma four right curly bracket , and cap u equals left curly bracket three comma four comma five comma six right curly bracket is $\|S\| \times \|T\| \times \|U\| = 3 \times 2 \times 4 = 24.$
Factorial: multiline equation line 1 n factorial equals n multiplication left parenthesis n minus one right parenthesis multiplication ellipsis multiplication one line 2 zero factorial equals one factorial equals one line 3 n factorial equals left parenthesis n minus one right parenthesis factorial left parenthesis n right parenthesis	4! = 4 × 3 × 2 × 1 = 24 4! = 3! × 4
Permutations: A data set of n objects has n! permutations	The data set of letters A, B, and C has 3! = 6 permutations: ABC, ACB, BAC, BCA, CAB, and CBA.
Combinations: The number of possible choices of k objects from a data set of n objects is	The number of 2-element subsets of data set {A, B, C, D, E} is

vector element 1 n element 2 k equals n factorial divided by k factorial left parenthesis n minus k right parenthesis factorial .	equation sequence part 1 vector element 1 five element 2 two equals part 2 five factorial divided by two factorial three factorial equals part 3 120 divided by left parenthesis two right parenthesis times left parenthesis six right parenthesis equals part 4 10 . The 10 subsets are: {A, B}, {A, C}, {A, D}, {A, E}, {B, C}, {B, D}, {B, E}, {C, D}, {C, E}, and {D, E}.

Probability and Sequences

PROBABILITY:

Concept	Definition, Notation, and Equations	Example: Rolling a die with 6 numbered sides once
Event	A set of outcomes of an experiment	The event of the outcome being an odd number is the set {1, 3, 5}.
Probability	The probability cap p of cap e of an event cap e is a number between 0 and 1, inclusive. If each outcome is equally likely, $P(E) = \frac{(\text{the number of possible outcomes in E})}{(\text{the total number of possible outcomes})}$.	If the 6 outcomes are equally likely, the probability of each outcome is $\frac{1}{6}$. The probability that the outcome is an odd number is equation sequence part 1 cap p times left parenthesis left curly bracket one comma three comma five right curly bracket right parenthesis equals part 2 absolute value of left curly bracket one comma three comma five right curly bracket divided by six equals part 3 three divided by six equals part 4 one divided by two .
Conditional Probability	The probability that cap e occurs if cap f occurs is cap p of cap e vertical line cap f equals absolute value of cap e intersection cap f divided by absolute value of cap f .	equation sequence part 1 cap p of left curly bracket one comma three comma five right curly bracket vertical line left curly bracket one comma two right curly bracket equals part 2 absolute value of left curly bracket one right curly bracket divided by absolute value of left curly bracket one comma two right curly bracket equals part 3 one divided by two

Not *E*	The set of outcomes not in event cap e: cap p of not cap e equals one minus cap p of cap e .	$P(\text{not}\{3\}) = \frac{6-1}{6} = \frac{5}{6}$
bold-italic cap e **and** bold-italic cap f	The set of outcomes in both cap e and F, that is, cap e intersection cap f ; equation sequence part 1 cap p of cap e and cap f equals part 2 cap p times left parenthesis cap e intersection cap f right parenthesis equals part 3 cap p of cap e vertical line cap f cap p of cap f .	For cap e equals left curly bracket one comma three comma five right curly bracket and $F = \{2, 3, 5\}$: multiline equation line 1 equation sequence part 1 cap p of cap e and cap f equals part 2 cap p times left parenthesis cap e intersection cap f right parenthesis equals part 3 cap p of left curly bracket three comma five right curly bracket equals part 4 line 2 equation sequence part 1 absolute value of left curly bracket three comma five right curly bracket divided by six equals part 2 two divided by six equals part 3 one divided by three
bold-italic cap e **or *F***	The set of outcomes in cap e or cap f or both, that is, $E \cup F$; cap p of cap e or cap f equals cap p of cap e plus cap p of cap f minus cap p of cap e and cap f .	For $E = \{1, 3, 5\}$ and cap f equals left curly bracket two comma three comma five right curly bracket

		: multiline equation line 1 cap p of cap e or cap f equals cap p of cap e plus cap p of cap f minus cap p of cap e and cap f line 2 equation sequence part 1 equals part 2 three divided by six plus three divided by six minus two divided by six equals part 3 four divided by six equals part 4 two divided by three full stop
Dependent and Independent Events	E is dependent on cap f if cap p of cap e vertical line cap f not equals cap p of cap e . E and cap f are independent if neither is dependent on the other. If E and cap f are independent, cap p of cap e and cap f equals cap p of cap e times cap p of cap f .	For $E = \{2, 4, 6\}$ and cap f equals left curly bracket five comma six right curly bracket : equation sequence part 1 cap p of cap e vertical line cap f equals part 2 cap p of cap e equals part 3 one divided by two , and $P(F\|E) = P(F) = \frac{1}{3}$, so cap e and cap f are independent. Thus equation sequence part 1 cap p of cap e times and times cap f equals part 2 cap p of cap e times cap p of cap f equals part 3 left parenthesis one divided by two right parenthesis times left parenthesis one divided by three right parenthesis equals part 4 one divided by six .
SEQUENCE: An ordered list of elements, starting with a first element. **Example:** The function a of n equals n squared plus left parenthesis n divided by five right parenthesis		

with the domain of all positive integers
n equals one comma two comma three, … is an infinite sequence a sub n.

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4.0 Quantitative Reasoning

4.0 Quantitative Reasoning

The GMAT™ exam's Quantitative Reasoning section tests your math skills in the areas reviewed in Chapter 3, "Math Review": Value, Order, Factors, Algebra, Equalities, Inequalities, Rates, Ratios, Percents, Statistics, Sets, Counting, Probability, Estimation, and Series. Quantitative Reasoning questions also test how well you reason mathematically, solve math problems, and interpret graphic data. All the math needed to answer the questions is generally taught in secondary school (or high school) math classes.

To answer a Quantitative Reasoning question, you pick one of five answer choices. First study the question to see what information it gives and what it's asking you to do with that information. Then scan the answer choices. If the problem seems simple, try to find the answer quickly. Then check your answer against the choices. If your answer isn't among the choices, or if the problem is complicated, think again about what the problem is asking you to do. Try to rule out some of the choices. If you still can't narrow down the answer to one choice, reread the question. It gives all the information you need to find the right answer. If you already know about the topic, don't use that knowledge to answer. Use only the information given.

You have 45 minutes to answer the 21 questions in the section. That's an average of about two minutes and nine seconds per question. You may run out of time if you spend too long on any one question. If you find yourself stuck on a question, just pick the answer that seems best, even if you're not sure. Guessing is not usually the best way to score high on the GMAT, but making an educated guess is better than not answering.

Below are question-solving tips, directions for the section, and sample questions with an answer key and answer explanations. These explanations also show strategies that may help you solve other Quantitative Reasoning questions.

4.1 Tips for Answering Quantitative Reasoning Questions

1. **Pace yourself.**

   Check the on-screen timer to see how much time you have left. Work carefully, but don't take too long double-checking an answer or struggling with a problem.

2. **Use the erasable notepad provided.**

   Solving a problem step by step on the notepad may help you avoid mistakes. If the problem doesn't show a diagram, drawing your own may help.

3. **Study each question to understand what it's asking.**

   Approach word problems one step at a time. Read each sentence. When useful, translate the problem into math expressions.

4. **Scan all the answer choices before working on the problem.**

   By scanning the answer choices, you can avoid finding the answer in the wrong form. For example, if the choices are all fractions like $\frac{1}{4}$, work out the answer as a fraction, not as a decimal like 0.25. Similarly, if the choices are all estimates, you can often take shortcuts. For example, you may be able to estimate by rounding 48% to 50% if every answer choice is an estimate.

5. **Don't waste time on a problem that's too hard for you.**

   Make your best guess. Then move on to the next question.

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4.2 Practice Questions

Solve the problem and pick the best answer choice given.

<u>Numbers:</u> All numbers used are real numbers.

Figures: **A figure in a Quantitative Reasoning question gives information useful in solving the problem. Figures are drawn as accurately as possible, except as noted. Lines shown as straight are straight. Lines that look jagged may also be straight. Points, angles, regions, etc., are positioned as shown. All figures are flat unless otherwise noted.**

Questions 1 to 82 — Difficulty: Easy

1. Working at a constant rate, a copy machine makes 20 copies of a one-page document per minute. If the machine works at this constant rate, how many hours does it take to make 4,800 copies of a one-page document?

   A. 4

   B. 5

   C. 6

   D. 7

   E. 8

2. If $x + y = 2$ and $x^2 + y^2 = 2$, what is the value of xy?

   A. –2

   B. –1

   C. 0

   D. 1

   E. 2

3. The sum S of the first n consecutive positive even integers is given by $S = n(n + 1)$. For what value of n is this sum equal to 110?

   A. 10

   B. 11

   C. 12

D. 13

E. 14

4. 6(87.30 + 0.65) − 5(87.30) =

A. 3.90

B. 39.00

C. 90.90

D. 91.20

E. 91.85

5. What is the value of $x^2yz - xyz^2$, if $x = -2$, $y = 1$, and $z = 3$?

A. 20

B. 24

C. 30

D. 32

E. 48

6. If T is a set of 35 consecutive integers, of which 17 are negative, what is the sum of all the integers in T?

A. –2

B. –1

C. 0

D. 1

E. 2

7. If $x > y$ and $y > z$, which of the following represents the greatest number?

A. $x - z$

B. $x - y$

C. $y - x$

D. $z - y$

E. $z - x$

8. 38, 69, 22, 73, 31, 47, 13, 82

Which of the following numbers is greater than three-fourths of the numbers but less than one-fourth of the numbers in the list above?

A. 56

B. 68

C. 69

D. 71

E. 73

9. A rug manufacturer produces rugs at a cost of $75 per rug. What is the manufacturer's gross profit from the sale of 150 rugs if $\frac{2}{3}$ of the rugs are sold for $150 per rug and the rest are sold for $200 per rug?

A. $10,350

B. $11,250

C. $13,750

D. $16,250

E. $17,800

10. The value of Maureen's investment portfolio has decreased by 5.8 percent since her initial investment in the portfolio. If her initial investment was $16,800, what is the current value of the portfolio?

A. $7,056.00

B. $14,280.00

C. $15,825.60

D. $16,702.56

E. $17,774.40

11. Company C produces toy trucks at a cost of $5.00 each for the first 100 trucks and $3.50 for each additional truck. If 500 toy trucks were produced by Company C and sold for $10.00 each, what was Company C's gross profit?

   A. $2,250

   B. $2,500

   C. $3,100

   D. $3,250

   E. $3,500

Division	Profit or Loss (in millions of dollars)				
	1991	1992	1993	1994	1995
A	1.1	(3.4)	1.9	2.0	0.6
B	(2.3)	5.5	(4.5)	3.9	(2.9)
C	10.0	(6.6)	5.3	1.1	(3.0)

12. The annual profit or loss for the three divisions of Company T for the years 1991 through 1995 are summarized in the table shown, where losses are enclosed in parentheses. For which division and which three consecutive years shown was the division's profit or loss for the three-year period closest to $0?

   A. Division A for 1991–1993

   B. Division A for 1992–1994

   C. Division B for 1991–1993

   D. Division B for 1993–1995

   E. Division C for 1992–1994

13. A computer appliance dealer bought a computer at a wholesale price of \$500 and then sold it at a 20 percent discount off the suggested retail price. If the dealer made a 12 percent profit on the wholesale price, what was the suggested retail price of the computer?

   A. \$560

   B. \$580

   C. \$660

   D. \$700

   E. \$730

14. Pat invested x dollars in a fund that paid 8 percent annual interest, compounded annually. Which of the following represents the value, in dollars, of Pat's investment plus interest at the end of 5 years?

   A. $5(0.08x)$

   B. $5(1.08x)$

   C. $[1 + 5(0.08)]x$

   D. $(1.08)^5 x$

   E. $(1.08x)^5$

15. There are five sales agents in a certain real estate office. One month Andy sold twice as many properties as Ellen, Bob sold 3 more than Ellen, Cary sold twice as many as Bob, and Dora sold as many as Bob and Ellen together. Who sold the most properties that month?

   A. Andy

   B. Bob

   C. Cary

   D. Dora

   E. Ellen

16. In a field day at a school, each child who competed in n events and scored a total of p points was given an overall score of $\frac{p}{n} + n$. Andrew competed in 1 event and scored 9 points. Jason competed in 3 events and scored 5, 6, and 7 points, respectively. What was the ratio of Andrew's overall score to Jason's overall score?

   A. $\frac{10}{23}$

   B. $\frac{7}{10}$

   C. $\frac{4}{5}$

   D. $\frac{10}{9}$

   E. $\frac{12}{7}$

17. A certain work plan for September requires that a work team, working every day, produce an average of 200 items per day. For the first half of the month, the team produced an average of 150 items per day. How many items per day must the team average during the second half of the month if it is to attain the average daily production rate required by the work plan?

   A. 225

   B. 250

   C. 275

   D. 300

   E. 350

18. A company sells radios for $15.00 each. It costs the company $14.00 per radio to produce 1,000 radios and $13.50 per radio to produce 2,000 radios. How much greater will the company's gross profit be

from the production and sale of 2,000 radios than from the production and sale of 1,000 radios?

A. \$500

B. \$1,000

C. \$1,500

D. \$2,000

E. \$2,500

19. Working at their respective constant rates, machine A makes 100 copies in 12 minutes and machine B makes 150 copies in 10 minutes. If these machines work simultaneously at their respective rates for 30 minutes, what is the total number of copies that they will produce?

A. 250

B. 425

C. 675

D. 700

E. 750

20. Company X mailed out a questionnaire. Sixty percent of those who received the questionnaire by mail responded. Company X needed 300 responses. What is the minimum number of questionnaires that Company X should have mailed in order to get the necessary responses?

A. 400

B. 420

C. 480

D. 500

E. 600

21. The toll T, in dollars, for a truck using a certain bridge is given by the formula $T = 1.50 + 0.50(x - 2)$, where x is the number of axles on

the truck. What is the toll for an 18-wheel truck that has 2 wheels on its front axle and 4 wheels on each of its other axles?

A. \$2.50

B. \$3.00

C. \$3.50

D. \$4.00

E. \$5.00

22. For what value of x between –4 and 4, inclusive, is the value of $x^2 - 10x + 16$ the greatest?

A. –4

B. –2

C. 0

D. 2

E. 4

23. If $x = -\frac{5}{8}$ and $y = -\frac{1}{2}$, what is the value of the expression $-2x - y^2$?

A. $-\frac{3}{2}$

B. –1

C. 1

D. $\frac{3}{2}$

E. $\frac{7}{4}$

24. At the opening of a trading day, stock X had a value of \$40.00 per share and stock Y had a value of \$25.00 per share. At the closing of the trading day, stock X had a value of \$41.00 per share and stock Y had a

value of \$28.50 per share. For this trading day, the percent increase in the value of stock Y was how many percentage points greater than the percent increase in the value of stock X?

A. 2.5

B. 6.25

C. 10

D. 11.5

E. 12.5

25. For positive integers a and b, the remainder when a is divided by b is equal to the remainder when b is divided by a. Which of the following could be a value of ab?

I. 24

II. 30

III. 36

A. II only

B. III only

C. I and II only

D. II and III only

E. I, II, and III

26. List S consists of the positive integers that are multiples of 9 and are less than 100. What is the median of the integers in S?

A. 36

B. 45

C. 49

D. 54

E. 63

27. A rope 20.6 meters long is cut into two pieces. If the length of one piece of rope is 2.8 meters shorter than the length of the other, what is

the length, in meters, of the longer piece of rope?

A. 7.5

B. 8.9

C. 9.9

D. 10.3

E. 11.7

28. If x and y are integers and $x - y$ is odd, which of the following must be true?

I. xy is even.

II. $x^2 + y^2$ is odd.

III. $(x + y)^2$ is even.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

29. On Monday, the opening price of a certain stock was $100 per share and its closing price was $110 per share. On Tuesday the closing price of the stock was 10 percent less than its closing price on Monday, and on Wednesday the closing price of the stock was 4 percent greater than its closing price on Tuesday. What was the approximate percent change in the price of the stock from its opening price on Monday to its closing price on Wednesday?

A. A decrease of 6%

B. A decrease of 4%

C. A decrease of 1%

D. An increase of 3%

E. An increase of 4%

30. $1 - 0.000001 =$

A. (1.01)(0.99)

B. (1.11)(0.99)

C. (1.001)(0.999)

D. (1.111)(0.999)

E. (1.0101)(0.0909)

31. In a certain history class of 17 juniors and seniors, each junior has written 2 book reports and each senior has written 3 book reports. If the 17 students have written a total of 44 book reports, how many juniors are in the class?

A. 7

B. 8

C. 9

D. 10

E. 11

32. $|-4|(|-20| - |5|) =$

A. −100

B. −60

C. 60

D. 75

E. 100

33. What is the greatest integer k for which 32.45×10^k is less than 1?

A. –2

B. 1

C. 0

D. 1

E. 2

34. How many integers x satisfy both $2 < x \leq 4$ and $0 \leq x \leq 3$?

A. 5

B. 4

C. 3

D. 2

E. 1

35. At the opening of a trading day at a certain stock exchange, the price per share of stock K was $8. If the price per share of stock K was $9 at the closing of the day, what was the percent increase in the price per share of stock K for that day?

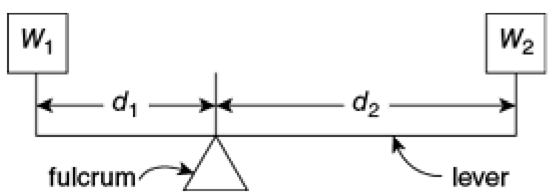
A. 1.4%

B. 5.9%

C. 11.1%

D. 12.5%

E. 23.6%



36. As shown in the diagram above, a lever resting on a fulcrum has weights of w_1 pounds and w_2 pounds, located d_1 feet and d_2 feet from the fulcrum. The lever is balanced and $w_1 d_1 = w_2 d_2$. Suppose w_1 is 50 pounds and w_2 is 30 pounds. If d_1 is 4 feet less than d_2, what is d_2, in feet?

A. 1.5

B. 2.5

C. 6

D. 10

E. 20

Day	Change in Dollars
Monday	$+1\frac{1}{2}$
Tuesday	$-\frac{3}{4}$
Wednesday	0
Thursday	$-\frac{1}{8}$
Friday	$+2\frac{1}{4}$

37. The table above shows the daily change in the price of a certain stock last week. What was the net change in dollars in the price of the stock for the week?

A. $-4\frac{5}{8}$

B. $-2\frac{7}{8}$

C. $+2\frac{7}{8}$

D. $+3\frac{3}{4}$

E. $-4\frac{5}{8}$

38. Of 120 hotel rooms rented one night, some were suites rented for \$115 each and the rest were double rooms rented for \$85 each. If the total revenue from the room rentals for this night was \$10,890, how many suites were rented?

A. 23

B. 35

C. 54

D. 94

E. 99

39. Last week Jack worked 70 hours and earned \$1,260. If he earned his regular hourly wage for the first 40 hours worked, $1\frac{1}{2}$ times his regular hourly wage for the next 20 hours worked, and 2 times his regular hourly wage for the remaining 10 hours worked, what was his regular hourly wage?

A. \$7.00

B. \$14.00

C. \$18.00

D. \$22.00

E. \$31.50

40. If $\frac{x}{4}$ is 2 more than $\frac{x}{8}$, then $x =$

A. 4

B. 8

C. 16

D. 32

E. 64

41. If $(2^x)(2^y) = 8$ and $(9^x)(3^y) = 81$, then $(x, y) =$

A. (1, 2)

B. (2, 1)

C. (1, 1)

D. (2, 2)

E. (1, 3)

42. A certain toll station on a highway has 7 tollbooths, and each tollbooth collects $0.75 from each vehicle that passes it. From 6 o'clock yesterday morning to 12 o'clock midnight, vehicles passed each of the tollbooths at the average rate of 4 vehicles per minute. Approximately how much money did the toll station collect during that time period?

A. $1,500

B. $3,000

C. $11,500

D. $23,000

E. $30,000

43. If $y = 2 + 2k$ and $y \neq 0$, then $\frac{1}{y} + \frac{1}{y} + \frac{1}{y} + \frac{1}{y} =$

A. $\frac{1}{8+8k}$

B. $\frac{2}{1+k}$

C. $\frac{1}{8+k}$

D. $\frac{4}{8+k}$

E. $\frac{4}{1+k}$

44. If $3^x - 3^{x-1} = 162$, then $x(x-1) =$

A. 12

B. 16

C. 20

D. 30

E. 81

45. The expression $n!$ is defined as the product of the integers from 1 through n. If p is the product of the integers from 100 through 299 and q is the product of the integers from 200 through 299, which of the following is equal to $\frac{p}{q}$?

A. 99!

B. 199!

C. $\frac{199!}{99!}$

D. $\frac{299!}{99!}$

E. $\frac{299!}{199!}$

46. A school club plans to package and sell dried fruit to raise money. The club purchased 12 containers of dried fruit, each containing $16\frac{3}{4}$ pounds. What is the maximum number of individual bags of dried fruit, each containing $\frac{1}{4}$ pounds, that can be sold from the dried fruit the club purchased?

A. 50

B. 64

C. 67

D. 768

E. 804

Height	Price
Less than 5 ft	$14.95
5 ft to 6 ft	$17.95
Over 6 ft	$21.95

47. A nursery sells fruit trees priced as shown in the chart above. In its inventory 54 trees are less than 5 feet in height. If the expected revenue from the sale of its entire stock is estimated at $2,450, approximately how much of this will come from the sale of trees that are at least 5 feet tall?

    A. $1,730

    B. $1,640

    C. $1,410

    D. $1,080

    E. $810

48. A certain bridge is 4,024 feet long. Approximately how many minutes does it take to cross this bridge at a constant speed of 20 miles per hour? (1 mile = 5,280 feet)

    A. 1

    B. 2

    C. 4

    D. 6

    E. 7

49. A purse contains 57 coins, all of which are nickels, dimes, or quarters. If the purse contains x dimes and 8 more nickels than dimes, which of the following gives the number of quarters the purse contains in terms of x?

A. $2x - 49$

B. $2x + 49$

C. $2x - 65$

D. $49 - 2x$

E. $65 - 2x$

50. The annual interest rate earned by an investment increased by 10 percent from last year to this year. If the annual interest rate earned by the investment this year was 11 percent, what was the annual interest rate last year?

A. 1%

B. 1.1%

C. 9.1%

D. 10%

E. 10.8%

51. A total of 5 liters of gasoline is to be poured into two empty containers with capacities of 2 liters and 6 liters, respectively, such that both containers will be filled to the same percent of their respective capacities. What amount of gasoline, in liters, must be poured into the 6-liter container?

A. $4\frac{1}{2}$

B. 4

C. $3\frac{3}{4}$

D. 3

E. $1\frac{1}{4}$

52. What is the larger of the 2 solutions of the equation $x^2 - 4x = 96$?

A. 8

B. 12

C. 16

D. 32

E. 100

$$x = \frac{1}{6}gt^2$$

53. In the formula shown, if g is a constant and $x = -6$ when $t = 2$, what is the value of x when $t = 4$?

A. –24

B. –20

C. –15

D. 20

E. 24

54. $\frac{(39{,}897)(0.0096)}{198.76}$ is approximately

A. 0.02

B. 0.2

C. 2

D. 20

E. 200

55. The “prime sum” of an integer n greater than 1 is the sum of all the prime factors of n, including repetitions. For example, the prime sum of 12 is 7, since 12 = 2 × 2 × 3 and 2 + 2 + 3 = 7. For which of the following integers is the prime sum greater than 35?

A. 440

B. 512

C. 620

D. 700

E. 750

56. Each machine at a toy factory assembles a certain kind of toy at a constant rate of one toy every 3 minutes. If 40 percent of the machines at the factory are to be replaced by new machines that assemble this kind of toy at a constant rate of one toy every 2 minutes, what will be the percent increase in the number of toys assembled in one hour by all the machines at the factory, working at their constant rates?

A. 20%

B. 25%

C. 30%

D. 40%

E. 50%

57. When a subscription to a new magazine was purchased for m months, the publisher offered a discount of 75 percent off the regular monthly price of the magazine. If the total value of the discount was equivalent to buying the magazine at its regular monthly price for 27 months, what was the value of m?

A. 18

B. 24

C. 30

D. 36

E. 48

58. At a garage sale, all of the prices of the items sold were different. If the price of a radio sold at the garage sale was both the 15th highest price and the 20th lowest price among the prices of the items sold, how many items were sold at the garage sale?

A. 33

B. 34

C. 35

D. 36

E. 37

59. Half of a large pizza is cut into 4 equal-sized pieces, and the other half is cut into 6 equal-sized pieces. If a person were to eat 1 of the larger pieces and 2 of the smaller pieces, what fraction of the pizza would remain uneaten?

A. $\frac{5}{12}$

B. $\frac{13}{24}$

C. $\frac{7}{12}$

D. $\frac{2}{3}$

E. $\frac{17}{24}$

60. If $a = 1 + \frac{1}{4} + \frac{1}{16} + \frac{1}{64}$ and $b = 1 + \frac{1}{4}a$, then what is the value of $a - b$?

A. $-\frac{85}{256}$

B. $-\frac{1}{256}$

C. $-\frac{1}{4}$

D. $\frac{125}{256}$

E. $\frac{169}{256}$

61. In a certain learning experiment, each participant had three trials and was assigned, for each trial, a score of either –2, –1, 0, 1, or 2. The participant's final score consisted of the sum of the first trial score, 2 times the second trial score, and 3 times the third trial score. If Anne received scores of 1 and –1 for her first two trials, not necessarily in that order, which of the following could NOT be her final score?

A. –4

B. –2

C. 1

D. 5

E. 6

62. For all positive integers m and v, the expression $m \Theta v$ represents the remainder when m is divided by v. What is the value of $((98 \Theta 33) \Theta 17) - (98 \Theta (33 \Theta 17))$?

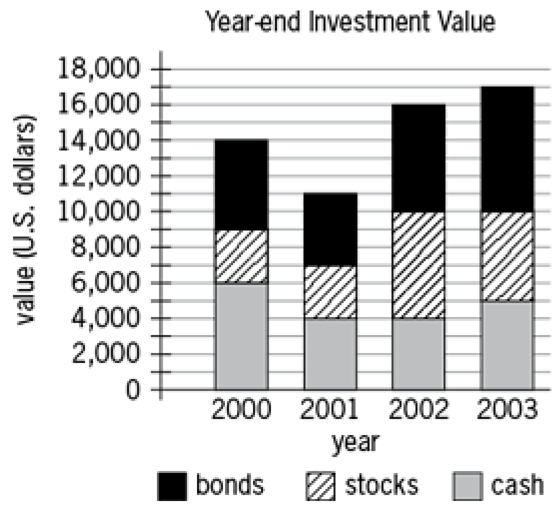
A. −10

B. −2

C. 8

D. 13

E. 17



63. The chart above shows year-end values for Darnella's investments. For just the stocks, what was the increase in value from year-end 2000 to year-end 2003?

A. $1,000

B. $2,000

C. $3,000

D. $4,000

E. $5,000

64. If the sum of the reciprocals of two consecutive odd integers is $\frac{12}{35}$, then the greater of the two integers is

A. 3

B. 5

C. 7

D. 9

E. 11

65. What is the sum of the odd integers from 35 to 85, inclusive?

A. 1,560

B. 1,500

C. 1,240

D. 1,120

E. 1,100

66. For all numbers a, b, c, and d, $\begin{vmatrix} a & b \\ c & d \end{vmatrix}$ is defined by the equation $\begin{vmatrix} a & b \\ c & d \end{vmatrix} = ad - cb$. Which of the following is equal to $\begin{vmatrix} s & t \\ 1 & 3 \end{vmatrix} - \begin{vmatrix} -t & 2 \\ s & 4 \end{vmatrix} + \begin{vmatrix} 2 & 2 \\ t & s \end{vmatrix}$?

A. $\begin{vmatrix} s & t \\ 1 & 5 \end{vmatrix}$

B. $\begin{vmatrix} s & t \\ 7 & 1 \end{vmatrix}$

C. $\begin{vmatrix} s & t \\ 5 & 7 \end{vmatrix}$

D. $\begin{vmatrix} s & -t \\ 1 & 5 \end{vmatrix}$

E. $\begin{vmatrix} s & -t \\ 1 & 7 \end{vmatrix}$

67. In a certain sequence, each term after the first term is one-half the previous term. If the tenth term of the sequence is between 0.0001 and 0.001, then the twelfth term of the sequence is between

A. 0.0025 and 0.025

B. 0.00025 and 0.0025

C. 0.000025 and 0.00025

D. 0.0000025 and 0.000025

E. 0.00000025 and 0.0000025

68. A certain drive-in movie theater has a total of 17 rows of parking spaces. There are 20 parking spaces in the first row and 21 parking spaces in the second row. In each subsequent row there are 2 more parking spaces than in the previous row. What is the total number of parking spaces in the movie theater?

A. 412

B. 544

C. 596

D. 632

E. 692

69. Ada and Paul received their scores on three tests. On the first test, Ada's score was 10 points higher than Paul's score. On the second test, Ada's score was 4 points higher than Paul's score. If Paul's average (arithmetic mean) score on the three tests was 3 points higher than Ada's average score on the three tests, then Paul's score on the third test was how many points higher than Ada's score?

A. 9

B. 14

C. 17

D. 23

E. 25

70. The price of a certain stock increased by 0.25 of 1 percent on a certain day. By what fraction did the price of the stock increase that day?

A. $\frac{1}{2{,}500}$

B. $\frac{1}{400}$

C. $\frac{1}{40}$

D. $\frac{1}{25}$

E. $\frac{1}{4}$

71. For each trip, a taxicab company charges \$4.25 for the first mile and \$2.65 for each additional mile or fraction thereof. If the total charge for a certain trip was \$62.55, how many miles at most was the trip?

A. 21

B. 22

C. 23

D. 24

E. 25

72. When 24 is divided by the positive integer n, the remainder is 4. Which of the following statements about n must be true?

I. n is even.

II. n is a multiple of 5.

III. n is a factor of 20.

A. III only

B. I and II only

C. I and III only

D. II and III only

E. I, II, and III

73. Terry needs to purchase some pipe for a plumbing job that requires pipes with lengths of 1 ft 4 in, 2 ft 8 in, 3 ft 4 in, 3 ft 8 in, 4 ft 8 in, 5 ft 8 in, and 9 ft 4 in. The store from which Terry will purchase the pipe sells pipe only in 10-ft lengths. If each 10-ft length can be cut into shorter pieces, what is the minimum number of 10-ft pipe lengths that Terry needs to purchase for the plumbing job?

(Note: 1 ft = 12 in)

A. 3

B. 4

C. 5

D. 6

E. 7

74. What is the thousandths digit in the decimal equivalent of $\frac{53}{5{,}000}$?

A. 0

B. 1

C. 3

D. 5

E. 6

75. If a equals the sum of the even integers from 2 to 20, inclusive, and b equals the sum of the odd integers from 1 to 19, inclusive, what is the value of $a - b$?

A. 1

B. 10

C. 19

D. 20

E. 21

76. If a, b, c, and d are consecutive even integers and $a < b < c < d$, then $a + b$ is how much less than $c + d$?

A. 2

B. 4

C. 6

D. 8

E. 10

77. A retailer sold an appliance for \$80. If the retailer's gross profit on the appliance was 25 percent of the retailer's cost for the appliance, how many dollars was the retailer's gross profit?

A. \$10

B. \$16

C. \$20

D. \$24

E. \$25

78. Beth has a collection of 8 boxes of clothing for a charity, and the average (arithmetic mean) number of pieces of clothing per box is c. If she replaces a box in the collection that contains 12 pieces of clothing with a box that contains 22 pieces of clothing, what is the average number of pieces of clothing per box for the new collection, in terms of c?

A. $c - \frac{5}{4}$

B. $c + \frac{5}{4}$

C. $8 - \frac{10}{c}$

D. $8 + \frac{10}{c}$

E. $8c - 10$

$$\frac{1 + 0.0001}{0.04 + 10}$$

79. The value of the expression above is closest to which of the following?

A. 0.0001

B. 0.001

C. 0.01

D. 1

E. 10

80. If $x + 1 = t$ and $t = 3 - x$, then $x =$

A. −2

B. −1

C. 0

D. 1

E. 2

81. If $x = kc$ and $y = kt$, then $y - x =$

A. $k(t - c)$

B. $k(c - t)$

C. $c(k - t)$

D. $t(k - c)$

E. $k(1 - t)$

82. If k is a positive even integer, which of the following must be an odd integer?

I. $k^2 - 3k + 4$

II. $k^5 + 3$

III. $7k - 7$

A. II only

B. III only

C. I and III only

D. II and III only

E. I, II, and III

Questions 83 to 151 — Difficulty: Medium

83. If $\frac{1}{2}$ the result obtained when 2 is subtracted from $5x$ is equal to the sum of 10 and $3x$, what is the value of x?

A. –22

B. –4

C. 4

D. 18

E. 22

84. If Car A took n hours to travel 2 miles and Car B took m hours to travel 3 miles, which of the following expresses the time it would take Car C, traveling at the average (arithmetic mean) of those rates, to travel 5 miles?

A. $\frac{10nm}{3n+2m}$

B. $\frac{3n+2m}{10(n+m)}$

C. $\frac{2n+3m}{5nm}$

D. $\frac{10(n+m)}{2n+3m}$

E. $\frac{5(n+m)}{2n+3m}$

85. If x, y, and k are positive and x is less than y, then $\frac{x+k}{y+k}$ is

A. 1

B. greater than $\frac{x}{y}$

C. equal to $\frac{x}{y}$

D. less than $\frac{x}{y}$

E. less than $\frac{x}{y}$ or greater than $\frac{x}{y}$, depending on the value of k

86. Consider the following set of inequalities: $p > q$, $s > r$, $q > t$, $s > p$, and $r > q$. Between which two quantities is no relationship established?

A. p and r

B. s and t

C. s and q

D. p and t

E. r and t

87. Carl averaged $2m$ miles per hour on a trip that took him h hours. If Ruth made the same trip in $\frac{2}{3}h$ hours, what was her average speed in miles per hour?

A. $\frac{1}{3}mh$

B. $\frac{2}{3}mh$

C. m

D. $\frac{3}{2}m$

E. $3m$

88. Of three persons, two take relish, two take pepper, and two take salt. The one who takes no salt takes no pepper, and the one who takes no pepper takes no relish. Which of the following statements must be true?

I. The person who takes no salt also takes no relish.

II. Any of the three persons who takes pepper also takes relish and salt.

III. The person who takes no relish is not one of those who takes salt.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

89. If the smaller of 2 consecutive odd integers is a multiple of 5, which of the following could NOT be the sum of these 2 integers?

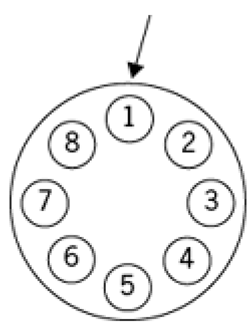
A. −8

B. 12

C. 22

D. 52

E. 252



90. Eight light bulbs numbered 1 through 8 are arranged in a circle as shown above. The bulbs are wired so that every third bulb, counting in a clockwise direction, flashes until all bulbs have flashed once. If the bulb numbered 1 flashes first, which numbered bulb will flash last?

A. 2

B. 3

C. 4

D. 6

E. 7

Closing Prices of Stock X
During a Certain Week
(in dollars)

Monday	Tuesday	Wednesday	Thursday	Friday
21	19	22	$24\frac{1}{2}$	23

91. A certain financial analyst defines the “volatility” of a stock during a given week to be the result of the following procedure: find the absolute value of the difference in the stock’s closing price for each pair of consecutive days in the week and then find the average (arithmetic mean) of these 4 values. What is the volatility of Stock X during the week shown in the table?

A. 0.50

B. 1.80

C. 2.00

D. 2.25

E. 2.50

92. If $y = \dfrac{|3x - 5|}{-x^2 - 3}$, for what value of x will the value of y be greatest?

A. -5

B. $-\dfrac{3}{5}$

C. 0

D. $\frac{3}{5}$

E. $\frac{5}{3}$

93. What values of x have a corresponding value of y that satisfies both $xy > 0$ and $xy = x + y$?

A. $x \leq -1$

B. $-1 < x \leq 0$

C. $0 < x \leq 1$

D. $x > 1$

E. All real numbers

94. Employee X's annual salary is \$12,000 more than half of Employee Y's annual salary. Employee Z's annual salary is \$15,000 more than half of Employee X's annual salary. If Employee X's annual salary is \$27,500, which of the following lists these three people in order of increasing annual salary?

A. Y, Z, X

B. Y, X, Z

C. Z, X, Y

D. X, Y, Z

E. X, Z, Y

$$C = \begin{cases} 0.10s, \text{ if } s \leq 60{,}000 \\ 0.10s + 0.04(s - 60{,}000), \text{ if } s > 60{,}000 \end{cases}$$

95. The formula above gives the contribution C, in dollars, to a certain profit-sharing plan for a participant with a salary of s dollars. How many more dollars is the contribution for a participant with a salary of \$70,000 than for a participant with a salary of \$50,000?

A. \$800

B. \$1,400

C. \$2,000

D. \$2,400

E. \$2,800

96. Next month, Ron and Cathy will each begin working part-time at $\frac{3}{5}$ of their respective current salaries. If the sum of their reduced salaries will be equal to Cathy's current salary, then Ron's current salary is what fraction of Cathy's current salary?

A. $\frac{1}{3}$

B. $\frac{2}{5}$

C. $\frac{1}{2}$

D. $\frac{3}{5}$

E. $\frac{2}{3}$

97. David and Ron are ordering food for a business lunch. David thinks that there should be twice as many sandwiches as there are pastries, but Ron thinks the number of pastries should be 12 more than one-fourth of the number of sandwiches. How many sandwiches should be ordered so that David and Ron can agree on the number of pastries to order?

A. 12

B. 16

C. 20

D. 24

E. 48

98. The cost of purchasing each box of candy from a certain mail order catalog is v dollars per pound of candy, plus a shipping charge of h dollars. How many dollars does it cost to purchase 2 boxes of candy, one containing s pounds of candy and the other containing t pounds of candy, from this catalog?

A. $h + stv$

B. $2h + stv$

C. $2hstv$

D. $2h + s + t + v$

E. $2h + v(s + t)$

99. If $x \neq -\frac{1}{2}$, then $\frac{6x^3 + 3x^2 - 8x - 4}{2x + 1} =$

A. $3x^2 + \frac{3}{2}x - 8$

B. $3x^2 + \frac{3}{2}x - 4$

C. $3x^2 - 4$

D. $3x - 4$

E. $3x + 4$

100. If $x^2 + bx + 5 = (x + c)^2$ for all numbers x, where b and c are positive constants, what is the value of b?

A. $\sqrt{5}$

B. $\sqrt{10}$

C. $2\sqrt{5}$

D. $2\sqrt{10}$

E. 10

101. Last year Shannon listened to a certain public radio station 10 hours per week and contributed $35 to the station. Of the following, which is closest to Shannon's contribution per minute of listening time last year?

A. $0.001

B. $0.010

C. $0.025

D. $0.058

E. $0.067

102. Each of the 20 employees at Company J is to receive an end-of-year bonus this year. Agnes will receive a larger bonus than any other employee, but only $500 more than Cheryl will receive. None of the employees will receive a smaller bonus than Cheryl. If the amount of money to be distributed in bonuses at Company J this year totals $60,000, what is the largest bonus Agnes can receive?

A. $3,250

B. $3,325

C. $3,400

D. $3,475

E. $3,500

103. Beth, Naomi, and Juan raised a total of $55 for charity. Naomi raised $5 less than Juan, and Juan raised twice as much as Beth. How much did Beth raise?

A. $9

B. $10

C. $12

D. $13

E. $15

104. The set of solutions for the equation $\left(x^2-25\right)^2=x^2-10x+25$ contains how many real numbers?

A. 0

B. 1

C. 2

D. 3

E. 4

105. An aerosol can is designed so that its bursting pressure, B, in pounds per square inch, is 120% of the pressure, F, in pounds per square inch, to which it is initially filled. Which of the following formulas expresses the relationship between B and F?

A. $B=1.2F$

B. $B=120F$

C. $B=1+0.2F$

D. $B=\dfrac{F}{1.2}$

E. $B=\dfrac{1.2}{F}$

106. The average (arithmetic mean) of the positive integers x, y, and z is 3. If $x<y<z$, what is the greatest possible value of z?

A. 5

B. 6

C. 7

D. 8

E. 9

107. The product of 3,305 and the 1-digit integer x is a 5-digit integer. The units (ones) digit of the product is 5 and the hundreds digit is y. If A is

the set of all possible values of x and B is the set of all possible values of y, then which of the following gives the members of A and B?

	$\underline{A}$	$\underline{B}$
(A)	{1, 3, 5, 7, 9}	{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
(B)	{1, 3, 5, 7, 9}	{1, 3, 5, 7, 9}
(C)	{3, 5, 7, 9}	{1, 5, 7, 9}
(D)	{5, 7, 9}	{1, 5, 7}
(E)	{5, 7, 9}	{1, 5, 9}

108. If x and y are integers such that $2 < x \leq 8$ and $2 < y \leq 9$, what is the maximum value of $\frac{1}{x} - \frac{x}{y}$?

A. $-3\frac{1}{8}$

B. 0

C. $\frac{1}{4}$

D. $\frac{5}{18}$

E. 2

109. Items that are purchased together at a certain discount store are priced at \$3 for the first item purchased and \$1 for each additional item purchased. What is the maximum number of items that could be purchased together for a total price that is less than \$30?

A. 25

B. 26

C. 27

D. 28

E. 29

110. What is the least integer z for which $(0.000125)(0.0025)(0.00000125) \times 10^z$ is an integer?

A. 18

B. 10

C. 0

D. −10

E. −18

111. The average (arithmetic mean) length per film for a group of 21 films is t minutes. If a film that runs for 66 minutes is removed from the group and replaced by one that runs for 52 minutes, what is the average length per film, in minutes, for the new group of films, in terms of t?

A. $t + \frac{2}{3}$

B. $t - \frac{2}{3}$

C. $21t + 14$

D. $t + \frac{3}{2}$

E. $t - \frac{3}{2}$

112. A garden center sells a certain grass seed in 5-pound bags at $13.85 per bag, 10-pound bags at $20.43 per bag, and 25-pound bags at $32.25 per bag. If a customer is to buy at least 65 pounds of the grass seed, but no more than 80 pounds, what is the least possible cost of the grass seed that the customer will buy?

A. $94.03

B. $96.75

C. $98.78

D. $102.07

E. $105.36

113. If $x = -|w|$, which of the following must be true?

A. $x = -w$

B. $x = w$

C. $x^2 = w$

D. $x^2 = w^2$

E. $x^3 = w^3$

114. A certain financial institution reported that its assets totaled $2,377,366.30 on a certain day. Of this amount, $31,724.54 was held in cash. Approximately what percent of the reported assets was held in cash on that day?

A. 0.00013%

B. 0.0013%

C. 0.013%

D. 0.13%

E. 1.3%

$$\begin{array}{r} AB \\ +\ BA \\ \hline AAC \end{array}$$

115. In the correctly worked addition problem shown, where the sum of the two-digit positive integers AB and BA is the three-digit integer AAC, and A, B, and C are different digits, what is the units digit of the integer AAC?

A. 9

B. 6

C. 3

D. 2

E. 0

116. The hard drive, monitor, and printer for a certain desktop computer system cost a total of $2,500. The cost of the printer and monitor together is equal to $\frac{2}{3}$ of the cost of the hard drive. If the cost of the printer is $100 more than the cost of the monitor, what is the cost of the printer?

A. $800

B. $600

C. $550

D. $500

E. $350

$$3r \leq 4s + 5$$
$$|\mathrm{s}| \leq 5$$

117. Given the inequalities above, which of the following CANNOT be the value of r?

A. −20

B. −5

C. 0

D. 5

E. 20

118. If m is an even integer, v is an odd integer, and $m > v > 0$, which of the following represents the number of even integers less than m and greater than v?

A. $\dfrac{m-v}{2}-1$

B. $\dfrac{m-v-1}{2}$

C. $\dfrac{m-v}{2}$

D. $m-v-1$

E. $m-v$

119. A positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9. If n is a positive integer, for which of the following values of k is $25\times10^n+k\times10^{2n}$ divisible by 9?

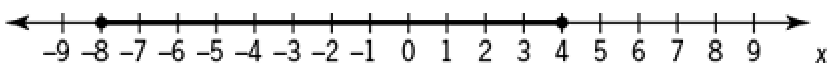
A. 9

B. 16

C. 23

D. 35

E. 47



120. On the number line, the shaded interval is the graph of which of the following inequalities?

A. $|x|\le4$

B. $|x|\le8$

C. $|x-2|\le4$

D. $|x-2|\le6$

E. $|x+2|\le6$

121. Last year members of a certain professional organization for teachers consisted of teachers from 49 different school districts, with an average (arithmetic mean) of 9.8 schools per district. Last year the

average number of teachers at these schools who were members of the organization was 22. Which of the following is closest to the total number of members of the organization last year?

A. 10^7

B. 10^6

C. 10^5

D. 10^4

E. 10^3

122. Of all the students in a certain dormitory, $\frac{1}{2}$ are first-year students and the rest are second-year students. If $\frac{4}{5}$ of the first-year students have not declared a major and if the fraction of second-year students who have declared a major is 3 times the fraction of first-year students who have declared a major, what fraction of all the students in the dormitory are second-year students who have not declared a major?

A. $\frac{1}{15}$

B. $\frac{1}{5}$

C. $\frac{4}{15}$

D. $\frac{1}{3}$

E. $\frac{2}{5}$

123. If the average (arithmetic mean) of x, y, and z is $7x$ and $x \neq 0$, what is the ratio of x to the sum of y and z?

A. 1:21

B. 1:20

C. 1:6

D. 6:1

E. 20:1

124. Jonah drove the first half of a 100-mile trip in x hours and the second half in y hours. Which of the following is equal to Jonah's average speed, in miles per hour, for the entire trip?

A. $\frac{50}{x+y}$

B. $\frac{100}{x+y}$

C. $\frac{25}{x}+\frac{25}{y}$

D. $\frac{50}{x}+\frac{50}{y}$

E. $\frac{100}{x}+\frac{100}{y}$

125. If the amount of federal estate tax due on an estate valued at $1.35 million is $437,000 plus 43 percent of the value of the estate in excess of $1.25 million, then the federal tax due is approximately what percent of the value of the estate?

A. 30%

B. 35%

C. 40%

D. 45%

E. 50%

$$7x+6y \leq 38{,}000$$
$$4x+5y \leq 28{,}000$$

126. A manufacturer wants to produce x balls and y boxes. Resource constraints require that x and y satisfy the inequalities shown. What is the maximum number of balls and boxes combined that can be produced given the resource constraints?

A. 5,000

B. 6,000

C. 7,000

D. 8,000

E. 10,000

127. If $\frac{3}{10^4} = x\%$, then $x =$

A. 0.3

B. 0.03

C. 0.003

D. 0.0003

E. 0.00003

128. What is the remainder when 3^{24} is divided by 5?

A. 0

B. 1

C. 2

D. 3

E. 4

129. José has a collection of 100 coins, consisting of nickels, dimes, quarters, and half-dollars. If he has a total of 35 nickels and dimes, a total of 45 dimes and quarters, and a total of 50 nickels and quarters, how many half-dollars does he have?

A. 15

B. 20

C. 25

D. 30

E. 35

130. David used part of $100,000 to purchase a house. Of the remaining portion, he invested $\frac{1}{3}$ of it at 4 percent simple annual interest and $\frac{2}{3}$ of it at 6 percent simple annual interest. If after a year the income from the two investments totaled $320, what was the purchase price of the house?

A. $96,000

B. $94,000

C. $88,000

D. $75,000

E. $40,000

131. A certain manufacturer sells its product to stores in 113 different regions worldwide, with an average (arithmetic mean) of 181 stores per region. If last year these stores sold an average of 51,752 units of the manufacturer's product per store, which of the following is closest to the total number of units of the manufacturer's product sold worldwide last year?

A. 10^6

B. 10^7

C. 10^8

D. 10^9

E. 10^{10}

132. Andrew started saving at the beginning of the year and had saved $240 by the end of the year. He continued to save and by the end of 2 years had saved a total of $540. Which of the following is closest to the percent increase in the amount Andrew saved during the second year compared to the amount he saved during the first year?

A. 11%

B. 25%

C. 44%

D. 56%

E. 125%

133. If x is a positive integer, r is the remainder when x is divided by 4, and R is the remainder when x is divided by 9, what is the greatest possible value of $r^2 + R$?

A. 25

B. 21

C. 17

D. 13

E. 11

134. Each of the nine digits 0, 1, 1, 4, 5, 6, 8, 8, and 9 is used once to form 3 three-digit integers. What is the greatest possible sum of the 3 integers?

A. 1,752

B. 2,616

C. 2,652

D. 2,775

E. 2,958

135. Given that $1^2 + 2^2 + 3^2 + \ldots + 10^2 = 385$, what is the value of $3^2 + 6^2 + 9^2 + \ldots + 30^2$?

A. 1,155

B. 1,540

C. 1,925

D. 2,310

E. 3,465

136. If water is leaking from a certain tank at a constant rate of 1,200 milliliters per hour, how many <u>seconds</u> does it take for 1 milliliter of water to leak from the tank?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. 2

D. 3

E. 20

137. When the positive integer k is divided by the positive integer n, the remainder is 11. If $\frac{k}{n} = 81.2$, what is the value of n?

A. 9

B. 20

C. 55

D. 70

E. 81

138. The total area of a certain continent is approximately 3.8×10^{16} square inches. Which of the following is closest to the area of the continent, in square miles? (1 square mile is approximately 4.0×10^{9} square inches.)

A. 6.7×10^{5}

B. 2.0×10^{6}

C. 9.5×10^{6}

D. 1.1×10^{7}

E. 9.5×10^{8}

139. If $xy = 1$, what is the value of $\frac{2(x+y)^2}{2(x-y)^2}$?

A. 2

B. 4

C. 8

D. 16

E. 32

140. Of the 20 members of a kitchen crew, 17 can use the meat-cutting machine, 18 can use the bread-slicing machine, and 15 can use both machines. If one member of the crew is to be chosen at random, what is the probability that the member chosen will be someone who <u>cannot</u> use either machine?

A. 0

B. $\frac{1}{10}$

C. $\frac{1}{7}$

D. $\frac{1}{4}$

E. $\frac{1}{3}$

141. Which of the following is an integer?

I. $\frac{12!}{6!}$

II. $\frac{12!}{8!}$

III. $\frac{12!}{7!5!}$

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

142. The function f is defined for all nonzero x by the equation $f(x) = x - \frac{1}{x}$. If $x \neq 0$, which of the following equals $f\left(\frac{1}{x}\right)$?

A. $f(x)$

B. $f(-x)$

C. $f\left(-\frac{1}{x}\right)$

D. $\frac{1}{f(x)}$

E. $-\frac{1}{f(x)}$

143. In the arithmetic sequence $t_1, t_2, t_3, \cdots, t_n, \cdots, t_1 = 23$ and $t_n = t_{n-1} - 3$ for each $n > 1$. What is the value of n when $t_n = -4$?

A. −1

B. 7

C. 10

D. 14

E. 20

144. How many seconds will it take for a car that is traveling at a constant rate of 45 miles per hour to travel a distance of 220 yards? (1 mile = 1,760 yards)

A. 8

B. 9

C. 10

D. 11

E. 12

145. A store’s selling price of $2,240 for a certain computer would yield a profit of 40 percent of the store’s cost for the computer. What selling price would yield a profit of 50 percent of the computer’s cost?

A. $2,400

B. $2,464

C. $2,650

D. $2,732

E. $2,800

146. If a certain coin is flipped, the probability that the coin will land heads up is $\frac{1}{2}$. If the coin is flipped 5 times, what is the probability that it will land heads up on the first 3 flips and not on the last 2 flips?

A. $\frac{3}{5}$

B. $\frac{1}{2}$

C. $\frac{1}{5}$

D. $\frac{1}{8}$

E. $\frac{1}{32}$

147. The operation $\otimes$ is defined for all nonzero numbers a and b by $a \otimes b = \frac{a}{b} - \frac{b}{a}$. If x and y are nonzero numbers, which of the following statements must be true?

I. $x \otimes xy = x(1 \otimes y)$

II. $x \otimes y = -(y \otimes x)$

III. $\frac{1}{x} \otimes \frac{1}{y} = y \otimes x$

A. I only

B. II only

C. III only

D. I and II

E. II and III

148. If each term in the sum $a_1 + a_2 + \cdots + a_n$ is either 7 or 77 and the sum equals 350, which of the following could be equal to n?

A. 38

B. 39

C. 40

D. 41

E. 42

149. (1.00001)(0.99999) – (1.00002)(0.99998) =

A. 0

B. 10^{-10}

C. $3(10^{-10})$

D. 10^{-5}

E. $3(10^{-5})$

150. A certain rectangular photograph has an area of $6\frac{1}{2}$ square feet. What is the area of the photograph in square inches? (1 foot = 12 inches)

A. 36

B. 78

C. 432

D. 864

E. 936

151. The toll for crossing a certain bridge is $0.75 each crossing. Drivers who frequently use the bridge may instead purchase a sticker each month for $13.00 and then pay only $0.30 each crossing during that month. If a particular driver will cross the bridge twice on each of x days next month and will not cross the bridge on any other day, what is the least value of x for which this driver can save money by using the sticker?

A. 14

B. 15

C. 16

D. 28

E. 29

Questions 152 to 207 — Difficulty: Hard

152. Two numbers differ by 2 and sum to S. Which of the following is the greater of the numbers in terms of S?

A. $\frac{S}{2} - 1$

B. $\frac{S}{2}$

C. $\frac{S}{2} + \frac{1}{2}$

D. $\dfrac{S}{2}+1$

E. $\dfrac{S}{2}+2$

153. If m is an integer and $m = 10^{32} - 32$, what is the sum of the digits of m?

A. 257

B. 264

C. 275

D. 284

E. 292

154. In a numerical table with 10 rows and 10 columns, each entry is either a 9 or a 10. If the number of 9s in the nth row is $n-1$ for each n from 1 to 10, what is the average (arithmetic mean) of all the numbers in the table?

A. 9.45

B. 9.50

C. 9.55

D. 9.65

E. 9.70

155. In 2004, the cost of 1 year-long print subscription to a certain newspaper was $4 per week. In 2005, the newspaper introduced a new rate plan for 1 year-long print subscription: $3 per week for the first 40 weeks of 2005 and $2 per week for the remaining weeks of 2005. How much less did 1 year-long print subscription to this newspaper cost in 2005 than in 2004?

A. $64

B. $78

C. $112

D. \$144

E. \$304

156. A positive integer n is a perfect number provided that the sum of all the positive factors of n, including 1 and n, is equal to $2n$. What is the sum of the reciprocals of all the positive factors of the perfect number 28?

A. $\frac{1}{4}$

B. $\frac{56}{27}$

C. 2

D. 3

E. 4

157. The infinite sequence $a_1, a_2, \ldots, a_n, \ldots$ is such that $a_1 = 2$, $a_2 = -3$, $a_3 = 5$, $a_4 = -1$, and $a_n = a_{n-4}$ for $n > 4$. What is the sum of the first 97 terms of the sequence?

A. 72

B. 74

C. 75

D. 78

E. 80

158. The sequence $a_1, a_2, \ldots a_n, \ldots$ is such that $a_n = 2a_{n-1} - x$ for all positive integers $n \geq 2$ and for a certain number x. If $a_5 = 99$ and $a_3 = 27$, what is the value of x?

A. 3

B. 9

C. 18

D. 36

E. 45

159. In a certain medical survey, 45 percent of the people surveyed had the type A antigen in their blood and 3 percent had both the type A antigen and the type B antigen. Which of the following is closest to the percent of those with the type A antigen who also had the type B antigen?

A. 1.35%

B. 6.67%

C. 13.50%

D. 15.00%

E. 42.00%

160. On a certain transatlantic crossing, 20 percent of a ship's passengers held round-trip tickets and also took their cars aboard the ship. If 60 percent of the passengers with round-trip tickets did <u>not</u> take their cars aboard the ship, what percent of the ship's passengers held round-trip tickets?

A. $33\frac{1}{3}\%$

B. 40%

C. 50%

D. 60%

E. $66\frac{2}{3}\%$

161. If x and k are integers and $(12^x)(4^{2x+1}) = (2^k)(3^2)$, what is the value of k?

A. 5

B. 7

C. 10

D. 12

E. 14

162. If S is the sum of the reciprocals of the 10 consecutive integers from 21 to 30, then S is between which of the following two fractions?

A. $\frac{1}{3}$ and $\frac{1}{2}$

B. $\frac{1}{4}$ and $\frac{1}{3}$

C. $\frac{1}{5}$ and $\frac{1}{4}$

D. $\frac{1}{6}$ and $\frac{1}{5}$

E. $\frac{1}{7}$ and $\frac{1}{6}$

163. For every even positive integer m, $f(m)$ represents the product of all even integers from 2 to m, inclusive. For example, $f(12) = 2 \times 4 \times 6 \times 8 \times 10 \times 12$. What is the greatest prime factor of $f(24)$?

A. 23

B. 19

C. 17

D. 13

E. 11

3, k, 2, 8, m, 3

164. The arithmetic mean of the list of numbers above is 4. If k and m are integers and $k \neq m$, what is the median of the list?

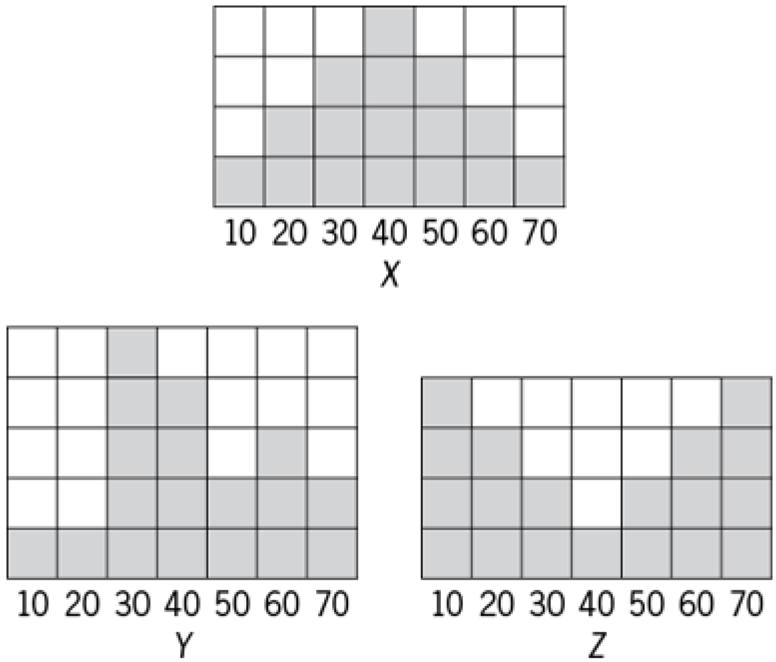
A. 2

B. 2.5

C. 3

D. 3.5

E. 4



165. If the variables X, Y, and Z take on only the values 10, 20, 30, 40, 50, 60, or 70 with frequencies indicated by the shaded regions above, for which of the frequency distributions is the mean equal to the median?

A. X only

B. Y only

C. Z only

D. X and Y

E. X and Z

$$\begin{aligned} 2x + y &= 12 \\ |y| &\leq 12 \end{aligned}$$

166. For how many ordered pairs (x, y) that are solutions of the system above are x and y both integers?

A. 7

B. 10

C. 12

D. 13

E. 14

167. The United States Mint produces coins in 1-cent, 5-cent, 10-cent, 25-cent, and 50-cent denominations. If a jar contains exactly 100 cents worth of these coins, which of the following could be the total number of coins in the jar?

I. 91

II. 81

III. 76

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

168. A certain university will select 1 of 7 candidates eligible to fill a position in the mathematics department and 2 of 10 candidates eligible to fill 2 identical positions in the computer science department. If none of the candidates is eligible for a position in both departments, how many different sets of 3 candidates are there to fill the 3 positions?

A. 42

B. 70

C. 140

D. 165

E. 315

169. A survey of employers found that during 1993 employment costs rose 3.5 percent, where employment costs consist of salary costs and fringe-benefit costs. If salary costs rose 3 percent and fringe-benefit costs rose 5.5 percent during 1993, then fringe-benefit costs represented what percent of employment costs at the beginning of 1993?

A. 16.5%

B. 20%

C. 35%

D. 55%

E. 65%

170. The subsets of the set $\{w, x, y\}$ are $\{w\}$, $\{x\}$, $\{y\}$, $\{w, x\}$, $\{w, y\}$, $\{x, y\}$, $\{w, x, y\}$, and { } (the empty subset). How many subsets of the set $\{w, x, y, z\}$ contain w?

A. Four

B. Five

C. Seven

D. Eight

E. Sixteen

171. The number $\sqrt{63 - 36\sqrt{3}}$ can be expressed as $x + y\sqrt{3}$ for some integers x and y. What is the value of xy?

A. –18

B. –6

C. 6

D. 18

E. 27

172. There are 10 books on a shelf, of which 4 are paperbacks and 6 are hardbacks. How many possible selections of 5 books from the shelf contain at least one paperback and at least one hardback?

A. 75

B. 120

C. 210

D. 246

E. 252

173. If x is to be chosen at random from the set {1, 2, 3, 4} and y is to be chosen at random from the set {5, 6, 7}, what is the probability that xy will be even?

A. $\frac{1}{6}$

B. $\frac{1}{3}$

C. $\frac{1}{2}$

D. $\frac{2}{3}$

E. $\frac{5}{6}$

174. The function f is defined for each positive three-digit integer n by $f(n) = 2^x\ 3^y\ 5^z$, where x, y, and z are the hundreds, tens, and units digits of n, respectively. If m and v are three-digit positive integers such that f of m equals nine times f of v, then m minus v equals

A. 8

B. 9

C. 18

D. 20

E. 80

175. If $10^{50} - 74$ is written as an integer in base 10 notation, what is the sum of the digits in that integer?

A. 424

B. 433

C. 440

D. 449

E. 467

176. A certain company that sells only cars and trucks reported that revenues from car sales in 1997 were down 11 percent from 1996 and revenues from truck sales in 1997 were up 7 percent from 1996. If total revenues from car sales and truck sales in 1997 were up 1 percent from 1996, what is the ratio of revenue from car sales in 1996 to revenue from truck sales in 1996?

A. 1:2

B. 4:5

C. 1:1

D. 3:2

E. 5:3

177. Becky rented a power tool from a rental shop. The rent for the tool was $12 for the first hour and $3 for each additional hour. If Becky paid a total of $27, excluding sales tax, to rent the tool, for how many hours did she rent it?

A. 5

B. 6

C. 9

D. 10

E. 12

178. If four less than seven minus x divided by three, which of the following must be true?

I. five less than x

II. absolute value of x plus three greater than two

III. negative left parenthesis x plus five right parenthesis is positive.

A. II only

B. III only

C. I and II only

D. II and III only

E. I, II, and III

179. On a certain day, a bakery produced a batch of rolls at a total production cost of $300. On that day, four divided by five of the rolls in the batch were sold, each at a price that was 50 percent greater than the average (arithmetic mean) production cost per roll. The remaining rolls in the batch were sold the next day, each at a price that was 20 percent less than the price of the day before. What was the bakery's profit on this batch of rolls?

A. $150

B. $144

C. $132

D. $108

E. $90

180. A set of numbers has the property that for any number t in the set, t plus two is in the set. If –1 is in the set, which of the following must also be in the set?

I. –3

II. 1

III. 5

A. I only

B. II only

C. I and II only

D. II and III only

E. I, II, and III

181. A couple decides to have 4 children. If they succeed in having 4 children and each child is equally likely to be a boy or a girl, what is the probability that they will have exactly 2 girls and 2 boys?

A. three divided by eight

B. one divided by four

C. three divided by 16

D. one divided by eight

E. one divided by 16

182. The closing price of Stock X changed on each trading day last month. The percent change in the closing price of Stock X from the first trading day last month to each of the other trading days last month was less than 50 percent. If the closing price on the second trading day last month was $10.00, which of the following CANNOT be the closing price on the last trading day last month?

A. $3.00

B. $9.00

C. $19.00

D. $24.00

E. $29.00

183. How many different positive integers are factors of 441?

A. 4

B. 6

C. 7

D. 9

E. 11

184. If n is a positive integer and n squared is divisible by 72, then the largest positive integer that must divide n is

A. 6

B. 12

C. 24

D. 36

E. 48

185. A certain grocery purchased x pounds of produce for p dollars per pound. If y pounds of the produce had to be discarded due to spoilage and the grocery sold the rest for s dollars per pound, which of the following represents the gross profit on the sale of the produce?

A. left parenthesis x minus y right parenthesis times s minus x times p

B. left parenthesis x minus y right parenthesis times p minus y times s

C. left parenthesis s minus p right parenthesis times y minus x times p

D. x times p minus y times s

E. left parenthesis x minus y right parenthesis times left parenthesis s minus p right parenthesis

186. If x comma y, and z are positive integers such that x is a factor of y, and x is a multiple of z, which of the following is NOT necessarily an integer?

A. x plus z divided by z

B. y plus z divided by x

C. x plus y divided by z

D. x times y divided by z

E. y times z divided by x

187. Running at their respective constant rates, Machine X takes 2 days longer to produce w widgets than Machine Y. At these rates, if the two machines together produce five divided by four times w widgets in 3 days, how many days would it take Machine X alone to produce two times w widgets?

A. 4

B. 6

C. 8

D. 10

E. 12

188. What is the greatest positive integer n such that five super n divides 10 factorial negative left parenthesis two right parenthesis times left parenthesis five factorial right parenthesis squared ?

A. 2

B. 3

C. 4

D. 5

E. 6

189. Yesterday, Candice and Sabrina trained for a bicycle race by riding around an oval track. They both began riding at the same time from the track’s starting point. However, Candice rode at a faster pace than Sabrina, completing each lap around the track in 42 seconds, while Sabrina completed each lap around the track in 46 seconds. How many laps around the track had Candice completed the next time that Candice and Sabrina were together at the starting point?

A. 21

B. 23

C. 42

D. 46

E. 483

190. If n equals nine factorial negative six super four, which of the following is the greatest integer k such that three super k is a factor of n?

A. 1

B. 3

C. 4

D. 6

E. 8

191. The integer 120 has many factorizations. For example, 120 = (2)(60), 120 = (3) (4)(10), and 120 = (–1)(–3)(4)(10). In how many of the factorizations of 120 are the factors consecutive integers in ascending order?

A. 2

B. 3

C. 4

D. 5

E. 6

192. Jorge's bank statement showed a balance that was $0.54 greater than what his records showed. He discovered that he had written a check for normal dollar times x full stop y times z and had recorded it as normal dollar times x full stop z times y, where each of x, y, and z represents a digit from 0 though 9. Which of the following could be the value of z?

A. 2

B. 3

C. 4

D. 5

E. 6

193. One side of a parking stall is defined by a straight stripe that consists of n painted sections of equal length with an unpainted section one divided by two as long between each pair of consecutive painted sections. The total length of the stripe from the beginning of the first

painted section to the end of the last painted section is 203 inches. If n is an integer and the length, in inches, of each unpainted section is an integer greater than 2, what is the value of n?

A. 5

B. 9

C. 10

D. 14

E. 29

194. This year, Henry will save a certain amount of his income, and he will spend the rest. Next year, Henry will have no income, but for each dollar that he saves this year, he will have one plus r dollars available to spend. In terms of r, what fraction of his income should Henry save this year so that next year the amount he has available to spend will be equal to half the amount that he spends this year?

A. one divided by r plus two

B. one divided by two times r plus two

C. one divided by three times r plus two

D. one divided by r plus three

E. one divided by two times r plus three

Machine	Consecutive Minutes Machine Is Off	Units of Power When On
A	17	15
B	14	18
C	11	12

195. At a certain factory, each of Machines A, B, and C is periodically on for exactly 1 minute and periodically off for a fixed number of

consecutive minutes. The table above shows that Machine A is on and uses 15 units of power every 18^{th} minute, Machine B is on and uses 18 units of power every 15^{th} minute, and Machine C is on and uses 12 units of power every 12^{th} minute. The factory has a backup generator that operates only when the total power usage of the 3 machines exceeds 30 units of power. What is the time interval, in minutes, between consecutive times the backup generator begins to operate?

A. 36

B. 63

C. 90

D. 180

E. 270

196. In a certain region, the number of children who have been vaccinated against rubella is twice the number who have been vaccinated against mumps. The number who have been vaccinated against both is twice the number who have been vaccinated only against mumps. If 5,000 have been vaccinated against both, how many have been vaccinated only against rubella?

A. 2,500

B. 7,500

C. 10,000

D. 15,000

E. 17,500

197. Three boxes of supplies have an average (arithmetic mean) weight of 7 kilograms and a median weight of 9 kilograms. What is the maximum possible weight, in kilograms, of the lightest box?

A. 1

B. 2

C. 3

D. 4

E. 5

Stock	Number of Shares
cap v	68
cap w	112
cap x	56
cap y	94
cap z	45

198. The table shows the number of shares of each of the 5 stocks owned by Mr. Sami. If Mr. Sami were to sell 20 shares of Stock cap x and buy 24 shares of Stock cap y, what would be the increase in the range of the numbers of shares of the 5 stocks owned by Mr. Sami?

A. 4

B. 6

C. 9

D. 15

E. 20

199. Last year, sales at Company cap x were 10% greater in February than in January, 15% less in March than in February, 20% greater in April than in March, 10% less in May than in April, and 5% greater in June than in May. In which month were sales closest to the sales in January?

A. February

B. March

C. April

D. May

E. June

200. If s and t are integers greater than 1 and each is a factor of the integer n, which of the following must be a factor of n super s times t?

I. s super t

II. left parenthesis s times t right parenthesis squared

III. s plus t

A. None

B. I only

C. II only

D. III only

E. I and II

201. How many 4-digit positive integers are there in which all 4 digits are even?

A. 625

B. 600

C. 500

D. 400

E. 256

202. If zero less than r less than one less than s less than two, which of the following must be less than 1?

I. r divided by s

II. r times s

III. s minus r

A. I only

B. II only

C. III only

D. I and II

E. I and III

203. Last month, 15 homes were sold in Town X. The average (arithmetic mean) sale price of the homes was $150,000 and the median sale price was $130,000. Which of the following statements must be true?

I. At least one of the homes was sold for more than $165,000.

II. At least one of the homes was sold for more than $130,000 and less than $150,000.

III. At least one of the homes was sold for less than $130,000.

A. I only

B. II only

C. III only

D. I and II

E. I and III

204. Pumps A, B, and C operate at their respective constant rates. Pumps A and B, operating simultaneously, can fill a certain tank in six divided by five hours; Pumps A and C, operating simultaneously, can fill the tank in three divided by two hours; and Pumps B and C, operating simultaneously, can fill the tank in 2 hours. How many hours does it take Pumps A, B, and C, operating simultaneously, to fill the tank?

A. one divided by three

B. one divided by two

C. two divided by three

D. five divided by six

E. 1

205. If n is a positive integer, then
left parenthesis negative two super n right parenthesis super negative two plus left parenthesis two super negative n right parenthesis squared equals

A. 0

B. two super negative two times n

C. two super two times n

D. two super negative two times n plus one

E. two super two times n plus one

206. Which of the following is equal to
left parenthesis two times root of order three over three divided by Square root of two right parenthesis cubed
?

A. three times Square root of two divided by two

B. three times Square root of two

C. six times Square root of two

D. 12

E. 12 times Square root of two

207. Dara ran on a treadmill that had a readout indicating the time remaining in her exercise session. When the readout indicated 24 min 18 sec, she had completed 10% of her exercise session. The readout indicated which of the following when she had completed 40% of her exercise session?

A. 10 min 48 sec

B. 14 min 52 sec

C. 14 min 58 sec

D. 16 min 6 sec

E. 16 min 12 sec

4.3 Answer Key

1. A
2. D
3. A
4. D
5. C
6. C
7. A
8. D
9. C
10. C
11. C
12. E
13. D
14. D
15. C
16. D
17. B
18. D
19. D
20. D
21. B
22. A
23. C
24. D
25. B

26. D
27. E
28. D
29. D
30. C
31. A
32. C
33. A
34. E
35. D
36. D
37. C
38. A
39. B
40. C
41. A
42. D
43. B
44. C
45. C
46. E
47. B
48. B
49. D
50. D
51. C

52. B
53. A
54. C
55. C
56. A
57. D
58. B
59. E
60. B
61. E
62. D
63. B
64. C
65. A
66. E
67. C
68. C
69. D
70. B
71. C
72. D
73. B
74. A
75. B
76. D
77. B

78. B
79. C
80. D
81. A
82. D
83. A
84. A
85. B
86. A
87. E
88. E
89. C
90. D
91. D
92. E
93. D
94. E
95. D
96. E
97. E
98. E
99. C
100. C
101. A
102. D
103. C

104. D

105. A

106. B

107. D

108. B

109. C

110. A

111. B

112. B

113. D

114. E

115. E

116. C

117. E

118. B

119. E

120. E

121. D

122. B

123. B

124. B

125. B

126. B

127. B

128. B

129. E

130. B
131. D
132. B
133. C
134. C
135. E
136. D
137. C
138. C
139. D
140. A
141. E
142. B
143. C
144. C
145. A
146. E
147. E
148. C
149. C
150. E
151. B
152. D
153. D
154. C
155. A

156. C
157. B
158. A
159. B
160. C
161. E
162. A
163. E
164. C
165. E
166. D
167. D
168. E
169. B
170. D
171. A
172. D
173. D
174. D
175. C
176. A
177. B
178. D
179. C
180. D
181. A

182. A
183. D
184. B
185. A
186. B
187. E
188. D
189. B
190. D
191. C
192. E
193. C
194. E
195. C
196. C
197. C
198. D
199. D
200. E
201. C
202. A
203. A
204. E
205. D
206. C
207. E

4.4 Answer Explanations

The following discussion is intended to familiarize you with the most efficient and effective approaches to the kinds of problems common to Problem-Solving questions. The particular questions in this chapter are generally representative of the kinds of problem-solving questions you will encounter on the GMAT exam. Remember that it is the problem-solving strategy that is important, not the specific details of a particular question.

Questions 1 to 82 — Difficulty: Easy

1. Working at a constant rate, a copy machine makes 20 copies of a one-page document per minute. If the machine works at this constant rate, how many hours does it take to make 4,800 copies of a one-page document?

   A. 4

   B. 5

   C. 6

   D. 7

   E. 8

   Arithmetic **Rate**

   The copy machine produces 20 copies of the one-page document each minute. Because there are 60 minutes in an hour, the constant rate of 20 copies per minute is equal to 60 × 20 = 1,200 copies per hour. With the machine working at this rate, the amount of time that it takes to produce 4,800 copies of the document is
   4800 c times o times p times i times e times s divided by 1200 c times o times p times i times e times s divided by h times o times u times r equals four hours
   .

   The correct answer is A.

2. If x plus y equals two and x squared plus y squared equals two, what is the value of x times y?

A. –2

B. –1

C. 0

D. 1

E. 2

Algebra **Second-Degree Equations**

x plus y	= 2	given
y	= two minus x	subtract x from both sides
x squared plus left parenthesis two minus x right parenthesis squared	= 2	substitute y equals two minus x into x squared plus y squared equals two
two times x squared minus four times x plus four	= 2	expand and combine like terms
two times x squared minus four times x plus two	= 0	subtract 2 from both sides
x squared minus two times x plus one	= 0	divide both sides by 2
left parenthesis x minus one right parenthesis times left parenthesis x minus one right parenthesis	= 0	factor
x	= 1	set each factor equal to 0
y	= 1	use x equals one and y equals two minus x
x times y	= 1	multiply 1 and 1

Alternatively, the value of x times y can be found by first squaring both sides of the equation x plus y equals two.

x plus y	= 2	given
left parenthesis x plus y right parenthesis squared	= 4	square both sides
sum with 3 summands x squared plus two times x times y plus y squared	= 4	expand and combine like terms
two plus two times x times y	= 4	replace x squared plus y squared with 2
two times x times y	= 2	subtract 2 from both sides
x times y	= 1	divide both sides by 2

The correct answer is D.

3. The sum cap s of the first n consecutive positive even integers is given by cap s equals n times left parenthesis n plus one right parenthesis. For what value of n is this sum equal to 110?

   A. 10

   B. 11

   C. 12

   D. 13

   E. 14

Algebra **Factoring**

Given that the sum of the first n even numbers is n times left parenthesis n plus one right parenthesis, the sum is equal to 110 when 110 equals n times left parenthesis n plus one right parenthesis. To find the value of n in this case, we need to find the two consecutive integers whose product is 110. These integers are 10 and 11; 10 × 11 = 110. The smaller of these numbers is n.

The correct answer is A.

4. 6(87.30 + 0.65) − 5(87.30) =

   A. 3.90

   B. 39.00

   C. 90.90

   D. 91.20

   E. 91.85

Arithmetic **Factors, Multiples, and Divisibility**

This question is most efficiently answered by distributing the 6 over 87.30 and 0.65, and then combining the terms that contain a factor of 87.30, as follows:

6(87.30 + 0.65) − 5(87.30) = 6(87.30) + 6(0.65) − 5 (87.30) = (6 − 5) 87.30 + 6(0.65) = 87.30 + 3.90 = 91.20

The correct answer is D.

5. What is the value of x squared times y times z minus x times y times z squared, if x equals negative two, y equals one, and z equals three?

   A. 20

   B. 24

   C. 30

   D. 32

   E. 48

Algebra **Operations on Integers**

Given that x equals negative two, y equals one, and z equals three, it follows by substitution that multiline equation row 1 x squared times y times z minus x times y times z squared equals left parenthesis negative two right parenthesis squared times left parenthesis one right parenthesis times left parenthesis three right parenthesis minus left parenthesis negative two right parenthesis times left parenthesis one right parenthesis times left parenthesis three squared right parenthesis row 2 equals left parenthesis four right parenthesis times left parenthesis one right parenthesis times left parenthesis three right parenthesis minus left parenthesis negative two right parenthesis times left parenthesis one right parenthesis times left parenthesis nine right parenthesis row 3 equals 12 minus left parenthesis negative 18 right parenthesis row 4 equals 12 plus 18 row 5 equals 30

The correct answer is C.

6. If cap t is a set of 35 consecutive integers, of which 17 are negative, what is the sum of all the integers in cap t?

   A. –2

   B. –1

   C. 0

   D. 1

   E. 2

   Arithmetic Properties of Numbers

   The only set of 35 consecutive integers of which exactly 17 are negative is {−17, −16, −15, …, 0, …, 15, 16, 17}. The sum of these integers is (−17 + 17) + (−16 + 16) + (−15 + 15) + … + 0 = 0 + 0 + 0 + … + 0 = 0.

   The correct answer is C.

7. If x greater than y and y greater than z, which of the following represents the greatest number?

   A. x minus z

   B. x minus y

C. y minus x

D. z minus y

E. z minus x

Algebra **Inequalities**

From x greater than y and y greater than z, it follows that x greater than z. These inequalities imply the following about the differences that are given in the answer choices:

Answer choice	Difference	Algebraic sign	Reason
A	x minus z	positive	x greater than z implies x minus z greater than zero
B	x minus y	positive	x greater than y implies x minus y greater than zero
C	y minus x	negative	x minus y greater than zero implies y minus x less than zero
D	z minus y	negative	y greater than z implies zero greater than z minus y
E	z minus x	negative	x minus z greater than zero implies z minus x less than zero

Since the expressions in A and B represent positive numbers and the expressions in C, D, and E represent negative numbers, the latter can be eliminated because every negative number is less than every positive number. To determine which of x minus z and x minus y is greater, consider the placement of points with coordinates x, y, and z on the number line.

image

The distance between x and z (that is, x minus z) is the sum of the distance between x and y (that is, x minus y) and the distance between y and z (that is, y minus z). Therefore, left parenthesis x minus z right parenthesis greater than left parenthesis x minus y right parenthesis
, which means that x minus z represents the greater of the numbers represented by left parenthesis x minus z right parenthesis and left parenthesis x minus y right parenthesis. Thus, x minus z represents the greatest of the numbers represented by the answer choices.

Alternatively,

y greater than z	given
negative y less than negative z	multiply both sides by –1
x minus y less than x minus z	add x to both sides

Thus, x minus z represents the greater of the numbers represented by left parenthesis x minus z right parenthesis and left parenthesis x minus y right parenthesis. Therefore, x minus z represents the greatest of the numbers represented by the answer choices.

The correct answer is A.

8. 38, 69, 22, 73, 31, 47, 13, 82

Which of the following numbers is greater than three-fourths of the numbers but less than one-fourth of the numbers in the list above?

A. 56

B. 68

C. 69

D. 71

E. 73

Arithmetic Order

Let’s rearrange the eight numbers in the list from least to greatest: 13, 22, 31, 38, 47, 69, 73, 82. A number greater than three-fourths of them will be greater than each of the first six but less than either of the last two. Thus, this number must be greater than 69 but less than 73. Among the answer options, only the number 71 meets this condition.

The correct answer is D.

9. A rug manufacturer produces rugs at a cost of $75 per rug. What is the manufacturer’s gross profit from the sale of 150 rugs if two divided by three of the rugs are sold for $150 per rug and the rest are sold for $200 per rug?

   A. $10,350

   B. $11,250

   C. $13,750

   D. $16,250

   E. $17,800

   Arithmetic Applied Problems; Proportions

   The gross profit from the sale of 150 rugs is equal to the revenue from the sale of the rugs minus the cost of producing them. For two divided by three of the 150 rugs—100 of them—the gross profit per rug is $150 − $75 = $75. For the remaining 50 rugs, the gross profit per rug is $200 − $75 = $125. The gross profit from the sale of the 150 rugs is therefore 100 × $75 + 50 × $125 = $13,750.

   The correct answer is C.

10. The value of Maureen's investment portfolio has decreased by 5.8 percent since her initial investment in the portfolio. If her initial investment was $16,800, what is the current value of the portfolio?

    A. $7,056.00

    B. $14,280.00

    C. $15,825.60

    D. $16,702.56

    E. $17,774.40

    Arithmetic **Percents**

    Maureen's initial investment was $16,800, and it has decreased by 5.8%. Its current value is therefore (100% − 5.8%) = 94.2% of $16,800, which is equal to 0.942 × $16,800. To make the multiplication simpler, this can be expressed as $(942 × 16.8). Thus multiplying, we obtain the result of $15,825.60.

    The correct answer is C.

11. Company C produces toy trucks at a cost of $5.00 each for the first 100 trucks and $3.50 for each additional truck. If 500 toy trucks were produced by Company C and sold for $10.00 each, what was Company C's gross profit?

    A. $2,250

    B. $2,500

    C. $3,100

    D. $3,250

    E. $3,500

    Arithmetic **Applied Problems**

    The company's gross profit on the 500 toy trucks is the company's revenue from selling the trucks minus the company's cost of producing the trucks. The revenue is (500)($10.00) = $5,000. The cost for the first 100 trucks is (100)($5.00) = $500, and the cost for the other 400 trucks is (400)($3.50) = $1,400 for a total cost of $500 + $1,400 = $1,900. Thus, the company's gross profit is $5,000 – $1,900 = $3,100.

The correct answer is C.

Division	Profit or Loss (in millions of dollars)				
	1991	1992	1993	1994	1995
cap a	1.1	(3.4)	1.9	2.0	0.6
cap b	(2.3)	5.5	(4.5)	3.9	(2.9)
cap c	10.0	(6.6)	5.3	1.1	(3.0)

12. The annual profit or loss for the three divisions of Company T for the years 1991 through 1995 are summarized in the table shown, where losses are enclosed in parentheses. For which division and which three consecutive years shown was the division's profit or loss for the three-year period closest to $0?

    A. Division cap a for 1991–1993

    B. Division cap a for 1992–1994

    C. Division cap b for 1991–1993

    D. Division cap b for 1993–1995

    E. Division cap c for 1992–1994

Arithmetic **Applied Problems**

For completeness, the table shows all 9 of the profit or loss amounts, in millions of dollars, for each of the 3 divisions and the 3 three-year periods.

	1991–1993	1992–1994	1993–1995
cap a	–0.4	0.5	4.5
cap b	–1.3	4.9	–3.5
cap c	8.7	**–0.2**	3.4

The correct answer is E.

13. A computer appliance dealer bought a computer at a wholesale price of $500 and then sold it at a 20 percent discount off the suggested retail price. If the dealer made a 12 percent profit on the wholesale price, what was the suggested retail price of the computer?

    A. $560

    B. $580

    C. $660

    D. $700

    E. $730

    Arithmetic Applied Problems

    The dealer's profit of 12 percent of the $500 wholesale price was 0.12 × $500 = $60. Thus, the dealer sold the computer for $500 + $60 = $560. Since this sale was at a 20 percent discount off the suggested retail price, $560 was 80 percent
    left parenthesis four divided by five right parenthesis of the suggested retail price. Therefore, the suggested retail price was
    normal dollar times 560 divided by 0.8 equals normal dollar times 700
    .

    The correct answer is D.

14. Pat invested x dollars in a fund that paid 8 percent annual interest, compounded annually. Which of the following represents the value, in dollars, of Pat’s investment plus interest at the end of 5 years?

   A. five times left parenthesis 0.08 times x right parenthesis

   B. five times left parenthesis 1.08 times x right parenthesis

   C. left square bracket one plus five times left parenthesis 0.08 right parenthesis right square bracket times x

   D. left parenthesis 1.08 right parenthesis super five times x

   E. left parenthesis 1.08 times x right parenthesis super five

   Arithmetic **Applied Problems**

   The formula for the compound interest earned on a principal x over n periods at percent rate r is
   left parenthesis one plus r divided by 100 right parenthesis super n times x minus x
   . For Pat’s investment, that means the compound interest in dollars over 5 years is
   left parenthesis 1.08 right parenthesis super five times x minus x. The total dollar value of Pat’s investment includes the principal x plus the compound interest, so the total dollar value after 5 years is
   left parenthesis left parenthesis 1.08 right parenthesis super five times x minus x right parenthesis plus x equals left parenthesis 1.08 right parenthesis super five times x
   .

   The correct answer is D.

15. There are five sales agents in a certain real estate office. One month Andy sold twice as many properties as Ellen, Bob sold 3 more than Ellen, Cary sold twice as many as Bob, and Dora sold as many as Bob and Ellen together. Who sold the most properties that month?

   A. Andy

   B. Bob

   C. Cary

D. Dora

E. Ellen

Algebra **Order**

Let x represent the number of properties that Ellen sold, where x greater than or equals zero. Then, since Andy sold twice as many properties as Ellen, two times x represents the number of properties that Andy sold. Bob sold 3 more properties than Ellen, so left parenthesis x plus three right parenthesis represents the number of properties that Bob sold. Cary sold twice as many properties as Bob, so
two times left parenthesis x plus three right parenthesis equals left parenthesis two times x plus six right parenthesis
represents the number of properties that Cary sold. Finally, Dora sold as many properties as Bob and Ellen combined, so
left square bracket left parenthesis x plus three right parenthesis plus x right square bracket equals left parenthesis two times x plus three right parenthesis
represents the number of properties that Dora sold. The following table summarizes these results.

Agent	Properties Sold
Andy	two times x
Bob	x plus three
Cary	two times x plus six
Dora	two times x plus three
Ellen	x

Since x greater than or equals zero, clearly two times x plus six exceeds x comma x plus three, two times x, and two times x plus three. Therefore, Cary sold the most properties.

The correct answer is C.

16. In a field day at a school, each child who competed in n events and scored a total of p points was given an overall score of p divided by n plus n. Andrew competed in 1 event and scored 9 points. Jason competed in 3 events and scored 5, 6, and 7 points, respectively. What was the ratio of Andrew's overall score to Jason's overall score?

    A. 10 divided by 23

    B. seven divided by 10

    C. four divided by five

    D. 10 divided by nine

    E. 12 divided by seven

    Algebra **Applied Problems; Substitution**

    Andrew participated in 1 event and scored 9 points, so his overall score was nine divided by one plus one equals 10. Jason participated in 3 events and scored 5 + 6 + 7 = 18 points, so his overall score was 18 divided by three plus three equals nine. The ratio of Andrew's overall score to Jason's overall score was 10 divided by nine.

    The correct answer is D.

17. A certain work plan for September requires that a work team, working every day, produce an average of 200 items per day. For the first half of the month, the team produced an average of 150 items per day. How many items per day must the team average during the second half of the month if it is to attain the average daily production rate required by the work plan?

    A. 225

    B. 250

    C. 275

    D. 300

E. 350

Arithmetic **Rate Problem**

The work plan requires that the team produce an average of 200 items per day in September. Because the team has only produced an average of 150 items per day in the first half of September, it has a shortfall of 200 − 150 = 50 items per day for the first half of the month. The team must make up for this shortfall in the second half of the month, which has an equal number of days as the first half of the month. The team must therefore produce in the second half of the month an average amount per day that is 50 items greater than the required average of 200 items per day for the entire month. This amount for the second half of September is 250 items per day.

The correct answer is B.

18. A company sells radios for $15.00 each. It costs the company $14.00 per radio to produce 1,000 radios and $13.50 per radio to produce 2,000 radios. How much greater will the company's gross profit be from the production and sale of 2,000 radios than from the production and sale of 1,000 radios?

A. $500

B. $1,000

C. $1,500

D. $2,000

E. $2,500

Arithmetic **Applied Problems**

If the company produces and sells 1,000 radios, its gross profit from the sale of these radios is equal to the total revenue from the sale of these radios minus the total cost. The total cost is equal to the number of radios produced multiplied by the production cost per radio: 1,000 × $15.00. The total revenue is equal to the number of radios sold multiplied by the selling price: 1,000 × $14.00. The gross profit in this case is therefore 1,000 × $15.00 − 1,000 × $14.00 = 1,000 × ($15.00 − $14.00) = 1,000 ($1.00) = $1,000. If 2,000 radios are produced and sold, the total cost is equal to 2,000 × \$13.50 and the total revenue is equal to 2,000 × \$15.00. The gross profit in this case is therefore 2,000 × \$15.00 − 2,000 × \$13.50 = 2,000 × (\$15.00 − \$13.50) = 2,000 × (\$1.50) = \$3,000. This profit of \$3,000 is \$2,000 greater than the gross profit of \$1,000 from producing and selling 1,000 radios.

The correct answer is D.

19. Working at their respective constant rates, machine A makes 100 copies in 12 minutes and machine B makes 150 copies in 10 minutes. If these machines work simultaneously at their respective rates for 30 minutes, what is the total number of copies that they will produce?

    A. 250

    B. 425

    C. 675

    D. 700

    E. 750

    Arithmetic Applied Problems

    Over 30 minutes, machine A makes
    100 copies prefix multiplication of left parenthesis 30 times minutes divided by 12 times minutes right parenthesis equals 250
    copies, while machine B makes
    150 times copies multiplication left parenthesis 30 times minutes divided by 10 times minutes right parenthesis equals 450 times copies
    . So together, the machines will make 250 + 450 = 700 copies in 30 minutes.

    The correct answer is D.

20. Company X mailed out a questionnaire. Sixty percent of those who received the questionnaire by mail responded. Company X needed 300 responses. What is the minimum number of questionnaires that Company X should have mailed in order to get the necessary responses?

    A. 400

B. 420

C. 480

D. 500

E. 600

Arithmetic **Percents**

Since 300, the number of responses received, was 60 percent of the number of copies of the questionnaire that were mailed, the number of copies mailed was 300 divided by 0.6 equals 500.

The correct answer is D.

21. The toll cap t, in dollars, for a truck using a certain bridge is given by the formula
cap t equals 1.50 plus 0.50 times left parenthesis x minus two right parenthesis
, where x is the number of axles on the truck. What is the toll for an 18-wheel truck that has 2 wheels on its front axle and 4 wheels on each of its other axles?

A. $2.50

B. $3.00

C. $3.50

D. $4.00

E. $5.00

Algebra **Operations on Rational Numbers**

The 18-wheel truck has 2 wheels on its front axle and 4 wheels on each of its other axles, and so if cap a represents the number of axles on the truck in addition to the front axle, then
two plus four times cap a equals 18, from which it follows that
four times cap a equals 16 and cap a equals four. Therefore, the total number of axles on the truck is
equation sequence part 1 one plus cap a equals part 2 one plus four equals part 3 five

. Then, using
cap t equals 1.50 plus 0.50 times left parenthesis x minus two right parenthesis
, where x is the number of axles on the truck and x equals five, it follows that
equation sequence part 1 cap t equals part 2 1.50 plus 0.50 times left parenthesis five minus two right parenthesis equals part 3 1.50 plus 1.50 equals part 4 3.00
. Therefore, the toll for the truck is $3.00.

The correct answer is B.

22. For what value of x between −4 and 4, inclusive, is the value of x squared minus 10 times x plus 16 the greatest?

A. −4

B. −2

C. 0

D. 2

E. 4

Algebra **Second-Degree Equations**

Given the expression x squared minus 10 times x plus 16, a table of values can be created for the corresponding function f of x equals x squared minus 10 times x plus 16 and the graph in the standard left parenthesis x comma y right parenthesis coordinate plane can be sketched by plotting selected points:

x	f of x
−4	72
−3	55
−2	40
−1	27
0	16
1	7
2	0
3	−5
4	−8
5	−9
6	−8
7	−5
8	0
9	7

image

It is clear from both the table of values and the sketch of the graph that as the value of x increases from −4 to 4, the values of x squared minus 10 times x plus 16 decrease. Therefore, the value of x squared minus 10 times x plus 16 is greatest when x equals negative four.

Alternatively, the given expression, x squared minus 10 times x plus 16, has the form sum with 3 summands a times x squared plus b times x plus c, where a equals one, b equals negative 10, and c equals 16. The graph in the standard left parenthesis x comma y right parenthesis coordinate plane of the corresponding function f of x equals sum with 3 summands a times x squared plus b times x plus c
is a parabola with vertex at x equals negative b divided by two times a, and so the vertex of the graph of f of x equals x squared minus 10 times x plus 16 is at equation sequence part 1 x equals part 2 negative left parenthesis negative 10 divided by two times left parenthesis one right parenthesis right parenthesis equals part 3 five full stop

Because a equals one and 1 is positive, this parabola opens upward and values of x squared minus 10 times x plus 16 decrease as x increases from −4 to 4. Therefore, the greatest value of x squared minus 10 times x plus 16 for all values of x between −4 and 4, inclusive, is at x equals negative four.

The correct answer is A.

23. If x equals negative five divided by eight and y equals negative one divided by two, what is the value of the expression negative two times x minus y squared?

A. negative three divided by two

B. negative one

C. 1

D. three divided by two

E. seven divided by four

Algebra **Fractions**

If x equals negative five divided by eight and
y equals negative one divided by two, then
equation sequence part 1 negative two times x minus y squared equals part 2 negative two times left parenthesis negative five divided by eight right parenthesis minus left parenthesis negative one divided by two right parenthesis squared equals part 3 five divided by four minus one divided by four equals part 4 four divided by four equals part 5 one
.

The correct answer is C.

24. At the opening of a trading day, stock X had a value of $40.00 per share and stock Y had a value of $25.00 per share. At the closing of the trading day, stock X had a value of $41.00 per share and stock Y had a value of $28.50 per share. For this trading day, the percent increase in the value of stock Y was how many percentage points greater than the percent increase in the value of stock X?

A. 2.5

B. 6.25

C. 10

D. 11.5

E. 12.5

Arithmetic **Percents**

Stock X's percentage increase in value was
equation sequence part 1 left parenthesis 41 minus 40 right parenthesis multiplication 100 percent divided by 40 equals part 2 100 percent divided by 40 equals part 3 2.5 percent
. Stock Y's was
equation sequence part 1 left parenthesis 28.5 minus 25 right parenthesis multiplication 100 percent divided by 25 equals part 2 350 percent divided by 25 equals part 3 14 percent

. Thus, the percentage increase in stock Y’s value was $(14 - 2.5)\% = 11.5$ percentage points greater than that in stock X’s value.

The correct answer is D.

25. For positive integers a and b, the remainder when a is divided by b is equal to the remainder when b is divided by a. Which of the following could be a value of a times b?

I. 24

II. 30

III. 36

A. II only

B. III only

C. I and II only

D. II and III only

E. I, II, and III

Arithmetic Properties of Integers

We are given that the remainder when a is divided by b is equal to the remainder when b is divided by a, and asked about possible values of a times b. We thus need to find what our given condition implies about a and b.

We consider two cases: a equals b and $a \neq b$.

If a equals b, then our given condition is trivially satisfied: the remainder when a is divided by a is equal to the remainder when a is divided by b. The condition thus allows that a be equal to b.

Now consider the case of a not equals b. Either $a < b$ or b less than a. Supposing that a less than b, the remainder when a is divided by b is simply a. (For example, if 7 is divided by 10, then the remainder is 7.) However, according to our given condition, this remainder, a, is also the remainder when b is divided by a, which is impossible. If b is divided by a, then the remainder must be less than a. (For example, for any number that is divided by 10, the remainder cannot be 10 or greater.) Similar reasoning applies if we suppose that $b < a$. This is also impossible.

We thus see that a must be equal to b, and consider the statements I, II, and III.

I. Factored in terms of prime numbers, 24 = 3 × 2 × 2 × 2. Because "3" occurs only once in the factorization, we see that there is no integer a such that $a \times a = 24$. Based on the reasoning above, we see that 24 cannot be a value of ab.

II. Factored in terms of prime numbers, 30 = 5 × 3 × 2. Because there is no integer a such that $a \times a = 30$, we see that 30 cannot be a value of ab.

III. Because 36 = 6 × 6, we see that 36 is a possible value of ab (with $a = b$).

The correct answer is B.

26. List S consists of the positive integers that are multiples of 9 and are less than 100. What is the median of the integers in S?

A. 36

B. 45

C. 49

D. 54

E. 63

Arithmetic **Series and Sequences**

In the set of positive integers less than 100, the greatest multiple of 9 is 99 (9 × 11) and the least multiple of 9 is 9 (9 × 1). The sequence of positive multiples of 9 that are less than 100 is therefore the sequence of numbers $9 \times k$, where k ranges from 1 through 11. The median of the numbers k from 1 through 11 is 6. Therefore, the median of the numbers $9 \times k$, where k ranges from 1 through 11, is 9 × 6 = 54.

The correct answer is D.

27. A rope 20.6 meters long is cut into two pieces. If the length of one piece of rope is 2.8 meters shorter than the length of the other, what is the length, in meters, of the longer piece of rope?

    A. 7.5

    B. 8.9

    C. 9.9

    D. 10.3

    E. 11.7

Algebra **First-Degree Equations**

If x represents the length of the longer piece of rope, then $x - 2.8$ represents the length of the shorter piece, where both lengths are in meters. The total length of the two pieces of rope is 20.6 meters so,

$x + (x - 2.8)$	=	20.6	given
$2x - 2.8$	=	20.6	add like terms
$2x$	=	23.4	add 2.8 to both sides
x	=	11.7	divide both sides by 2

Thus, the length of the longer piece of rope is 11.7 meters.

The correct answer is E.

28. If x and y are integers and $x - y$ is odd, which of the following must be true?

    I. xy is even.

    II. $x^2 + y^2$ is odd.

III. $(x+y)^2$ is even.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

Arithmetic Properties of Numbers

We are given that x and y are integers and that $x - y$ is odd, and then asked, for various operations on x and y, whether the results of the operations are odd or even. It is therefore useful to determine, given that $x - y$ is odd, whether x and y are odd or even. If both x and y are even—that is, divisible by 2—then

$$x - y = 2m - 2n = 2(m - n)$$

for integers m and n. We thus see if both x and y are even then $x - y$ cannot be odd. And because $x - y$ *is* odd, we see that x and y cannot both be even. Similarly, if both x and y are odd, then, for integers j and k, $x = 2j + 1$ and $y = 2k + 1$. Therefore,

$$x - y = (2j + 1) - (2k + 1)$$

. The ones cancel, and we are left with

$$x - y = 2j - 2k = 2(j - k)$$

. Because $2(j - k)$ would be even, x and y cannot both be odd if $x - y$ is odd. It follows from all of this that one of x or y must be even and the other odd.

Now consider the statements I through III.

I. If one of x or y is even, then one of x or y is divisible by 2. It follows that xy is divisible by 2 and that xy is even.

II. Given that a number x or y is odd—not divisible by 2—we know that its product with itself is not divisible by 2 and is therefore odd. On the other hand, given that a number x or y is even, we know that its product with itself *is* divisible by 2 and is therefore even. The sum $x^2 + y^2$ is therefore the sum of an even number and an odd number. In such a case, the sum can be written as $(2m) + (2n + 1) = 2(m + n) + 1$, with m and n integers. It follows that $x^2 + y^2$ is not divisible by 2 and is therefore odd.

III. We know that one of x or y is even and the other is odd. We can therefore see from the discussion of statement II that $x + y$ is odd, and then also see, from the discussion of statement II, that the product of $x + y$ with itself, $(x + y)^2$, is odd.

The correct answer is D.

29. On Monday, the opening price of a certain stock was \$100 per share and its closing price was \$110 per share. On Tuesday the closing price of the stock was 10 percent less than its closing price on Monday, and on Wednesday the closing price of the stock was 4 percent greater than its closing price on Tuesday. What was the approximate percent change in the price of the stock from its opening price on Monday to its closing price on Wednesday?

A. A decrease of 6%

B. A decrease of 4%

C. A decrease of 1%

D. An increase of 3%

E. An increase of 4%

Arithmetic **Percents**

The closing share price on Tuesday was 10% less than the closing price on Monday, \$110. 10% of \$110 is equal to 0.1 × \$110 = \$11, so the closing price on Tuesday was \$110 − \$11 = \$99. The closing price on Wednesday was 4% greater than this: \$99 + (0.04 × \$99) = \$99 + \$3.96 = \$102.96. This value, \$102.96, is 2.96% greater than \$100, the opening price on Monday. The percentage change from the opening share price on Monday is therefore an increase of approximately 3%, which is the closest of the available answers to an increase of 2.96%.

The correct answer is D.

30. 1 − 0.000001 =

A. (1.01) (0.99)

B. (1.11) (0.99)

C. (1.001) (0.999)

D. (1.111) (0.999)

E. (1.0101) (0.0909)

Arithmetic **Place Value**

The task in this question is to find among the available answers the expression that is equal to 1 − 0.000001 = 0.999999. In the case of answer choice C, the first of the two factors, (1.001), is equal to 1 + 0.001. One may therefore observe that (1.001)(0.999) = (1 + 0.001)(0.999) = 0.999 + 0.000999 = 0.999999. Answer choice C is therefore a correct answer.

For answer choice A, (1.01)(0.99) = (1 + 0.01)(0.99) = 0.9999. This answer choice is therefore incorrect. For answer choice B, (1.11)(0.99) = (1 + 0.1 + 0.01)(0.99) = 0.99 + 0.099 + 0.0099 = 1.0989. This answer choice is therefore incorrect. For answer choice D, (1.111)(0.999) = 0.999 + 0.0999 + 0.00999 + 0.000999 = 1.109889. This answer choice is therefore incorrect. For answer choice E, (1.0101)(0.909) = 0.909 + 0.00909 + 0.0000909 = 0.9181809. This answer choice is therefore incorrect.

The correct answer is C.

31. In a certain history class of 17 juniors and seniors, each junior has written 2 book reports and each senior has written 3 book reports. If the 17 students have written a total of 44 book reports, how many juniors are in the class?

A. 7

B. 8

C. 9

D. 10

E. 11

Algebra **Simultaneous Equations**

Letting j and s, respectively, represent the juniors and seniors in the class, it is given that j plus s equals 17 or s equals 17 minus j. Also, since it is given that each junior has written 2 book reports and each senior has written 3 book reports for a total of 44 book reports, it follows that two times j plus three times s equals 44 or two times j plus three times left parenthesis 17 minus j right parenthesis equals 44

. Therefore,

equation sequence part 1 j equals part 2 three times left parenthesis 17 right parenthesis minus 44 equals part 3 seven

.

The correct answer is A.

32. absolute value of negative four times left parenthesis absolute value of negative 20 minus absolute value of five right parenthesis equals

A. −100

B. −60

C. 60

D. 75

E. 100

Arithmetic **Absolute Value**

equation sequence part 1 absolute value of negative four times left parenthesis absolute value of negative 20 minus absolute value of five right parenthesis equals part 2 four times left parenthesis 20 minus five right parenthesis equals part 3 four multiplication 15 equals part 4 60

The correct answer is C.

33. What is the greatest integer k for which 32.45 multiplication 10 super k is less than 1?

A. –2

B. 1

C. 0

D. 1

E. 2

Arithmetic **Operations with Integers**

multirelation 32.45 multiplication 10 super negative two equals 0.3245 less than one
, but $32.45 \times 10^{-1} = 3.245 > 1$. Thus, –2 is the greatest integer k for which 32.45 multiplication 10 super k less than one.

The correct answer is A.

34. How many integers x satisfy both $2 < x \leq 4$ and zero less than or equals x less than or equals three?

A. 5

B. 4

C. 3

D. 2

E. 1

Arithmetic **Inequalities**

The integers that satisfy
multirelation two less than x less than or equals four are 3 and 4. The integers that satisfy
zero less than or equals x less than or equals three are 0, 1, 2, and 3. The only integer that satisfies both
multirelation two less than x less than or equals four and
zero less than or equals x less than or equals three is 3, and so there is only one integer that satisfies both
multirelation two less than x less than or equals four and
zero less than or equals x less than or equals three.

The correct answer is E.

35. At the opening of a trading day at a certain stock exchange, the price per share of stock K was $8. If the price per share of stock K was $9 at the closing of the day, what was the percent increase in the price per share of stock K for that day?

    A. 1.4%

    B. 5.9%

    C. 11.1%

    D. 12.5%

    E. 23.6%

    Arithmetic Percents

    An increase from $8 to $9 represents an increase of
    left parenthesis nine minus eight divided by eight multiplication 100 right parenthesis percent equals 100 divided by eight percent equals 12.5 percent
    .

    The correct answer is D.

image

36. As shown in the diagram above, a lever resting on a fulcrum has weights of w sub one pounds and w sub two pounds, located

d sub one feet and d sub two feet from the fulcrum. The lever is balanced and
w sub one times d sub one equals w sub two times d sub two. Suppose w sub one is 50 pounds and w sub two is 30 pounds. If d sub one is 4 feet less than d sub two, what is d sub two, in feet?

A. 1.5

B. 2.5

C. 6

D. 10

E. 20

Algebra **First-Degree Equations; Substitution**

Given w sub one times d sub one equals w sub two times d sub two, w sub one equals 50 comma normal times w sub two equals 30, and d sub one equals d sub two minus four, it follows that $50(d_2 - 4) = 30d_2$, and so

50 times left parenthesis d sub two minus four right parenthesis	=	30 times d sub two	given
50 times d sub two minus 200	=	30 times d sub two	distributive principle
20 times d sub two	=	200	add 200 minus 30 times d sub two to both sides
d sub two	=	10	divide both sides by 20

The correct answer is D.

Day	Change in Dollars
Monday	prefix plus of
Tuesday	negative three divided by four
Wednesday	0
Thursday	negative one divided by eight
Friday	prefix plus of

37. The table above shows the daily change in the price of a certain stock last week. What was the net change in dollars in the price of the stock for the week?

   A. negative

   B. negative

   C. prefix plus of

   D. prefix plus of

   E. negative

   Arithmetic **Interpretation of Tables**

   The net change in dollars is
   equation sequence part 1 three divided by two minus three divided by four minus one divided by eight plus nine divided by four equals part 2 12 minus six minus one plus 18 divided by eight equals part 3 23 divided by eight equals
   . So, that’s an increase of dollars.

   The correct answer is C.

38. Of 120 hotel rooms rented one night, some were suites rented for $115 each and the rest were double rooms rented for $85 each. If the total

revenue from the room rentals for this night was $10,890, how many suites were rented?

A. 23

B. 35

C. 54

D. 94

E. 99

Algebra **Simultaneous Equations**

Let x be the number of suites rented. Then
equation sequence part 1 normal dollar times 10,890 equals part 2 normal dollar times 115 times x plus normal dollar times 85 times left parenthesis 120 minus x right parenthesis equals part 3 normal dollar times 115 times x plus normal dollar times 10,200 minus normal dollar times 85 times x
, so normal dollar times 690 equals normal dollar times 30 times x. Thus, x equals 23.

The correct answer is A.

39. Last week Jack worked 70 hours and earned $1,260. If he earned his regular hourly wage for the first 40 hours worked, times his regular hourly wage for the next 20 hours worked, and 2 times his regular hourly wage for the remaining 10 hours worked, what was his regular hourly wage?

A. $7.00

B. $14.00

C. $18.00

D. $22.00

E. $31.50

Algebra **First-Degree Equations**

If w represents Jack's regular hourly wage, then Jack's earnings for the week can be represented by the sum of the following amounts, in

dollars: 40 times w (his earnings for the first 40 hours he worked), left parenthesis 20 right parenthesis times left parenthesis 1.5 times w right parenthesis
(his earnings for the next 20 hours he worked), and
left parenthesis 10 right parenthesis times left parenthesis two times w right parenthesis
(his earnings for the last 10 hours he worked). Therefore,

sum with 3 summands 40 times w plus left parenthesis 20 right parenthesis times left parenthesis 1.5 times w right parenthesis plus left parenthesis 10 right parenthesis times left parenthesis two times w right parenthesis	= 1,260	given
90 times w	= 1,260	add like terms
w	= 14	divide both sides by 90

Jack’s regular hourly wage was $14.00.

The correct answer is B.

40. If x divided by four is 2 more than x divided by eight, then x equals

A. 4

B. 8

C. 16

D. 32

E. 64

Algebra **First-Degree Equations**

If x divided by four equals two plus x divided by eight, then we can multiply both sides of the equation by 8 to find that two times x equals 16 plus x, so $x = 16$.

The correct answer is C.

41. If left parenthesis two super x right parenthesis times left parenthesis two super y right parenthesis equals eight and left parenthesis nine super x right parenthesis times left parenthesis three super y right parenthesis equals 81, then left parenthesis x comma y right parenthesis equals

   A. (1, 2)

   B. (2, 1)

   C. (1, 1)

   D. (2, 2)

   E. (1, 3)

Algebra **Exponents**

If equation sequence part 1 left parenthesis two super x right parenthesis times left parenthesis two super y right parenthesis equals part 2 eight equals part 3 two cubed, then x plus y equals three. Of the five answer options, the only ones satisfying this equation are option A, in which left parenthesis x comma y right parenthesis equals left parenthesis one comma two right parenthesis, and option B, in which left parenthesis x comma y right parenthesis equals left parenthesis two comma one right parenthesis. Since

multirelation left parenthesis nine super x right parenthesis times left parenthesis three super y right parenthesis equals 81 equals left parenthesis nine super one right parenthesis times left parenthesis three squared right parenthesis not equals left parenthesis nine squared right parenthesis times left parenthesis three super one right parenthesis
, we find that x equals one and y equals two, so
left parenthesis x comma y right parenthesis equals left parenthesis one comma two right parenthesis
.

The correct answer is A.

42. A certain toll station on a highway has 7 tollbooths, and each tollbooth collects $0.75 from each vehicle that passes it. From 6 o’clock yesterday morning to 12 o’clock midnight, vehicles passed each of the tollbooths at the average rate of 4 vehicles per minute. Approximately how much money did the toll station collect during that time period?

   A. $1,500

   B. $3,000

   C. $11,500

   D. $23,000

   E. $30,000

Arithmetic **Rate Problem**

On average, 4 vehicles pass each tollbooth every minute. There are 7 tollbooths at the station, and each passing vehicle pays $0.75. Therefore, the average rate, per minute, at which money is collected by the toll station is
equation sequence part 1 normal dollar times left parenthesis seven multiplication four multiplication 0.75 right parenthesis equals part 2 normal dollar times left parenthesis seven multiplication four multiplication three divided by four right parenthesis equals part 3 normal dollar times left parenthesis seven multiplication three right parenthesis equals part 4 normal dollar times 21
. From 6:00 a.m. through midnight there are 18 hours. And because 18 hours is equal to 18 × 60 minutes, from 6:00 a.m. through midnight there are 1,080 minutes. The total amount of money collected by the toll station during this period is therefore 1,080 × $21 = $22,680, which is approximately $23,000.

The correct answer is D.

43. If y equals two plus two times k and y not equals zero, then sum with 4 summands one divided by y plus one divided by y plus one divided by y plus one divided by y equals

A. one divided by eight plus eight times k

B. two divided by one plus k

C. one divided by eight plus k

D. four divided by eight plus k

E. four divided by one plus k

Algebra **Equations**

If multirelation y equals two plus two times k not equals zero, then equation sequence part 1 sum with 4 summands one divided by y plus one divided by y plus one divided by y plus one divided by y equals part 2 four divided by y equals part 3 four divided by two plus two times k equals part 4 two divided by one plus k
.

The correct answer is B.

44. If three super x minus three super x minus one equals 162, then x times left parenthesis x minus one right parenthesis equals

   A. 12

   B. 16

   C. 20

   D. 30

   E. 81

Algebra **Exponents**

If
equation sequence part 1 three super x minus three super x minus one equals part 2 162 equals part 3 two times left parenthesis 81 right parenthesis equals part 4 two times left parenthesis three super four right parenthesis equals part 5 three times left parenthesis three super four right parenthesis minus one times left parenthesis three super four right parenthesis equals part 6 three super five minus three super four
, then x equals five. Thus,
equation sequence part 1 x times left parenthesis x minus one right parenthesis equals part 2 five times left parenthesis four right parenthesis equals part 3 20
.

The correct answer is C.

45. The expression n factorial is defined as the product of the integers from 1 through n. If p is the product of the integers from 100 through 299 and q is the product of the integers from 200 through 299, which of the following is equal to p divided by q?

   A. 99!

   B. 199!

   C. 199 factorial divided by 99 factorial

   D. 299 factorial divided by 99 factorial

   E. 299 factorial divided by 199 factorial

Arithmetic **Series and Sequences**

The number p is equal to $100 \times 101 \times 102 \times \ldots \times 299$ and the number q is equal to $200 \times 201 \times 202 \times \ldots \times 299$. The number p divided by q is thus equal to

100 multiplication 101 multiplication 102 multiplication ellipsis multiplication 299 divided by 200 multiplication 201 multiplication 202 multiplication ellipsis multiplication 299 equals 100 multiplication 101 multiplication 102 multiplication ellipsis multiplication 199 multiplication 200 multiplication 201 multiplication 202 multiplication ellipsis multiplication 299 divided by 200 multiplication 201 multiplication 202 multiplication ellipsis multiplication 299

. Canceling $200 \times 201 \times 202 \times \ldots \times 299$ from the numerator and the denominator, we see that

p divided by q equals 100 multiplication 101 multiplication 102 multiplication full stop full stop full stop multiplication 199

. Note that the multiplication in this expression for p divided by q begins with 100 (the smallest of the numbers being multiplied), whereas the multiplication in

n factorial equals one multiplication two multiplication three multiplication full stop full stop full stop multiplication n

begins with 1. Starting with 199! as our numerator, we thus need to find a denominator that will cancel the undesired elements of the multiplication (in 199!). This number is $1 \times 2 \times 3 \times \ldots \times 99 = 99!$ That is,

equation sequence part 1 p divided by q equals part 2 100 multiplication 101 multiplication 102 multiplication full stop full stop full stop multiplication 199 equals part 3 one multiplication two multiplication three multiplication ellipsis multiplication 99 multiplication 100 multiplication 101 multiplication 102 multiplication ellipsis multiplication 199 divided by one multiplication two multiplication three multiplication ellipsis multiplication 99 equals part 4 199 factorial divided by 99 factorial

.

The correct answer is C.

46. A school club plans to package and sell dried fruit to raise money. The club purchased 12 containers of dried fruit, each containing pounds. What is the maximum number of individual bags of dried fruit, each containing one divided by four pounds, that can be sold from the dried fruit the club purchased?

A. 50

B. 64

C. 67

D. 768

E. 804

Arithmetic **Applied Problems; Operations with Fractions**

The 12 containers, each containing pounds of dried fruit, contain a total of
equation sequence part 1 left parenthesis 12 right parenthesis times left parenthesis right parenthesis equals part 2 left parenthesis 12 right parenthesis times left parenthesis 67 divided by four right parenthesis equals part 3 left parenthesis three right parenthesis times left parenthesis 67 right parenthesis equals part 4 201
pounds of dried fruit, which will make
equation sequence part 1 201 divided by one divided by four equals part 2 left parenthesis 201 right parenthesis times left parenthesis four right parenthesis equals part 3 804
individual bags that can be sold.

The correct answer is E.

Height	Price
Less than 5 ft	$14.95
5 ft to 6 ft	$17.95
Over 6 ft	$21.95

47. A nursery sells fruit trees priced as shown in the chart above. In its inventory 54 trees are less than 5 feet in height. If the expected revenue from the sale of its entire stock is estimated at $2,450, approximately how much of this will come from the sale of trees that are at least 5 feet tall?

   A. $1,730

   B. $1,640

   C. $1,410

   D. $1,080

   E. $810

Arithmetic **Applied Problems**

If the nursery sells its entire stock of trees, it will sell the 54 trees that are less than 5 feet in height at the price per tree of $14.95 shown in the chart. The expected revenue from the sale of the trees that are less than 5 feet tall is therefore 54 × $14.95 = $807.30. The revenue from the sale of the trees that are at least 5 feet tall is thus equal to the total revenue from the sale of the entire stock of trees minus $807.30. The revenue from the sale of the entire stock of trees is estimated at $2,450. Based on this estimate, the revenue from the sale of the trees that are at least 5 feet tall will be $2,450 − $807.30 = $1,642.70, which is approximately $1,640.

The correct answer is B.

48. A certain bridge is 4,024 feet long. Approximately how many minutes does it take to cross this bridge at a constant speed of 20 miles per hour? (1 mile = 5,280 feet)

   A. 1

   B. 2

   C. 4

   D. 6

   E. 7

Arithmetic **Applied Problems**

First, convert 4,024 feet to miles since the speed is given in miles per hour:
4,024 ft multiplication one mi divided by 5,280 ft equals 4,024 divided by 5,280 mi
. Now, divide by 20 mph:
4,024 divided by 5,280 equation sequence part 1 mi division 20 mi divided by one hr equals part 2 4,024 mi divided by 5,280 multiplication one hr divided by 20 mi equals part 3 4,024 hr divided by left parenthesis 5,280 right parenthesis times left parenthesis 20 right parenthesis
.

Last, convert
4,024 hr divided by left parenthesis 5,280 right parenthesis times left parenthesis 20 right parenthesis
to minutes:
multirelation 4,024 hr divided by left parenthesis 5,280 right parenthesis times left parenthesis 20 right parenthesis multiplication 60 min divided by one hr equals left parenthesis 4,024 right parenthesis times left parenthesis 60 right parenthesis min divided by left parenthesis 5,280 right parenthesis times left parenthesis 20 right parenthesis almost equals 4,000 divided by 5,000 multiplication 60 divided by 20 min full stop Then comma 4,000 divided by 5,000 multiplication 60 divided by 20 min equals 0.8 multiplication three min almost equals two min
. Thus, at a constant speed of 20 miles per hour, it takes approximately 2 minutes to cross the bridge.

The correct answer is B.

49. A purse contains 57 coins, all of which are nickels, dimes, or quarters. If the purse contains x dimes and 8 more nickels than dimes, which of the following gives the number of quarters the purse contains in terms of x?

   A. two times x minus 49

   B. two times x plus 49

   C. two times x minus 65

D. 49 minus two times x

E. 65 minus two times x

Algebra **First-Degree Equations**

Letting cap q be the number of quarters, there are left parenthesis x plus eight right parenthesis nickels, x dimes, and cap q quarters for a total of 57 coins.

sum with 3 summands left parenthesis x plus eight right parenthesis plus x plus cap q	=	57	given
sum with 3 summands two times x plus eight plus cap q	=	57	combine like terms
cap q	=	49 minus two times x	subtract two times x plus eight from both sides

The correct answer is D.

50. The annual interest rate earned by an investment increased by 10 percent from last year to this year. If the annual interest rate earned by the investment this year was 11 percent, what was the annual interest rate last year?

A. 1%

B. 1.1%

C. 9.1%

D. 10%

E. 10.8%

Arithmetic **Percents**

If cap l is the annual interest rate last year, then the annual interest rate this year is 10% greater than cap l, or 1.1 times cap l. It is

given that 1.1 times cap l equals 11 percent. Therefore,
equation sequence part 1 cap l equals part 2 11 percent divided by 1.1 equals part 3 10 percent
. (Note that if the given information had been that the investment increased by *10 percentage points*, then the equation would have been cap l plus 10 percent equals 11 percent.) **The correct answer is D.**

51. A total of 5 liters of gasoline is to be poured into two empty containers with capacities of 2 liters and 6 liters, respectively, such that both containers will be filled to the same percent of their respective capacities. What amount of gasoline, in liters, must be poured into the 6-liter container?

   A.

   B. 4

   C.

   D. 3

   E.

   Algebra Ratio and Proportion

   If x represents the amount, in liters, of gasoline poured into the 6-liter container, then five minus x represents the amount, in liters, of gasoline poured into the 2-liter container. After the gasoline is poured into the containers, the 6-liter container will be filled to
   left parenthesis x divided by six multiplication 100 right parenthesis percent
   of its capacity and the 2-liter container will be filled to
   left parenthesis five minus x divided by two multiplication 100 right parenthesis percent
   of its capacity. Because these two percents are equal,

x divided by six	=	five minus x divided by two	given
two times x	=	six times left parenthesis five minus x right parenthesis	multiply both sides by 12
two times x	=	30 minus six times x	use distributive property
eight times x	=	30	add six times x to both sides
x	=		divide both sides by 8

Therefore, liters of gasoline must be poured into the 6-liter container.

The correct answer is C.

52. What is the larger of the 2 solutions of the equation x squared minus four times x equals 96?

A. 8

B. 12

C. 16

D. 32

E. 100

Algebra **Second-Degree Equations**

It is given that x squared minus four times x equals 96, or x squared minus four times x minus 96 equals zero, or left parenthesis x minus 12 right parenthesis times left parenthesis x plus eight right parenthesis equals zero. Therefore, x equals 12 or x equals negative eight, and the larger of these two numbers is 12.

Alternatively, from x squared minus four times x equals 96 it follows that x times left parenthesis x minus four right parenthesis equals 96. By inspection, the left side is either the product of 12 and 8, where the value of x is 12, or the product of −8 and −12, where the value of x is −8, and the larger of these two values of x is 12.

The correct answer is B.

x equals one divided by six times g times t squared

53. In the formula shown, if g is a constant and x equals negative six when t equals two, what is the value of x when t equals four?

A. –24

B. –20

C. –15

D. 20

E. 24

Algebra Formulas

Since x equals negative six when t equals two, it follows that
negative six equals left parenthesis one divided by six right parenthesis times left parenthesis g right parenthesis times left parenthesis four right parenthesis
so
equation sequence part 1 g equals part 2 three divided by two times left parenthesis negative six right parenthesis equals part 3 negative nine
. Then, when t equals four,
equation sequence part 1 x equals part 2 one divided by six times left parenthesis negative nine right parenthesis times left parenthesis 16 right parenthesis equals part 3 negative 24
.

The correct answer is A.

54. left parenthesis 39,897 right parenthesis times left parenthesis 0.0096 right parenthesis divided by 198.76
is approximately

A. 0.02

B. 0.2

C. 2

D. 20

E. 200

Arithmetic **Estimation**

multirelation left parenthesis 39,897 right parenthesis times left parenthesis 0.0096 right parenthesis divided by 198.76 almost equals left parenthesis 40,000 right parenthesis times left parenthesis 0.01 right parenthesis divided by 200 equals left parenthesis 200 right parenthesis times left parenthesis 0.01 right parenthesis equals two

The correct answer is C.

55. The “prime sum” of an integer n greater than 1 is the sum of all the prime factors of n, including repetitions. For example, the prime sum of 12 is 7, since 12 = 2 × 2 × 3 and 2 + 2 + 3 = 7. For which of the following integers is the prime sum greater than 35?

A. 440

B. 512

C. 620

D. 700

E. 750

Arithmetic **Properties of Numbers**

A. Since 440 = 2 × 2 × 2 × 5 × 11, the prime sum of 440 is 2 + 2 + 2 + 5 + 11 = 22, which is not greater than 35.

B. Since 512 = 2^9, the prime sum of 512 is 9(2) = 18, which is not greater than 35.

C. Since 620 = 2 × 2 × 5 × 31, the prime sum of 620 is 2 + 2 + 5 + 31 = 40, which is greater than 35.

Because there can be only one correct answer, D and E need not be checked. However, for completeness,

D. Since 700 = 2 × 2 × 5 × 5 × 7, the prime sum of 700 is 2 + 2 + 5 + 5 + 7 = 21, which is not greater than 35.

E. Since 750 = 2 × 3 × 5 × 5 × 5, the prime sum of 750 is 2 + 3 + 5 + 5 + 5 = 20, which is not greater than 35.

The correct answer is C.

56. Each machine at a toy factory assembles a certain kind of toy at a constant rate of one toy every 3 minutes. If 40 percent of the machines at the factory are to be replaced by new machines that assemble this kind of toy at a constant rate of one toy every 2 minutes, what will be the percent increase in the number of toys assembled in one hour by all the machines at the factory, working at their constant rates?

A. 20%

B. 25%

C. 30%

D. 40%

E. 50%

Arithmetic **Applied Problems; Percents**

Let n be the total number of machines working. Currently, it takes each machine 3 minutes to assemble 1 toy, so each machine assembles 20 toys in 1 hour and the total number of toys assembled in 1 hour by all the current machines is $20n$. It takes each new machine 2 minutes to assemble 1 toy, so each new machine assembles 30 toys in 1 hour. If 60% of the machines assemble 20 toys each hour and 40% assemble 30 toys each hour, then the total number of toys produced by the machines each hour is

$$(0.60n)(20) + (0.40n)(30) = 24n$$

. The percent increase in hourly production is
24 times n minus 20 times n divided by 20 times n equals one divided by five
or 20%.

The correct answer is A.

57. When a subscription to a new magazine was purchased for m months, the publisher offered a discount of 75 percent off the regular monthly price of the magazine. If the total value of the discount was equivalent to buying the magazine at its regular monthly price for 27 months, what was the value of m?

A. 18

B. 24

C. 30

D. 36

E. 48

Algebra **Percents**

Let cap p represent the regular monthly price of the magazine. The discounted monthly price is then 0.75 times cap p. Paying this price for m months is equivalent to paying the regular price for 27 months. Therefore, 0.75 times m times cap p equals 27 times cap p, and so 0.75 times m equals 27. It follows that
equation sequence part 1 m equals part 2 27 divided by 0.75 equals part 3 36
.

The correct answer is D.

58. At a garage sale, all of the prices of the items sold were different. If the price of a radio sold at the garage sale was both the 15^{th} highest price and the 20^{th} lowest price among the prices of the items sold, how many items were sold at the garage sale?

A. 33

B. 34

C. 35

D. 36

E. 37

Arithmetic **Operations with Integers**

If the price of the radio was the 15th highest price, there were 14 items that sold for prices higher than the price of the radio. If the price of the radio was the 20th lowest price, there were 19 items that sold for prices lower than the price of the radio. Therefore, the total number of items sold is 14 + 1 + 19 = 34.

The correct answer is B.

59. Half of a large pizza is cut into 4 equal-sized pieces, and the other half is cut into 6 equal-sized pieces. If a person were to eat 1 of the larger pieces and 2 of the smaller pieces, what fraction of the pizza would remain uneaten?

A. five divided by 12

B. 13 divided by 24

C. seven divided by 12

D. two divided by three

E. 17 divided by 24

Arithmetic **Operations with Fractions**

Each of the 4 equal-sized pieces represents one divided by eight of the whole pizza since each slice is one divided by four of one divided by two of the pizza. Each of the 6 equal-sized pieces represents one divided by 12 of the whole pizza since each slice is one divided by six of one divided by two of the pizza. The fraction of the pizza remaining after a person eats one of the larger pieces and 2 of the smaller pieces is

equation sequence part 1 one minus left square bracket one divided by eight plus two times left parenthesis one divided by 12 right parenthesis right square bracket equals part 2 one minus left parenthesis one divided by eight plus one divided by six right parenthesis equals part 3 one minus six plus eight divided by 48 equals part 4 one minus seven divided by 24 equals part 5 17 divided by 24
.

The correct answer is E.

60. If
a equals sum with 4 summands one plus one divided by four plus one divided by 16 plus one divided by 64
and b equals one plus one divided by four times a, then what is the value of a minus b?

A. negative 85 divided by 256

B. negative one divided by 256

C. negative one divided by four

D. 125 divided by 256

E. 169 divided by 256

Arithmetic **Operations with Fractions**

Given that
a equals sum with 4 summands one plus one divided by four plus one divided by 16 plus one divided by 64
, it follows that
one divided by four times a equals sum with 4 summands one divided by four plus one divided by 16 plus one divided by 64 plus one divided by 256
and so
b equals sum with 5 summands one plus one divided by four plus one divided by 16 plus one divided by 64 plus one divided by 256
. Then

equation sequence part 1 a minus b equals part 2 left parenthesis sum with 4 summands one plus one divided by four plus one divided by 16 plus one divided by 64 right parenthesis minus left parenthesis sum with 5 summands one plus one divided by four plus one divided by 16 plus one divided by 64 plus one divided by 256 right parenthesis equals part 3 negative one divided by 256
.

The correct answer is B.

61. In a certain learning experiment, each participant had three trials and was assigned, for each trial, a score of either –2, –1, 0, 1, or 2. The participant’s final score consisted of the sum of the first trial score, 2 times the second trial score, and 3 times the third trial score. If Anne received scores of 1 and –1 for her first two trials, not necessarily in that order, which of the following could NOT be her final score?

   A. –4

   B. –2

   C. 1

   D. 5

   E. 6

Arithmetic **Applied Problems**

If x represents Anne’s score on the third trial, then Anne’s final score is either
equation sequence part 1 sum with 3 summands one plus two times left parenthesis negative one right parenthesis plus three times x equals part 2 sum with 3 summands three times x minus one times or minus one plus two times left parenthesis one right parenthesis plus three times x equals part 3 three times x plus one
, where x can have the value –2, –1, 0, 1, or 2. The following table shows Anne’s final score for each possible value of x.

x	three times x minus one	three times x plus one
–2	–7	–5
–1	–4	–2
0	–1	1
1	2	4
2	5	7

Among the answer choices, the only one not found in the table is 6.

The correct answer is E.

62. For all positive integers m and v, the expression m times normal cap theta times v represents the remainder when m is divided by v. What is the value of ((98 Θ 33) Θ 17) − (98 Θ (33 Θ 17))?

A. −10

B. −2

C. 8

D. 13

E. 17

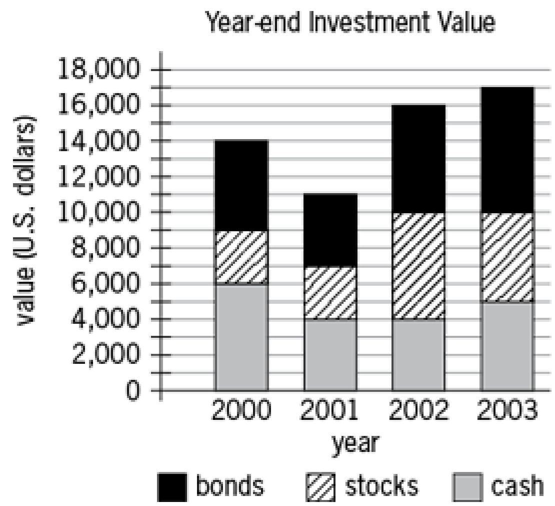
Arithmetic **Operations with Integers**

First, for ((98 Θ 33) Θ 17), determine 98 Θ 33, which equals 32, since 32 is the remainder when 98 is divided by 33 (98 = 2(33) + 32). Then, determine 32 Θ 17, which equals 15, since 15 is the remainder when 32 is divided by 17 (32 = 1(17) + 15). Thus, ((98 Θ 33) Θ 17) = 15.

Next, for (98 Θ (33 Θ 17)), determine 33 Θ 17, which equals 16, since 16 is the remainder when 33 is divided by 17 (33 = 1(17) + 16). Then, determine 98 Θ 16, which equals 2, since 2 is the remainder when 98 is divided by 16 (98 = 6(16) + 2). Thus, (98 Θ (33 Θ 17)) = 2.

Finally, ((98 Θ 33) Θ 17 − (98 Θ (33 Θ 17)) = 15 − 2 = 13.

The correct answer is D.



63. The chart above shows year-end values for Darnella's investments. For just the stocks, what was the increase in value from year-end 2000 to year-end 2003?

   A. $1,000

   B. $2,000

   C. $3,000

   D. $4,000

   E. $5,000

Arithmetic **Interpretation of Graphs**

From the graph, the year-end 2000 value for stocks is 9,000 − 6,000 = 3,000 and the year-end 2003 value for stocks is 10,000 − 5,000 = 5,000. Therefore, for just the stocks, the increase in value from year-end 2000 to year-end 2003 is 5,000 − 3,000 = 2,000.

The correct answer is B.

64. If the sum of the reciprocals of two consecutive odd integers is 12 divided by 35, then the greater of the two integers is

A. 3

B. 5

C. 7

D. 9

E. 11

Arithmetic **Operations with Fractions**

The sum of the reciprocals of 2 integers, a and b, is one divided by a plus one divided by b equals a plus b divided by a times b
. Therefore, since 12 divided by 35 is the sum of the reciprocals of 2 consecutive odd integers, the integers must be such that their sum is a multiple of 12 and their product is the same multiple of 35 so that the fraction reduces to 12 divided by 35. Considering the simplest case where a plus b equals 12 and a times b equals 35, it is easy to see that the integers are 5 and 7 since 5 and 7 are the only factors of 35 that are consecutive odd integers. The larger of these is 7.

Algebraically, if a is the greater of the two integers, then
b equals a minus two and
multiline equation row 1 a plus left parenthesis a minus two right parenthesis divided by a times left parenthesis a minus two right parenthesis equals 12 divided by 35 row 2 two times a minus two divided by a times left parenthesis a minus two right parenthesis equals 12 divided by 35 row 3 35 times left parenthesis two times a minus two right parenthesis equals 12 times a times left parenthesis a minus two right parenthesis row 4 70 times a minus 70 equals 12 times a squared minus 24 times a row 5 zero equals 12 times a squared minus 94 times a plus 70 row 6 zero equals two times left parenthesis six times a minus five right parenthesis times left parenthesis a minus seven right parenthesis

Thus, six times a minus five equals zero, so
a equals five divided by six, or a minus seven equals zero, so
a equals seven. Since a must be an integer, it follows that
a equals seven.

The correct answer is C.

65. What is the sum of the odd integers from 35 to 85, inclusive?

A. 1,560

B. 1,500

C. 1,240

D. 1,120

E. 1,100

Arithmetic Operations on Integers

The odd integers from 35 through 85 form an arithmetic sequence with first term 35 and each subsequent term 2 more than the preceding term. Thus the sum 35 + 37 + 39 + … + 85 can be found as follows:

1^{st} term	35	=	35		
2^{nd} term	37	=	35	+	1(2)
3^{rd} term	39	=	35	+	2(2)
4^{th} term	41	=	35	+	3(2)
…	…	…	…	…	…
26^{th} term	85	=	35	+	25(2)
	Sum	=	35(26)	+	(1 + 2 + 3 + … + 25)(2)
		=	35(26)	+	left parenthesis 25 right parenthesis times left parenthesis 26 right parenthesis divided by two (2) see note below
		=	910	+	650
		=	1,560		

Note that if
s equals sum with variable number of summands one plus two plus three plus ellipsis plus 25
, then

two times s equals left parenthesis sum with variable number of summands one plus two plus three plus ellipsis plus 25 right parenthesis plus left parenthesis sum with variable number of summands 25 plus 24 plus 23 plus ellipsis plus one right parenthesis
, and so
equation sequence part 1 two times s equals part 2 sum with variable number of summands left parenthesis one plus 25 right parenthesis plus left parenthesis two plus 24 right parenthesis plus left parenthesis three plus 23 right parenthesis plus ellipsis plus left parenthesis 25 plus one right parenthesis equals part 3 left parenthesis 25 right parenthesis times left parenthesis 26 right parenthesis
. Therefore,
s equals left parenthesis 25 right parenthesis times left parenthesis 26 right parenthesis divided by two
.

Alternatively, to determine the number of odd integers from 35 to 85, inclusive, consider that 3 of them (35, 37, and 39) have tens digit 3. Half of the integers with tens digit 4 are odd, so 5 of the odd integers between 35 and 85, inclusive, have tens digit 4. Similarly, 5 of the odd integers between 35 and 85, inclusive, have tens digit 5; 5 have tens digit 6; and 5 have tens digit 7. Finally, 3 have tens digit 8 (81, 83, and 85), and so the number of odd integers between 35 and 85, inclusive, is $3 + 5 + 5 + 5 + 5 + 3 = 26$. Now, let
cap s equals sum with variable number of summands 35 plus 37 plus 39 plus ellipsis plus 85
. Then,
cap s equals sum with variable number of summands 85 plus 83 plus 81 plus ellipsis plus 35
, and it follows that
equation sequence part 1 two times cap s equals part 2 sum with variable number of summands left parenthesis 35 plus 85 right parenthesis plus left parenthesis 37 plus 83 right parenthesis plus left parenthesis 39 plus 81 right parenthesis plus ellipsis plus left parenthesis 85 plus 35 right parenthesis equals part 3 left parenthesis 120 right parenthesis times left parenthesis 26 right parenthesis
. Thus,

equation sequence part 1 cap s equals part 2 sum with variable number of summands 35 plus 37 plus 39 plus ellipsis plus 85 equals part 3 left parenthesis 120 right parenthesis times left parenthesis 26 right parenthesis divided by two equals part 4 1,560
.

The correct answer is A.

66. For all numbers a, b, c, and d,
vector element 1 a b element 2 c d is defined by the equation
vector element 1 a b element 2 c d equals a times d minus c times b.
Which of the following is equal to
vector element 1 s t element 2 one three minus vector element 1 negative t two element 2 s four plus vector element 1 two two element 2 t s
?

A. vector element 1 s t element 2 one five

B. vector element 1 s t element 2 seven one

C. vector element 1 s t element 2 five seven

D. vector element 1 s negative t element 2 one five

E. vector element 1 s negative t element 2 one seven

Algebra Formulas

First, expand the given expression using the given definition.

equation sequence part 1 vector element 1 s t element 2 one three minus vector element 1 negative t two element 2 s four plus vector element 1 two two element 2 t s equals part 2 left parenthesis three times s minus t right parenthesis minus left parenthesis negative four times t minus two times s right parenthesis plus left parenthesis two times s minus two times t right parenthesis equals part 3 sum with 4 summands three times s minus t plus four times t plus two times s plus two times s minus two times t equals part 4 seven times s plus t

Next, compare the result, seven times s plus t, with the expanded versions of the answer choices.

A
vector element 1 s t element 2 one five equals five times s minus t (not correct)

B
vector element 1 s t element 2 seven one equals s minus seven times t (not correct)

C
vector element 1 s t element 2 five seven equals seven times s minus five times t (not correct)

D
vector element 1 s negative t element 2 one five equals five times s plus t (not correct)

E
vector element 1 s negative t element 2 one seven equals seven times s plus t (correct)

The correct answer is E.

67. In a certain sequence, each term after the first term is one-half the previous term. If the tenth term of the sequence is between 0.0001 and 0.001, then the twelfth term of the sequence is between

A. 0.0025 and 0.025

B. 0.00025 and 0.0025

C. 0.000025 and 0.00025

D. 0.0000025 and 0.000025

E. 0.00000025 and 0.0000025

Arithmetic Sequences

Let a sub n represent the n super th term of the sequence. It is given that each term after the first term is one divided by two the previous term and that 0.0001 less than a sub 10 less than 0.001. Then for

a sub 11 comma 0.0001 divided by two less than a sub 11 less than 0.001 divided by two

, or

0.00005 less than a sub 11 less than 0.0005 full stop times For times a sub 12 comma 0.00005 divided by two less than times a sub 12 less than 0.0005 divided by two comma or times 0.000025 less than a sub 12 less than 0.00025

. Thus, the twelfth term of the sequence is between 0.000025 and 0.00025.

The correct answer is C.

68. A certain drive-in movie theater has a total of 17 rows of parking spaces. There are 20 parking spaces in the first row and 21 parking spaces in the second row. In each subsequent row there are 2 more parking spaces than in the previous row. What is the total number of parking spaces in the movie theater?

    A. 412

    B. 544

    C. 596

    D. 632

    E. 692

    Arithmetic **Operations on Integers**

Row	Number of parking spaces		
1^{st} row	20		
2^{nd} row	21		
3^{rd} row	21	+	1(2)
4^{th} row	21	+	2(2)
…	…	…	…
17^{th} row	21 +		15(2)

Then, letting cap s represent the total number of parking spaces in the theater,

multiline equation row 1 cap s equals sum with 3 summands 20 plus left parenthesis 16 right parenthesis times left parenthesis 21 right parenthesis plus left parenthesis sum with variable number of summands one plus two plus three plus ellipsis plus 15 right parenthesis times left parenthesis two right parenthesis row 2 equals sum with 3 summands 20 plus 336 plus left parenthesis 15 right parenthesis times left parenthesis 16 right parenthesis divided by two times left parenthesis two right parenthesis see note below row 3 equals 356 plus 240 row 4 equals 596

Note that if
s equals sum with variable number of summands one plus two plus three plus ellipsis plus 15
, then
two times s equals left parenthesis sum with variable number of summands one plus two plus three plus ellipsis plus 15 right parenthesis plus left parenthesis sum with variable number of summands 15 plus 14 plus 13 plus ellipsis plus one right parenthesis

, and so
equation sequence part 1 two times s equals part 2 sum with variable number of summands left parenthesis one plus 15 right parenthesis plus left parenthesis two plus 14 right parenthesis plus left parenthesis three plus 13 right parenthesis plus ellipsis plus left parenthesis 15 plus one right parenthesis equals part 3 left parenthesis 15 right parenthesis times left parenthesis 16 right parenthesis
. Therefore,
s equals left parenthesis 15 right parenthesis times left parenthesis 16 right parenthesis divided by two
.

The correct answer is C.

69. Ada and Paul received their scores on three tests. On the first test, Ada's score was 10 points higher than Paul's score. On the second test, Ada's score was 4 points higher than Paul's score. If Paul's average (arithmetic mean) score on the three tests was 3 points higher than Ada's average score on the three tests, then Paul's score on the third test was how many points higher than Ada's score?

A. 9

B. 14

C. 17

D. 23

E. 25

Algebra **Statistics**

Let a sub one, a sub two, and a sub three be Ada's scores on the first, second, and third tests, respectively, and let p sub one, p sub two, and p sub three be Paul's scores on the first, second, and third tests, respectively. Then, Ada's average score is
sum with 3 summands a sub one plus a sub two plus a sub three divided by three
and Paul's average score is
sum with 3 summands p sub one plus p sub two plus p sub three divided by three

. But Paul’s average score is 3 points higher than Ada’s average score, so
sum with 3 summands p sub one plus p sub two plus p sub three divided by three equals sum with 3 summands a sub one plus a sub two plus a sub three divided by three plus three
. Also, it is given that a sub one equals p sub one plus 10 and a sub two equals p sub two plus four, so by substitution,
sum with 3 summands p sub one plus p sub two plus p sub three divided by three equals sum with 3 summands left parenthesis p sub one plus 10 right parenthesis plus left parenthesis p sub two plus four right parenthesis plus a sub three divided by three plus three
. Then
sum with 3 summands p sub one plus p sub two plus p sub three equals sum with 4 summands left parenthesis p sub one plus 10 right parenthesis plus left parenthesis p sub two plus four right parenthesis plus a sub three plus nine
and so p sub three equals a sub three plus 23. On the third test, Paul’s score was 23 points higher than Ada’s score.

The correct answer is D.

70. The price of a certain stock increased by 0.25 of 1 percent on a certain day. By what fraction did the price of the stock increase that day?

A. one divided by 2,500

B. one divided by 400

C. one divided by 40

D. one divided by 25

E. one divided by four

Arithmetic **Percents**

It is given that the price of a certain stock increased by 0.25 of 1 percent on a certain day. This is equivalent to an increase of one divided by four of one divided by 100, which is left parenthesis one divided by four right parenthesis times left parenthesis one divided by 100 right parenthesis

, and
left parenthesis one divided by four right parenthesis times left parenthesis one divided by 100 right parenthesis equals one divided by 400
.

The correct answer is B.

71. For each trip, a taxicab company charges $4.25 for the first mile and $2.65 for each additional mile or fraction thereof. If the total charge for a certain trip was $62.55, how many miles at most was the trip?

A. 21

B. 22

C. 23

D. 24

E. 25

Arithmetic Applied Problems

Subtracting the charge for the first mile leaves a charge of $62.55 – $4.25 = $58.30 for the miles after the first mile. Divide this amount by $2.65 to find the number of miles to which $58.30 corresponds: 58.30 divided by 2.65 equals 22 miles. Therefore, the total number of miles is at most 1 (the first mile) added to 22 (the number of miles after the first mile), which equals 23.

The correct answer is C.

72. When 24 is divided by the positive integer n, the remainder is 4. Which of the following statements about n must be true?

I. n is even.

II. n is a multiple of 5.

III. n is a factor of 20.

A. III only

B. I and II only

C. I and III only

D. II and III only

E. I, II, and III

Arithmetic **Properties of Numbers**

Since the remainder is 4 when 24 is divided by the positive integer n and the remainder must be less than the divisor, it follows that 24 equals q times n plus four for some positive integer q and four less than n, or q times n equals 20 and n greater than four. It follows that n equals five, or n equals 10, or n equals 20 since these are the only factors of 20 that exceed 4.

I. n is not necessarily even. For example, n could be 5.

II. n is necessarily a multiple of 5 since the value of n is either 5, 10, or 20.

III. n is a factor of 20 since 20 equals q times n for some positive integer q.

The correct answer is D.

73. Terry needs to purchase some pipe for a plumbing job that requires pipes with lengths of 1 ft 4 in, 2 ft 8 in, 3 ft 4 in, 3 ft 8 in, 4 ft 8 in, 5 ft 8 in, and 9 ft 4 in. The store from which Terry will purchase the pipe sells pipe only in 10-ft lengths. If each 10-ft length can be cut into shorter pieces, what is the minimum number of 10-ft pipe lengths that Terry needs to purchase for the plumbing job?

(Note: 1 ft = 12 in)

A. 3

B. 4

C. 5

D. 6

E. 7

Arithmetic **Operations with Integers; Measurement Conversion**

The 7 lengths of pipe Terry needs total 30 feet plus 8 inches, which means Terry will need to buy at least 4 pipes, each 10 feet long. Four pipes will suffice if Terry cuts pieces of the following lengths:

1st pipe: 9 feet 4 inches (with 8 inches left)

2nd pipe: 5 feet 8 inches and 3 feet 8 inches (with 8 inches left)

3rd pipe: 4 feet 8 inches, 3 feet 4 inches, and 1 foot 4 inches (with 8 inches left)

4th pipe: 2 feet 8 inches (with 7 feet 4 inches left)

The correct answer is B.

74. What is the thousandths digit in the decimal equivalent of 53 divided by 5,000?

A. 0

B. 1

C. 3

D. 5

E. 6

Arithmetic Place Value

equation sequence part 1 53 divided by 5,000 equals part 2 106 divided by 10,000 equals part 3 0.0106
and the thousandths digit is 0.

The correct answer is A.

75. If a equals the sum of the even integers from 2 to 20, inclusive, and b equals the sum of the odd integers from 1 to 19, inclusive, what is the value of a minus b?

A. 1

B. 10

C. 19

D. 20

E. 21

Arithmetic **Series and Sequences**

multiline equation row 1 a equals sum with variable number of summands two plus four plus six plus ellipsis plus 20 row 2 b equals sum with variable number of summands one plus three plus five plus ellipsis plus 19 row 3 a minus b equals sum with variable number of summands left parenthesis two minus one right parenthesis plus left parenthesis four minus three right parenthesis plus left parenthesis six minus five right parenthesis plus ellipsis plus left parenthesis 20 minus 19 right parenthesis row 4 equals sum with variable number of summands one plus one plus one plus ellipsis plus one

Since there are 10 even numbers in the expression for a above—note that the expression for a can be written as 2(1 + 2 + 3 + … + 10)—it follows that there are 10 occurrences of 1 in the expression for a minus b, and hence equation sequence part 1 a minus b equals part 2 10 multiplication one equals part 3 10
.

The correct answer is B.

76. If a, b, c, and d are consecutive even integers and a less than b less than c less than d, then a plus b is how much less than c plus d?

A. 2

B. 4

C. 6

D. 8

E. 10

Algebra **Series and Sequences; Simplifying Algebraic Expressions**

Since a, b, c, and d are consecutive even integers in this order, we have b equals a plus two,

equation sequence part 1 c equals part 2 b plus two equals part 3 a plus four
, and
equation sequence part 1 d equals part 2 c plus two equals part 3 a plus six
. Therefore, the amount by which a plus b is less than
c plus d is equation sequence part 1 left parenthesis c plus d right parenthesis minus left parenthesis a plus b right parenthesis equals part 2 left parenthesis sum with 4 summands a plus four plus a plus six right parenthesis minus left parenthesis sum with 3 summands a plus a plus two right parenthesis equals part 3 eight
.

The correct answer is D.

77. A retailer sold an appliance for $80. If the retailer's gross profit on the appliance was 25 percent of the retailer's cost for the appliance, how many dollars was the retailer's gross profit?

A. $10

B. $16

C. $20

D. $24

E. $25

Algebra Applied Problems; First-Degree Equations

Let normal dollar times cap c be the cost of the appliance. The appliance sold for $80 and the profit was 25% of the cost. Since selling price equals cost plus profit, it follows that
equation sequence part 1 80 equals part 2 cap c plus 0.25 times cap c equals part 3 1.25 times cap c
. Dividing both sides by 1.25 (equivalently, dividing both sides by five divided by four), we have cap c equals 64. Therefore, the profit is

equation sequence part 1 0.25 times left parenthesis normal dollar times cap c right parenthesis equals part 2 one divided by four multiplication normal dollar times 64 equals part 3 normal dollar times 16

.

The correct answer is B.

78. Beth has a collection of 8 boxes of clothing for a charity, and the average (arithmetic mean) number of pieces of clothing per box is c. If she replaces a box in the collection that contains 12 pieces of clothing with a box that contains 22 pieces of clothing, what is the average number of pieces of clothing per box for the new collection, in terms of c?

   A. c minus five divided by four

   B. c plus five divided by four

   C. eight minus 10 divided by c

   D. eight plus 10 divided by c

   E. eight times c minus 10

Algebra **Average**

Before the replacement, the total number of pieces of clothing in the 8 boxes was eight times c. After the replacement, the total number of pieces of clothing in the 8 boxes was eight times c minus 12 plus 22 equals eight times c plus 10. Therefore, the average number of pieces of clothing per box for the new collection is equation sequence part 1 eight times c plus 10 divided by eight equals part 2 eight times c divided by eight plus 10 divided by eight equals part 3 c plus five divided by four

.

The correct answer is B.

one plus 0.0001 divided by 0.04 plus 10

79. The value of the expression above is closest to which of the following?

A. 0.0001

B. 0.001

C. 0.01

D. 1

E. 10

Arithmetic **Estimation**

multirelation one plus 0.0001 divided by 0.04 plus 10 equals 1.0001 divided by 10.04 almost equals one divided by 10 equals 0.1 Alternatively (and less heuristically), because a small relative change (i.e., a small percentage change) in the value of the numerator of a fraction produces a small relative change in the value of the fraction, and similarly for the denominator, it follows that multirelation one plus 0.0001 divided by 0.04 plus 10 almost equals one divided by 0.04 plus 10 almost equals one divided by 10 equals 0.1

The correct answer is C.

80. If x plus one equals t and t equals three minus x, then x equals

A. –2

B. –1

C. 0

D. 1

E. 2

Algebra **First-Degree Equations**

x plus one = three minus x	substitution
x plus x = 3 − 1	add x minus one to both sides
two times x = 2	combine like terms
x = 1	divide both sides by 2

The correct answer is D.

81. If x equals k times c and y equals k times t, then y minus x equals

A. k times left parenthesis t minus c right parenthesis

B. k times left parenthesis c minus t right parenthesis

C. c times left parenthesis k minus t right parenthesis

D. t times left parenthesis k minus c right parenthesis

E. k times left parenthesis l minus t right parenthesis

Algebra Simplifying Algebraic Expressions

y minus x equals k times t minus k times c	given
y minus x equals k times left parenthesis t minus c right parenthesis	distributive property

The correct answer is A.

82. If k is a positive even integer, which of the following must be an odd integer?

I. k squared minus three times k plus four

II. k super five plus three

III. seven times k minus seven

A. II only

B. III only

C. I and III only

D. II and III only

E. I, II, and III

Arithmetic Properties of Integers

The table below shows some results for adding, subtracting, and multiplying even and odd integers. For example, an even integer plus an even integer is an even integer (for example, 4 + 8 equals 12, which is even) and an odd integer times an odd integer is an odd integer (for example, 3 × 7 equals 21, which is odd).

even	±	even	is	even
even	±	odd	is	odd
odd	±	odd	is	even
even	×	even	is	even
even	×	odd	is	even
odd	×	odd	is	odd

Now apply these results to the expressions in I, II, and III when k is an even integer.

I. k squared minus three times k plus four is even − even + even, which is **EVEN**.

II. k super five plus three is even + odd, which is **ODD**.

III. seven k minus seven is even – odd, which is **ODD**.

The correct answer is D.

Questions 83 to 151 — Difficulty: Medium

83. If one divided by two the result obtained when 2 is subtracted from five times x is equal to the sum of 10 and three times x, what is the value of x?

    A. –22

    B. –4

    C. 4

    D. 18

    E. 22

Algebra **First-Degree Equations**

The result obtained when 2 is subtracted from five times x is five times x minus two, and the sum of 10 and three times x is 10 plus three times x.

Therefore, it is given that one divided by two of five times x minus two is equal to 10 plus three times x, or one divided by two times left parenthesis five times x minus two right parenthesis equals 10 plus three times x
.

one divided by two times left parenthesis five times x minus two right parenthesis	=	10 plus three times x	given
five times x minus two	=	20 plus six times x	multiply both sides by 2
–22	=	x	subtract both five times x and 20 from both sides

The correct answer is A.

84. If Car A took n hours to travel 2 miles and Car B took m hours to travel 3 miles, which of the following expresses the time it would take Car C, traveling at the average (arithmetic mean) of those rates, to travel 5 miles?

A. 10 times n times m divided by three times n plus two times m

B. three times n plus two times m divided by 10 times left parenthesis n plus m right parenthesis

C. two times n plus three times m divided by five times n times m

D. 10 times left parenthesis n plus m right parenthesis divided by two times n plus three times m

E. five times left parenthesis n plus m right parenthesis divided by two times n plus three times m

Algebra **Applied Problems**

This is a rate problem that can be solved by several applications of the formula

rate × time = distance.

Let r sub cap a and r sub cap b be the rates, respectively and in miles per hour, of Car A and Car B. Then omitting units for simplicity, for Car A this formula becomes
r sub cap a multiplication n equals two, or
r sub cap a equals two divided by n, and for Car B this formula becomes r sub cap b multiplication m equals three, or
r sub cap b equals three divided by m. Thus, the average of the two rates is
equation sequence part 1 one divided by two times left parenthesis r sub cap a plus r sub cap b right parenthesis equals part 2 one divided by two times left parenthesis two divided by n plus three divided by m right parenthesis equals part 3 one divided by two times left parenthesis two times m plus three times n divided by m times n right parenthesis equals part 4 three times n plus two times m divided by two times m times n
. Therefore, if t is the desired time, in hours, that Car C traveled, then

the above formula for Car C becomes
three times n plus two times m divided by two times m times n multiplication t equals five
, or
equation sequence part 1 t equals part 2 five multiplication two times m times n divided by three times n plus two times m equals part 3 10 times m times n divided by three times n plus two times m
.

The correct answer is A.

85. If x comma y, and k are positive and x is less than y, then x plus k divided by y plus k is

A. 1

B. greater than x divided by y

C. equal to x divided by y

D. less than x divided by y

E. less than x divided by y or greater than x divided by y, depending on the value of k

Algebra Ratios

x	<	y	given
k times x	<	k times y	multiply by positive k
x times y plus k times x	<	x times y plus k times y	add x times y
x times left parenthesis y plus k right parenthesis	<	y times left parenthesis x plus k right parenthesis	factor
x	<	y times left parenthesis x plus k right parenthesis divided by y plus k	divide by positive y plus k
x divided by y	<	x plus k divided by y plus k	divide by positive y

Thus, x plus k divided by y plus k greater than x divided by y.

The correct answer is B.

86. Consider the following set of inequalities:
p greater than q comma s greater than r comma q greater than t comma s greater than p
, and r greater than q. Between which two quantities is no relationship established?

A. p and r

B. s and t

C. s and q

D. p and t

E. r and t

Algebra **Order**

Using r greater than q and q greater than t gives r greater than t, so a relationship is established between r and t. The correct answer is NOT E.

Using p greater than q and q greater than t gives p greater than t, so a relationship is established between p and t. The correct answer is NOT D.

Using s greater than r and r greater than q gives s greater than q, so a relationship is established between s and q. The correct answer is NOT C.

Using s greater than r, r greater than q, and q greater than t gives s greater than t, so a relationship is established between s and t. The correct answer is NOT B.

Alternately, the diagram below shows the given relationships and does not establish a relationship between p and r.

image

The correct answer is A.

87. Carl averaged two times m miles per hour on a trip that took him h hours. If Ruth made the same trip in two divided by three times h hours, what was her average speed in miles per hour?

A. one divided by three times m times h

B. two divided by three times m times h

C. m

D. three divided by two times m

E. three times m

Algebra **Applied Problems**

Using distance = rate × time, the distance Carl traveled on the trip was two times m times h miles. Using rate equals distance divided by time, Ruth's rate was equation sequence part 1 two times m times h divided by two divided by three times h equals part 2 three divided by two times left parenthesis two times m right parenthesis equals part 3 three times m .

The correct answer is E.

88. Of three persons, two take relish, two take pepper, and two take salt. The one who takes no salt takes no pepper, and the one who takes no pepper takes no relish. Which of the following statements must be true?

I. The person who takes no salt also takes no relish.

II. Any of the three persons who takes pepper also takes relish and salt.

III. The person who takes no relish is not one of those who takes salt.

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

Arithmetic Sets (Venn Diagrams)

Although this problem can be solved by the use of a Venn diagram, it is probably simpler to use ordinary reasoning. The single person who takes no salt takes no pepper, and the single person who takes no pepper takes no relish, so exactly one person does not take any of the three. Thus, each of the other two people take all three. The table below shows these results where Person 1 does not take any of the three and Persons 2 and 3 each take all three.

Person	Relish	Pepper	Salt
1	no	no	no
2	yes	yes	yes
3	yes	yes	yes

The only person who takes no salt is Person 1, who also takes no relish, so I must be true.

The only people who take pepper are Persons 2 and 3, and each of them also takes relish and salt, so II must be true.

The only person who takes no relish is Person 1, who is not a person who takes salt, so III must be true.

The correct answer is E.

89. If the smaller of 2 consecutive odd integers is a multiple of 5, which of the following could NOT be the sum of these 2 integers?

A. –8

B. 12

C. 22

D. 52

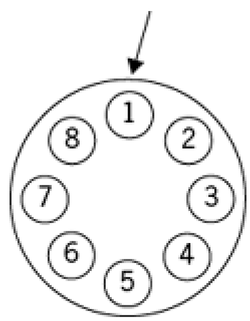
E. 252

Algebra **Operations with Integers**

Since the smaller of the 2 consecutive odd integers is a multiple of 5, let it be represented by five times n for some integer n. Then the other odd integer can be represented by five times n plus two. The sum of these two integers is 10 times n plus two. The sum is –8 when n equals negative one and five times n equals left parenthesis five right parenthesis times left parenthesis negative one right parenthesis is odd. The sum is 12 when n equals one and

five times n equals left parenthesis five right parenthesis times left parenthesis one right parenthesis
is odd. The sum is 22 when n equals two, but
five times n equals left parenthesis five right parenthesis times left parenthesis two right parenthesis
is not odd. There is no need to check the D and E because it has been determined that 22 cannot be the sum of the 2 consecutive odd integers. For completeness, the sum is 52 when n equals five and
five times n equals left parenthesis five right parenthesis times left parenthesis five right parenthesis
is odd. The sum is 252 when n equals 25 and
five times n equals left parenthesis five right parenthesis times left parenthesis 25 right parenthesis
is odd.

The correct answer is C.



90. Eight light bulbs numbered 1 through 8 are arranged in a circle as shown above. The bulbs are wired so that every third bulb, counting in a clockwise direction, flashes until all bulbs have flashed once. If the bulb numbered 1 flashes first, which numbered bulb will flash last?

A. 2

B. 3

C. 4

D. 6

E. 7

Arithmetic Properties of Integers

The easiest way to do this problem might be by just counting every third bulb going clockwise around the circle starting at Bulb 1, which flashes, skipping 2 bulbs and getting to Bulb 4, which flashes, skipping 2 bulbs and getting to Bulb 7, which flashes, skipping 2 bulbs and getting to Bulb 2, which flashes, skipping 2 bulbs and getting to Bulb 5, which flashes, skipping 2 bulbs and getting to Bulb 8, which flashes, skipping 2 bulbs and getting to Bulb 3, which flashes, and finally skipping 2 bulbs and getting to Bulb 6, which flashes. Now, all 8 bulbs have flashed once and the last one to flash was Bulb 6.

The correct answer is D.

Closing Prices of Stock X
During a Certain Week
(in dollars)

Monday	Tuesday	Wednesday	Thursday	Friday
21	19	22		23

91. A certain financial analyst defines the “volatility” of a stock during a given week to be the result of the following procedure: find the absolute value of the difference in the stock’s closing price for each pair of consecutive days in the week and then find the average (arithmetic mean) of these 4 values. What is the volatility of Stock X during the week shown in the table?

A. 0.50

B. 1.80

C. 2.00

D. 2.25

E. 2.50

Arithmetic Statistics

The volatility of Stock X during the week is the average of the 4 values associated with the 4 pairs of consecutive days during the week.

multiline equation line 1 sum with 4 summands left parenthesis Mon ampersand Tue right parenthesis plus left parenthesis Tue ampersand Wed right parenthesis plus left parenthesis Wed ampersand Thu right parenthesis plus left parenthesis Thu ampersand Fri right parenthesis divided by four line 2 equals sum with 4 summands absolute value of 19 minus 21 plus absolute value of 22 minus 19 plus absolute value of 24.5 minus 22 plus absolute value of 23 minus 24.5 divided by four line 3 equation sequence part 1 equals part 2 sum with 4 summands absolute value of negative two plus absolute value of three plus absolute value of 2.5 plus absolute value of negative 1.5 divided by four equals part 3 sum with 4 summands two plus three plus 2.5 plus 1.5 divided by four equals part 4 nine divided by four equals part 5 2.25

The correct answer is D.

92. If
y equals absolute value of three times x minus five divided by negative x squared minus three
, for what value of x will the value of y be greatest?

A. –5

B. negative three divided by five

C. 0

D. three divided by five

E. five divided by three

Algebra Functions; Absolute Value

Since the absolute value of any real number is greater than or equal to zero, it follows that
absolute value of three times x minus five greater than or equals zero
. Also, for any real number x we have
x squared greater than or equals zero, and hence
negative x squared less than or equals zero. Subtracting 3 from both sides of the last inequality gives
negative x squared minus three less than or equals negative three.

Therefore, the numerator of the expression for y is greater than or equal to zero and the denominator of the expression for y is negative. It follows that the value of y cannot be greater than 0. However, the value of y is equal to 0 when
absolute value of three times x minus five equals zero, or
three times x minus five equals zero, or
x equals five divided by three. Therefore, the value of x for which the value of y is greatest (i.e., when y equals zero) is
x equals five divided by three.

The correct answer is E.

93. What values of x have a corresponding value of y that satisfies both x times y greater than zero and x times y equals x plus y?

A. x less than or equals negative one

B. multirelation negative one less than x less than or equals zero

C. multirelation zero less than x less than or equals one

D. x greater than one

E. All real numbers

Algebra Equations; Inequalities

First, use x times y equals x plus y to solve for y in terms of x.

x times y	=	x plus y	given
x times y minus y	=	x	subtract y from both sides
y times left parenthesis x minus one right parenthesis	=	x	factor
y	=	x divided by x minus one	divide both sides by x minus one

Note that the division by x minus one requires x not equals one, and thus x equals one needs to be considered separately. However, if x equals one, then x times y equals x plus y becomes y equals one plus y, which is not true for any value of y. Using y equals x divided by x minus one, it follows that the inequality x times y greater than zero is equivalent to x squared divided by x minus one greater than zero. Since x squared greater than or equals zero for each value of x, the quotient x squared divided by x minus one can only be positive when x not equals zero and x minus one is positive, or when x greater than one.

Alternatively, the correct answer can be found by eliminating the incorrect answers, which can be accomplished by considering the endpoints of the intervals given in the answer choices.

Case 1: If x equals negative one, then x times y equals x plus y becomes negative y equals negative one plus y, or y equals one divided by two. However, in this case x times y equals left parenthesis negative one right parenthesis times left parenthesis one divided by two right parenthesis is negative, and thus x times y greater than zero is not true. Therefore, the answer cannot be A or E.

Case 2: If x equals zero, then x times y equals zero, and thus x times y greater than zero is not true. Therefore, the answer cannot be B or E.

Case 3: If x equals one, then x times y equals x plus y becomes y equals one plus y, which is not true for any value of y. Therefore, the answer cannot be C or E.

Since the answer cannot be A, B, C, or E, it follows that the answer is D.

The correct answer is D.

94. Employee X’s annual salary is $12,000 more than half of Employee Y’s annual salary. Employee Z’s annual salary is $15,000 more than half of Employee X’s annual salary. If Employee X’s annual salary is $27,500, which of the following lists these three people in order of increasing annual salary?

   A. Y, Z, X

   B. Y, X, Z

   C. Z, X, Y

   D. X, Y, Z

   E. X, Z, Y

Algebra **First-Degree Equations**

Letting x comma y, and z represent the annual salary, in dollars, of Employee X, Employee Y, and Employee Z, respectively, the following information is given:

1. x equals 12,000 plus y divided by two
2. x equals one times 5,000 plus x divided by two
3. x equals 27,500

From (1) and (3), it follows that
27,500 equals 12,000 plus y divided by two or
equation sequence part 1 y equals part 2 two times left parenthesis 27,500 minus 12,000 right parenthesis equals part 3 31,000
. From (2) and (3), it follows that
equation sequence part 1 z equals part 2 15,000 plus 27,500 divided by two equals part 3 28,750
. Therefore, x less than z less than y.

The correct answer is E.

95. cap c equals case statement case 1 0.10 times s comma if s less than or equals 60,000 case 2 0.10 times s plus 0.04 times left parenthesis s minus 60,000 right parenthesis comma if s greater than 60,000

The formula above gives the contribution cap c, in dollars, to a certain profit-sharing plan for a participant with a salary of s dollars. How many more dollars is the contribution for a participant with a salary of $70,000 than for a participant with a salary of $50,000?

A. $800

B. $1,400

C. $2,000

D. $2,400

E. $2,800

Algebra **Applied Problems; Formulas**

For a participant with a salary of $70,000, equation sequence part 1 cap c equals part 2 0.1 times left parenthesis normal dollar times 70,000 right parenthesis plus 0.04 times left parenthesis normal dollar times 70,000 minus normal dollar times 60,000 right parenthesis equals part 3 normal dollar times 7,000 plus normal dollar times 400 equals part 4 normal dollar times 7,400 . For a participant with a salary of normal dollar times 50,000 comma equation sequence part 1 cap c equals part 2 0.1 times left parenthesis normal dollar times 50,000 right parenthesis equals part 3 normal dollar times 5,000 . The difference is normal dollar times 7,400 minus normal dollar times 5,000 equals normal dollar times 2,400 .

The correct answer is D.

96. Next month, Ron and Cathy will each begin working part-time at three divided by five of their respective current salaries. If the sum of their reduced salaries will be equal to Cathy's current salary, then Ron's current salary is what fraction of Cathy's current salary?

A. one divided by three

B. two divided by five

C. one divided by two

D. three divided by five

E. two divided by three

Algebra **First-Degree Equations**

Letting cap r and cap c, respectively, represent Ron’s and Cathy’s current salaries, it is given that
three divided by five times cap r plus three divided by five times cap c equals cap c
. It follows that
three divided by five times cap r equals two divided by five times cap c
and
equation sequence part 1 cap r equals part 2 five divided by three times left parenthesis two divided by five times cap c right parenthesis equals part 3 two divided by three times cap c
.

The correct answer is E.

97. David and Ron are ordering food for a business lunch. David thinks that there should be twice as many sandwiches as there are pastries, but Ron thinks the number of pastries should be 12 more than one-fourth of the number of sandwiches. How many sandwiches should be ordered so that David and Ron can agree on the number of pastries to order?

A. 12

B. 16

C. 20

D. 24

E. 48

Algebra **Simultaneous Equations**

Let cap s be the number of sandwiches that should be ordered and let cap p be the number of pastries that should be ordered. Then David desires cap s equals two times cap p and Ron desires cap p equals 12 plus one divided by four times cap s.

cap s equals two times cap p	given
cap s equals two times left parenthesis 12 plus one divided by four times cap s right parenthesis	cap p equals 12 plus one divided by four times cap s
cap s equals 24 plus one divided by two times cap s	distributive law
one divided by two times cap s equals 24	subtract one divided by two times cap s from both sides
cap s equals 48	multiply both sides by 2

The correct answer is E.

98. The cost of purchasing each box of candy from a certain mail order catalog is v dollars per pound of candy, plus a shipping charge of h dollars. How many dollars does it cost to purchase 2 boxes of candy, one containing s pounds of candy and the other containing t pounds of candy, from this catalog?

 A. h plus s times t times v

 B. two times h plus s times t times v

 C. two times h times s times t times v

 D. sum with 4 summands two times h plus s plus t plus v

 E. two times h plus v times left parenthesis s plus t right parenthesis

Algebra Formulas

The cost, in dollars, to purchase the 2 boxes of candy is the sum of 2 shipping charges and the cost of s plus t pounds of candy.

multiline equation row 1 cost equals left parenthesis two times shipping charges right parenthesis plus left parenthesis v right parenthesis times left parenthesis s plus t right parenthesis row 2 cost equals two times left parenthesis h right parenthesis plus left parenthesis v right parenthesis times left parenthesis s plus t right parenthesis row 3 cost equals two times h plus v times left parenthesis s plus t right parenthesis

The correct answer is E.

99. If x not equals negative one divided by two, then six times x cubed plus three times x squared minus eight times x minus four divided by two times x plus one equals

   A. three times x squared plus three divided by two times x minus eight

   B. three times x squared plus three divided by two times x minus four

   C. three times x squared minus four

   D. three times x minus four

   E. three times x plus four

Algebra **Factoring**

multiline equation line 1 six times x cubed plus three times x squared minus eight times x minus four divided by two times x plus one	=	multiline equation line 1 left parenthesis six times x cubed plus three times x squared right parenthesis minus left parenthesis eight times x plus four right parenthesis divided by two times x plus one	group
	=	multiline equation line 1 three times x squared times left parenthesis two times x plus one right parenthesis minus four times left parenthesis two times x plus one right parenthesis divided by two times x plus one	factor
	=	multiline equation line 1 left parenthesis three times x squared minus four right parenthesis times left parenthesis two times x plus one right parenthesis divided by two times x plus one	factor
	=	multiline equation line 1 three times x squared minus four	cancel since x not equals one divided by two

Alternatively, sometimes it is easier or quicker to test one-variable expressions for equality by substituting a convenient value for the variable and eliminating answer choices for which the value of the expression in that answer choice does not equal the value of the given expression. For example, choose x equals zero, since calculations for

x equals zero are minimal. Then, as shown in the table below,
equation sequence part 1 six times x cubed plus three times x squared minus eight times x minus four divided by two times x plus one equals part 2 negative four divided by one equals part 3 negative four
, but
three times x squared plus three divided by two times x minus eight equals negative eight
and three times x plus four equals four, neither of which equals –4, so answer choices A and E can be eliminated. Another convenient value to choose for x is 1. There is no need to evaluate answer choices A and E at 1 since they have already been eliminated. As shown, when
x equals one comma six times x cubed plus three times x squared minus eight times x minus four divided by two times x plus one equals negative one
, but
multirelation three times x squared plus three divided by two times x minus four equals one divided by two not equals negative one
, so answer choice B can be eliminated. A third convenient value for x is –1. There is no need to evaluate answer choices A, B, and E at –1 since they have already been eliminated. As shown, when
x equals negative one comma six times x cubed plus three times x squared minus eight times x minus four divided by two times x plus one equals negative one
, but
multirelation three times x minus four equals negative seven not equals negative one
, so answer choice D can be eliminated. Note that, if
x equals negative one had been chosen initially, A, B, D, and E would have been eliminated immediately since
multirelation three times x squared plus three divided by two times x minus eight equals negative not equals negative one
,

multirelation three times x squared plus three divided by two times x minus four equals negative not equals negative one comma multirelation three times x minus four equals negative seven not equals negative one
, and
multirelation three times x plus four equals one not equals negative one
.

		x equals zero	x equals one	x equals negative one
	six times x cubed plus three times x squared minus eight times x minus four divided by two times x plus one	–4	–1	–1
A	three times x squared plus three divided by two times x minus eight	–8		
B	three times x squared plus three divided by two times x minus four	–4	one divided by two	
C	three times x squared minus four	–4	–1	–1
D	three times x minus four	–4	–1	–7
E	three times x plus four	4		

The correct answer is C.

100. If
sum with 3 summands x squared plus b times x plus five equals left parenthesis x plus c right parenthesis squared
for all numbers x, where b and c are positive constants, what is the value of b?

   A. Square root of five

   B. Square root of 10

   C. two times Square root of five

   D. two times Square root of 10

   E. 10

Algebra **Second-Degree Equations**

Given that
sum with 3 summands x squared plus b times x plus five equals left parenthesis x plus c right parenthesis squared
, since
left parenthesis x plus c right parenthesis squared equals sum with 3 summands x squared plus two times c times x plus c squared
, it follows that five equals c squared and b equals two times c. The possible values of c are negative Square root of five and Square root of five, but since c is positive, c equals Square root of five and
equation sequence part 1 b equals part 2 two times c equals part 3 two times Square root of five
.

The correct answer is C.

101. Last year Shannon listened to a certain public radio station 10 hours per week and contributed $35 to the station. Of the following, which is closest to Shannon's contribution per minute of listening time last year?

   A. $0.001

   B. $0.010

C. $0.025

D. $0.058

E. $0.067

Arithmetic **Measurement Conversion**

Since there are 52 weeks in 1 year and 60 minutes in 1 hour, 10 hours per week is equivalent to (10)(52)(60) = 31,200 minutes per year. Shannon’s $35 contribution is then 35 divided by 31,200 dollars per minute, which is closest to $0.001 per minute.

The correct answer is A.

102. Each of the 20 employees at Company J is to receive an end-of-year bonus this year. Agnes will receive a larger bonus than any other employee, but only $500 more than Cheryl will receive. None of the employees will receive a smaller bonus than Cheryl. If the amount of money to be distributed in bonuses at Company J this year totals $60,000, what is the largest bonus Agnes can receive?

A. $3,250

B. $3,325

C. $3,400

D. $3,475

E. $3,500

Algebra **Applied Problems**

Since the total amount of the bonuses is fixed, the largest possible bonus that Agnes can receive will occur when the total amount received by the 19 employees other than Agnes is the smallest possible. Let cap a be the bonus, in dollars, that Agnes receives. Then, in dollars, Cheryl will receive left parenthesis cap a minus 500 right parenthesis, and each of the remaining 18 employees will receive between left parenthesis cap a minus 500 right parenthesis and cap a. Therefore, the total amount received by the 19 employees other than

Agnes is smallest when each of these 19 employees receives $(A - 500)$ dollars.

$19(A - 500) + A$	=	60,000	total of 20 bonuses is $60,000
$19A - 9{,}500 + A$	=	60,000	distributive law
$20A - 9{,}500$	=	60,000	combine like terms
$20A$	=	69,500	add 9,500 to both sides
A	=	3,475	divide both sides by 20

The correct answer is D.

103. Beth, Naomi, and Juan raised a total of $55 for charity. Naomi raised $5 less than Juan, and Juan raised twice as much as Beth. How much did Beth raise?

A. $9

B. $10

C. $12

D. $13

E. $15

Algebra **Simultaneous Equations**

Let B, N, and J be the amounts raised, respectively and in dollars, by Beth, Naomi, and Juan.

$B+N+J$	= 55	given
cap n	= cap j minus five	given
cap j	= two times cap b	given
cap j	= $55-B-N$	subtract cap b plus cap n from both sides of first equation
J	= 55 minus cap b minus left parenthesis cap j minus five right parenthesis	cap n equals cap j minus five
two times cap b	= $55-B-(2B-5)$	$J=2B$
cap b	= 12	solve for cap b

The correct answer is C.

104. The set of solutions for the equation $\left(x^2-25\right)^2=x^2-10x+25$ contains how many real numbers?

A. 0

B. 1

C. 2

D. 3

E. 4

Algebra Second-Degree Equations

left parenthesis x squared minus 25 right parenthesis squared	=	x squared minus 10 times x plus 25	given
left parenthesis x plus five right parenthesis squared times left parenthesis x minus five right parenthesis squared	=	left parenthesis x minus five right parenthesis squared	factor
$(x+5)^2(x-5)^2-(x-5)^2$	=	0	subtract left parenthesis x minus five right parenthesis squared
left parenthesis x minus five right parenthesis squared times left square bracket left parenthesis x plus five right parenthesis squared minus one right square bracket	=	0	factor
$(x-5)^2[(x+5)-1][(x+5)+1]$	=	0	factor
left parenthesis x minus five right parenthesis squared times left parenthesis x plus four right parenthesis times left parenthesis x plus six right parenthesis	=	0	subtraction, addition

Thus, the solution set of
left parenthesis x squared minus 25 right parenthesis squared equals x squared minus 10 times x plus 25
contains 3 real numbers: 5, –4, and –6.

The correct answer is D.

105. An aerosol can is designed so that its bursting pressure, cap b, in pounds per square inch, is 120% of the pressure, cap f, in pounds per square inch, to which it is initially filled. Which of the following formulas expresses the relationship between B and cap f?

A. cap b equals 1.2 times cap f

B. cap b equals 120 times cap f

C. cap b equals one plus 0.2 times cap f

D. $B = \frac{F}{1.2}$

E. cap b equals 1.2 divided by cap f

Algebra **Formulas; Percents**

We are given that cap b is 120% of F, so
cap b equals left parenthesis 120 percent right parenthesis times cap f
, or cap b equals 1.2 times cap f. Note that both cap b and cap f are given in pounds per square inch, so there are no unit conversions involved.

The correct answer is A.

106. The average (arithmetic mean) of the positive integers x, y, and z is 3. If x less than y less than z, what is the greatest possible value of z?

A. 5

B. 6

C. 7

D. 8

E. 9

Algebra **Inequalities**

It is given that
sum with 3 summands x plus y plus z divided by three equals three,

or $x + y + z = 9$, or
z equals nine plus left parenthesis negative x minus y right parenthesis
. It follows that the greatest possible value of z occurs when
negative x minus y equals negative left parenthesis x plus y right parenthesis
has the greatest possible value, which occurs when $x + y$ has the least possible value. Because x and y are different positive integers, the least possible value of x plus y occurs when x equals one and y equals two. Therefore, the greatest possible value of z is 9 – 1 – 2 = 6.

The correct answer is B.

107. The product of 3,305 and the 1-digit integer x is a 5-digit integer. The units (ones) digit of the product is 5 and the hundreds digit is y. If cap a is the set of all possible values of x and cap b is the set of all possible values of y, then which of the following gives the members of A and cap b?

	cap a low line	cap b low line
(A)	{1, 3, 5, 7, 9}	{0, 1, 2, 3, 4, 5, 6, 7, 8, 9}
(B)	{1, 3, 5, 7, 9}	{1, 3, 5, 7, 9}
(C)	{3, 5, 7, 9}	{1, 5, 7, 9}
(D)	{5, 7, 9}	{1, 5, 7}
(E)	{5, 7, 9}	{1, 5, 9}

Arithmetic **Properties of Numbers**

Since the products of 3,305 and 1, 3,305 and 2, and 3,305 and 3 are the 4-digit integers 3,305, 6,610, and 9,915, respectively, it follows that x must be among the 1-digit integers 4, 5, 6, 7, 8, and 9. Also, since

the units digit of the product of 3,305 and x is 5, it follows that x cannot be 4 (product has units digit 0), 6 (product has units digit 0), or 8 (product has units digit 0). Therefore,
cap a equals left curly bracket five comma seven comma nine right curly bracket
. The possibilities for y will be the hundreds digits of the products (3,305)(5) = 16,525, (3,305)(7) = 23,135, and (3,305)(9) = 29,745. Thus, y can be 5, 1, or 7, and so B = {1, 5, 7}.

The correct answer is D.

108. If x and y are integers such that $2 < x \leq 8$ and multirelation two less than y less than or equals nine, what is the maximum value of one divided by x minus x divided by y?

A. $-3\frac{1}{8}$

B. 0

C. one divided by four

D. five divided by 18

E. 2

Algebra Inequalities

Because x and y are both positive, the maximum value of $\frac{1}{x} - \frac{x}{y}$ will occur when the value of $\frac{1}{x}$ is maximum and the value of $\frac{x}{y}$ is minimum. The value of one divided by x is maximum when the value of x is minimum or when x equals three. The value of $\frac{x}{y}$ is minimum when the value of x is minimum (or when x equals three) and the value of y is maximum (or when y equals nine). Thus, the maximum value of one divided by x minus x divided by y is $\frac{1}{3} - \frac{3}{9} = 0$.

The correct answer is B.

109. Items that are purchased together at a certain discount store are priced at $3 for the first item purchased and $1 for each additional item purchased. What is the maximum number of items that could be purchased together for a total price that is less than $30?

A. 25

B. 26

C. 27

D. 28

E. 29

Arithmetic Applied Problems

After the first item is purchased, $29.99 – $3.00 = $26.99 remains to purchase the additional items. Since the price for each of the additional items is $1.00, a maximum of 26 additional items could be purchased. Therefore, a maximum of 1 + 26 = 27 items could be purchased for less than $30.00.

The correct answer is C.

110. What is the least integer z for which
left parenthesis 0.000125 right parenthesis times left parenthesis 0.0025 right parenthesis times left parenthesis 0.00000125 right parenthesis multiplication 10 super z
is an integer?

A. 18

B. 10

C. 0

D. −10

E. −18

Arithmetic Decimals

Considering each of the three decimal numbers in parentheses separately, we know that 0.000125×10^6 is the integer 125, 0.0025×10^4 is the integer 25, and 0.00000125×10^8 is the integer 125. We thus know that $(0.000125) \times 10^6 \times (0.0025) \times 10^4 \times (0.00000125) \times 10^8 = (0.000125)(0.0025)(0.00000125) \times 10^6 \times 10^4 \times 10^8 = (0.000125)(0.0025)(0.00000125) \times 10^{6+4+8} = (0.000125)(0.0025)(0.00000125) \times 10^{18}$ is the integer $125 \times 25 \times 125$. We therefore know that if $z = 18$
, then
left parenthesis 0.000125 right parenthesis times left parenthesis 0.0025 right parenthesis times left parenthesis 0.00000125 right parenthesis multiplication 10 super z
is an integer.

Now, if the product $125 \times 25 \times 125$ were divisible by 10, then for at least one integer z less than 18, $(0.000125)(0.0025)(0.00000125) \times 10^z$ would be an integer. However, each of the three numbers being multiplied in the product $125 \times 25 \times 125$ is odd (not divisible by 2). We thus know that $125 \times 25 \times 125$ is not divisible by 2 and is therefore odd. Because only even numbers are divisible by 10, we know that $125 \times 25 \times 125$ is not divisible by 10. We thus know that 18 is the *least* integer z such that
left parenthesis 0.000125 right parenthesis times left parenthesis 0.0025 right parenthesis times left parenthesis 0.00000125 right parenthesis multiplication 10 super z
is an integer.

Note that it is not necessary to perform the multiplication $125 \times 25 \times 125$.

The correct answer is A.

111. The average (arithmetic mean) length per film for a group of 21 films is t minutes. If a film that runs for 66 minutes is removed from the group and replaced by one that runs for 52 minutes, what is the average length per film, in minutes, for the new group of films, in terms of t?

A. $t + \frac{2}{3}$

B. t minus two divided by three

C. $21t + 14$

D. t plus three divided by two

E. t minus three divided by two

Arithmetic **Statistics**

Let S denote the sum of the lengths, in minutes, of the 21 films in the original group. Since the average length is t minutes, it follows that $\frac{S}{21} = t$. If a 66-minute film is replaced by a 52-minute film, then the sum of the lengths of the 21 films in the resulting group is cap s minus 66 plus 52 equals cap s minus 14. Therefore, the average length of the resulting 21 films is equation sequence part 1 cap s minus 14 divided by 21 equals part 2 cap s divided by 21 minus 14 divided by 21 equals part 3 t minus two divided by three full stop

The correct answer is B.

112. A garden center sells a certain grass seed in 5-pound bags at $13.85 per bag, 10-pound bags at $20.43 per bag, and 25-pound bags at $32.25 per bag. If a customer is to buy at least 65 pounds of the grass seed, but no more than 80 pounds, what is the least possible cost of the grass seed that the customer will buy?

A. $94.03

B. $96.75

C. $98.78

D. $102.07

E. $105.36

Arithmetic **Applied Problems**

Let x represent the amount of grass seed, in pounds, the customer is to buy. It follows that 65 less than or equals x less than or equals 80. Since the grass seed is available in only 5-pound, 10-pound, and 25-

pound bags, then the customer must buy either 65, 70, 75, or 80 pounds of grass seed. Because the seed is more expensive per pound for smaller bags, the customer should minimize the number of the smaller bags and maximize the number of 25-pound bags to incur the least possible cost for the grass seed. The possible purchases are given in the table below.

x	Number of 25-pound bags	Number of 10-pound bags	Number of 5-pound bags	Total cost
65	2	1	1	\$98.78
70	2	2	0	\$105.36
75	3	0	0	\$96.75
80	3	0	1	\$110.60

The least possible cost is then 3(\$32.25) = \$96.75.

The correct answer is B.

113. If x equals negative absolute value of w, which of the following must be true?

A. x equals negative w

B. x equals w

C. $x^2 = w$

D. x squared equals w squared

E. x cubed equals w cubed

Algebra **Absolute Value**

Squaring both sides of $x = -|w|$ gives $x^2 = (-|w|)^2$, or equation sequence part 1 x squared equals part 2 absolute value of w squared equals part 3 w squared
.

Alternatively, if left parenthesis x comma w right parenthesis is equal to either of the pairs (−1, 1) or (−1, −1), then x equals negative absolute value of w is true. However, each of the answer choices except x squared equals w squared is false for at least one of these two pairs.

The correct answer is D.

114. A certain financial institution reported that its assets totaled \$2,377,366.30 on a certain day. Of this amount, \$31,724.54 was held in cash. Approximately what percent of the reported assets was held in cash on that day?

A. 0.00013%

B. 0.0013%

C. 0.013%

D. 0.13%

E. 1.3%

Arithmetic Percents; Estimation

The requested percent can be estimated by converting the values into scientific notation.

31,724.54 divided by 2,377,366.30	value as fraction
equals 3.172454 multiplication 10 super four divided by 2.37736630 multiplication 10 super six	convert to scientific notation
equals 3.172454 divided by 2.37736630 multiplication 10 super four divided by 10 super six	arithmetic property of fractions
equals 3.172454 divided by 2.37736630 multiplication 10 super negative two	subtract exponents
almost equals three divided by two multiplication 10 super negative two	approximate
$= 1.5 \times 10^{-2}$	convert to decimal fraction
$= 0.015$	multiply
$= 1.5\%$	convert to percent

A more detailed computation would show that 1.3% is a better approximation. However, in order to select the best value from the values given as answer choices, the above computation is sufficient.

The correct answer is E.

cap a times cap b

plus cap b times cap a

---

cap a times cap a times cap c

115. In the correctly worked addition problem shown, where the sum of the two-digit positive integers cap a times cap b and cap b times cap a is the three-digit integer cap a times cap a times cap c, and cap a,

cap b, and cap c are different digits, what is the units digit of the integer cap a times cap a times cap c?

A. 9

B. 6

C. 3

D. 2

E. 0

Arithmetic Place Value

Determine the value of cap c.

It is given that
left parenthesis 10 times cap a plus cap b right parenthesis plus left parenthesis 10 times cap b plus cap a right parenthesis equals sum with 3 summands 100 times cap a plus 10 times cap a plus cap c
or
11 times cap a plus 11 times cap b equals 110 times cap a plus cap c. Thus, 11 times cap b minus 99 times cap a equals cap c, or
11 times left parenthesis cap b minus nine times cap a right parenthesis equals cap c
. Therefore, cap c is divisible by 11, and 0 is the only digit that is divisible by 11.

The correct answer is E.

116. The hard drive, monitor, and printer for a certain desktop computer system cost a total of $2,500. The cost of the printer and monitor together is equal to two divided by three of the cost of the hard drive. If the cost of the printer is $100 more than the cost of the monitor, what is the cost of the printer?

A. $800

B. $600

C. $550

D. $500

E. $350

Algebra **Simultaneous Equations**

Letting d, m, and p, respectively, represent the cost of the hard drive, monitor, and printer, the following equations are given:

1. sum with 3 summands d plus m plus p equals 2,500
2. p plus m equals two divided by three times d
3. p equals m plus 100

Using (2) and substituting two divided by three times d for m plus p in (1) gives five divided by three times d equals 2,500, from which d equals 1,500. Then from (1), p plus m equals 1,000, but m equals p minus 100 from (2), so two times p minus 100 equals 1,000 comma two times p equals 1,100
, and p equals 550.

The correct answer is C.

multiline inequality row 1 three times r less than or equals four times s plus five row 2 absolute value of s less than or equals five

117. Given the inequalities above, which of the following CANNOT be the value of r?

A. −20

B. −5

C. 0

D. 5

E. 20

Algebra **Inequalities**

Since absolute value of s less than or equals five, it follows that negative five less than or equals s less than or equals five. Therefore, negative 20 less than or equals four times s less than or equals 20,

and hence
negative 15 less than or equals four times s plus five less than or equals 25
. Since three times r less than or equals four times s plus five (given) and four times s plus five less than or equals 25 (end of previous sentence), it follows that three times r less than or equals 25. Among the answer choices, three times r less than or equals 25 is false only for r equals 20.

The correct answer is E.

118. If m is an even integer, v is an odd integer, and m greater than v greater than zero, which of the following represents the number of even integers less than m and greater than v?

  A. m minus v divided by two minus one

  B. m minus v minus one divided by two

  C. m minus v divided by two

  D. m minus v minus one

  E. m minus v

Arithmetic Properties of Numbers

Since there is only one correct answer, one method of solving the problem is to choose values for m and v and determine which of the expressions gives the correct number for these values. For example, if m equals six and v equals one, then there are 2 even integers less than 6 and greater than 1, namely the even integers 2 and 4. As the table below shows, m minus v minus one divided by two is the only expression given that equals 2.

multiline equation row 1 m minus v divided by two minus one equals 1.5 row 2 m minus v minus one divided by two equals two row 3 m minus v divided by two equals 2.5 row 4 m minus v minus one equals four row 5 m minus v equals five

To solve this problem it is not necessary to show that m minus v minus one divided by two always gives the correct number of even integers. However, one way this can be done is by the following method, first shown for a specific example and then shown in general. For the specific example, suppose v equals 15 and m equals 144. Then a list—call it the first list—of the even integers greater than v and less than m is 16, 18, 20, …, 140, 142. Now subtract 14 (chosen so that the second list will begin with 2) from each of the integers in the first list to form a second list, which has the same number of integers as the first list: 2, 4, 6, …, 128. Finally, divide each of the integers in the second list (all of which are even) by 2 to form a third list, which also has the same number of integers as the first list: 1, 2, 3, …, 64. Since the number of integers in the third list is 64, it follows that the number of integers in the first list is 64. For the general situation, the first list is the following list of even integers: v plus one, v plus three, v plus five, …, m minus four, m minus two. Now subtract the even integer v minus one from (i.e., add negative v plus one to) each of the integers in the first list to obtain the second list: 2, 4, 6, …, m minus v minus three, m minus v minus one. (Note, for example, that m minus four minus left parenthesis v minus one right parenthesis equals m minus v minus three .) Finally, divide each of the integers (all of which are even) in the second list by 2 to obtain the third list: 1, 2, 3, …, m minus v minus three divided by two, m minus v minus one divided by two. Since the number of integers in the third list is m minus v minus one divided by two, it follows that the number of integers in the first list is m minus v minus one divided by two.

The correct answer is B.

119. A positive integer is divisible by 9 if and only if the sum of its digits is divisible by 9. If n is a positive integer, for which of the following values of k is
25 multiplication 10 super n plus k multiplication 10 super two times n
divisible by 9?

A. 9

B. 16

C. 23

D. 35

E. 47

Arithmetic **Properties of Numbers**

Since n can be any positive integer, let n equals two. Then 25 multiplication 10 super n equals 2,500, so its digits consist of the digits 2 and 5 followed by two digits of 0. Also, k multiplication 10 super two times n equals k multiplication 10,000, so its digits consist of the digits of k followed by four digits of 0. Therefore, the digits of left parenthesis 25 multiplication 10 super n right parenthesis plus left parenthesis k multiplication 10 super two times n right parenthesis consist of the digits of k followed by the digits 2 and 5, followed by two digits of 0. The table below shows this for n equals two and k equals 35:
multiline equation row 1 25 multiplication 10 super n equals 2,500 row 2 35 multiplication 10 super two times n equals 350,000 row 3 left parenthesis 25 multiplication 10 super n right parenthesis plus left parenthesis 35 multiplication 10 super two times n right parenthesis equals 352,500

Thus, when n equals two, the sum of the digits of left parenthesis 25 multiplication 10 super n right parenthesis plus left parenthesis k multiplication 10 super two times n right parenthesis will be 2 + 5 = 7 plus the sum of the digits of k. Of the answer choices, this sum of digits is divisible by 9 only for k equals 47, which gives 2 + 5 + 4 + 7 = 18. It can also be verified that, for each positive integer n, the only such answer choice is k equals 47, although this additional verification is not necessary to obtain the correct answer.

The correct answer is E.

image

120. On the number line, the shaded interval is the graph of which of the following inequalities?

A. absolute value of x less than or equals four

B. absolute value of x less than or equals eight

C. absolute value of x minus two less than or equals four

D. absolute value of x minus two less than or equals six

E. absolute value of x plus two less than or equals six

Algebra Inequalities; Absolute Value

The midpoint of the interval from −8 to 4, inclusive, is negative eight plus four divided by two equals negative two and the length of the interval from −8 to 4, inclusive, is 4−(−8) = 12, so the interval consists of all numbers within a distance of 12 divided by two equals six from −2. Using an inequality involving absolute values, this can be described by absolute value of x minus left parenthesis negative two right parenthesis less than or equals six, or absolute value of x plus two less than or equals six.

Alternatively, the inequality negative eight less than or equals x less than or equals four can be written as the conjunction negative eight less than or equals x and x less than or equals four. Rewrite this conjunction so that the lower value, −8, and the upper value, 4, are shifted to values that have the same magnitude. This can be done by adding 2 to each side of each inequality, which gives negative six less than or equals x plus two and x plus two less than or equals six. Thus, x plus two lies between −6 and 6, inclusive, and it follows that absolute value of x plus two less than or equals six.

The correct answer is E.

121. Last year members of a certain professional organization for teachers consisted of teachers from 49 different school districts, with an average (arithmetic mean) of 9.8 schools per district. Last year the average number of teachers at these schools who were members of the organization was 22. Which of the following is closest to the total number of members of the organization last year?

A. 10^7

B. 10^6

C. 10^5

D. 10^4

E. 10^3

Arithmetic **Statistics**

There are 49 school districts and an average of 9.8 schools per district, so the number of schools is (49)(9.8) ≈ (50)(10) = 500. There are approximately 500 schools and an average of 22 teachers at each school, so the number of teachers is approximately (500)(22) ≈ (500)(20) = 10,000 = 10^4.

The correct answer is D.

122. Of all the students in a certain dormitory, one divided by two are first-year students and the rest are second-year students. If four divided by five of the first-year students have not declared a major and if the fraction of second-year students who have declared a major is 3 times the fraction of first-year students who have declared a major, what fraction of all the students in the dormitory are second-year students who have not declared a major?

A. one divided by 15

B. one divided by five

C. four divided by 15

D. one divided by three

E. two divided by five

Arithmetic Applied Problems

Consider the table below in which cap t represents the total number of students in the dormitory. Since one divided by two of the students are first-year students and the rest are second-year students, it follows that one divided by two of the students are second-year students, and so the totals for the first-year and second-year columns are both 0.5 times cap t. Since four divided by five of the first-year students have not declared a major, it follows that the middle entry in the first-year column is

four divided by five times left parenthesis 0.5 times cap t right parenthesis equals 0.4 times cap t

and the first entry in the first-year column is

0.5 times cap t minus 0.4 times cap t equals 0.1 times cap t. Since the fraction of second-year students who have declared a major is 3 times the fraction of first-year students who have declared a major, it follows that the first entry in the second-year column is

three times left parenthesis 0.1 times cap t right parenthesis equals 0.3 times cap t

and the second entry in the second-year column is

0.5 times cap t minus 0.3 times cap t equals 0.2 times cap t. Thus, the fraction of students that are second-year students who have not declared a major is

equation sequence part 1 0.2 times cap t divided by cap t equals part 2 0.2 equals part 3 one divided by five

.

	First-year	Second-year	Total
Declared major	0.1 times cap t	0.3 times cap t	0.4 times cap t
Not declared major	0.4 times cap t	0.2 times cap t	0.6 times cap t
Total	0.5 times cap t	0.5 times cap t	cap t

The correct answer is B.

123. If the average (arithmetic mean) of x, y, and z is seven times x and x not equals zero, what is the ratio of x to the sum of y and z?

A. 1:21

B. 1:20

C. 1:6

D. 6:1

E. 20:1

Algebra **Ratio and Proportion**

Given that the average of x comma y, and z is seven times x, it follows that
sum with 3 summands x plus y plus z divided by three equals seven times x
, or sum with 3 summands x plus y plus z equals 21 times x, or y plus z equals 20 times x. Dividing both sides of the last equation by 20 times left parenthesis y plus z right parenthesis gives one divided by 20 equals x divided by y plus z, so the ratio of x to the sum of y and z is 1:20.

The correct answer is B.

124. Jonah drove the first half of a 100-mile trip in x hours and the second half in y hours. Which of the following is equal to Jonah's average speed, in miles per hour, for the entire trip?

A. 50 divided by x plus y

B. 100 divided by x plus y

C. 25 divided by x plus 25 divided by y

D. 50 divided by x plus 50 divided by y

E. 100 divided by x plus 100 divided by y

Algebra **Applied Problems**

Using average speed equals total distance divided by total time, it follows that Jonah's average speed for his entire 100-mile trip is 100 divided by x plus y.

The correct answer is B.

125. If the amount of federal estate tax due on an estate valued at $1.35 million is $437,000 plus 43 percent of the value of the estate in excess of $1.25 million, then the federal tax due is approximately what percent of the value of the estate?

A. 30%

B. 35%

C. 40%

D. 45%

E. 50%

Arithmetic **Percents; Estimation**

The amount of tax divided by the value of the estate is

left square bracket 0.437 plus left parenthesis 0.43 right parenthesis times left parenthesis 1.35 minus 1.25 right parenthesis right square bracket mittion divided by 1.35 mittion	value as fraction
equals 0.437 plus left parenthesis 0.43 right parenthesis times left parenthesis 0.1 right parenthesis divided by 1.35	arithmetic
equation sequence part 1 equals part 2 0.48 divided by 1.35 equals part 3 48 divided by 135	arithmetic

By long division, 48 divided by 135 is approximately 35.6, so the closest answer choice is 35%.

Alternatively, 48 divided by 135 can be estimated by multirelation 48 divided by 136 equals six divided by 17 almost equals six divided by 18 equals one divided by three almost equals 33 percent
, so the closest answer choice is 35%. Note that 48 divided by 135 is greater than 48 divided by 136, and six divided by 17 is greater than six divided by 18, so the correct value is greater than 33%, which rules out 30% being the closest.

The correct answer is B.

multiline inequality row 1 seven times x plus six times y less than or equals 38,000 row 2 four times x plus five times y less than or equals 28,000

126. A manufacturer wants to produce x balls and y boxes. Resource constraints require that x and y satisfy the inequalities shown. What is the maximum number of balls and boxes combined that can be produced given the resource constraints?

A. 5,000

B. 6,000

C. 7,000

D. 8,000

E. 10,000

Algebra **Inequalities**

We are to determine the maximum value of x plus y given the inequalities above. Note that if cap a less than or equals cap b and cap c less than or equals cap d, then we can "add inequalities" to obtain cap a plus cap c less than or equals cap b plus cap d, since (roughly speaking) the sum of two smaller numbers is less than the sum of two larger numbers. Adding the inequalities shown above gives left parenthesis seven times x plus six times y right parenthesis plus left parenthesis four times x plus five times y right parenthesis less than or equals 38,000 plus 28,000
, or 11 times x plus 11 times y less than or equals 66,000. Dividing both sides of this last inequality by 11 gives
x plus y less than or equals 6,000. Therefore, the values of x plus y are at most 6,000, and hence the maximum value of x plus y is *at most* 6,000.

The fact that the maximum value of x plus y is *equal* to 6,000 follows from the fact that the system of simultaneous equations seven times x plus six times y equals 38,000 and four times x plus five times y equals 28,000 has a solution, which in turn follows from the fact that these two equations correspond to a pair of nonparallel lines in the standard left parenthesis x comma y right parenthesis coordinate plane. In particular, the pair x equals 2,000 and y equals 4,000 satisfy both the two inequalities and the equation x plus y equals 6,000.

The correct answer is B.

127. If three divided by 10 super four equals x percent, then x equals

A. 0.3

B. 0.03

C. 0.003

D. 0.0003

E. 0.00003

Arithmetic **Percents**

Given that three divided by 10 super four equals x percent, and writing x percent as x divided by 100, it follows that three divided by 10 super four equals x divided by 100. Multiplying both sides by 100 gives equation sequence part 1 x equals part 2 300 divided by 10 super four equals part 3 300 divided by 10,000 equals part 4 three divided by 100 equals part 5 0.03

.

The correct answer is B.

128. What is the remainder when 3^{24} is divided by 5?

A. 0

B. 1

C. 2

D. 3

E. 4

Arithmetic **Properties of Numbers**

A pattern in the units digits of the numbers 3, $3^2 = 9$, $3^3 = 27$, $3^4 = 81$, $3^5 = 243$, etc., can be found by observing that the units digit of a product of two integers is the same as the units digit of the product of the units digit of the two integers. For example, the units digit of $3^5 = 3 \times 3^4 = 3 \times 81$ is 3 since the units digit of 3×1 is 3, and the units digit of $3^6 = 3 \times 3^5 = 3 \times 243$ is 9 since the units digit of 3×3 is 9. From this it follows that the units digit of the powers of 3 follow the pattern 3, 9, 7, 1, 3, 9, 7, 1, etc., with a units digit of 1 for 3^4, 3^8, 3^{12}, …, 3^{24}, …. Therefore, the units digit of 3^{24} is 1. Thus, 3^{24} is 1 more than a multiple of 10, and hence 3^{24} is 1 more than a multiple of 5, and so the remainder when 3^{24} is divided by 5 is 1.

The correct answer is B.

129. José has a collection of 100 coins, consisting of nickels, dimes, quarters, and half-dollars. If he has a total of 35 nickels and dimes, a total of 45 dimes and quarters, and a total of 50 nickels and quarters, how many half-dollars does he have?

A. 15

B. 20

C. 25

D. 30

E. 35

Algebra **Simultaneous Equations**

Letting n, d, q, and h, respectively, represent the numbers of nickels, dimes, quarters, and half-dollars José has, determine the value of h.

The following are given:

1. $n + d + q + h = 100$
2. $n + d = 35$
3. $d + q = 45$
4. $n + q = 50$

Adding (2), (3), and (4) gives

$$2n + 2d + 2q = 130$$

or $n + d + q = 65$. Subtracting this equation from (1) gives

$$h = 100 - 65 = 35$$

.

The correct answer is E.

130. David used part of $100,000 to purchase a house. Of the remaining portion, he invested $\frac{1}{3}$ of it at 4 percent simple annual interest and $\frac{2}{3}$ of it at 6 percent simple

annual interest. If after a year the income from the two investments totaled $320, what was the purchase price of the house?

A. $96,000

B. $94,000

C. $88,000

D. $75,000

E. $40,000

Algebra **Applied Problems; Percents**

Let x be the amount, in dollars, that David used to purchase the house. Then David invested
left parenthesis 100,000 minus x right parenthesis dollars,
one divided by three at 4% simple annual interest and
two divided by three at 6% simple annual interest. After one year the total interest, in dollars, on this investment was
one divided by three times left parenthesis 100,000 minus x right parenthesis times left parenthesis 0.04 right parenthesis plus two divided by three times left parenthesis 100,000 minus x right parenthesis times left parenthesis 0.06 right parenthesis equals 320
. Solve this equation to find the value of x.

$\frac{1}{3}(100{,}000-x)(0.04)+\frac{2}{3}(100{,}000-x)(0.06)$	$= 320$	given
$(100{,}000-x)(0.04)+2(100{,}000-x)(0.06)$	$= 960$	multiply both sides by 3
$4{,}000-0.04x+12{,}000-0.12x$	$= 960$	distributive property
$16{,}000-0.16x$	$= 960$	combine like terms
$16{,}000-960$	$= 0.16x$	add $0.16x-960$ to both sides
$100{,}000-6{,}000$	$= x$	divide both sides by 0.16
$94{,}000$	$= x$	

Therefore, the purchase price of the house was $94,000.

The correct answer is B.

131. A certain manufacturer sells its product to stores in 113 different regions worldwide, with an average (arithmetic mean) of 181 stores per region. If last year these stores sold an average of 51,752 units of the manufacturer’s product per store, which of the following is closest

to the total number of units of the manufacturer’s product sold worldwide last year?

A. 10^6

B. 10^7

C. 10^8

D. 10^9

E. 10^{10}

Arithmetic **Estimation**

multiline equation row 1 left parenthesis 113 right parenthesis times left parenthesis 181 right parenthesis times left parenthesis 51,752 right parenthesis almost equals left parenthesis 100 right parenthesis times left parenthesis 200 right parenthesis times left parenthesis 50,000 right parenthesis row 2 equals 10 squared multiplication left parenthesis two multiplication 10 squared right parenthesis multiplication left parenthesis five multiplication 10 super four right parenthesis row 3 equals left parenthesis two multiplication five right parenthesis multiplication 10 super sum with 3 summands two plus two plus four row 4 equals 10 super one multiplication 10 super eight equals 10 super nine

The correct answer is D.

132. Andrew started saving at the beginning of the year and had saved \$240 by the end of the year. He continued to save and by the end of 2 years had saved a total of \$540. Which of the following is closest to the percent increase in the amount Andrew saved during the second year compared to the amount he saved during the first year?

A. 11%

B. 25%

C. 44%

D. 56%

E. 125%

Arithmetic **Percents**

Andrew saved $240 in the first year and $540 – $240 = $300 in the second year. The percent increase in the amount Andrew saved in the second year compared to the amount he saved in the first year is left parenthesis 300 minus 240 divided by 240 multiplication 100 right parenthesis percent equals left parenthesis 60 divided by 240 multiplication 100 right parenthesis percent equals left parenthesis one divided by four multiplication 100 right parenthesis percent equals 25 percent
.

The correct answer is B.

133. If x is a positive integer, r is the remainder when x is divided by 4, and cap r is the remainder when x is divided by 9, what is the greatest possible value of r squared plus cap r?

A. 25

B. 21

C. 17

D. 13

E. 11

Arithmetic **Properties of Integers**

If r is the remainder when the positive integer x is divided by 4, then multirelation zero less than or equals r less than four, so the maximum value of r is 3. If cap r is the remainder when the positive integer x is divided by 9, then multirelation zero less than or equals cap r less than nine, so the maximum value of cap r is 8. Thus, the maximum value of r squared plus cap r is three squared plus eight equals 17.

The correct answer is C.

134. Each of the nine digits 0, 1, 1, 4, 5, 6, 8, 8, and 9 is used once to form 3 three-digit integers. What is the greatest possible sum of the 3 integers?

A. 1,752

B. 2,616

C. 2,652

D. 2,775

E. 2,958

Arithmetic **Place Value**

To create 3 three-digit numbers using each of the digits 0, 1, 1, 4, 5, 6, 8, 8, and 9 and having the maximum possible sum, the greatest three digits must be in hundreds place, the next greatest three in tens place, and the three smallest digits in units place. The sum will then be $(9 + 8 + 8)(100) + (4 + 5 + 6)(10) + (0 + 1 + 1) = 25(100) + 15(10) + 2 = 2{,}500 + 150 + 2 = 2{,}652$.

The correct answer is C.

135. Given that $1^2 + 2^2 + 3^2 + \ldots + 10^2 = 385$, what is the value of $3^2 + 6^2 + 9^2 + \ldots + 30^2$?

A. 1,155

B. 1,540

C. 1,925

D. 2,310

E. 3,465

Arithmetic **Series and Sequences**

We can use the fact that each term of the second series is $3^2 = 9$ times greater than the corresponding term of the first series to find the sum of the second series.

$3^2 + 6^2 + 9^2 + \ldots + 30^2$

$= (3 \cdot 1)^2 + (3 \cdot 2)^2 + (3 \cdot 3)^2 + \ldots + (3 \cdot 10)^2$

$= (3^2 \cdot 1^2) + (3^2 \cdot 2^2) + (3^2 \cdot 3^2) + \ldots + (3^2 \cdot 10^2)$

$= 3^2(1^2 + 2^2 + 3^2 + \ldots + 10^2) = 9(385) = 3{,}465$

The correct answer is E.

136. If water is leaking from a certain tank at a constant rate of 1,200 milliliters per hour, how many <u>seconds</u> does it take for 1 milliliter of water to leak from the tank?

A. one divided by three

B. one divided by two

C. 2

D. 3

E. 20

Arithmetic Measurement Conversion; Rate Problem

We are given that 1,200 milliliters leak in 1 hour, or 3,600 seconds. Because the leaking is at a constant rate, it takes one divided by 1,200 as long for 1 milliliter to leak, and so 1 milliliter leaks in 3,600 divided by 1,200 equals three times seconds.

The correct answer is D.

137. When the positive integer k is divided by the positive integer n, the remainder is 11. If k divided by n equals 81.2, what is the value of n?

A. 9

B. 20

C. 55

D. 70

E. 81

Arithmetic Properties of Integers

When the positive integer k is divided by the positive integer n, there exist unique positive integers q (the quotient) and r (the remainder) such that k equals q times n plus r, where multirelation zero less than or equals r less than n. For example, when 43 is divided by 5, we have 43 = (8)(5) + 3 and

multirelation zero less than or equals three less than five. Dividing both sides of k equals q times n plus r by n and using r equals 11 gives
k divided by n equals sum with 3 summands q plus r divided by n times q plus 11 divided by n
. We are given that
equation sequence part 1 k divided by n equals part 2 81.2 equals part 3 81 plus two divided by 10
, so q plus 11 divided by n equals 81 plus two divided by 10. Therefore, q equals 81 and
11 divided by n equals two divided by 10, or n equals 55.

The correct answer is C.

138. The total area of a certain continent is approximately 3.8×10^{16} square inches. Which of the following is closest to the area of the continent, in square miles? (1 square mile is approximately 4.0×10^{9} square inches.)

A. 6.7×10^{5}

B. 2.0×10^{6}

C. 9.5×10^{6}

D. 1.1×10^{7}

E. 9.5×10^{8}

Arithmetic Measurement Conversion

The following calculations make use of the given approximations.

4.0 multiplication 10 super nine times in squared	≈	one times mi squared	given
one in squared	≈	one divided by 4.0 multiplication 10 super nine times mi squared	divide both sides by 4.0×10^9
3.8 multiplication 10 super 16 in squared	≈	3.8 multiplication 10 super 16 divided by 4.0 multiplication 10 super nine times mi squared	multiply both sides by 3.8×10^{16}
3.8 multiplication 10 super 16 in squared	≈	3.8 divided by 4.0 multiplication 10 super 16 divided by 10 super nine times mi squared	use a divided by b multiplication c divided by d equals a times c divided by b times d
3.8 multiplication 10 super 16 in squared	≈	19 divided by 20 multiplication 10 super 16 minus nine mi squared	use a super b divided by a super c equals a super b minus c
3.8 multiplication 10 super 16 in squared	≈	0.95 multiplication 10 super seven mi squared	
3.8 multiplication 10 super 16 in squared	≈	9.5 multiplication 10 super six mi squared	

The correct answer is C.

139. If x times y equals one, what is the value of two super left parenthesis x plus y right parenthesis squared divided by two super left parenthesis x minus y right parenthesis squared
?

A. 2

B. 4

C. 8

D. 16

E. 32

Algebra **Exponents; Simplifying Algebraic Expressions**

Using the property
two super a divided by two super b equals two super a minus b,
two super left parenthesis x plus y right parenthesis squared divided by two super left parenthesis x minus y right parenthesis squared equals two super left parenthesis x plus y right parenthesis squared minus left parenthesis x minus y right parenthesis squared
. Also, since
equation sequence part 1 left parenthesis x plus y right parenthesis squared minus left parenthesis x minus y right parenthesis squared equals part 2 left parenthesis sum with 3 summands x squared plus two times x times y plus y squared right parenthesis minus left parenthesis x squared minus two times x times y plus y squared right parenthesis equals part 3 four times x times y
and x times y equals one, it follows that
equation sequence part 1 two super left parenthesis x plus y right parenthesis squared divided by two super left parenthesis x minus y right parenthesis squared equals part 2 two super four times x times y equals part 3 two super four multiplication one equals part 4 16
.

The correct answer is D.

140. Of the 20 members of a kitchen crew, 17 can use the meat-cutting machine, 18 can use the bread-slicing machine, and 15 can use both machines. If one member of the crew is to be chosen at random, what is the probability that the member chosen will be someone who cannot use either machine?

A. 0

B. one divided by 10

C. one divided by seven

D. one divided by four

E. one divided by three

Arithmetic **Probability; Sets (Venn diagrams)**

Since 17 members can use the meat-cutting machine and 15 members can use both machines, 17 − 15 = 2 members can use only the meat-cutting machine. Similarly, since 18 members can use the bread-slicing machine and 15 members can use both machines, 18 − 15 = 3 members can use only the bread-slicing machine. The Venn diagram below shows these values, where x is the number of members who cannot use either machine.

image

Since there is a total of 20 members, it follows that sum with 4 summands two plus 15 plus three plus x equals 20, or x equals zero. Therefore, the probability that the member chosen cannot use either machine is zero divided by 20 equals zero.

The correct answer is A.

141. Which of the following is an integer?

I. 12 factorial divided by six factorial

II. 12 factorial divided by eight factorial

III. 12 factorial divided by seven factorial five factorial

A. I only

B. II only

C. III only

D. I and II only

E. I, II, and III

Arithmetic **Arithmetic Operations**

I. multiline equation row 1 12 factorial divided by six factorial equals factorial six factorial multiplication seven multiplication eight multiplication nine multiplication 10 multiplication 11 multiplication 12 divided by factorial six factorial row 2 equals seven multiplication eight multiplication nine multiplication 10 multiplication 11 multiplication 12 comma row 3 which is an integer full stop

II. multiline equation row 1 12 factorial divided by eight factorial equals factorial eight factorial multiplication nine multiplication 10 multiplication 11 multiplication 12 divided by factorial eight factorial row 2 equals nine multiplication 10 multiplication 11 multiplication 12 comma row 3 which is an integer full stop

III. multiline equation row 1 12 factorial divided by seven factorial five factorial equals factorial seven factorial multiplication eight multiplication nine multiplication 10 multiplication 11 multiplication 12 divided by factorial seven factorial multiplication one multiplication two multiplication multiplication multiplication 34 multiplication five row 2 equals eight multiplication nine multiplication 10 multiplication 11 divided by multiplication multiplication 25 equals eight multiplication nine multiplication 11 comma row 3 which is an integer full stop

Alternatively for III, note that
12 factorial divided by seven factorial five factorial equals vector element 1 12 element 2 seven
is the number of combinations of 12 objects taken 7 at a time, and thus
12 factorial divided by seven factorial five factorial is an integer.

Therefore, the expression in each of I, II, and III is an integer.

The correct answer is E.

142. The function f is defined for all nonzero x by the equation f of x equals x minus one divided by x. If x not equals zero, which of the following equals f of one divided by x?

A. f of x

B. f of negative x

C. f of negative one divided by x

D. one divided by f of x

E. negative one divided by f of x

Algebra Functions

multiline equation line 1 equation sequence part 1 f of one divided by x equals part 2 one divided by x minus one divided by one divided by x equals part 3 one divided by x minus x line 2 equation sequence part 1 f of negative x equals part 2 negative x minus one divided by negative x equals part 3 negative x plus one divided by x equals part 4 one divided by x minus x

Therefore, f of one divided by x equals f of negative x.

Alternatively, evaluate all of the expressions for an appropriate value of x, such as x equals two.

multiline equation row 1 f of one divided by x equals f of one divided by two equals one divided by two minus two equals negative three divided by two row 2 f of x equals f of two equals two minus one divided by two equals three divided by two row 3 f of negative x equals f of negative two equals negative two plus one divided by two equals negative three divided by two row 4 f of negative one divided by x equals f of negative one divided by two equals negative one divided by two plus two equals three divided by two row 5 one divided by f of x equals one divided by f of two equals one divided by two minus one divided by two equals two divided by three row 6 negative one divided by f of x equals negative one divided by f of two equals negative one divided by two minus one divided by two equals negative two divided by three

The correct answer is B.

143. In the arithmetic sequence t sub one comma t sub two comma t sub three comma ellipsis comma t sub n comma times ellipsis comma t sub one equals 23 and t sub n equals t sub n minus one minus three for each n greater than one. What is the value of n when t sub n equals negative four?

A. −1

B. 7

C. 10

D. 14

E. 20

Algebra **Series and Sequences**

From the given information, it follows that if we subtract 3 a total of n minus one times from t sub one equals 23, then we will obtain t sub n equals negative four. Therefore, n satisfies the equation 23 minus left parenthesis n minus one right parenthesis times left parenthesis three right parenthesis equals negative four
. Now solve for n.

23 minus left parenthesis n minus one right parenthesis times left parenthesis three right parenthesis	=	negative four	from above
23 minus three times n plus three	=	negative four	distributive property
negative three times n	=	negative 30	combine like terms
n	=	10	divide both sides by −3

Alternatively, subtract 3 from 23 until −4 is obtained and count the number of subtractions. One more than this number is the value of n such that t sub n equals negative four.

The correct answer is C.

144. How many seconds will it take for a car that is traveling at a constant rate of 45 miles per hour to travel a distance of 220 yards? (1 mile = 1,760 yards)

A. 8

B. 9

C. 10

D. 11

E. 12

Arithmetic Rate Problem; Measurement Conversion

First, convert 45 miles per hour to yards per second. We know that 1 hour is equal to 3,600 seconds giving us the fraction 45 divided by 3,600. Divide both sides of the fraction by 3,600 to get the miles per second which equals one divided by 80 miles in one second. We know that 1 mile is equal to 1,760 yards so the nest step is 1,760 multiplication one divided by 80. Therefore, it will take the car

1 second to travel 22 yards; hence it will take 10 seconds for the car to travel 220 yards.

Alternatively, let t be the number of seconds it will take for the car to travel 220 yards.

220 yd	=	left parenthesis 45 mi divided by hr right parenthesis times left parenthesis t secant right parenthesis	distance = rate × time
left parenthesis 220 yd right parenthesis times left parenthesis one mi divided by 1,760 yd right parenthesis	=	left parenthesis 45 mi divided by hr right parenthesis times left parenthesis one hr divided by 3,600 secant right parenthesis times left parenthesis t secant right parenthesis	1 mi = 1,760 yd 1 hr = 3,600 sec
220 divided by 1,760 mi	=	left parenthesis 45 divided by 3,600 t right parenthesis mi	cancel units
two multiplication 11 multiplication 10 divided by 11 multiplication 16 multiplication 10	=	left parenthesis 45 divided by 45 multiplication 80 right parenthesis times t	factor
one divided by eight	=	t divided by 80	cancel common factors
t	=	10	multiply both sides by 80

The correct answer is C.

145. A store’s selling price of $2,240 for a certain computer would yield a profit of 40 percent of the store’s cost for the computer. What selling price would yield a profit of 50 percent of the computer’s cost?

A. $2,400

B. $2,464

C. $2,650

D. $2,732

E. $2,800

Algebra **Applied Problems**

Let normal dollar times cap c be the store’s cost for the computer. Then the store’s profit from selling the computer would be normal dollar times left parenthesis 2,240 minus cap c right parenthesis
, which is equal to 40 percent of the store’s cost.

2,240 minus cap c	=	0.4 times cap c	given
2,240	=	1.4 times cap c	add cap c to both sides
cap c	=	1,600	divide both sides by 1.4

Thus, the store’s cost for the computer is $1,600. Therefore, to yield a profit of 50 percent of the computer’s cost, the selling price would have to be 1.5($1,600) = $2,400.

The correct answer is A.

146. If a certain coin is flipped, the probability that the coin will land heads up is one divided by two. If the coin is flipped 5 times, what is the probability that it will land heads up on the first 3 flips and not on the last 2 flips?

A. three divided by five

B. one divided by two

C. one divided by five

D. one divided by eight

E. one divided by 32

Arithmetic **Probability**

Because the 5 flips of the coin are independent of each other, the probability of the coin landing heads up on the first 3 flips and tails up (i.e., not heads up) on the last 2 flips is

cap p of cap h multiplication cap p of cap h multiplication cap p of cap h multiplication cap p of cap t multiplication cap p of cap t , where cap p of cap h and cap p of cap t are the probabilities, respectively, that a single flip of the coin lands heads up and tails up. Since cap p of cap t equals one minus cap p of cap h and cap p of cap h equals one divided by two, it follows that cap p of cap t equals one divided by two.

Therefore, the requested probability is one divided by two multiplication one divided by two multiplication one divided by two multiplication one divided by two multiplication one divided by two equals one divided by 32 .

The correct answer is E.

147. The operation ⊗ is defined for all nonzero numbers a and b by a circled times b equals a divided by b minus b divided by a. If x and y are nonzero numbers, which of the following statements must be true?

I. x circled times x times y equals x times left parenthesis one circled times y right parenthesis

II. x circled times y equals negative left parenthesis y circled times x right parenthesis

III. one divided by x circled times one divided by y equals y circled times x

A. I only

B. II only

C. III only

D. I and II

E. II and III

Algebra **Simplifying Algebraic Expressions**

I. CAN BE FALSE: Let x equals two and y equals three. Then one divided by y minus y equals negative eight divided by three and
x times left parenthesis one divided by y minus y right parenthesis equals negative 16 divided by three
, so
x circled times x times y equals x times left parenthesis one circled times y right parenthesis
can be false.

II. multiline equation row 1 MUST BE TRUE colon x circled times y equals x divided by y minus y divided by x equals negative left parenthesis y divided by x minus x divided by y right parenthesis row 2 equals negative left parenthesis y circled times x right parenthesis

III. MUST BE TRUE:

equation sequence part 1 one divided by x circled times one divided by y equals part 2 one divided by x divided by one divided by y minus one divided by y divided by one divided by x equals part 3 y divided by x minus x divided by y equals part 4 y circled times x

The correct answer is E.

148. If each term in the sum
sum with variable number of summands a sub one plus a sub two plus ellipsis plus a sub n
is either 7 or 77 and the sum equals 350, which of the following could be equal to n?

A. 38

B. 39

C. 40

D. 41

E. 42

Arithmetic Operations with Integers

Since
equation sequence part 1 sum with variable number of summands a sub one plus a sub two plus ellipsis plus a sub n equals part 2 350 equals part 3 seven multiplication 50
, if each term in the sum were a 7, then n would equal 50. But the only answer options are the consecutive integers from 38 to 42. So, suppose
equation sequence part 1 a sub one equals part 2 77 equals part 3 seven multiplication 11
, while each of the other terms is a 7. Then
equation sequence part 1 sum with variable number of summands a sub one plus a sub two plus ellipsis plus a sub n equals part 2 seven multiplication 50 equals part 3 seven multiplication left parenthesis 11 plus 39 right parenthesis equals part 4 left parenthesis seven multiplication 11 right parenthesis plus left parenthesis seven multiplication 39 right parenthesis
. In that case, in addition to a sub one equals 77, the sum has 39 other terms, each equal to
seven colon a sub two comma ellipsis comma a sub 40. Thus,
equation sequence part 1 n equals part 2 one plus 39 equals part 3 40
.

The correct answer is C.

149. (1.00001)(0.99999) – (1.00002)(0.99998) =

A. 0

B. 10^{-10}

C. $3(10^{-10})$

D. 10^{-5}

E. $3(10^{-5})$

Arithmetic Equations

(1.00001)(0.99999) − (1.00002)(0.99998) = (1 + 0.00001)(0.99999) − (1 + 0.00002)(0.99998) = (0.99999 + 0.00069999) − (0.99998 + 0.0000199996) = 0.9999999999 − 0.9999999996 = 0.0000000003 = $3(10^{-10})$.

The correct answer is C.

150. A certain rectangular photograph has an area of square feet. What is the area of the photograph in square inches? (1 foot = 12 inches)

A. 36

B. 78

C. 432

D. 864

E. 936

Arithmetic Rational Numbers

Since 1 foot is 12 inches, 1 square foot is 12^2 = 144 square inches. So, the photograph's area of 6.5 square feet is 6.5 × 144 = 936 square inches.

The correct answer is E.

151. The toll for crossing a certain bridge is $0.75 each crossing. Drivers who frequently use the bridge may instead purchase a sticker each month for $13.00 and then pay only $0.30 each crossing during that month. If a particular driver will cross the bridge twice on each of x days next month and will not cross the bridge on any other day, what is the least value of x for which this driver can save money by using the sticker?

A. 14

B. 15

C. 16

D. 28

E. 29

Algebra **First-Degree Equations**

We need to find the least whole-number value of x for which 0.75 times left parenthesis two times x right parenthesis greater than 13 plus 0.3 times left parenthesis two times x right parenthesis
; that is, the least whole number value for which
multirelation left parenthesis 1.5 minus 0.6 right parenthesis times x equals 0.9 times x greater than 13
, and thus for which
multirelation x greater than 130 divided by nine equals.

This value is 15.

The correct answer is B.

Questions 152 to 207 — Difficulty: Hard

152. Two numbers differ by 2 and sum to cap s. Which of the following is the greater of the numbers in terms of cap s?

A. cap s divided by two minus one

B. cap s divided by two

C. cap s divided by two plus one divided by two

D. cap s divided by two plus one

E. cap s divided by two plus two

Algebra **First-Degree Equations**

Let x represent the greater of the two numbers that differ by 2. Then, x minus two represents the lesser of the two numbers. The two numbers sum to cap s, so
x plus left parenthesis x minus two right parenthesis equals cap s. It follows that two times x minus two equals cap s, or
two times x equals cap s plus two, or
x equals cap s divided by two plus one.

The correct answer is D.

153. If m is an integer and m equals 10 super 32 minus 32, what is the sum of the digits of m?

A. 257

B. 264

C. 275

D. 284

E. 292

Arithmetic **Arithmetic Operations**

When written in standard base 10 notation, 10^{32} is the digit 1 followed by 32 digits of 0. Now consider the following subtractions and the digit pattern they suggest, a pattern which is easily seen to continue.

$$100 - 32 = 68$$

$$1{,}000 - 32 = 968$$

$$10{,}000 - 32 = 9{,}968$$

$$100{,}000 - 32 = 99{,}968$$

$$1{,}000{,}000 - 32 = 999{,}968$$

Using this digit pattern, m equals 10 super 32 minus 32 in standard base 10 notation consists of $32 - 2 = 30$ occurrences of the digit 9 followed by the digits 6 and 8. Therefore, the sum of the digits of m is $30(9) + 6 + 8 = 270 + 14 = 284$.

The correct answer is D.

154. In a numerical table with 10 rows and 10 columns, each entry is either a 9 or a 10. If the number of 9s in the n th row is n minus one for

each n from 1 to 10, what is the average (arithmetic mean) of all the numbers in the table?

A. 9.45

B. 9.50

C. 9.55

D. 9.65

E. 9.70

Arithmetic **Operations with Integers**

There are (10)(10) = 100 entries in the table. In rows 1, 2, 3, …, 10, the number of 9s is 0, 1, 2, …, 9, respectively, giving a total of 0 + 1 + 2 + … + 9 = 45 entries with a 9. This leaves a total of 100 – 45 = 55 entries with a 10. Therefore, the sum of the 100 entries is 45(9) + 55(10) = 405 + 550 = 955, and the average of the 100 entries is 955 divided by 100 equals 9.55

The correct answer is C.

155. In 2004, the cost of 1 year-long print subscription to a certain newspaper was $4 per week. In 2005, the newspaper introduced a new rate plan for 1 year-long print subscription: $3 per week for the first 40 weeks of 2005 and $2 per week for the remaining weeks of 2005. How much less did 1 year-long print subscription to this newspaper cost in 2005 than in 2004?

A. $64

B. $78

C. $112

D. $144

E. $304

Arithmetic **Applied Problems**

The cost, in dollars, of 1 year-long print subscription in 2004 was 52(4) = 208 and the cost, in dollars, of 1 year-long print subscription in

2005 was 40(3) + 12(2) = 120 + 24 = 144. Therefore, the cost in 2005 was less than the cost in 2004 by 208 − 144 = 64 dollars.

The correct answer is A.

156. A positive integer n is a perfect number provided that the sum of all the positive factors of n, including 1 and n, is equal to two times n. What is the sum of the reciprocals of all the positive factors of the perfect number 28?

A. one divided by four

B. 56 divided by 27

C. 2

D. 3

E. 4

Arithmetic **Properties of Integers**

The factors of 28 are 1, 2, 4, 7, 14, and 28. Therefore, the sum of the reciprocals of the factors of 28 is equation sequence part 1 sum with 6 summands one divided by one plus one divided by two plus one divided by four plus one divided by seven plus one divided by 14 plus one divided by 28 equals part 2 sum with 6 summands 28 divided by 28 plus 14 divided by 28 plus seven divided by 28 plus four divided by 28 plus two divided by 28 plus one divided by 28 equals part 3 sum with 6 summands 28 plus 14 plus seven plus four plus two plus one divided by 28 equals part 4 56 divided by 28 equals part 5 two full stop

The correct answer is C.

157. The infinite sequence
a sub one comma a sub two comma ellipsis comma a sub n comma ellipsis
is such that
a sub one equals two comma a sub two equals negative three,
a sub three equals five comma a sub four equals negative one, and
a sub n equals a sub n minus four for n greater than four. What is the sum of the first 97 terms of the sequence?

A. 72

B. 74

C. 75

D. 78

E. 80

Arithmetic **Sequences and Series**

Because a sub n equals a sub n minus four for n greater than four, it follows that the terms of the sequence repeat in groups of 4 terms:

Values for n	Values for a sub n
1, 2, 3, 4	2, –3, 5, –1
5, 6, 7, 8	2, –3, 5, –1
9, 10, 11, 12	2, –3, 5, –1
13, 14, 15, 16	2, –3, 5, –1

Thus, since 97 = 24(4) + 1, the sum of the first 97 terms can be grouped into 24 groups of 4 terms each, with one remaining term, which allows the sum to be easily found:

multiline equation line 1 sum with 3 summands left parenthesis sum with 4 summands a sub one plus a sub two plus a sub three plus a sub four right parenthesis plus left parenthesis sum with 4 summands a sub five plus a sub six plus a sub seven plus a sub eight right parenthesis plus ellipsis postfix plus line 2 left parenthesis sum with 4 summands a sub 93 plus a sub 94 plus a sub 95 plus a sub 96 right parenthesis plus a sub 97 multiline equation line 1 equals sum with 3 summands left parenthesis two minus three plus five minus one right parenthesis plus left parenthesis two minus three plus five minus one right parenthesis plus ellipsis postfix plus line 2 left parenthesis two minus three plus five minus one right parenthesis plus two

multiline equation line 1 equation sequence part 1 equals part 2 24 times left parenthesis two minus three plus five minus one right parenthesis plus two equals part 3 24 times left parenthesis three right parenthesis plus two equals part 4 74

The correct answer is B.

158. The sequence a sub one comma a sub two comma ellipsis comma a sub n comma ellipsis
is such that a sub n equals two times a sub n minus one minus x for all positive integers n greater than or equals two and for a certain number x. If a sub five equals 99 and a sub three equals 27, what is the value of x?

A. 3

B. 9

C. 18

D. 36

E. 45

Algebra **Sequences and Series**

An expression for a sub five that involves x can be obtained using a sub three equals 27 and applying the equation a sub n equals two times a sub n minus one minus x twice, once for n equals four and once for n equals five.

a sub four equals two times a sub three minus x	using a sub n equals two times a sub n minus one minus x for n equals four
equals two times left parenthesis 27 right parenthesis minus x	using a sub three equals 27
a sub five equals two times a sub four minus x	using a sub n equals two times a sub n minus one minus x for n equals four
equals two times left square bracket two times left parenthesis 27 right parenthesis minus x right square bracket minus x	using a sub four equals two times left parenthesis 27 right parenthesis minus x
equals four times left parenthesis 27 right parenthesis minus three times x	combine like terms

Therefore, using a sub five equals 99, we have

99 equals four times left parenthesis 27 right parenthesis minus three times x	given
three times x equals four times left parenthesis 27 right parenthesis minus 99	adding left parenthesis three times x minus 99 right parenthesis to both sides
x equals four times left parenthesis nine right parenthesis minus 33	dividing both sides by 3
x equals three	arithmetic

The correct answer is A.

159. In a certain medical survey, 45 percent of the people surveyed had the type A antigen in their blood and 3 percent had both the type A antigen and the type B antigen. Which of the following is closest to the percent of those with the type A antigen who also had the type B antigen?

A. 1.35%

B. 6.67%

C. 13.50%

D. 15.00%

E. 42.00%

Arithmetic Applied Problems; Percents

Let n be the total number of people surveyed. Then, the proportion of the people who had type A who also had type B is

equation sequence part 1 left parenthesis three percent right parenthesis times n divided by left parenthesis 45 percent right parenthesis times n equals part 2 three divided by 45 equals part 3 one divided by 15
, which as a percent is approximately 6.67%. Note that by using one divided by 15 equals one divided by three multiplication one divided by five
, which equals one divided by three of 20%, we can avoid dividing by a 2-digit integer.

The correct answer is B.

160. On a certain transatlantic crossing, 20 percent of a ship's passengers held round-trip tickets and also took their cars aboard the ship. If 60 percent of the passengers with round-trip tickets did <u>not</u> take their cars aboard the ship, what percent of the ship's passengers held round-trip tickets?

   A. percent

   B. 40%

   C. 50%

   D. 60%

   E. percent

Arithmetic **Percents**

Since the number of passengers on the ship is immaterial, let the number of passengers on the ship be 100 for convenience. Let x be the number of passengers that held round-trip tickets. Then, since 20 percent of the passengers held a round-trip ticket and took their cars aboard the ship, 0.20(100) = 20 passengers held round-trip tickets and took their cars aboard the ship. The remaining passengers with round-trip tickets did not take their cars aboard, and they represent 0.6 times x (that is, 60 percent of the passengers with round-trip tickets). Thus 0.6 times x plus 20 equals x, from which it follows that 20 equals 0.4 times x, and so x equals 50. The percent of

passengers with round-trip tickets is, then,
50 divided by 100 equals 50 percent.

The correct answer is C.

161. If x and k are integers and
left parenthesis 12 super x right parenthesis times left parenthesis four super two times x plus one right parenthesis equals left parenthesis two super k right parenthesis times left parenthesis three squared right parenthesis
, what is the value of k?

A. 5

B. 7

C. 10

D. 12

E. 14

Arithmetic Exponents

Rewrite the expression on the left so that it is a product of powers of 2 and 3.

$$\begin{aligned}(12^x)(4^{2x+1}) &= [(3 \cdot 2^2)^x][(2^2)^{2x+1}] \\ &= (3^x)[(2^2)^x][2^{2(2x+1)}] \\ &= (3^x)(2^{2x})(2^{4x+2}) \\ &= (3^x)(2^{6x+2})\end{aligned}$$

Then, since $(12^x)(4^{2x+1}) = (2^k)(3^2)$, it follows that $(3^x)(2^{6x+2}) = (2^k)(3^2) = (3^2)(2^k)$, so $x = 2$ and $k = 6x + 2$. Substituting 2 for x gives $k = 6(2) + 2 = 14$.

The correct answer is E.

162. If S is the sum of the reciprocals of the 10 consecutive integers from 21 to 30, then S is between which of the following two

fractions?

A. one divided by three and one divided by two

B. one divided by four and one divided by three

C. one divided by five and one divided by four

D. one divided by six and one divided by five

E. one divided by seven and one divided by six

Arithmetic Estimation

The value of
sum with variable number of summands one divided by 21 plus one divided by 22 plus one divided by 23 plus ellipsis plus one divided by 30
is LESS than
sum with variable number of summands one divided by 20 plus one divided by 20 plus one divided by 20 plus ellipsis plus one divided by 20
(10 numbers added), which equals
10 times left parenthesis one divided by 20 right parenthesis equals one divided by two
, and GREATER than
sum with variable number of summands one divided by 30 plus one divided by 30 plus one divided by 30 plus ellipsis plus one divided by 30
(10 numbers added), which equals
10 times left parenthesis one divided by 30 right parenthesis equals one divided by three
.

Therefore, the value of
sum with variable number of summands one divided by 21 plus one divided by 22 plus one divided by 23 plus ellipsis plus one divided by 30
is between one divided by three and one divided by two.

The correct answer is A.

163. For every even positive integer m comma f of m represents the product of all even integers from 2 to m, inclusive. For example, f of 12 equals two multiplication four multiplication six multiplication eight multiplication 10 multiplication 12
. What is the greatest prime factor of f of 24?

A. 23

B. 19

C. 17

D. 13

E. 11

Arithmetic Properties of Integers

Rewriting
f of 24 equals two multiplication four multiplication six multiplication eight multiplication 10 multiplication 12 multiplication 14 multiplication ellipsis multiplication 20 multiplication 22 multiplication 24
as 2 × 4 × 2(3) × 8 × 2(5) × 12 × 2(7) × … × 20 × 2(11) × 24 shows that all of the prime numbers from 2 through 11 are factors of f of 24
. The next prime number is 13, but 13 is not a factor of f of 24
because none of the even integers from 2 through 24 has 13 as a factor. Therefore, the largest prime factor of f of 24 is 11.

The correct answer is E.

three comma k comma two comma eight comma m comma three

164. The arithmetic mean of the list of numbers above is 4. If k and m are integers and k not equals m what is the median of the list?

A. 2

B. 2.5

C. 3

D. 3.5

E. 4

Arithmetic **Statistics**

Since the
arithmetic mean equals sum of values divided by number of values, then
sum with 6 summands three plus k plus two plus eight plus m plus three divided by six equals four
, and so
sum with 3 summands 16 plus k plus m divided by six equals four comma sum with 3 summands 16 plus k plus m equals 24 comma k plus m equals eight
. Since k not equals m, then either k less than four and m greater than four or k greater than four and m less than four. Because k and m are integers, either k less than or equals three and m greater than or equals five or k greater than or equals five and m less than or equals three.

Case (i): If k less than or equals two, then m greater than or equals six and the six integers in ascending order are k comma two comma three comma three, m, 8 or k, 2, 3, 3, 8, m. The two middle integers are both 3 so the median is three plus three divided by two equals three.

Case (ii): If k equals three, then m equals five and the six integers in ascending order are two comma k comma three comma three, m, 8. The two middle integers are both 3 so the median is three plus three divided by two equals three.

Case (iii): If k equals five, then m equals three and the six integers in ascending order are 2, m, 3, 3, k, 8. The two middle integers are both 3 so the median is three plus three divided by two equals three.

Case (iv): If k greater than or equals six, then m less than or equals two and the six integers in ascending order are m, 2, 3, 3, k, 8 or m, 2, 3, 3, 8, k. The two middle integers are both 3 so the median is three plus three divided by two equals three.

The correct answer is C.

image

165. If the variables cap x, cap y, and cap z take on only the values 10, 20, 30, 40, 50, 60, or 70 with frequencies indicated by the shaded regions above, for which of the frequency distributions is the mean equal to the median?

A. cap x only

B. cap y only

C. cap z only

D. cap x and cap y

E. cap x and cap z

Arithmetic **Statistics**

The frequency distributions for both cap x and cap z are symmetric about 40, and thus both cap x and cap z have mean = median = 40. Therefore, any answer choice that does not include both cap x and cap z can be eliminated. This leaves only answer choice E.

The correct answer is E.

multiline equation line 1 two times x plus y equals 12 line 2 absolute value of y less than or equals 12

166. For how many ordered pairs left parenthesis x comma y right parenthesis that are solutions of the system above are x and y both integers?

A. 7

B. 10

C. 12

D. 13

E. 14

Algebra **Absolute Value**

From absolute value of y less than or equals 12, if y must be an integer, then y must be in the set cap s equals left curly bracket prefix plus minus of 12 comma prefix plus minus of 11 comma prefix plus minus of 10 comma ellipsis comma prefix plus minus of three comma prefix plus minus of two comma prefix plus minus of one comma zero right curly bracket .

Since two times x plus y equals 12, then x equals 12 minus y divided by two. If x must be an integer, then

$12 - y$ must be divisible by 2; that is, $12 - y$ must be even. Since 12 is even, $12 - y$ is even if and only if y is even. This eliminates all odd integers from S, leaving only the even integers $\pm 12, \pm 10, \pm 8, \pm 6, \pm 4, \pm 2$, and 0. Thus, there are 13 possible integer y-values, each with a corresponding integer x-value and, therefore, there are 13 ordered pairs (x, y), where x and y are both integers, that solve the system.

The correct answer is D.

167. The United States Mint produces coins in 1-cent, 5-cent, 10-cent, 25-cent, and 50-cent denominations. If a jar contains exactly 100 cents worth of these coins, which of the following could be the total number of coins in the jar?

I. 91

II. 81

III. 76

A. I only

B. II only

C. III only

D. I and III only

E. I, II, and III

Arithmetic **Operations with Integers**

Letting p, n, d, q, and h, respectively, represent the numbers of pennies (1-cent coins), nickels (5-cent coins), dimes (10-cent coins), quarters (25-cent coins), and half-dollars (50-cent coins) with a total worth of 100 cents, it follows that $p + 5n + 10d + 25q + 50h = 100$

. Then
equation sequence part 1 p equals part 2 100 minus left parenthesis sum with 4 summands five times n plus 10 times d plus 25 times q plus 50 times h right parenthesis equals part 3 five times left parenthesis 20 minus n minus two times d minus five times q minus 10 times h right parenthesis
, so p must be a multiple of 5.

I. If the jar contained 90 pennies and 1 dime, the total number of coins would be 90 + 1 = 91 and the coins would be worth 90 + 10 = 100 cents.

II. For the jar to contain 81 coins with a total worth of 100 cents, there could be 80 pennies, but then the one coin remaining would have to amount to 20 cents to make the coins' total worth 100 cents. This is not possible since none of the coins is a 20-cent coin. If there were 75 pennies, then the remaining 6 coins would have to amount to 25 cents. This is not possible because 6 coins of the next smallest denomination would be worth 30 cents. If there were 70 pennies, then the remaining 11 coins would have to amount to 30 cents. This is not possible because 11 coins of the next smallest denomination would be worth 55 cents. Continuing in this manner shows that it is not possible for the jar to contain 81 coins with a total worth of 100 cents.

III. If the jar contained 70 pennies and 6 nickels, the total number of coins would be 70 + 6 = 76 and the coins would be worth 70 + 30 = 100 cents.

The correct answer is D.

168. A certain university will select 1 of 7 candidates eligible to fill a position in the mathematics department and 2 of 10 candidates eligible to fill 2 identical positions in the computer science department. If none of the candidates is eligible for a position in both departments, how many different sets of 3 candidates are there to fill the 3 positions?

A. 42

B. 70

C. 140

D. 165

E. 315

Arithmetic **Elementary Combinatorics**

To fill the position in the math department, 1 candidate will be selected from a group of 7 eligible candidates, and so there are 7 sets of 1 candidate each to fill the position in the math department. To fill the positions in the computer science department, any one of the 10 eligible candidates can be chosen for the first position and any of the remaining 9 eligible candidates can be chosen for the second position, making a total of 10 × 9 = 90 sets of 2 candidates to fill the computer science positions. But this number includes the set in which Candidate A was chosen to fill the first position and Candidate B was chosen to fill the second position as well as the set in which Candidate B was chosen for the first position and Candidate A was chosen for the second position. These sets are not different essentially since the positions are identical and in both sets Candidates A and B are chosen to fill the 2 positions. Therefore, there are 90 divided by two equals 45 sets of 2 candidates to fill the computer science positions. Then, using the multiplication principle, there are 7 × 45 = 315 different sets of 3 candidates to fill the 3 positions.

The correct answer is E.

169. A survey of employers found that during 1993 employment costs rose 3.5 percent, where employment costs consist of salary costs and fringe-benefit costs. If salary costs rose 3 percent and fringe-benefit costs rose 5.5 percent during 1993, then fringe-benefit costs represented what percent of employment costs at the beginning of 1993?

A. 16.5%

B. 20%

C. 35%

D. 55%

E. 65%

Algebra; Arithmetic First-Degree Equations; Percents

Let cap e represent employment costs, cap s represent salary costs, and cap f represent fringe-benefit costs. Then cap e equals cap s plus cap f. An increase of 3 percent in salary costs and a 5.5 percent increase in fringe-benefit costs resulted in a 3.5 percent increase in employment costs. Therefore 1.03 times cap s plus 1.055 times cap f equals 1.035 times cap e. But, cap e equals cap s plus cap f, so equation sequence part 1 1.03 times cap s plus 1.055 times cap f equals part 2 1.035 times left parenthesis cap s plus cap f right parenthesis equals part 3 1.035 times cap s plus 1.035 times cap f
.

Combining like terms gives left parenthesis 1.055 minus 1.035 right parenthesis times cap f equals left parenthesis 1.035 minus 1.03 right parenthesis times cap s or 0.02 times cap f equals 0.005 times cap s. Then, equation sequence part 1 cap s equals part 2 0.02 divided by 0.005 times cap f equals part 3 four times cap f
. Thus, since cap e equals cap s plus cap f, it follows that equation sequence part 1 cap e equals part 2 four times cap f plus cap f equals part 3 five times cap f
. Then, cap f as a percent of cap e is equation sequence part 1 cap f divided by cap e equals part 2 cap f divided by five times cap f equals part 3 one divided by five equals part 4 20 percent
.

The correct answer is B.

170. The subsets of the set left curly bracket w comma x comma y right curly bracket are left curly bracket w right curly bracket comma left curly bracket x right curly bracket comma left curly bracket y right curly bracket comma left curly bracket w comma x right curly bracket comma left curly bracket w comma y right curly bracket comma left curly bracket x comma y right curly bracket comma left curly bracket w comma x comma y right curly bracket

, and left curly bracket right curly bracket (the empty subset). How many subsets of the set left curly bracket w comma x comma y comma z right curly bracket contain w?

A. Four

B. Five

C. Seven

D. Eight

E. Sixteen

Arithmetic Sets

As shown in the table, the subsets of left curly bracket w comma x comma y comma z right curly bracket can be organized into two columns, those subsets of left curly bracket w comma x comma y comma z right curly bracket that do not contain w (left column) and the corresponding subsets of left curly bracket w comma x comma y comma z right curly bracket that contain w (right column), and each of these collections has the same number of sets. Therefore, there are 8 subsets of left curly bracket w comma x comma y comma z right curly bracket that contain w.

subsets not containing w	subsets containing w
left curly bracket right curly bracket	left curly bracket w right curly bracket
left curly bracket x right curly bracket	left curly bracket w comma x right curly bracket
left curly bracket y right curly bracket	left curly bracket w comma y right curly bracket
left curly bracket z right curly bracket	left curly bracket w comma z right curly bracket
left curly bracket x comma y right curly bracket	left curly bracket w comma x comma y right curly bracket
left curly bracket x comma z right curly bracket	left curly bracket w comma x comma z right curly bracket
left curly bracket y comma z right curly bracket	left curly bracket w comma y comma z right curly bracket
left curly bracket x comma y comma z right curly bracket	left curly bracket w comma x comma y comma z right curly bracket

The correct answer is D.

171. The number Square root of 63 minus 36 times Square root of three can be expressed as x plus y times Square root of three for some integers x and y. What is the value of x times y?

A. –18

B. –6

C. 6

D. 18

E. 27

Algebra **Operations on Radical Expressions**

Squaring both sides of
Square root of 63 minus 36 times Square root of three equals x plus y times Square root of three
gives
equation sequence part 1 63 minus 36 times Square root of three equals part 2 sum with 3 summands x squared plus two times x times y times Square root of three plus three times y squared equals part 3 left parenthesis x squared plus three times y squared right parenthesis plus left parenthesis two times x times y right parenthesis times Square root of three
, which implies that negative 36 equals two times x times y, or x times y equals negative 18. Indeed, if negative 36 not equals two times x times y, or equivalently, if 36 plus two times x times y not equals zero, then we could write Square root of three as a quotient of the two integers 63 minus x squared minus three times y squared and 36 plus two times x times y, which is not possible because Square root of three is an irrational number. To be more explicit,
63 minus 36 times Square root of three equals sum with 3 summands x squared plus two times x times y times Square root of three plus three times y squared
implies

63 minus x squared minus three times y squared equals left parenthesis 36 plus two times x times y right parenthesis times Square root of three
, and if 36 plus two times x times y not equals zero, then we could divide both sides of the equation
63 minus x squared minus three times y squared equals left parenthesis 36 plus two times x times y right parenthesis times Square root of three
by 36 plus two times x times y to get
63 minus x squared minus three times y squared divided by 36 plus two times x times y equals Square root of three
.

The correct answer is A.

172. There are 10 books on a shelf, of which 4 are paperbacks and 6 are hardbacks. How many possible selections of 5 books from the shelf contain at least one paperback and at least one hardback?

A. 75

B. 120

C. 210

D. 246

E. 252

Arithmetic Elementary Combinatorics

The number of selections of 5 books containing at least one paperback and at least one hardback is equal to cap t minus cap n, where cap t is the total number of selections of 5 books and cap n is the number of selections that do not contain both a paperback and a hardback. The value of T is

$$\begin{aligned} \binom{10}{5} &= \frac{10!}{5!(10-5)!} = \frac{(6)(7)(8)(9)(10)}{(1)(2)(3)(4)(5)} \\ &= (7)(2)(9)(2) = 252. \end{aligned}$$

To find the value of N, first note that no selection of 5 books can contain all paperbacks, since there are only 4 paperback books. Thus, the value of N is equal to the number of selections of 5 books that contain all hardbacks, which is equal to 6 since there are 6 ways that a single hardback can be left out when choosing the 5 hardback books. It follows that the number of selections of 5 books containing at least one paperback and at least one hardback is $T - N = 252 - 6 = 246$

.

The correct answer is D.

173. If x is to be chosen at random from the set {1, 2, 3, 4} and y is to be chosen at random from the set {5, 6, 7}, what is the probability that xy will be even?

   A. $\frac{1}{6}$

   B. $\frac{1}{3}$

   C. $\frac{1}{2}$

   D. $\frac{2}{3}$

   E. $\frac{5}{6}$

Arithmetic; Algebra Probability; Concepts of Sets

By the principle of multiplication, since there are 4 elements in the first set and 3 elements in the second set, there are (4)(3) = 12 possible products of x times y, where x is chosen from the first set and y is chosen from the second set. These products will be even EXCEPT when both x and y are odd. Since there are 2 odd numbers in the first set and 2 odd numbers in the second set, there are (2)(2) = 4 products of x and y that are odd. This means that the remaining 12 − 4 = 8 products are even. Thus, the probability that x times y is even is 18 divided by 12 equals two divided by three.

The correct answer is D.

174. The function f is defined for each positive three-digit integer n by f of n equals two super x three super y five super z, where x comma y, and z are the hundreds, tens, and units digits of n, respectively. If m and v are three-digit positive integers such that f of m equals nine times f of v, then m minus v equals

A. 8

B. 9

C. 18

D. 20

E. 80

Algebra Place Value

Let the hundreds, tens, and units digits of m be cap a comma cap b, and cap c, respectively; and let the hundreds, tens, and units digits of v be a comma b, and c, respectively. From f of m equals nine times f of v it follows that equation sequence part 1 two super cap a times three super cap b times five super cap c equals part 2 nine times left parenthesis two super a times three super b times five super c right parenthesis equals part 3 three squared times left parenthesis two super a times three super b times five super c right parenthesis equals part 4 two super a times three super b plus two times five super c

. Therefore, cap a equals a, cap b equals b plus two, and cap c equals c. Now calculate m minus v.

m minus v	= left parenthesis sum with 3 summands 100 times cap a plus 10 times cap b plus cap c right parenthesis minus left parenthesis sum with 3 summands 100 times a plus 10 times b plus c right parenthesis	place value property
	= left parenthesis sum with 3 summands 100 times a plus 10 times left parenthesis b plus two right parenthesis plus c right parenthesis minus left parenthesis sum with 3 summands 100 times a plus 10 times b plus c right parenthesis	obtained above
	= 10 times left parenthesis b plus two right parenthesis minus 10 times b	combine like terms
	= 10 times b plus 20 minus 10 times b	distributive property
	= 20	combine like terms

The correct answer is D.

175. If $10^{50} – 74$ is written as an integer in base 10 notation, what is the sum of the digits in that integer?

A. 424

B. 433

C. 440

D. 449

E. 467

Arithmetic **Properties of Numbers**

$10^2 - 74$	=	100 – 74	=	26
$10^3 - 74$	=	1,000 – 74	=	926
$10^4 - 74$	=	10,000 – 74	=	9,926
$10^5 - 74$	=	100,000 – 74	=	99,926
$10^6 - 74$	=	1,000,000 – 74	=	999,926

From the table above it is clear that $10^{50} - 74$ in base 10 notation will be 48 digits of 9 followed by the digits 2 and 6. Therefore, the sum of the digits of $10^{50} - 74$ is equal to 48(9) + 2 + 6 = 440.

The correct answer is C.

176. A certain company that sells only cars and trucks reported that revenues from car sales in 1997 were down 11 percent from 1996 and revenues from truck sales in 1997 were up 7 percent from 1996. If total revenues from car sales and truck sales in 1997 were up 1 percent from 1996, what is the ratio of revenue from car sales in 1996 to revenue from truck sales in 1996?

A. 1:2

B. 4:5

C. 1:1

D. 3:2

E. 5:3

Algebra; **Arithmetic First-Degree Equations; Percents**

Let cap c sub 96 and cap c sub 97 represent revenues from car sales in 1996 and 1997, respectively, and let cap t sub 96 and cap t sub 97 represent revenues from truck sales in 1996 and 1997,

respectively. A decrease of 11 percent in revenue from car sales from 1996 to 1997 can be represented as
left parenthesis one minus 0.11 right parenthesis times cap c sub 96 equals cap c sub 97
, and a 7 percent increase in revenue from truck sales from 1996 to 1997 can be represented as
left parenthesis one plus 0.07 right parenthesis times cap t sub 96 equals cap t sub 97
.

An overall increase of 1 percent in revenue from car and truck sales from 1996 to 1997 can be represented as
cap c sub 97 plus cap t sub 97 equals left parenthesis one plus 0.01 right parenthesis times left parenthesis cap c sub 96 plus cap t sub 96 right parenthesis
.

Then, by substitution of expressions for cap c sub 97 and cap t sub 97 that were derived above,
left parenthesis one minus 0.11 right parenthesis times cap c sub 96 plus left parenthesis one plus 0.07 right parenthesis times cap t sub 96 equals left parenthesis one plus 0.01 right parenthesis times left parenthesis cap c sub 96 plus cap t sub 96 right parenthesis
and so
0.89 times cap c sub 96 plus 1.07 times cap t sub 96 equals 1.01 times left parenthesis cap c sub 96 plus cap t sub 96 right parenthesis
or
0.89 times cap c sub 96 plus 1.07 times cap t sub 96 equals 1.01 times cap c sub 96 plus 1.01 times cap t sub 96
.

Then, combining like terms gives
left parenthesis 1.07 minus 1.01 right parenthesis times cap t sub 96 equals left parenthesis 1.01 minus 0.89 right parenthesis times cap c sub 96
or 0.06 times cap t sub 96 equals 0.12 times cap c sub 96. Thus
equation sequence part 1 cap c sub 96 divided by cap t sub 96 equals part 2 0.06 divided by 0.12 equals part 3 one divided by two

. The ratio of revenue from car sales in 1996 to revenue from truck sales in 1996 is 1:2.

The correct answer is A.

177. Becky rented a power tool from a rental shop. The rent for the tool was $12 for the first hour and $3 for each additional hour. If Becky paid a total of $27, excluding sales tax, to rent the tool, for how many hours did she rent it?

A. 5

B. 6

C. 9

D. 10

E. 12

Arithmetic Applied Problems

Becky paid a total of $27 to rent the power tool. She paid $12 to rent the tool for the first hour and $27 – $12 = $15 to rent the tool for the additional hours at the rate of $3 per additional hour. It follows that she rented the tool for 15 divided by three equals five additional hours and a total of 1 + 5 = 6 hours.

The correct answer is B.

178. If four less than seven minus x divided by three which of the following must be true?

I. five less than x

II. absolute value of x plus three greater than two

III. negative left parenthesis x plus five right parenthesis is positive.

A. II only

B. III only

C. I and II only

D. II and III only

E. I, II, and III

Algebra **Inequalities**

Given that four less than seven minus x divided by three, it follows that 12 less than seven minus x. Then five less than negative x or, equivalently, x less than negative five.

I. If four less than seven minus x divided by three, then x less than negative five. If five less than x were true, then, by combining five less than x and x less than negative five, it would follow that five less than negative five, which cannot be true. Therefore, it is not the case that, if four less than seven minus x divided by three, then Statement I must be true. In fact, Statement I is never true.

II. If four less than seven minus x divided by three, then x less than negative five, and it follows that x plus three less than negative two. Since negative two less than zero, then x plus three less than zero and
absolute value of x plus three equals negative left parenthesis x plus three right parenthesis
. If x plus three less than negative two, then negative left parenthesis x plus three right parenthesis greater than two
and by substitution,
absolute value of x plus three greater than two. Therefore, Statement II must be true for every value of x such that x less than negative five. Therefore, Statement II must be true if four less than seven minus x divided by three.

III. If four less than seven minus x divided by three, then x less than negative five and x plus five less than zero. But if x plus five less than zero, then it follows that negative left parenthesis x plus five right parenthesis greater than zero
and so negative left parenthesis x plus five right parenthesis is

positive. Therefore Statement III must be true if four less than seven minus x divided by three.

The correct answer is D.

179. On a certain day, a bakery produced a batch of rolls at a total production cost of $300. On that day, four divided by five of the rolls in the batch were sold, each at a price that was 50 percent greater than the average (arithmetic mean) production cost per roll. The remaining rolls in the batch were sold the next day, each at a price that was 20 percent less than the price of the day before. What was the bakery's profit on this batch of rolls?

A. $150

B. $144

C. $132

D. $108

E. $90

Arithmetic Applied Problems

Let n be the number of rolls in the batch and p be the average production price, in dollars, per roll. Then the total cost of the batch is n times p equals 300 dollars, and the total revenue from selling the rolls in the batch is equation sequence part 1 left parenthesis four divided by five times n right parenthesis times left parenthesis 1.5 times p right parenthesis plus left parenthesis one divided by five times n right parenthesis times left parenthesis 0.8 right parenthesis times left parenthesis 1.5 times p right parenthesis equals part 2 left parenthesis four divided by five times n right parenthesis times left parenthesis three divided by two times p right parenthesis plus left parenthesis one divided by five times n right parenthesis times left parenthesis four divided by five right parenthesis times left parenthesis three divided by two times p right parenthesis equals part 3 left parenthesis six divided by five plus six divided by 25 right parenthesis times n times p equals part 4 left parenthesis 36 divided by 25 right parenthesis times n times p
. Therefore, the profit from selling the rolls in the batch is

equation sequence part 1 left parenthesis 36 divided by 25 right parenthesis times n times p minus n times p equals part 2 left parenthesis 11 divided by 25 right parenthesis times n times p equals part 3 left parenthesis 11 divided by 25 right parenthesis times left parenthesis 300 right parenthesis
dollars = 132 dollars.

The correct answer is C.

180. A set of numbers has the property that for any number t in the set, t plus two is in the set. If –1 is in the set, which of the following must also be in the set?

I. –3

II. 1

III. 5

A. I only

B. II only

C. I and II only

D. II and III only

E. I, II, and III

Arithmetic **Properties of Numbers**

It is given that –1 is in the set and, if t is in the set, then t plus two is in the set.

I. Since {–1, 1, 3, 5, 7, 9, 11, …} contains –1 and satisfies the property that if t is in the set, then t plus two is in the set, it is not true that –3 must be in the set.

II. Since –1 is in the set, $-1 + 2 = 1$ is in the set. Therefore, it must be true that 1 is in the set.

III. Since –1 is in the set, $-1 + 2 = 1$ is in the set. Since 1 is in the set, $1 + 2 = 3$ is in the set. Since 3 is in the set, $3 + 2 = 5$ is in the set. Therefore, it must be true that 5 is in the set.

The correct answer is D.

181. A couple decides to have 4 children. If they succeed in having 4 children and each child is equally likely to be a boy or a girl, what is the probability that they will have exactly 2 girls and 2 boys?

A. three divided by eight

B. one divided by four

C. three divided by 16

D. one divided by 18

E. one divided by 16

Arithmetic Probability

Representing the birth order of the 4 children as a sequence of 4 letters, each of which is B for boy and G for girl, there are 2 possibilities (B or G) for the first letter, 2 for the second letter, 2 for the third letter, and 2 for the fourth letter, making a total of two super four equals 16 sequences. The table below categorizes some of these 16 sequences.

# of boys	# of girls	Sequences	# of sequences
0	4	GGGG	1
1	3	BGGG, GBGG, GGBG, GGGB	4
3	1	GBBB, BGBB, BBGB, BBBG	4
4	0	BBBB	1

The table accounts for 1 + 4 + 4 + 1 = 10 sequences. The other 6 sequences will have 2Bs and 2Gs. Therefore the probability that the couple will have exactly 2 boys and 2 girls is six divided by 16 equals three divided by eight. For the mathematically inclined, if it is assumed that a couple has a fixed

number of children, that the probability of having a girl each time is p, and that the sex of each child is independent of the sex of the other children, then the number of girls, x, born to a couple with n children is a random variable having the binomial probability distribution. The probability of having exactly x girls born to a couple with n children is given by the formula vector element 1 n element 2 x times p super x times left parenthesis one minus p right parenthesis super n minus x
. For the problem at hand, it is given that each child is equally likely to be a boy or a girl, and so p equals one divided by two. Thus, the probability of having exactly 2 girls born to a couple with 4 children is equation sequence part 1 vector element 1 four element 2 two times left parenthesis one divided by two right parenthesis squared times left parenthesis one divided by two right parenthesis squared equals part 2 four factorial divided by two factorial two factorial times left parenthesis one divided by two right parenthesis squared times left parenthesis one divided by two right parenthesis squared equals part 3 left parenthesis six right parenthesis times left parenthesis one divided by four right parenthesis times left parenthesis one divided by four right parenthesis equals part 4 six divided by 16 equals part 5 three divided by eight full stop

The correct answer is A.

182. The closing price of Stock X changed on each trading day last month. The percent change in the closing price of Stock X from the first trading day last month to each of the other trading days last month was less than 50 percent. If the closing price on the second trading day last month was $10.00, which of the following CANNOT be the closing price on the last trading day last month?

A. $3.00

B. $9.00

C. $19.00

D. $24.00

E. $29.00

Arithmetic **Applied Problems; Percents**

Let cap p be the first-day closing price, in dollars, of the stock. It is given that the second-day closing price was left parenthesis one plus n percent right parenthesis times cap p equals 10, so cap p equals 10 divided by one plus n percent, for some value of n such that negative 50 less than n less than 50. Therefore, cap p is between 10 divided by one plus 0.50 almost equals 6.67 and 10 divided by one minus 0.50 equals 20. Hence, if cap q is the closing price, in dollars, of the stock on the last day, then cap q is between (0.50)(6.67) ≈ 3.34 (50% decrease from the lowest possible first-day closing price) and (1.50)(20) = 30 (50% increase from the greatest possible first-day closing price). The only answer choice that gives a number of dollars not between 3.34 and 30 is the first answer choice.

The correct answer is A.

183. How many different positive integers are factors of 441?

A. 4

B. 6

C. 7

D. 9

E. 11

Arithmetic **Properties of Numbers**

Like any integer, 441 has itself and 1 as factors. To find all its other positive factors, we first use prime factorization to write 441 as a product of its prime factors: 441 = 3 × 3 × 7 × 7. Then we can group those two 3s and two 7s in various ways to find all their products, which will be the other positive factors of 441. That is, 441 = 3 × (3 × 7 × 7) = (3 × 3) × (7 × 7) = (3 × 7) × (3 × 7) = (3 × 3 × 7) × 7. This shows that 441 has 7 different positive integer factors other than itself and 1. These are 3, 7, 3 × 3 = 9, 3 × 7 = 21, 7 × 7 = 49, 3 × 3 × 7 = 63, and 3 × 7 × 7 = 147. So altogether, 441 has 9 different positive integers as factors: 1, 3, 7, 9, 21, 49, 63, 147, and 441.

The correct answer is D.

184. If n is a positive integer and n squared is divisible by 72, then the largest positive integer that must divide n is

A. 6

B. 12

C. 24

D. 36

E. 48

Arithmetic **Properties of Numbers**

Since n squared is divisible by 72, n squared equals 72 times k for some positive integer k. Since n squared equals 72 times k, then 72 times k must be a perfect square. Since 72 times k equals left parenthesis two cubed right parenthesis times left parenthesis three squared right parenthesis times k
, then k equals two times m squared for some positive integer m in order for 72 times k to be a perfect square. Then,
equation sequence part 1 n squared equals part 2 72 times k equals part 3 left parenthesis two cubed right parenthesis times left parenthesis three squared right parenthesis times left parenthesis two times m squared right parenthesis equals part 4 left parenthesis two super four right parenthesis times left parenthesis three squared right parenthesis times m squared equals part 5 left square bracket left parenthesis two squared right parenthesis times left parenthesis three right parenthesis times left parenthesis m right parenthesis right square bracket squared
, and
n equals left parenthesis two squared right parenthesis times left parenthesis three right parenthesis times left parenthesis m right parenthesis

. The positive integers that MUST divide n are 1, 2, 3, 4, 6, and 12. Therefore, the largest positive integer that must divide n is 12.

The correct answer is B.

185. A certain grocery purchased x pounds of produce for p dollars per pound. If y pounds of the produce had to be discarded due to spoilage and the grocery sold the rest for s dollars per pound, which of the following represents the gross profit on the sale of the produce?

   A. left parenthesis x minus y right parenthesis times s minus x times p

   B. left parenthesis x minus y right parenthesis times p minus y times s

   C. left parenthesis s minus p right parenthesis times y minus x times p

   D. x times p minus y times s

   E. left parenthesis x minus y right parenthesis times left parenthesis s minus p right parenthesis

Algebra **Simplifying Algebraic Expressions; Applied Problems**

Since the grocery bought x pounds of produce for p dollars per pound, the total cost of the produce was x times p dollars. Since y pounds of the produce was discarded, the grocery sold x minus y pounds of produce at the price of s dollars per pound, yielding a total revenue of left parenthesis x minus y right parenthesis times s dollars. Then, the grocery's gross profit on the sale of the produce is its total revenue minus its total cost or
left parenthesis x minus y right parenthesis times s minus x times p dollars.

The correct answer is A.

186. If x, y, and z are positive integers such that x is a factor of y, and x is a multiple of z, which of the following is NOT necessarily an integer?

A. $\dfrac{x+z}{z}$

B. $\dfrac{y+z}{x}$

C. $\dfrac{x+y}{z}$

D. x times y divided by z

E. y times z divided by x

Arithmetic Properties of Numbers

Since the positive integer x is a factor of y, then y equals k times x for some positive integer k. Since x is a multiple of the positive integer z, then $x = mz$ for some positive integer m.

Substitute these expressions for x and/or y into each answer choice to find the one expression that is NOT necessarily an integer.

A. $\dfrac{x+z}{z} = \dfrac{mz+z}{z} = \dfrac{(m+1)\,z}{z} = m+1$, which MUST be an integer

B. equation sequence part 1 y plus z divided by x equals part 2 y divided by x plus z divided by x equals part 3 k times x divided by x plus z divided by m times z equals part 4 k plus one divided by m
, which NEED NOT be an integer

Because only one of the five expressions need not be an integer, the expressions given in C, D, and E need not be tested. However, for completeness,

C. equation sequence part 1 x plus y divided by z equals part 2 m times z plus k times x divided by z equals part 3 m times z plus k of m times z divided by z equals part 4 m times z times left parenthesis one plus k right parenthesis divided by z equals part 5 m times left parenthesis one plus k right parenthesis
, which MUST be an integer

D. $\frac{xy}{z} = \frac{(mz)\,y}{z} = my$, which MUST be an integer

E. equation sequence part 1 y times z divided by x equals part 2 left parenthesis k times x right parenthesis times left parenthesis z right parenthesis divided by x equals part 3 k times z
, which MUST be an integer

The correct answer is B.

187. Running at their respective constant rates, Machine X takes 2 days longer to produce w widgets than Machine Y. At these rates, if the two machines together produce five divided by four times w widgets in 3 days, how many days would it take Machine X alone to produce $2w$ widgets?

A. 4

B. 6

C. 8

D. 10

E. 12

Algebra **Applied Problems**

If x, where $x > 2$, represents the number of days Machine X takes to produce w widgets, then Machine Y takes $x - 2$ days to produce w widgets. It follows that Machines X and Y can produce
w divided by x and $\frac{w}{x-2}$ widgets, respectively, in 1 day and together they can produce
w divided by x plus w divided by x minus two widgets in 1 day. Since it is given that, together, they can produce
five divided by four times w widgets in 3 days, it follows that,

together, they can produce $\frac{1}{3}\left(\frac{5}{4}w\right) = \frac{5}{12}w$ widgets in 1 day. Thus, multiline equation row 1 w divided by x plus w divided by x minus two equals five divided by 12 times w row 2 left parenthesis one divided by x plus one divided by x minus two right parenthesis times w equals five divided by 12 times w row 3 left parenthesis one divided by x plus one divided by x minus two right parenthesis equals five divided by 12 row 4 12 times x times left parenthesis x minus two right parenthesis times left parenthesis one divided by x plus one divided by x minus two right parenthesis equals 12 times x times left parenthesis x minus two right parenthesis times left parenthesis five divided by 12 right parenthesis row 5 12 times left square bracket left parenthesis x minus two right parenthesis plus x right square bracket equals five times x times left parenthesis x minus two right parenthesis row 6 12 times left parenthesis two times x minus two right parenthesis equals five times x times left parenthesis x minus two right parenthesis row 7 24 times x minus 24 equals five times x squared minus 10 times x row 8 zero equals five times x squared minus 34 times x plus 24 row 9 zero equals left parenthesis five times x minus four right parenthesis times left parenthesis x minus six right parenthesis row 10 x equals four divided by five times or times six

Therefore, since x greater than two, it follows that $x = 6$. Machine X takes 6 days to produce w widgets and 2(6) = 12 days to produce two times w widgets.

The correct answer is E.

188. What is the greatest positive integer n such that five super n divides $10! - (2)(5!)^2$?

A. 2

B. 3

C. 4

D. 5

E. 6

Arithmetic **Properties of Numbers; Exponents**

The greatest positive integer n such that 5^n divides a given integer is the number of factors of 5 in the prime factorization of the given integer. By repeated identification of common factors, the indicated difference can be factored sufficiently to determine the number of factors of 5 in its prime factorization. In the computations that follow, we have used the equalities 10! = (5!)(6)(7)(8)(9)(10) and 5! = (2)(3)(4)(5).

$10! - (2)(5!)^2$

$= (5! \cdot 6 \cdot 7 \cdot 8 \cdot 9 \cdot 10) - (2 \cdot 5! \cdot 5!)$

$= (5!)(6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 - 2 \cdot 5!)$

$= (5!)(6 \cdot 7 \cdot 8 \cdot 9 \cdot 10 - 2 \cdot 2 \cdot 3 \cdot 4 \cdot 5)$

$= (5!)(2^4)(3 \cdot 7 \cdot 9 \cdot 10 - 3 \cdot 5)$

$= (5!)(2^4)(3)(7 \cdot 9 \cdot 10 - 5)$

$= (5!)(2^4)(3)(5)(7 \cdot 9 \cdot 2 - 1)$

$= (5!)(2^4)(3)(5)(63 \cdot 2 - 1)$

$= (5!)(2^4)(3)(5)(126 - 1)$

$= (5!)(2^4)(3)(5)(125)$

Since there is exactly 1 factor of 5 in 5! = (2)(3)(4)(5), no factors of 5 in either 2^4 or 3, exactly 1 factor of 5 in 5, and exactly 3 factors of 5 in $125 = 5^3$, it follows that there are 1 + 1 + 3 = 5 factors of 5 in $10! - (2)(5!)^2 = (5!)(2^4)(3)(5)(125)$.

The correct answer is D.

189. Yesterday, Candice and Sabrina trained for a bicycle race by riding around an oval track. They both began riding at the same time from the track's starting point. However, Candice rode at a faster pace than Sabrina, completing each lap around the track in 42 seconds, while Sabrina completed each lap around the track in 46 seconds. How many laps around the track had Candice completed the next time that Candice and Sabrina were together at the starting point?

A. 21

B. 23

C. 42

D. 46

E. 483

Arithmetic **Applied Problems; Properties of Integers**

Let C and S be the number of laps around the track, respectively, whenever Candice and Sabrina were together again at the starting point. Since Candice completes each lap in 42 seconds, Candice had been riding for a total of $42C$ seconds, and since Sabrina completes each lap in 46 seconds, Sabrina had been riding for a total of $46S$ seconds. Because they had been riding for the same total amount of time, we have $42C = 46S$, or $21C = 23S$, where C and S are positive integers.

Because 23 is a prime number that divides the product of 21 and C (note that 23 divides $23S$ and $23S = 21C$), it follows that 23 divides 21 (not true) or 23 divides C, and hence 23 divides C. Also, because 3 is a prime number that divides the product of 23 and S (note that 3 divides $21C$ and $21C = 23S$), it follows that 3 divides 23 (not true) or 3 divides S, and hence 3 divides S. Finally, because 7 is a prime number that divides the product of 23 and S (note that 7 divides $21C$ and $21C = 23S$), it follows that 7 divides 23 (not true) or 7 divides S, and hence 7 divides S

It follows that C is a multiple of 23 and S is a multiple of both 3 and 7. The least positive integer values for C and S with this property are $C = 23$ and $S = 3 \times 7 = 21$. Therefore, the next time after beginning that Candice and Sabrina were together at the starting point, Candice had completed 23 laps and Sabrina had completed 21 laps.

The correct answer is B.

190. If $n = 9! - 6^4$, which of the following is the greatest integer k such that 3^k is a factor of n?

A. 1

B. 3

C. 4

D. 6

E. 8

Arithmetic **Properties of Integers**

The following charts isolate and count the occurrences of 2 and of 3 in the factorizations of 9! = (9)(8)(7)(6)(5)(4)(3)(2) and 6^4 = (6)(6)(6)(6).

9!	=	(9)	(8)	(7)	(6)	(5)	(4)	(3)	(2)
Occurrences of 2		0	3	0	1	0	2	0	1
Occurrences of 3		2	0	0	1	0	0	1	0

6^4	=	(6)	(6)	(6)	(6)
Occurrences of 2		1	1	1	1
Occurrences of 3		1	1	1	1

So, $9! - 6^4 = (2^7 \cdot 3^4 \cdot 5 \cdot 7) - (2^4 \cdot 3^4) = (2^4 \cdot 3^4)(2^3 \cdot 5 \cdot 7 - 1) = (2^4 \cdot 3^4)(279) = (2^4 \cdot 3^4)(3^2 \cdot 31) = 2^4 \cdot 3^6 \cdot 31$, where 31 is prime. Therefore $k = 6$.

Alternatively, express $n = 9! - 6^4$ as $(9)(8)(7)(6)(5)(4)(3)(2)(1) - 6^4 = [(9)(8)](7)(6)(5)(4)[(3)(2)](1) - 6^4$. Then factor (9)(8) as (36)(2) = $(6^2)(2)$ and multiply (3)(2) to get 6. Factoring 6^4 from $n = (6^2)(2)(7)(6)(5)(4)(6)(1) - 6^4$ gives $n = 6^4\left[(2)(7)(5)(4)(1) - 1\right] = 6^4(279)$. It follows that 6^4 has 4 factors of 3 and 279 has 2 additional factors of 3 since 279 = $(3^2)(31)$, so the greatest integer k such that 3^k is a factor of n is 4 + 2 = 6.

The correct answer is D.

191. The integer 120 has many factorizations. For example, 120 = (2)(60), 120 = (3)(4)(10), and 120 = (–1)(–3)(4)(10). In how many of the factorizations of 120 are the factors consecutive integers in ascending order?

A. 2

B. 3

C. 4

D. 5

E. 6

Arithmetic Properties of Integers

All of the positive factors of 120 listed in ascending order are 1, 2, 3, 4, 5, 6, 8, 10, 12, 15, 20, 24, 30, 40, 60, and 120. The negative factors of 120 listed in ascending order are –120, –60, –40, –30, –24, –20, –15, –12, –10, –8, –6, –5, –4, –3, –2, and –1. Examining these lists for groups of consecutive factors whose product is 120 gives (1)(2)(3)(4)(5), (2)(3)(4)(5), (4)(5)(6), and (–5)(–4)(–3)(–2).

The correct answer is C.

192. Jorge's bank statement showed a balance that was $0.54 greater than what his records showed. He discovered that he had written a check for $\$x.yz$ and had recorded it as $\$x.zy$, where each of x, y, and z represents a digit from 0 though 9. Which of the following could be the value of z?

A. 2

B. 3

C. 4

D. 5

E. 6

Arithmetic Place Value

Since the amount Jorge recorded for the check $(\$x.zy)$ was \$0.54 more than the actual amount of the check ($\$x.yz$), it follows that $\$x.zy - \$x.yz = \$0.54$. This is equivalent to $x + \frac{z}{10} + \frac{y}{100} - \left(x + \frac{y}{10} + \frac{z}{100}\right) = \frac{54}{100}$. Then $\frac{10z + y}{100} - \frac{10y + z}{100} = \frac{54}{100}$ or, equivalently, $10z + y - (10y + z) = 54$. It follows that $9z - 9y = 54$ or $z - y = 6$. Since y and z are digits, the possible values of y and z, respectively, are 0 and 6, 1 and 7, 2 and 8, and 3 and 9. Of the possible values of z, only 6 is given as one of the answer choices.

The correct answer is E.

193. One side of a parking stall is defined by a straight stripe that consists of n painted sections of equal length with an unpainted section $\frac{1}{2}$ as long between each pair of consecutive painted sections. The total length of the stripe from the beginning of the first painted section to the end of the last painted section is 203 inches. If n is an integer and the length, in inches, of each unpainted section is an integer greater than 2, what is the value of n?

A. 5

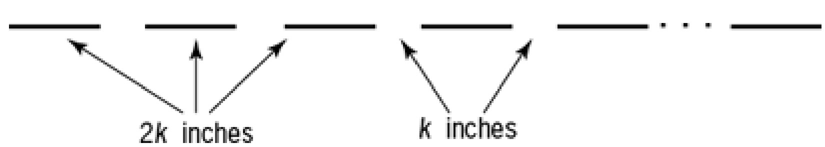
B. 9

C. 10

D. 14

E. 29

Algebra Applied Problems



The figure above is a schematic diagram of the parking stall's painted sections, where each painted section has length $2k$ inches and each unpainted section has length k inches. Since there is a total of n painted sections and a total of $(n-1)$ unpainted sections, it follows that $n(2k)+(n-1)k=203$, or $(3n-1)k=203$. Also, since n and k are positive integers with $n \geq 2$, and 203 = (7)(29) is the only factorization of 203 with integer factors greater than or equal to 2, we have two cases.

Case 1: $3n-1=7$ and $k=29$. In this case we have $n=\frac{8}{3}$.

Case 2: $3n-1=29$ and $k=7$. In this case we have $n=10$.

Because $\frac{8}{3}$ is not an integer, we discard Case 1, and hence $n=10$.

The correct answer is C.

194. This year, Henry will save a certain amount of his income, and he will spend the rest. Next year, Henry will have no income, but for each dollar that he saves this year, he will have $1+r$ dollars available to spend. In terms of r, what fraction of his income should Henry save this year so that next year the amount he has available to spend will be equal to half the amount that he spends this year?

   A. one divided by r plus two
   B. one divided by two times r plus two
   C. one divided by three times r plus two
   D. one divided by r plus three
   E. one divided by two times r plus three

Algebra Sets

Let x be the number of dollars Henry should save this year and y be his total income in dollars this year. Then for him to have an amount available to spend next year equal to half the amount he spends this year, x times left parenthesis one plus r right parenthesis must equal y minus x divided by two.

Thus, left parenthesis two plus two times r right parenthesis times x must equal y minus x, so y must equal left parenthesis three plus two times r right parenthesis times x. Therefore, the fraction of his income he must save this year is equation sequence part 1 x divided by y equals part 2 x divided by left parenthesis three plus two times r right parenthesis times x equals part 3 one divided by two times r plus three
.

The correct answer is E.

Machine	Consecutive Minutes Machine Is Off	Units of Power When On
A	17	15
B	14	18
C	11	12

195. At a certain factory, each of Machines A, B, and C is periodically on for exactly 1 minute and periodically off for a fixed number of consecutive minutes. The table above shows that Machine A is on and uses 15 units of power every 18th minute, Machine B is on and uses 18 units of power every 15th minute, and Machine C is on and uses 12 units of power every 12th minute. The factory has a backup generator that operates only when the total power usage of the 3 machines exceeds 30 units of power. What is the time interval, in minutes, between consecutive times the backup generator begins to operate?

   A. 36

   B. 63

   C. 90

D. 180

E. 270

Arithmetic Applied Problems

The given table shows that the backup generator will not operate when only one of the machines is operating, since none of the machines uses more than 30 units of power. The table below shows the power usage when more than one machine is operating at the same time.

Machines	Units of power when on	Backup generator
A & B	15 + 18 = 33	On
B & C	18 + 12 = 30	Off
C & A	12 + 15 = 27	Off
A & B & C	15 + 18 + 12 = 45	On

Thus, the backup generator will be on whenever Machines A and B are both on, this being true regardless of whether Machine C is on. We are given that Machine A is on for 1 minute every 18 minutes and Machine B is on for 1 minute every 15 minutes. Therefore, if Machines A and B are both on for a certain minute, then the following are the minutes when these machines are again on.

Minutes when Machine A is on: 1st, 19th, 37th, 55th, 73rd, **91st**, 109th, …

Minutes when Machine B is on: 1st, 16th, 31st, 46th, 61st, 76th, **91st**, 106th, …

Therefore, the next time Machines A and B are both on is the 91st minute, which is 90 minutes after the first minute.

Alternatively, Machine A is on every 18 = (2)(3^2) minutes and Machine B is on every 15 = (3)(5) minutes, so the machines are both on every $(2)(3^2)(5) = 90$ minutes (least common multiple of 18 and 15).

The correct answer is C.

196. In a certain region, the number of children who have been vaccinated against rubella is twice the number who have been vaccinated against mumps. The number who have been vaccinated against both is twice the number who have been vaccinated against mumps. If 5,000 have been vaccinated against both, how many have been vaccinated against rubella only?

   A. 2,500

   B. 7,500

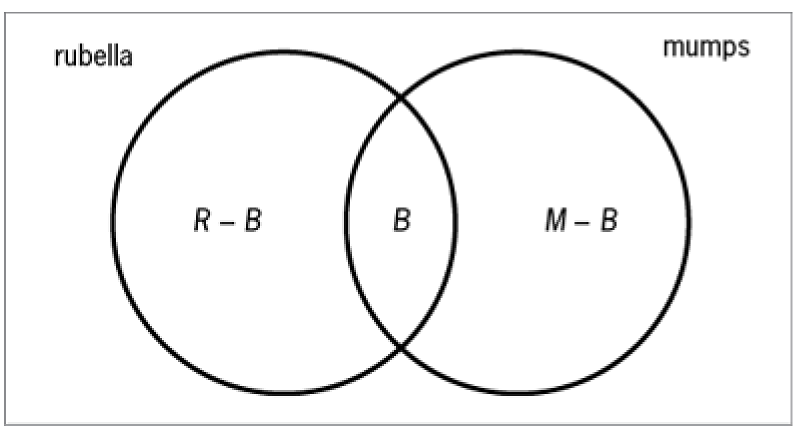
   C. 10,000

   D. 15,000

   E. 17,500

Algebra Simultaneous Equations; Sets (Venn Diagrams)

Let R, M, and B be the number, respectively, of children who have been vaccinated against rubella, against mumps, and against both. Then the number of children who have been vaccinated against rubella only is $R - B$ and the number of children who have been vaccinated against mumps only is $M - B$, as shown in the Venn diagram below.



From the given information, we have (i) $R = 2M$, (ii) $B = 2(M - B)$, and (iii) $B = 5{,}000$. Substituting the value of B

given in (iii) into (ii) gives $5{,}000 = 2(M - 5{,}000)$, or $M = 7{,}500$. Using this value of M in (i) gives $R = 2(7{,}500) = 15{,}000$. Therefore, the number vaccinated against rubella only is $R - B = 15{,}000 - 5{,}000 = 10{,}000$.

The correct answer is C.

197. Three boxes of supplies have an average (arithmetic mean) weight of 7 kilograms and a median weight of 9 kilograms. What is the maximum possible weight, in kilograms, of the lightest box?

   A. 1

   B. 2

   C. 3

   D. 4

   E. 5

   Algebra **Statistics**

   Let x, y, and z be the weights, in kilograms, of the boxes, where $x \leq y \leq z$. Since the average weight of the boxes is 7 kilograms and the median weight of the boxes is 9 kilograms, we have $x + y + z = 3(7) = 21$ and $y = 9$. Hence, $x + 9 + z = 21$, or $z = 12 - x$. Since $z = 12 - x$, $z \geq y$, and $y = 9$, it follows that $12 - x \geq 9$, or $x \leq 3$. Therefore, the maximum possible value of x, which is the maximum possible weight of the lightest box in kilograms, is less than or equal to 3. Since $x = 3$ is a possible value of x, which can be seen by considering $x = 3$, $y = 9$, and $z = 9$, the maximum possible weight of the lightest box is equal to 3 kilograms.

   The correct answer is C.

Stock	Number of Shares
V	68
W	112
X	56
Y	94
Z	45

198. The table shows the number of shares of each of the 5 stocks owned by Mr. Sami. If Mr. Sami were to sell 20 shares of Stock X and buy 24 shares of Stock Y, what would be the increase in the range of the numbers of shares of the 5 stocks owned by Mr. Sami?

A. 4

B. 6

C. 9

D. 15

E. 20

Arithmetic Statistics

The table below shows the numbers of shares of the 5 stocks before and after the change in the numbers of shares of Stocks X and Y.

Stock	Number of Shares (before)	Number of Shares (after)
V	68	68
W	112	112
X	56	36
Y	94	118
Z	45	45

Before the change in the number of shares of Stocks X and Y, the range was 112 − 45 = 67. After the change in the number of shares of Stocks X and Y, the range was 118 − 36 = 82. Therefore, the increase in the range is 82 − 67 = 15.

The correct answer is D.

199. Last year, sales at Company X were 10% greater in February than in January, 15% less in March than in February, 20% greater in April than in March, 10% less in May than in April, and 5% greater in June than in May. In which month were sales closest to the sales in January?

A. February

B. March

C. April

D. May

E. June

Arithmetic **Percents; Estimation**

Let N be the sales at Company X in January. The table gives numerical expressions for sales in each of the months from January through June. For example, since sales in March were 15% less than sales in February, sales in March can be obtained by multiplying sales in February by 0.85.

Month	Sales
January	N
February	$(1.1)N$
March	$(0.85)(1.1)N$
April	$(1.2)(0.85)(1.1)N$
May	$(0.9)(1.2)(0.85)(1.1)N$
June	$(1.05)(0.9)(1.2)(0.85)(1.1)N$

The month after January with sales closest to sales in January will be the month after January having sales closest to N. To this end, the following two results will be helpful.

a. $(0.85)(1.1) = 0.935$

b. $(0.85)(1.2) = 1.02$

From (a) it follows that sales in January were closer to sales in March than to sales in February, since 1 is closer to 0.935 than to 1.1. Hence, the month with sales closest to January is NOT February. From (b) it follows that sales in January were closer to sales in February than to sales in April, since $(1.2)(0.85)(1.1) = (1.02)(1.1) > 1.1$. Hence, the month with sales closest to January is NOT April. For May, because $(0.9)(1.1) = (1 - 0.1)(1 + 0.1) = 1^2 - (0.1)^2 = 0.99$, it follows from (b) that $(0.9)(1.2)(0.85)(1.1) = (0.99)(1.02) = (0.99)(1 + 0.02)$, which is 0.99 plus a little less than 0.02 (0.99×0.02 is 99% of 0.02).

Thus, sales in January were closer to sales in May (between N and $1.02N$) than to sales in March (sales were $0.935N$). Hence, the month with sales closest to January is NOT March. Finally, because sales in June were greater than sales in May, and both were greater than N, sales in January were closer to sales in May than to sales in June. Hence, the month with sales closest to January is NOT June. Therefore, the month with sales closest to January is May.

Alternatively, multiply the indicated products and determine which is closest to N.

Month	Sales
January	N
February	$1.1N$
March	$0.935N$
April	$1.122N$
May	$1.0098N$
June	$1.06029N$

The correct answer is D.

200. If s and t are integers greater than 1 and each is a factor of the integer n, which of the following must be a factor of n^{st}?

I. s^t

II. $(st)^2$

III. $s+t$

A. None

B. I only

C. II only

D. III only

E. I and II

Algebra **Factoring; Simplifying Algebraic Expressions**

Since each of s and t is a factor of n, it follows that there are positive integers a and b such that $n = as$ and $n = bt$.

I. MUST BE A FACTOR: nst divided by st is an integer, since $\frac{n^{st}}{s^t} = \frac{(as)^{st}}{s^t} = \frac{a^{st} s^{st}}{s^t} = a^{st} s^{st-t} = a^{st} s^{t(s-1)}$ and each of a^{st} and $st(s-1)$ is an integer. Note that $st(s-1)$ is an integer because s is greater than 1, and hence the exponent $t(s-1)$ is not negative.

II. MUST BE A FACTOR: Since $s \geq 2$ and $t \geq 2$, we have $st \geq 4$, or $st - 4 \geq 0$. Hence, n^{st-4} is an integer because the exponent $st - 4$ is not negative. Therefore, because $n^{st} = n^4 n^{st-4}$, it follows that n^4 is a factor of n^{st}. Also, because $n^4 = (n)(n)(n)(n) = (as)(as)(bt)(bt) = (ab)^2 (st)^2$, it follows that $(st)^2$ is a factor of n^4. Finally, because $(st)^2$ is a factor of n^4 and n^4 is a factor of n^{st}, it follows that $(st)^2$ is a factor of n^{st}.

III. CAN FAIL TO BE A FACTOR. Let $s = 2$, $t = 3$, and $n = 6$. Then s and t are integers greater than 1 and each is a factor of n. However, $s + t = 5$ is not a factor of $n^{st} = 6^6$.

The correct answer is E.

201. How many 4-digit positive integers are there in which all 4 digits are even?

A. 625

B. 600

C. 500

D. 400

E. 256

Arithmetic **Elementary Combinatorics**

There are 4 possibilities for the left-most digit (2, 4, 6, and 8) and 5 possibilities for each of the 3 remaining digits (0, 2, 4, 6, and 8). Therefore, the number of 4-digit positive integers in which all 4 digits are even is 4 × 5 × 5 × 5 = 500.

The correct answer is C.

202. If $0 < r < 1 < s < 2$, which of the following must be less than 1?

I. $\frac{r}{s}$

II. rs

III. $s - r$

A. I only

B. II only

C. III only

D. I and II

E. I and III

Arithmetic **Inequalities**

I. MUST BE LESS THAN 1: We have $0 < r < s$ from the given information. Dividing throughout by the positive number s gives $0 < \frac{r}{s} < 1$.

II. CAN FAIL TO BE LESS THAN 1: If $r = 0.9$ and $s = 1.9$, then $0 < r < 1 < s < 2$ and $rs = 1.71$ is not less than 1.

III. CAN FAIL TO BE LESS THAN 1: If $r = 0.1$ and $s = 1.9$, then $0 < r < 1 < s < 2$ and $s - r = 1.8$ is not less than 1.

The correct answer is A.

203. Last month, 15 homes were sold in Town X. The average (arithmetic mean) sale price of the homes was \$150,000 and the median sale price was \$130,000. Which of the following statements must be true?

I. At least one of the homes was sold for more than \$165,000.

II. At least one of the homes was sold for more than \$130,000 and less than \$150,000.

III. At least one of the homes was sold for less than \$130,000.

A. I only

B. II only

C. III only

D. I and II

E. I and III

Arithmetic **Statistics**

I. MUST BE TRUE: Since the average sale price of the 15 homes was \$150,000, the total of the 15 sale prices was 15(\$150,000) = \$2,250,000. Also, since the median sale price of the 15 homes was \$130,000, 7 of the homes each had a sale price less than or equal to \$130,000, 7 of the homes each had a sale price greater than or equal to \$130,000, and the remaining home had a sale price equal to \$130,000.

$$\underbrace{\leq \$130{,}000 \text{ each}}_{7 \text{ homes}} \; \underbrace{= \$130{,}000}_{1 \text{ home}} \; \underbrace{\geq \$130{,}000 \text{ each}}_{7 \text{ homes}}$$

If none of the homes was sold for more than \$165,000, then the total of the 15 sale prices would be less than or equal to 7(\$130,000) + \$130,000 + 7(\$165,000) = \$2,195,000, which is less than \$2,250,000, contradicting what we know is the total of the 15 sale prices. Therefore, it is not possible for none of the homes to be sold for more than \$165,000, and thus it must be true that at least one of the homes was sold for more than \$165,000.

II. CAN BE FALSE: If 13 of the sale prices were each \$130,000 and 2 of the sale prices were each \$280,000, then the median of the sale prices would be \$130,000 and the average of the sale prices would be $\frac{13(\$130{,}000) + 2(\$280{,}000)}{15} = \$150{,}000.$ However, in this example none of the homes was sold for more than \$130,000 and less than \$150,000.

III. CAN BE FALSE: If 13 of the sale prices were each \$130,000 and the 2 remaining sale prices were \$140,000 and \$420,000, then the median of the sale prices would be \$130,000 and the average of the sale prices would be

$$\frac{13(\$130{,}000) + \$140{,}000 + \$420{,}000}{15} = \$150{,}000\cdot$$

However, in this example none of the homes was sold for less than \$130,000.

The correct answer is A.

204. Pumps A, B, and C operate at their respective constant rates. Pumps A and B, operating simultaneously, can fill a certain tank in $\frac{6}{5}$ hours; Pumps A and C, operating simultaneously, can fill the tank in $\frac{3}{2}$ hours; and Pumps B and C, operating simultaneously, can fill the tank in 2 hours. How many hours does it take Pumps A, B, and C, operating simultaneously, to fill the tank?

A. $\frac{1}{3}$

B. $\frac{1}{2}$

C. $\frac{2}{3}$

D. $\frac{5}{6}$

E. 1

Algebra **Work Problem**

Let t_A, t_B, and t_C be the number of hours, respectively, for Pumps A, B, and C operating alone to fill the tank, and let t be the number of hours for Pumps A, B, and C operating simultaneously to fill the tank.

Since Pumps A and B, operating simultaneously, can fill the tank in $\frac{6}{5}$ hours, it follows that $\frac{1}{\mathrm{t_A}} + \frac{1}{\mathrm{t_B}} = \frac{1}{\frac{6}{5}} = \frac{5}{6}$, and similarly for the other two pairs of pumps we are given information about.

$$\frac{1}{t_A} + \frac{1}{t_B} = \frac{5}{6}$$
$$\frac{1}{t_A} + \frac{1}{t_C} = \frac{2}{3}$$
$$\frac{1}{t_B} + \frac{1}{t_C} = \frac{1}{2}$$

Adding these 3 equations gives $2\left(\frac{1}{t_A} + \frac{1}{t_B} + \frac{1}{t_C}\right) = \frac{5}{6} + \frac{2}{3} + \frac{1}{2} = 2$, or $\frac{1}{t_A} + \frac{1}{t_B} + \frac{1}{t_C} = 1$. Since $\frac{1}{t_A} + \frac{1}{t_B} + \frac{1}{t_C} = \frac{1}{t}$, it follows that $\frac{1}{t} = 1$, or $t = 1$.

The correct answer is E.

205. If n is a positive integer, then $(-2^n)^{-2} + (2^{-n})^2 =$

A. 0

B. 2^{-2n}

C. 2^{2n}

D. 2^{-2n+1}

E. 2^{2n+1}

Algebra **Simplifying Algebraic Expressions**

$(-2^n)^{-2} + (2^{-n})^2$	given
$= \dfrac{1}{(-2^n)^2} + \left(\dfrac{1}{2^n}\right)^2$	use $a^{-b} = \dfrac{1}{a^b}$
$= \dfrac{1}{(-1)^2(2^n)^2} + \left(\dfrac{1}{2^n}\right)^2$	$-2^n = (-1)(2^n)$ and use $(ab)^c = a^c b^c$
$= \dfrac{1}{(2^n)^2} + \dfrac{1}{(2^n)^2}$	$(-1)^2 = 1$ and use $\left(\dfrac{a}{b}\right)^c = \dfrac{a^c}{b^c}$
$= \dfrac{2}{(2^n)^2}$	adding fractions with a common denominator
$= \dfrac{2}{2^{2n}}$	use $(a^b)^c = a^{bc}$
$= 2^{1-2n}$	2 = 2^1 and use $\dfrac{a^b}{a^c} = a^{b-c}$
$= 2^{-2n+1}$	$1 - 2n = -2n + 1$

The correct answer is D.

206. Which of the following is equal to $\left(\frac{2\sqrt[3]{3}}{\sqrt{2}}\right)^3$?

A. $\frac{3\sqrt{2}}{2}$

B. $3\sqrt{2}$

C. $6\sqrt{2}$

D. 12

E. $12\sqrt{2}$

Arithmetic **Operations on Radical Expressions**

$$\left(\frac{2\sqrt[3]{3}}{\sqrt{2}}\right)^3 = \frac{\left(2\sqrt[3]{3}\right)^3}{\left(\sqrt{2}\right)^3} \qquad \text{use } \left(\frac{a}{b}\right)^c = \frac{a^c}{b^c}$$

$$= \frac{(2)^3\left(\sqrt[3]{3}\right)^3}{\left(\sqrt{2}\right)\left(\sqrt{2}\right)\left(\sqrt{2}\right)} \qquad \text{use } (ab)^c = a^c b^c$$

$$= \frac{(8)(3)}{2\sqrt{2}} \qquad \left(\sqrt{2}\right)\left(\sqrt{2}\right) = 2 \text{ and } \left(\sqrt[3]{3}\right)^3 = 3$$

$$= \frac{12}{\sqrt{2}} \qquad \text{reduce}$$

$$= \frac{12}{\sqrt{2}} \times \frac{\sqrt{2}}{\sqrt{2}} \qquad \text{multiply by 1}$$

$$= \frac{12\sqrt{2}}{2} \qquad \left(\sqrt{2}\right)\left(\sqrt{2}\right) = 2$$

$$= 6\sqrt{2} \qquad \text{reduce}$$

The correct answer is C.

207. Dara ran on a treadmill that had a readout indicating the time remaining in her exercise session. When the readout indicated 24 min 18 sec, she had completed 10% of her exercise session. The readout indicated which of the following when she had completed 40% of her exercise session?

A. 10 min 48 sec

B. 14 min 52 sec

C. 14 min 58 sec

D. 16 min 6 sec

E. 16 min 12 sec

Arithmetic Rate Problem

Since 24 min 18 sec (or 1,458 sec) represents 90% of the total time for the exercise session, it follows that the total time for the exercise session was $\frac{1,458}{0.9}$ sec = 1,620 sec.

Thus, when she had completed 40% of the exercise session, the readout would have indicated 60% of the total time, which is 0.6(1,620 sec) = 972 sec = 16 min 12 sec.

The correct answer is E.

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5.0 GMAT™ Official Guide Quantitative Review Question Index

5.0 GMAT™ Official Guide Quantitative Review Question Index

The Quantitative Review Question Index is organized by the section, difficulty level, and then by mathematical concept. The question number, page number, and answer explanation page number are listed so that questions within the book can be quickly located.

Quantitative Reasoning — Chapter 4 – Page 74

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Absolute Value	32	300212	82	120
Easy	Applied Problems	11	300199	79	113
Easy	Applied Problems	12	300189	79	113
Easy	Applied Problems	13	300332	79	113
Easy	Applied Problems	14	300333	80	114
Easy	Applied Problems	18	300188	80	115
Easy	Applied Problems	19	300334	80	115
Easy	Applied Problems	47	300193	84	123
Easy	Applied Problems	48	300216	84	124
Easy	Applied Problems	61	300100	86	127
Easy	Applied Problems	71	300207	88	131
Easy	Applied Problems; First-Degree Equations	77	300291	89	133
Easy	Applied Problems; Operations with Fractions	46	300147	84	123

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Applied Problems; Percents	56	300191	85	126
Easy	Applied Problems; Proportions	9	300173	79	112
Easy	Applied Problems; Substitution	16	300104	80	114
Easy	Average	78	300292	89	133
Easy	Equations	43	300342	84	122
Easy	Estimation	54	300105	85	125
Easy	Estimation	79	300293	89	133
Easy	Exponents	41	300341	83	122
Easy	Exponents	44	300343	84	122
Easy	Factoring	3	300198	78	110
Easy	Factors, Multiples, and Divisibility	4	300208	78	111
Easy	First-Degree Equations	27	300250	81	118
Easy	First-Degree Equations	39	300247	83	121

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
easy	First-Degree Equations	40	300340	83	122
Easy	First-Degree Equations	49	300213	85	124
Easy	First-Degree Equations	80	300294	89	134
Easy	First-Degree Equations; Substitution	36	300134	83	120
Easy	Formulas	53	300120	85	125
Easy	Formulas	66	300260	87	129
Easy	Fractions	23	300146	81	117
Easy	Inequalities	7	300242	79	111
Easy	Inequalities	34	300145	82	120
Easy	Interpretation of Graphs	63	300129	87	128
Easy	Interpretation of Tables	37	300338	83	121
Easy	Operations on Integers	5	300220	78	111
Easy	Operations on Integers	65	300228	87	129
Easy	Operations on Integers	68	300214	87	130

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Operations on Rational Numbers	21	300218	81	116
Easy	Operations with Fractions	59	300112	86	127
Easy	Operations with Fractions	60	300136	86	127
Easy	Operations with Fractions	64	300226	87	128
Easy	Operations with Integers	33	300337	82	120
Easy	Operations with Integers	58	300102	86	126
Easy	Operations with Integers	62	300168	86	128
Easy	Operations with Integers; Measurement Conversion	73	300155	88	132
Easy	Order	8	300331	79	112
Easy	Order	15	300209	80	114
Easy	Percents	10	300258	79	112

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Percents	20	300335	80	115
Easy	Percents	24	300336	81	117
Easy	Percents	29	300170	82	119
Easy	Percents	35	300285	82	120
Easy	Percents	50	300116	85	124
Easy	Percents	57	300101	86	126
Easy	Percents	70	300232	88	131
Easy	Place Value	30	300175	82	119
Easy	Place Value	74	300117	88	132
Easy	Properties of Integers	25	300182	81	117
Easy	Properties of Integers	82	300296	89	134
Easy	Properties of Numbers	6	300330	78	111
Easy	Properties of Numbers	28	300178	81	118
Easy	Properties of Numbers	55	300174	85	126
Easy	Properties of Numbers	72	300119	88	131

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Rate	1	300203	78	110
Easy	Rate Problem	17	300239	80	115
Easy	Rate Problem	42	300181	84	122
Easy	Ratio and Proportion	51	300252	85	125
Easy	Second-Degree Equations	2	300107	78	110
Easy	Second-Degree Equations	22	300131	81	116
Easy	Second-Degree Equations	52	300130	85	125
Easy	Sequences	67	300172	87	130
Easy	Series and Sequences	26	300186	81	118
Easy	Series and Sequences	45	300185	84	123
Easy	Series and Sequences	75	300289	89	132
Easy	Series and Sequences; Simplifying Algebraic Expressions	76	300290	89	133

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Easy	Simplifying Algebraic Expressions	81	300295	89	134
Easy	Simultaneous Equations	31	300202	82	119
Easy	Simultaneous Equations	38	300339	83	121
Easy	Statistics	69	300103	88	131
Medium	Absolute Value	113	300118	94	144
Medium	Applied Problems	84	300160	90	135
Medium	Applied Problems	87	300161	90	136
Medium	Applied Problems	102	300283	93	141
Medium	Applied Problems	109	300187	94	143
Medium	Applied Problems	112	300204	94	144
Medium	Applied Problems	122	300284	95	148
Medium	Applied Problems	124	300221	96	148
Medium	Applied Problems	145	300306	98	155

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Applied Problems; Formulas	95	300272	92	139
Medium	Applied Problems; Percents	130	300245	96	150
Medium	Arithmetic Operations	141	300302	98	154
Medium	Decimals	110	300171	94	143
Medium	Equations; Inequalities	93	300269	91	138
Medium	Equations	149	300345	99	156
Medium	Estimation	131	300286	96	151
Medium	Exponents; Simplifying Algebraic Expressions	139	300300	97	153
Medium	Factoring	99	300277	92	140
Medium	First-Degree Equations	83	300159	90	134
Medium	First-Degree Equations	94	300271	92	139
Medium	First-Degree Equations	96	300273	92	139
Medium	First-Degree Equations	151	300347	99	157

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Formulas	98	300276	92	140
Medium	Formulas; Percents	105	300140	93	142
Medium	Functions	142	300303	98	154
Medium	Functions; Absolute Value	92	300267	91	137
Medium	Inequalities	106	300176	93	142
Medium	Inequalities	108	300244	93	143
Medium	Inequalities	117	300241	95	146
Medium	Inequalities	126	300114	96	149
Medium	Inequalities; Absolute Value	120	300268	95	147
Medium	Measurement Conversion	101	300282	93	141
Medium	Measurement Conversion	138	300299	97	153
Medium	Measurement Conversion; Rate Problem	136	300297	97	152

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Operations with Integers	89	300165	91	137
Medium	Operations with Integers	148	300344	99	156
Medium	Order	86	300163	90	135
Medium	Percents	127	300229	96	149
Medium	Percents	132	300154	97	151
Medium	Percents; Estimation	114	300210	94	145
Medium	Percents; Estimation	125	300222	96	149
Medium	Place Value	115	300287	94	145
Medium	Place Value	134	300223	97	152
Medium	Probability	146	300307	98	155
Medium	Probability; Sets (Venn Diagrams)	140	300301	98	153
Medium	Properties of Integers	90	300166	91	137
Medium	Properties of Integers	133	300261	97	151

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Properties of Integers	137	300298	97	152
Medium	Properties of Numbers	107	300133	93	143
Medium	Properties of Numbers	118	300111	95	146
Medium	Properties of Numbers	119	300192	95	147
Medium	Properties of Numbers	128	300179	96	150
Medium	Ratio and Proportion	123	300280	95	148
Medium	Rate Problem; Measurement Conversion	144	300305	98	155
Medium	Rational Numbers	150	300346	99	156
Medium	Ratios	85	300162	90	135
Medium	Second-Degree Equations	100	300279	92	141
Medium	Second-Degree Equations	104	300127	93	142
Medium	Series and Sequences	135	300141	97	152
Medium	Series and Sequences	143	300304	98	154

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Medium	Sets (Venn Diagrams)	88	300164	90	136
Medium	Simplifying Algebraic Expressions	147	300308	99	156
Medium	Simultaneous Equations	97	300275	92	139
Medium	Simultaneous Equations	103	300126	93	142
Medium	Simultaneous Equations	116	300255	94	145
Medium	Simultaneous Equations	129	300196	96	150
Medium	Statistics	91	300266	91	137
Medium	Statistics	111	300262	94	144
Medium	Statistics	121	300115	95	147
Hard	Absolute Value	166	300201	101	161
Hard	Applied Problems	155	300139	100	158
Hard	Applied Problems	177	300135	103	165
Hard	Applied Problems	179	300137	103	166

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Applied Problems	187	300263	104	169
Hard	Applied Problems	193	300157	105	171
Hard	Applied Problems	195	300156	105	172
Hard	Applied Problems; Percents	159	300113	100	159
Hard	Applied Problems; Percents	182	300183	103	167
Hard	Applied Problems; Properties of Integers	189	300143	104	170
Hard	Arithmetic Operations	153	300167	99	157
Hard	Elementary Combinatorics	168	300197	101	162
Hard	Elementary Combinatorics	172	300180	102	164
Hard	Elementary Combinatorics	201	300314	106	175
Hard	Estimation	162	300106	100	160
Hard	Exponents	161	300205	100	160

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Factoring; Simplifying Algebraic Expressions	200	300313	106	175
Hard	First-Degree Equations	152	300158	99	157
Hard	First-Degree Equations; Percents	169	300233	102	163
Hard	First-Degree Equations; Percents	176	300281	102	165
Hard	Inequalities	178	300243	103	166
Hard	Inequalities	202	300315	106	176
Hard	Operations on Radical Expressions	171	300153	102	163
Hard	Operations on Radical Expressions	206	300319	107	177
Hard	Operations with Integers	154	300190	99	158
Hard	Operations with Integers	167	300234	101	162
Hard	Percents	160	300219	100	159
Hard	Percents; Estimation	199	300312	106	174

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Place Value	174	300211	102	164
Hard	Place Value	192	300138	105	171
Hard	Probability	181	300215	103	167
Hard	Probability; Concepts of Sets	173	300206	102	164
Hard	Properties of Integers	156	300264	100	158
Hard	Properties of Integers	163	300274	101	160
Hard	Properties of Integers	190	300148	104	170
Hard	Properties of Integers	191	300144	105	171
Hard	Properties of Numbers	175	300237	102	165
Hard	Properties of Numbers	180	300230	103	166
Hard	Properties of Numbers	183	300348	104	168
Hard	Properties of Numbers	184	300253	104	168
Hard	Properties of Numbers	186	300152	104	168
Hard	Properties of Numbers; Exponents	188	300142	104	169

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Rate Problem	207	300320	107	178
Hard	Sequences and Series	157	300108	100	158
Hard	Sequences and Series	158	300288	100	159
Hard	Sets	170	300195	102	163
Hard	Sets	194	300349	105	172
Hard	Simplifying Algebraic Expressions	205	300318	107	177
Hard	Simplifying Algebraic Expressions; Applied Problems	185	300151	104	168
Hard	Simultaneous Equations; Sets (Venn Diagrams)	196	300309	106	173
Hard	Statistics	164	300257	101	161
Hard	Statistics	165	300217	101	161
Hard	Statistics	197	300310	106	173
Hard	Statistics	198	300311	106	174
Hard	Statistics	203	300316	107	176

Difficulty	Concept	Question #	Question ID #	Page	Answer Explanation Page
Hard	Work Problem	204	300317	107	177

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Appendix A Answer Sheet

Quantitative Reasoning Answer Sheet

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